Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 — UNIT 2 · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

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· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_02 · UNIT 2

1 UE26ME142A · Unit_02: UNIT 2

This unit master organizes source material for Engineering Mechanics 1, Unit_02, officially titled UNIT 2. It includes 15 video transcripts and 18 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26ME142A, Unit_02 (UNIT 2). Bucket folder present: yes. Videos: 15; slide decks: 18.

EV0000001

2 UE26ME142A_Unit 2_Class 14_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 14_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

Graphs and diagrams



Figure G000001. UE26ME142A Unit 2 Class 14 Lecture PPT · source slide page 1.

EV0000259

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Engineering Mechanics

Statics

EIGHTH Edition
SI Version

J. L. Meriam and L. G. Kraige

Unit 2
Chapter 3
Equilibrium

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Figure G000002. UE26ME142A Unit 2 Class 14 Lecture PPT · source slide page 2.

EV0000262

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ENGINEERING MECHANICS STATICS

Equilibrium

When a body is in equilibrium, the resultant of all forces acting on it is zero. Thus, the resultant force R and the resultant couple M are both zero, and we have the equilibrium equations

$R = \Sigma F = 0$

$M = \Sigma M = 0$

Free-body diagram, which is a diagrammatic representation of the isolated system treated as a single body.

Free-body diagram is the most important single step in the solution of problems in mechanics.

Figure G000003. UE26ME142A Unit 2 Class 14 Lecture PPT · source slide page 3.

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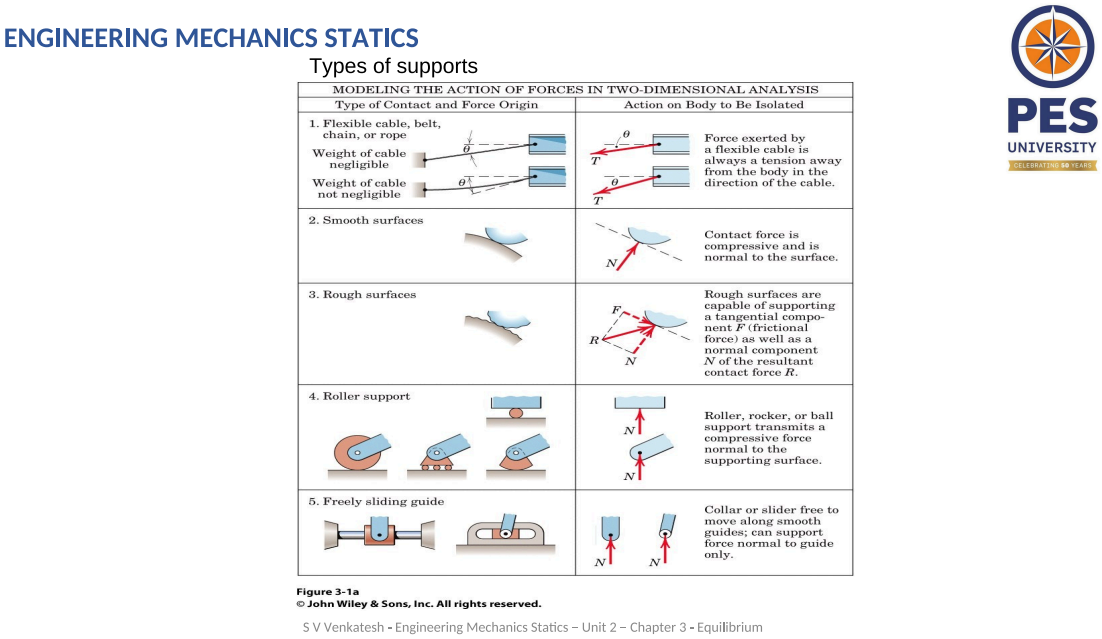


Figure G000004. UE26ME142A Unit 2 Class 14 Lecture PPT · source slide page 4.

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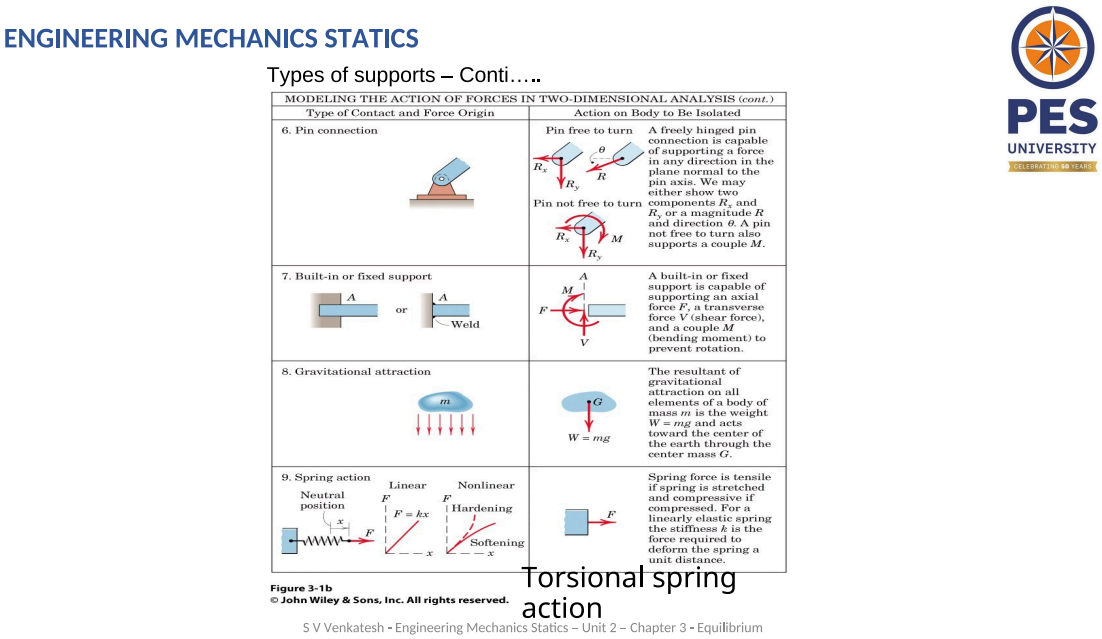


Figure G000005. UE26ME142A Unit 2 Class 14 Lecture PPT · source slide page 5.

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ENGINEERING MECHANICS STATICS

Free body diagram

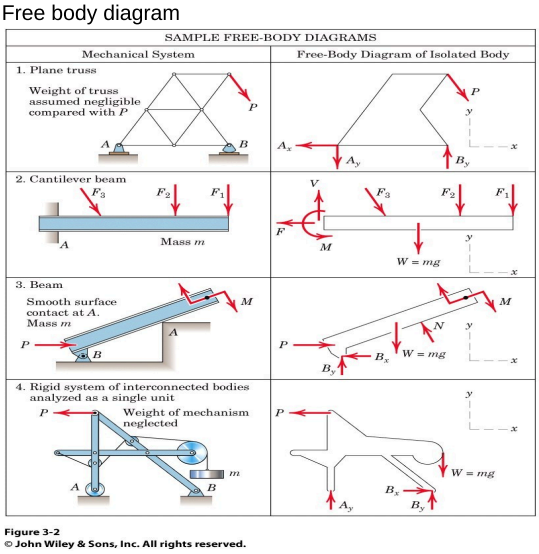


Figure 3-2
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S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

Figure G000006. UE26ME142A Unit 2 Class 14 Lecture PPT · source slide page 6.

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ENGINEERING MECHANICS STATICS

Free body diagram – Conti……..

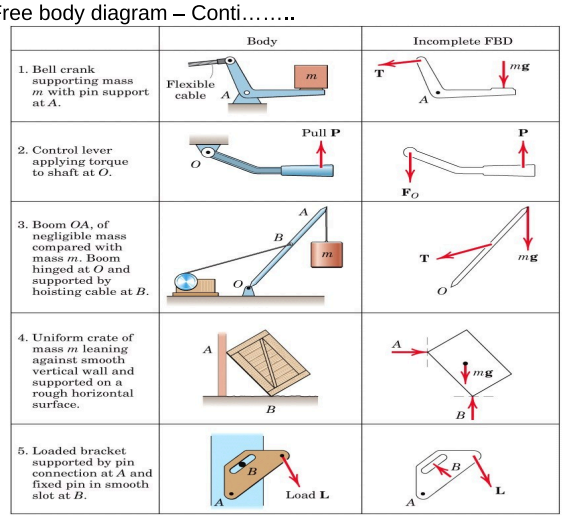


Figure 3-A
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S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

Figure G000007. UE26ME142A Unit 2 Class 14 Lecture PPT · source slide page 7.

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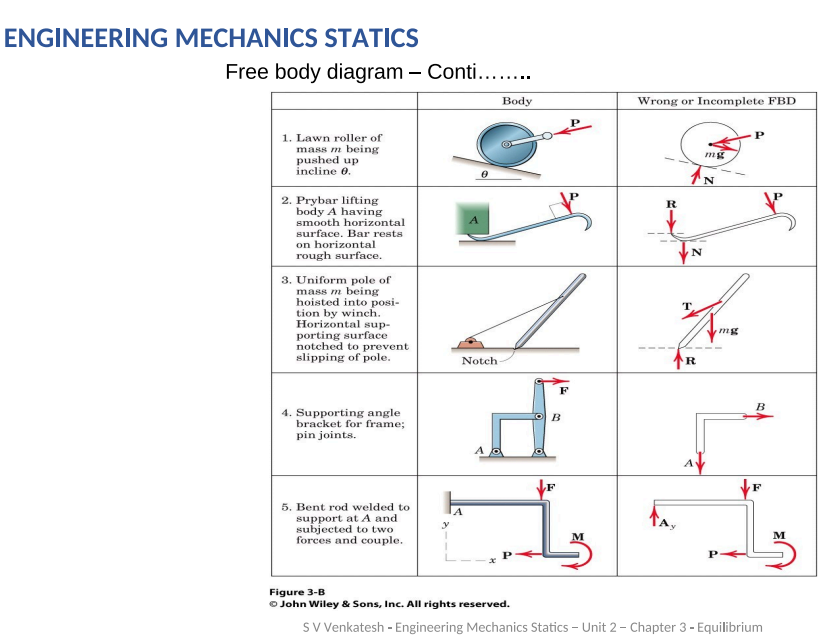


Figure G000008. UE26ME142A Unit 2 Class 14 Lecture PPT · source slide page 8.

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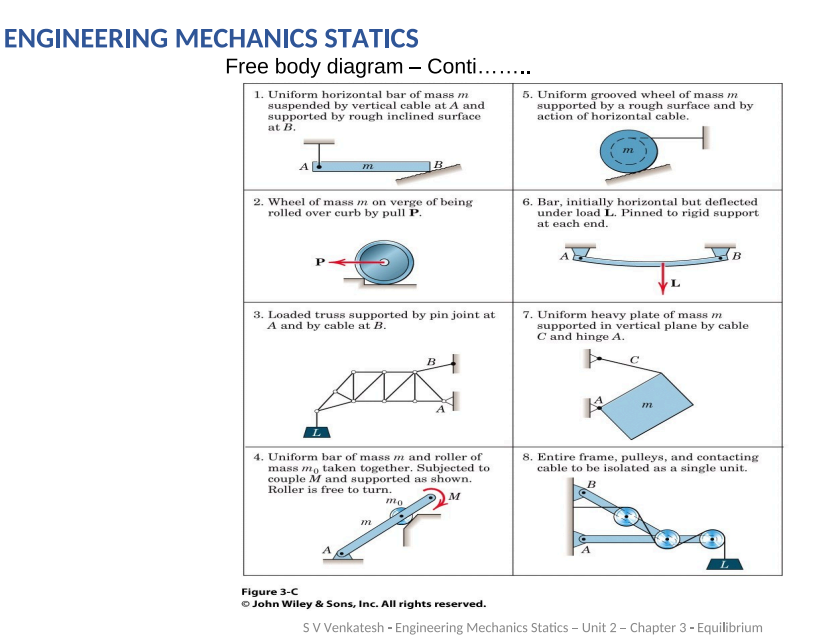


Figure G000009. UE26ME142A Unit 2 Class 14 Lecture PPT · source slide page 9.

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THANK YOU

Dr. S V Venkatesh
Civil Engineering
Department of S & H

Figure G000010. UE26ME142A Unit 2 Class 14 Lecture PPT · source slide page 10.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000258 · EV0000259 · EV0000260
2. Source slide page 2 (extracted text): Engineering Mechanics Statics EIGHTH Edition SI Version J. L. Meriam and L. G. Kraige Unit 2 Chapter 3 Equilibrium Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000261 · EV0000262 · EV0000263
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS Equilibrium When a body is in equilibrium, the resultant of all forces acting on it is zero. Thus, the resultant force R and the resultant couple M are both zero, and we have the equilibrium equations R = ΣF = 0 M = ΣM = 0 Free-body diagram, which is a diagrammatic representation of the isolated system treated as a single body. Free-body diagram is the most important single step in the solution of problems in mechanics. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000264 · EV0000265 · EV0000266
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS Types of supports S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000267 · EV0000268 · EV0000269
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS Types of supports – Conti..... Torsional spring action S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000270 · EV0000271 · EV0000272

6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS Free body diagram S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000273 · EV0000274 · EV0000275
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS Free body diagram – Conti..... S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000276 · EV0000277 · EV0000278
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS Free body diagram – Conti..... S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000279 · EV0000280 · EV0000281
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS STATICS Free body diagram – Conti..... S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 10 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000282
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SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.612] Dear student, welcome to Engineering Mechanics Static classes. My name is Dr. SV Venkatish from Civil Engineering and the Department of Science and Humanities PS University. So far, we had discussed Unit 1. [00:00:18.084] Two chapters which are there in unit one, chapter one and chapter two. In chapter one, we had discussed a few of the definitions and the units of forces. In chapter two, we had four topics. One as resolution of forces. [00:00:37.252] In that you have rectangular components, N and T components, then axes which are not at 90 degrees, then the difference between projection and components. On that. That's topic one under chapter two. The second topic was on movement. [00:00:55.172] We discussed moment, varingance principle, and we have solved problems on moment also. That's second topic. The third topic was on couple. We have seen what is the definition of couple, characteristics of couple, classification of couple, effects of couple, and very importantly, what do you mean by force couple system? [00:01:15.076] And then we have solved problems on that. The last topic in unit one, chapter two, is resulted. [00:01:23.716] We have discussed what is resulted, how do we determine the characteristics of resultant, four characteristics that is magnitude, direction, sense, and point of application. To determine the point of application, we have used principle of moments, okay, and then we have solved problems on that. [00:01:43.204] So that ends unit 1 and unit 1 is required for us to understand unit 2 as well as unit 4. Okay. To certain extent unit 3 also. Okay. So with the assuming that [00:02:03.716] You have the background of unit one. We will now get into unit two. And the prescribed textbook that we are using is engineering mechanics statics. Very importantly, it is the eighth edition, which is in SI version. [00:02:20.836] And the author is Miriam Craig and adopted by Bolton. Okay, from this textbook, we will we have Unit 2 and Chapter 3. There's one more in this other than equilibrium, we also have trusses. We will discuss that later when we take up that topic. [00:02:40.772] So first we will discuss about chapter three, which is equilibrium. So let's now define what is equilibrium. What is equilibrium? If the resultant force and the resultant movement acting on a body on which there are several [00:03:00.228] Forces and moments acting, that is a system of forces are acting. If the

resultant force and moment both of them are zero, then we say that the body is in equilibrium. So by saying that, let us look at what is the equations of equilibrium. [00:03:19.076] So equation of equilibrium is the resultant force, that is the algebraic sum of all the forces acting on the body should be equal to zero. So with this, what is that we are ensuring is that there is no translatory motion of the body. [00:03:38.564] So when we take a body and when we apply a force onto this, as I have explained in unit one, the effect of the force can be to make this body move from one place to another place, that is translatory motion. The second [00:03:56.580] Application of the force would be a rotatory motion which will make this body to rotate. It won't move. But you can also have a combination of transmitter motion as well as rotatory motion. So if I take or if this equation is satisfied. [00:04:14.436] It ensures that there is no transmitter emotion. Why I'm saying this? Because as you could see, the net force because of a couple is zero. But what is the effect of a couple? It makes the body to rotate. For equilibrium. [00:04:30.212] It should not get displaced, it should not rotate also. So this one will ensure that there is no displacement. [00:04:37.988] But if we take this, m is equal to 0, or algebraic sum of all the moments of all the forces at any point, at any point should be equal to 0. This will ensure that there is no rotatory motion. [00:04:55.524] Okay, so this is what is the definition of equilibrium and the equations of equilibrium that is to be satisfied for us to call a body is in equilibrium. Okay. Now what are the other things that we need to understand to

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:14.404] Ensure or to show that the body is in equilibrium or to determine the forces which makes the body to be in equilibrium. Let us look at that. What is that? The first thing is a free body diagram. We have already discussed about free body diagram. [00:05:31.524] But since it comes exclusively in this chapter, I am again arbitrating or bringing the concept of a free body diagram. As the name suggests, free body diagram is a diagram in which we will have a diagram. [00:05:49.508] Such that there is no body in that. The diagram will have the forces, the characteristics of forces which are acting on the body, and we were going to remove that body because we are not interested in the body. [00:06:07.716] What we are interested is what are the forces that is acting on the body, what is the effect of that. That is what is called as a free body diagram, as I suggest, as the name itself suggests, we are making the diagram free of the body. So we are removing the body and having the diagram and looking at the diagram. [00:06:27.684] Okay, with the forces, their magnitude, their position, their direction, their sense, and the point of application. Position means the point of application. So free body diagram is the most important single step. [00:06:45.828] In the solution of problems in mechanics, if you know how to draw the free body diagram, fifty percent of the problem is solved when it comes to equilibrium chapter. [00:06:59.748] But that free body diagram should be clear and sufficient for you to determine the unknowns. Okay. Now the other thing that we should know is what are the different types of supports. [00:07:16.388] A few of these supports we have already discussed in our unit one. The first type of support is a flexible cable or belt or chain or rope. Now we have seen what is this flexible belt or cable or chain or slow or rope. [00:07:35.140] It is a body which can support tensile forces. When I say tensile forces, if you have a rope and if you try to pull it, okay, once it is tight, afterwards it will not move. [00:07:51.076] Okay, two equal and opposite forces, external forces acting on a body would constitute a tensile force. For this type of support where you have a flexible cable or a bolt, or sorry, belt, or a chain, or a rope. [00:08:09.700] It can resist tensile forces, and that is what is indicated here. So you have a cable or

a belt or a belt or rope or chain which is attached to this one, the other end is attached here. [00:08:25.220] So this, if I try to pull this it will not move because of this oh. [00:08:30.052] And there's a tension here, the internal forces will be acting towards each other and that is what is indicated here. If I cut it here and look at this, I will get the force to be this. Similarly here, if I cut it, I will get the force to be acting here. There will be only one force with no line of action. [00:08:46.820] If you are looking at what is the force acting on the in the row, then it is internal. If you are looking at what is the force causing the tension is the external force. The second one is a smooth surface. A smooth surface is one in which you will have one reaction with known line of action. [00:09:05.572] Which will always act perpendicular to the surface. This is the surface [00:09:11.492] And this is the reaction that we get and this angle is always ninety degrees and the sense is going to be opposite to the body weight that is acting. The body weight will be acting here, so you will have the normal reaction acting here. [00:09:27.716] There won't be any friction when it comes to smooth surface. Or the frictional force that develops because of the smooth surface would be quite less when compared to the force that is acting. So we neglect that force. So it is only one force that is acting. [00:09:46.340] The next category is a rough surface. You have already studied about this. If the roughness of the body which are in contact with each other is quite large. [00:09:58.628] Then it will prevent the body from moving tangential to the surface, and that we call it as frictional force. In addition to this frictional force, we will also have a normal reaction. So normal reaction will always act perpendicular to the surface, the frictional force will act along the surface or tangential.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:18.884] to the normal. We can represent these two by one resultant force, but the direction of this resultant force depends upon the magnitude of F1N. Okay. [00:10:33.316] The next category of the type of support is a roller support. Now look at how the rollers are represented here. Four different ways of representing the rollers. Now in this case they have shown all the rollers to be on the horizontal line. [00:10:49.956] You can also have a roller vertical or inclined also. Okay? Now for roller always there will be one reaction. [00:10:59.428] with known line of action, which will always be acting perpendicular to the line of rollers. In this case the line of roller is horizontal in all the four cases. So reaction will be perpendicular that is vertical and upwards also. [00:11:15.748] Okay, you can have this roller opposite. Okay, mirror image of this, then you will have the reaction acting downwards. Okay? You can also have vertical if the rollers are to the left, then you will have the support reaction acting to the right. If you have the rollers [00:11:31.492] To the right, then you will have the reaction to the left. Okay. The next one is a freely sliding guide. In this case, you will have one reaction only. [00:11:45.956] In this case, you can see that the sliding guide can move horizontally, it cannot move vertically. So you will have one reaction. The reaction can be upwards or downwards. But [00:11:59.172] Along the horizontal, it can move only this much. Similarly, it can move only this much on the right side and left side. [00:12:06.468] Okay. After which, if it moves till here, then it cannot move. Then you will have a reaction in the horizontal direction also. That is why it is called as freely sliding train. Only till that position. [00:12:20.196] The horizontal support will not be there. After that, you will have the horizontal support. Okay. But however, here what is indicated for this position is there will be only one normal reaction, normal reacting vertically. Okay. Continuing the different types of supports, the next one comes. [00:12:37.892] Pin connection, look at how the pin connection is indicated. There is a clear difference between a pin connection and a

roller connection. In case of rollers, you will have roller here and then you will have a the surface. Whereas pin connection, directly the triangle is resting on the surface. [00:12:56.996] So in case of pin connection, it is also called as a hinge support. The best example that I can give you is the door. The door frame or the door itself is fixed to the frame of the door. Okay, by what do you mean by hinges? [00:13:14.404] Hinges are the best example for pin connection. In case of hinges, you will have two reaction. No, you cannot move the door upwards or downwards. Okay, you cannot move the door inwards. [00:13:28.548] Toward the frame, or you cannot pull it out of the frame, but you can rotate the door. That is permitted in case of the door. That is what the hint support does. It will provide you two reactions. It will permit you to rotate. Okay. [00:13:48.676] In the plane opposite to this two reaction that is coming. If this is X and Y, it will rotate along the Z axis. Okay? No, you have a door with a stopper. Doe stopper, we are we know the door stopper. [00:14:06.020] Now when we apply this doorstopper, what will happen? Because the doorstopper, it will prevent the door from closing. Okay? Then we will have the moment also. That is why you can see here. Pin not free to turn. [00:14:22.212] Where you will have restriction for rotation. Okay. The next one is a fixed or a built-end support. Now you can see how the representation for fixed support are. Okay these are the representation for fixed support. This is welded connection, this is fixed. [00:14:40.228] So in this case you will have all the three reactions, one vertical, one horizontal under momentum. The sense of this, even here the sense of this reactions can be upwards or downwards, right side or left side, clockwise or anti-clockwise based on the force that is acting on the body. [00:14:59.300] Then we have also looked at gravitational force earlier. Always the gravitational force will be acting downwards. Okay, and it will be equal to mass times the acceleration due to gravity, and always we take acceleration due to gravity as 9.81 itself.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:17.892] Okay, you cannot round it off to ten because if you round it off to ten then you will not get the correct answer what is mentioned in the textbook. So we should take it as nine point eight one itself. Okay. Next other one important type of support is a spring connection. [00:15:35.748] The spring, in case of spring connection, the spring will allow it to displace or it will allow it to elongate or compress, but with certain force. So you will have the stiffness of the spring. [00:15:55.684] Which defines what should be the force that you need to apply for unit elongation or compression. Okay. That is what is called as stiffness K. So if you want to find out the force for any. [00:16:13.284] uh elongation or compression that I have to give, then that we can find it as F is equal to K times the elongation or compression. Okay. [00:16:23.524] So, this are the different types of supports. Along with this, we also have one more, what is called as torsional spring action, where instead of a linear spring, this is a linear spring where you can elongate or compress. In case of torsional spring, [00:16:41.572] You can you have to apply a movement so that it rotates. So their stiffness will be equal to rotation or the moment required for unit rotation. That is m by theta. Stiffness. K [00:16:59.268] Yes, KS or KT torsional spring is equal to what is the moment required for an unit displacement. Okay, these are the different types of support that we have. Now let's look at the free body diagram. So there are a few examples that I have taken here where we have a body on which the force is acting. [00:17:19.716] Now this is the free body diagram. You can see here the body as such is removed, only the outer edges are intact. Then the supports replaced by their respective reactions. Here it is an inch support, so you have two reactions, here it is a roller support, so you have one reaction. [00:17:38.724] Similarly, for the other three also they have given you you can go

through this and try to understand. If you don't understand, you can always come and talk to me later. Now here in this table, they have given you an incomplete free body diagram. [00:17:54.308] First one now here you can see there is a force acting here. This is a cable, so there will be one reaction and they have indicated that that tension. Here they have kept a mass here. So because of this mass there will be a force which is m times g. Here you have a [00:18:11.172] Hinge support which is not indicated here. So you will have two reactions, one vertical and one horizontal. Similarly, there are few more four examples that is given, which is the free word diagram is incomplete. You have to identify what is incomplete here. [00:18:26.468] And according to that you should complete the free body diagram. I hope you can do that. Next, this table gives you a free body diagram, which is wrong or incomplete. You can see here it is wrong or incomplete. I'll explain the first one. [00:18:45.188] You have a roller inclined on an inclined surface, or kept on an inclined surface, there's a force of P that is applied. So they are given a wrong free body diagram here. What is wrong? The force P is acting here, that is exactly right, P is correct. [00:19:03.972] If this is a smooth surface, so you have one reaction normal to the surface, that is also correct. But you can see here they have given Mg. Mg is the force because of the sulfate of the roller. So always the sulphate is going to be acting vertically down. [00:19:21.764] That is because of gravitation. So here they have indicated it in some angle, that is wrong. It should act vertically down. Okay. So you can try for the rest four, and if there are any difficulties, you can come back to me. Okay. Right. Now here there are six different diagrams that is given. [00:19:42.404] You have to draw the free body diagram for all this, sorry, eight diagrams, all this eight diagrams. I will explain the first one. Now in this what you have, you have a member here, then you have a support here, and if you read the diagram, it says uniform horizontal bar of mosum.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:01.892] Suspended by vertical cable at A and supported by rough inclined surface at B. So this is a rough inclined surface. This is a cable. So if I want to draw the free wide ion, I will represent this mass by a single line. [00:20:16.932] Then this I will remove it and replace it by a force acting upwards because it's tension, the internal forces will be acting upwards. Whereas here there will be a normal reaction acting upwards, then there will be a frictional force acting along this surface and it is upwards. [00:20:34.020] Okay. So the rest of the things I request you to kindly go through it and draw yourself. If you have any doubts you can meet. [00:20:42.276] We will stop it here. We'll continue in the next class. Thank you very much. My name is Dr. Sv Vength from Civil Engineering under Department of Science and Humanities, PS University. Thank you. Let's meet in the next class. This is class number eighteen. Next we will have class number nineteen. Thank you.

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Full transcript: 3158 normalized words, 76 timestamped segments, 1259.947 seconds duration. Low-confidence segments flagged: 0.

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3 UE26ME142A_Unit 2_Class 15_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 15_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000022 · EV0000023

Graphs and diagrams



Figure G000011. UE26ME142A Unit 2 Class 15 Lecture PPT · source slide page 1.

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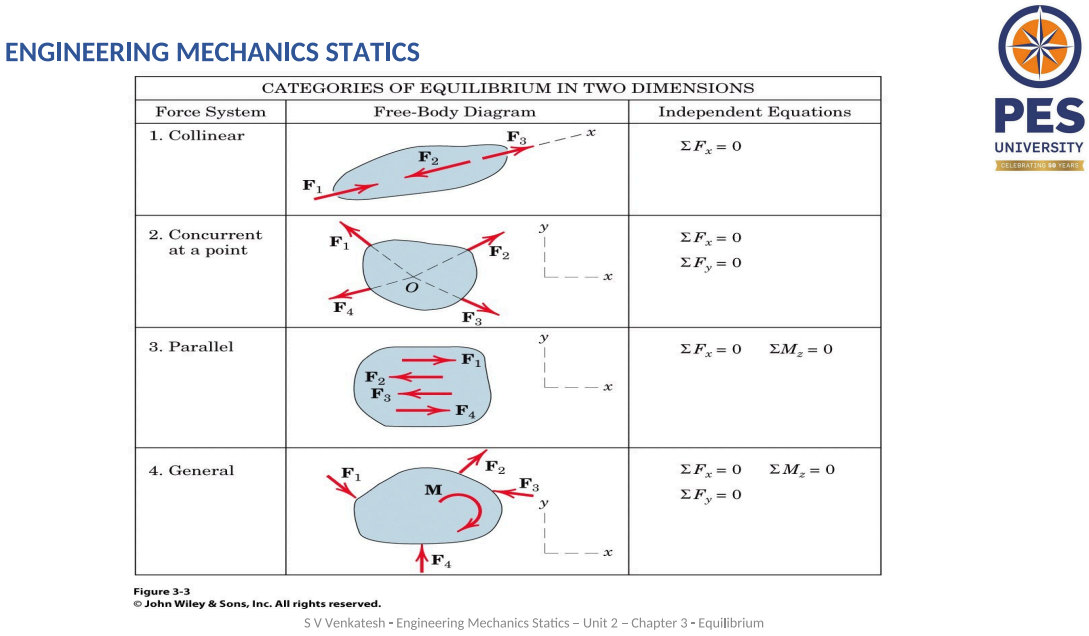


Figure G000012. UE26ME142A Unit 2 Class 15 Lecture PPT · source slide page 2.

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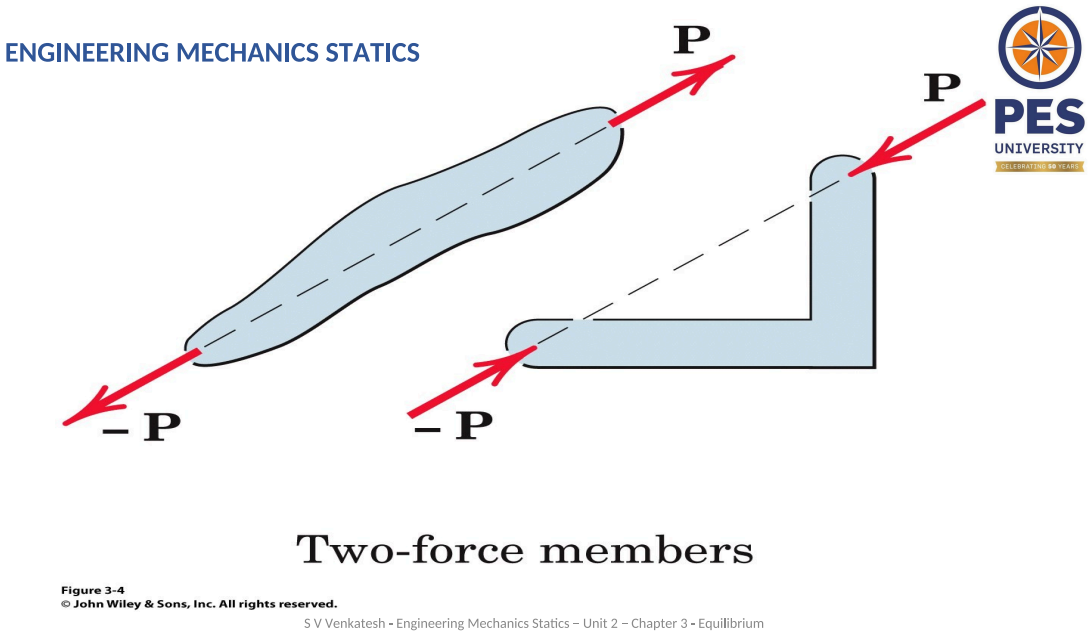


Figure G000013. UE26ME142A Unit 2 Class 15 Lecture PPT · source slide page 3.

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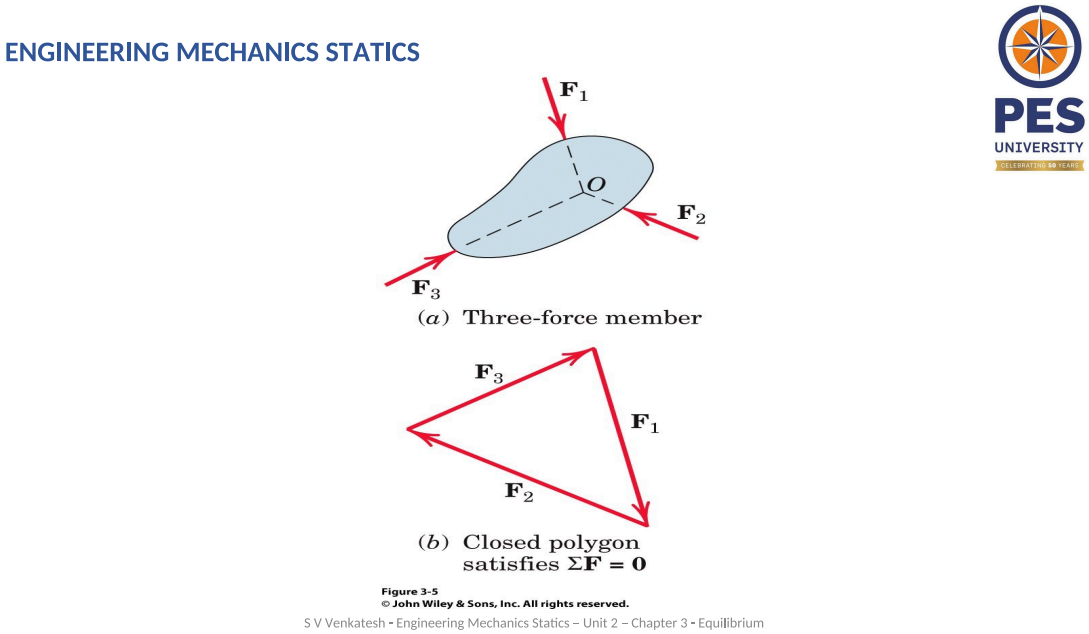


Figure G000014. UE26ME142A Unit 2 Class 15 Lecture PPT · source slide page 4.

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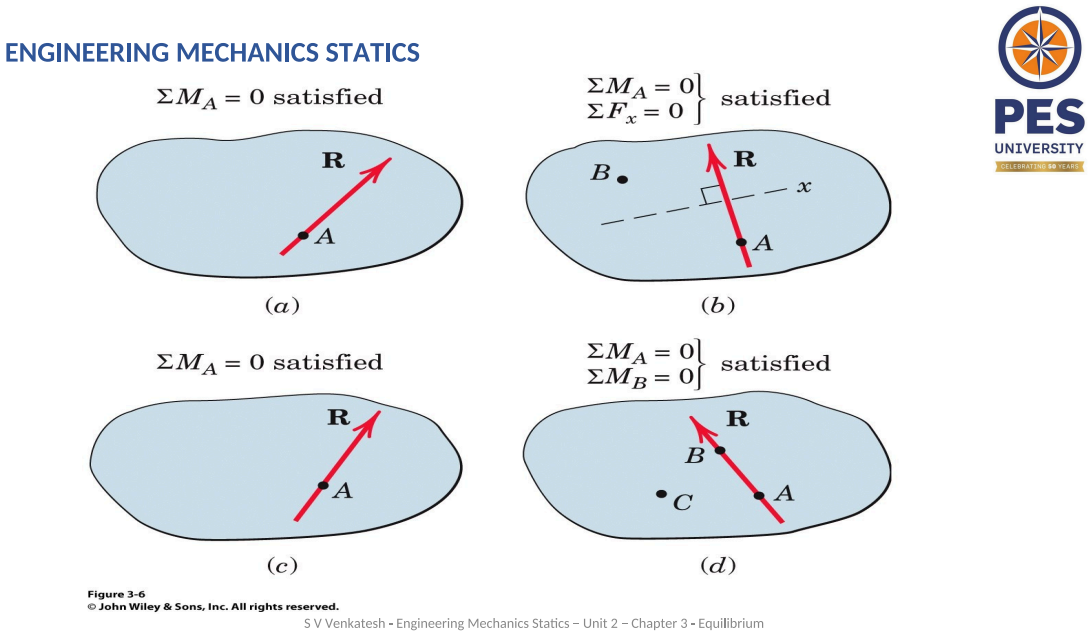


Figure G000015. UE26ME142A Unit 2 Class 15 Lecture PPT · source slide page 5.

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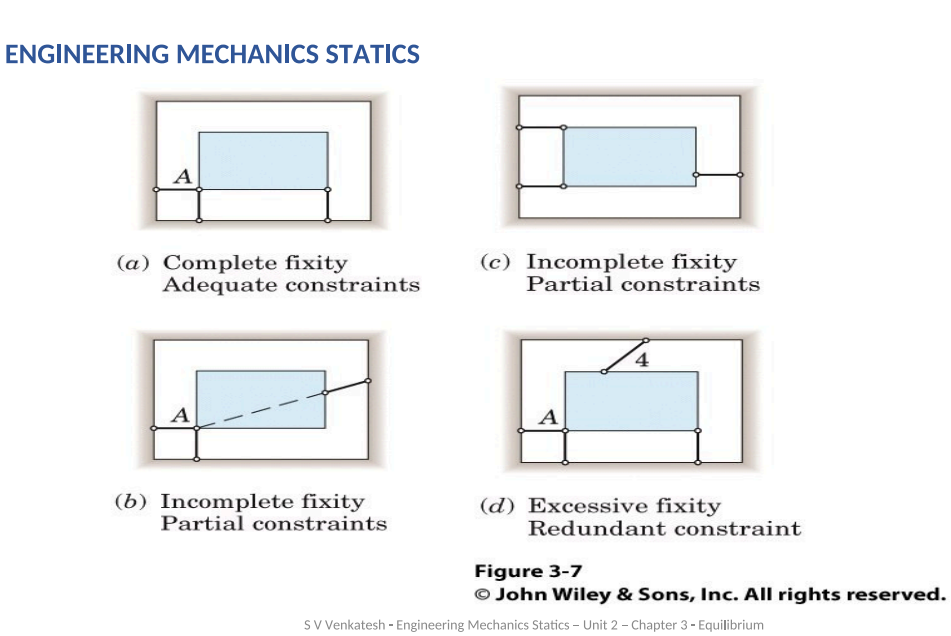


Figure G000016. UE26ME142A Unit 2 Class 15 Lecture PPT · source slide page 6.

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Figure G000017. UE26ME142A Unit 2 Class 15 Lecture PPT · source slide page 7.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000284 · EV0000285 · EV0000286
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 3 (OCR) P ENGINEERINGMECHANICSSTATICS P PES UNIVERSITY CELEBRATING50YEARS EP P Two-force members Figure3-4 Johnwiley&Sons,Inc.Allrightsreserved. SVVenkatesh-EngineeringMechanicsStatics-Unit2-Chapter3-Equilibrium EV0000287 · EV0000288 · EV0000289
3. Source slide page 3: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000290 · EV0000291 · EV0000292
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000293
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 6 (OCR) ENGINEERINGMECHANICSSTATICS PES UNIVERSITY CELEBRATING50YEARS (a) Completefixity (c) Incomplete fixity Adequateconstraints Partialconstraints (6) Incomplete fixity (d) Excessive fixity Partial constraints Redundant constraint Figure3-7 JohnWiley&Sons,Inc.Allrightsreserved. S V Venkatesh - Engineering Mechanics Statics - Unit 2 - Chapter 3 - Equilibrium ## Page / slide 7 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000294
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000295
7. Source slide page 7: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000296

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.484] Dear student, welcome to Engineering Mechanics Statics classes. My name is Dr. Svend Kitish from Civil Engineering under Department of Science and Humanities, PS University. This is class number nineteen. And this is class number two. [00:00:19.972] Under unit two. Under unit two, in class one, that is class number eighteen, we had discussed about the definition of equilibrium. What are the equity that need to be considered for a body to be in equilibrium? [00:00:36.900] Then we have looked at few of the concepts or definitions, or what is that is required for us to know to understand equilibrium. For example, a free body diagram. How do we draw a free body diagram? And various different [00:00:55.844] Tables were given where how to draw the free body diagram was explained and there were some diagrams which was explained which was wrong and how do we correct that also was explained. Along with [00:01:11.620] What are the different types of supports that we should know? And what are the reactions that is going to be acting because of those supports were written in the first class of unit 2. Today we will be continuing in this class further. [00:01:29.828] Now let's look at what are the category of equilibriums based on the system of forces that's going to be acting. So this table provides you different system of forces that can act on a coplanar body or coplanar force system. [00:01:49.348] The first one is collinear. So in this case, there are several forces that's going

to be acting along the same line of action. In this case, it is taken that the forces is acting along X axis. Okay, there are three forces that is given. [00:02:08.292] So if the body is in equilibrium, then the algebraic sum of all the forces acting along this axis X should be equal to 0. If this equation is satisfied, then we can easily say [00:02:24.612] That the body is in equilibrium. We don't have to look at anything else. Okay. For example, summation of all the forces perpendicular to x-axis. Why? Because there is no force acting perpendicular to it. Then movement about any point is equal to zero, because if the net force acting is zero. [00:02:44.420] Then that would not produce any movement except if it is a couple. But in this case, since all the forces are acting along the same axis, that would not produce any couple. I hope that is clear. So only one equation is sufficient. [00:03:03.236] To specify whether the body is in equilibrium or not. If the algebra sum of all the forces is equal to zero, the body will be in equilibrium. Okay. The second category of the force system is concurrent at a point. What do you mean by concurrent? [00:03:22.852] The point of application of all the forces when extended would meet at the same point. For example, in this case, there are four forces. So if we extend this four forces, all of them would come and meet at the point O only. Now if I take x-axis and y-axis. [00:03:42.244] Like this, whatever is indicated. And if I resolve these forces along this axis X and Y, and if I could get the algebra sum of all the forces along X to be equal to 0, and the algebraic sum of all the forces along y to be equal to 0. [00:04:01.380] That means the result is 0. If this is proved, then I don't have to go and take the other equation that is sigma m equals to 0. So only these two equations is sufficient to say that. [00:04:19.492] The body is in equilibrium. That is because the point of application of all these forces meet at the same point O. So when they meet at the same point O, the resultant will also act at this point. But the resultant is equal to zero because summation of all the forces along X is 0, along Y is also 0. [00:04:39.364] So, since there is no resultant that is acting, you can say that the body is in equilibrium. That is only when there is concurrent, the forces are concurrent. Okay. The next category is a parallel force system. So a simple example is taken here. [00:04:58.020] Now all the forces are considered to be parallel to the x-axis. There are four forces acting. Now for us to say that the body is in equilibrium, sum of all the forces acting along X should be equal to 0. Now if that is the case.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:18.116] Yes, the displacement may not be there, but it can produce a moment. So to ensure that it will not produce a rotation or it will not have a moment, we should also take that summation of [00:05:36.260] The forces or the summation of the moment of all the forces about the z-axis about any point should be equal to zero. If that is also satisfied, then this body would be in equilibrium. Okay. [00:05:53.924] Right. The next category is a general category where you have several force system. You can also have an applied moment. The forces when it is extended may not meet at the same point. In such case, we have to have all these three equations of equilibrium. [00:06:13.028] Which should be satisfied, that is algebraic sum of all the forces acting along yes. Algebic sum of all the forces acting along Y? [00:06:22.500] And the moment about the z-axis about any point should be equal to zero. If it satisfies all these three equations, then we can say that this body is in equilibrium. So, in general, if I have to say that the body is in equilibrium. [00:06:42.500] the component of all the forces about the algebraic sum of the components of all the forces about x axis should be zero. Then it means that there is no displacement along X. If [00:06:54.468] The algebraic sum of all the components of all the forces about y axis is 0, then I can also say that the displacement along y is 0. And if the

movement of all the forces and the couple or the moment [00:07:10.468] About any point along the z axis is equal to zero, then there is no rotation. So that indicates that the body is in equilibrium. So essentially if a body has to be in equilibrium, it has to satisfy all these three equations. [00:07:26.340] And these three equations are called as the equations of equilibrium for a body which is considered to be a two dimensional [00:07:36.964] We are not looking at three dimensional. If it is three dimensional, we will have additional equations that needs to be proved or additional things that needs to be proved. In addition to all these three, we will have another three equations. [00:07:49.520] That is algebraic sum of the components of all the forces about the z axis also should be equal to zero. Then algebraic sum of the moments about x axis should be zero. Then algebraic sum of all the moments about the y axis also should be equal to zero. So that makes it six equation. [00:08:06.724] And six will be the equation of equilibrium for a three dimensional. But if it is two dimensional and if we take the body to be in X and Y axis, then these are the three equations of equilibrium, which has to be satisfied for us to say that the body is in equilibrium. Okay. Next continuation of this. [00:08:27.108] If we have a two force members, how do we say whether the body is in equilibrium or not? If it is two force members, if it is collinear, and if one force is opposite to the other, plus [00:08:44.132] If the magnitude of the force is the same, then we can easily say that the body is in equilibrium. That is, if I take the algebraic sum of all the forces along this axis, okay, then I can easily say that the body is in equilibrium. Okay. [00:09:00.932] Next, if I take a three force system, something like this, which is concurrent, then if this follows the law of triangle of forces as indicated here, then I can easily say that the body is in equilibrium. That means [00:09:21.124] If you take the two forces on the adjacent sides of a parallelogram, if the other force forms the other side of the parallelogram, other side of the triangle taken in the opposite order, then we can easily say that the body is in equilibrium. [00:09:40.612] Now let us look at what are the other equations that we can have. Say for example I take a body, there are several forces acting, and if I find that there is a result, if it exists, if there is a result and that is acting. [00:09:58.500] Along this axis, and I also find that the movement about point A is zero. Okay. Now I can easily say that the body is in equilibrium provided the resultant force is zero. Now we can also look at this.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:18.276] If we take an axis perpendicular to the resultant force if it exists, okay, and if the summation of all the forces along this axis is 0. [00:10:32.420] And we also find that the moment about the point A, which is on this result and axis to be equal to zero. And if [00:10:45.540] The movement about any point B where A and B is not perpendicular to R, if that is also equal to 0, then I can easily say that the body is in equilibrium. So what are the other conditions that need to be [00:11:00.304] Consider here or the equations of equilibrium. Other equations of equilibrium for such thing is your summation of all the forces about the x-axis should be equal to zero. The moment if a resultant exists. [00:11:16.420] Along that line of action of the resultant, M A should be equal to 0. And also, if you take any point such that AB is not perpendicular to R, and if the movement about B is also equal to 0. [00:11:33.444] Then I can easily say that the body is in equilibrium. Another condition that we take that if there exists a result, something like this, and if we found that the movement about point A is 0. [00:11:49.572] And if I take some other point B and even there the moment because of all the forces is zero, that means that the resultant has to act along this A B line only. [00:12:02.148] In addition to this, we take another point C where this AB or B C are not perpendicular to R, either

this or this, and if we find that the moment of this about C is also 0. [00:12:18.660] Then also we can say that the body is in equilibrium. So first thing is I have told you that sigma f x should be equal to 0, sigma f y should be equal to 0, and the sigma about the z axis about any point should be equal to 0. These are equations of equilibrium. [00:12:35.236] Other than this, if I don't want to take, then I can also take this condition where I will be proving sigma fx equal to 0, sigma is equal to 0, where a is taken as the point along the line of action of the resultant if it exists. [00:12:52.740] And also, if I can prove that sigma m b is equal to 0, where AB is not perpendicular to the resultant, if it exists, then I can say that the body is in equilibrium. Or if I take something like this, if there exists a resultant, and if I find that [00:13:12.644] About the point the moment is 0 where A is on the line of action of the resultant, and also if I take some other point B where AB is not perpendicular to R, and if I find that sigma M B is also 0, then your resultant if it takes it, it has to act along AB. [00:13:30.756] And also I will take a point C where A C or A B C are not perpendicular to this resultant or this resultant. And if I find that movement about C because of this is also zero. [00:13:46.788] Then I can easily say that the body is in equilibrium. This is these are the alternate equations of equipment that I can take. Right. Next, let's look at what are the fixity and adequate constraints that is required for me to say that the body is in equilibrium. [00:14:06.052] Now there is an example that is taken here. Now you can see that this body has a constraint here as well as here as well as here. Okay? So the weight of this body is going to be acting downwards, or if there is some forces acting on this. [00:14:24.804] If I could resolve all the forces along X and Y, which I could do, then this constraint or this support will balance all the forces acting along the X axis, then there won't be any displacement along X. [00:14:40.932] Then if the all the forces, the component of the force that is acting along y, if this support or constraint balances along with this of all this vertical force that is acting on the body, then there won't be any displacement along the y-axis. [00:14:59.236] So there is no displacement along X, there is no displacement along Y. Now, if I take this point A, and if the moment about this point, because of all the forces that is acting along with this uh support reaction value.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:19.812] Will be equal to zero. That means there is no rotation that is going to occur. Okay, then I can say that this is having complete fixity and there is adequate constraint. [00:15:36.836] Okay, whether I take A or I take B or any other point, the moment has to be equal to zero. Along with the support reaction values. Then I can say that it is it has got a complete fixity and the constraint is also sufficient for complete fixity. [00:15:54.660] Or for the equations of equilibrium to be valid. Now if I take this, okay. Now there is one support here, one support here, there is one more support here, but this support, the line of action of this support, would actually pass through A. [00:16:14.532] Now what happens? This may prevent the translatory motion along X, this may prevent the rotatory motion translatory motion along Y, but since this is acting along the line of action passes through A itself. [00:16:32.996] Now moment about A because of this would not come into picture. So that means this body now can or able to rotate. So there is incomplete fixity, but there is partial constraint. [00:16:50.916] Okay, so it does not allow it to move along X or displace along X, displays along Y, but it is allowing it to rotate. Similarly, if I take this one, now in this case we have the fixity. [00:17:06.692] Along this three, all this three is producing the displacement along X to be 0. Whereas when I apply a force here, vertical force, there is no constraint. It does not prevent it from displacing along Y. So it has got incomplete fixity. [00:17:26.340] Or a partial constraint because it

allows the displacement to occur a long way. Now coming to the fourth diagram. Okay. So here you have one fixity here, one support here, one support here, one support here. Along with this, there is one more support here. This [00:17:45.860] Whether it passes through A or it does not pass through A, it does not matter. But if you compare this diagram and this diagram, only this support, this support and this support should have been enough for complete fixity and adequate constraint. [00:18:04.484] But here we have excess fixity. So there is a redundant constraint. One extra support is there. Whatever is required for equilibrium. More than that, we have provided a support. There is one redundant constraint that is existing. So this is what you should know. [00:18:24.836] before we get into the problem solving mode. So the support should be sufficient for us to apply the equations of equilibrium, which should not be excessive also. If it is excessive then you will have a redundant constraint. [00:18:41.380] Then the sufficient should be sufficient so that there is no translatory motion or rotatory motion. [00:18:49.892] If anyone is possible to occur, then we cannot say that the body is in equilibrium. And how do I how can how I can prove that? By using the or if it doesn't satisfy the equations of equilibrium, I can easily say that there is no adequate. [00:19:05.988] constraint or I can say that it is partially constrained. Okay. So with this we have come to the end of today's class. We will continue in the next class that is class number 20. That would be the class number three under unit two. [00:19:25.412] So thank you very much. I am Doctor Sv Vengtesh. We will meet in the next class. Thank you.

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Full transcript: 2853 normalized words, 68 timestamped segments, 1171.803967 seconds duration. Low-confidence segments flagged: 0.

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4 UE26ME142A_Unit 2_Class 16_AVSummary

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Graphs and diagrams



Figure G000018. UE26ME142A Unit 2 Class 16 Lecture PPT · source slide page 1.

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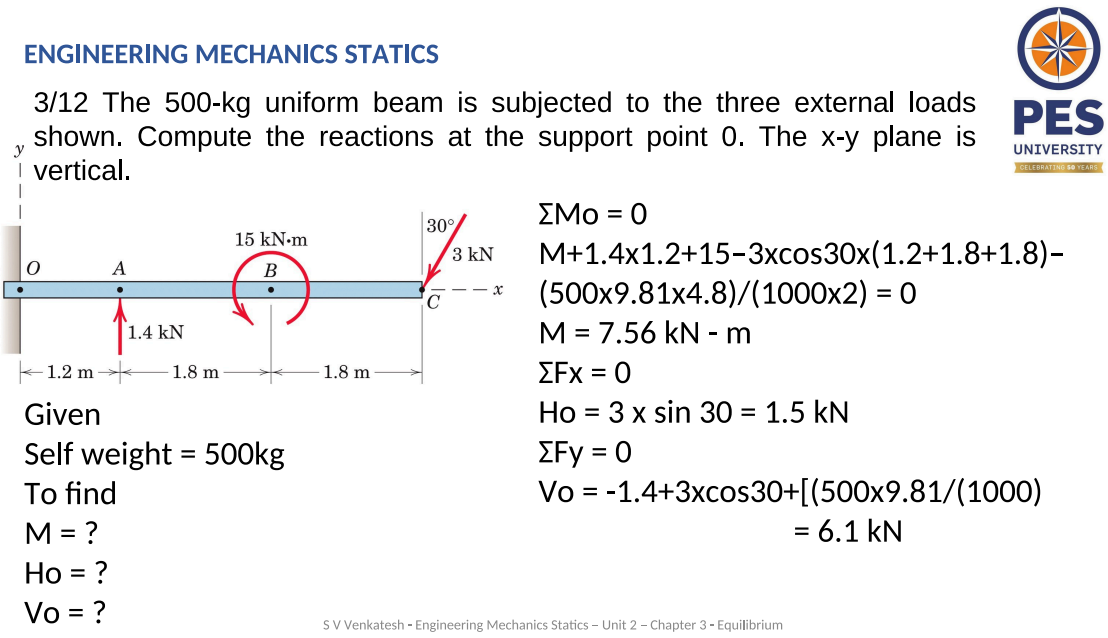


Figure G000019. UE26ME142A Unit 2 Class 16 Lecture PPT · source slide page 2.

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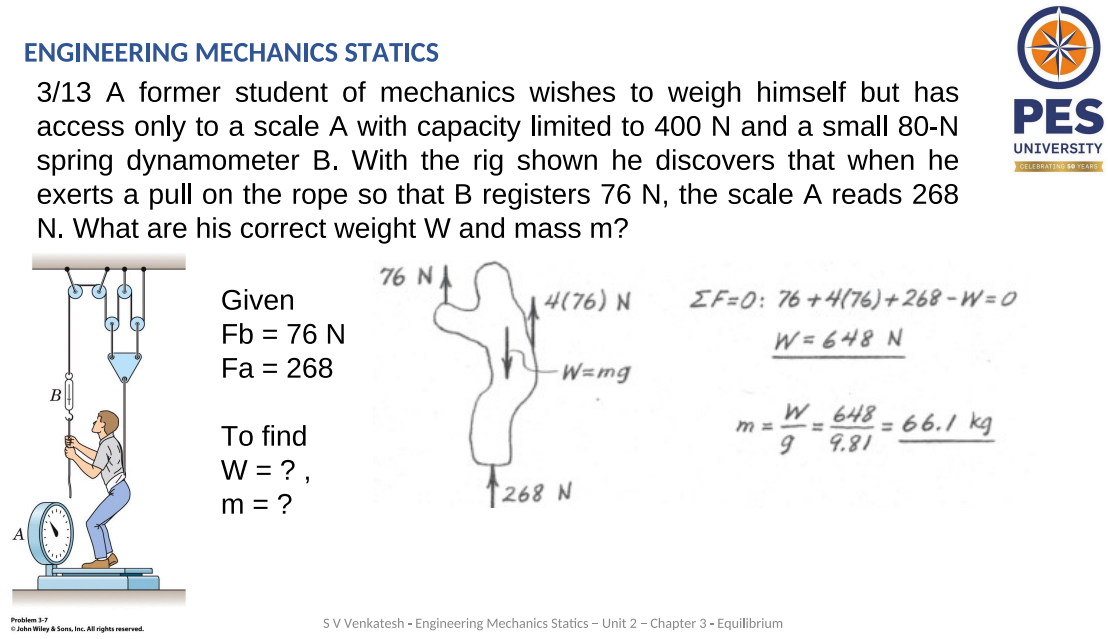


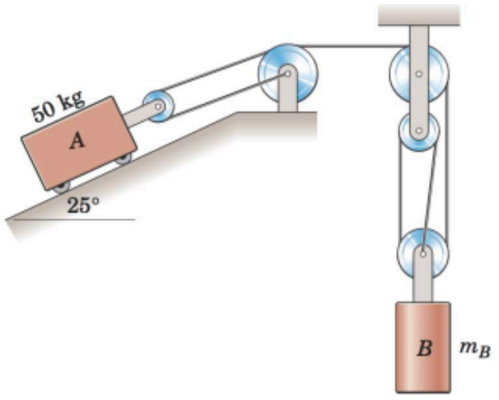
Figure G000020. UE26ME142A Unit 2 Class 16 Lecture PPT · source slide page 3.

EV0000304

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ENGINEERING MECHANICS STATICS

3/15 What mass m_B will cause the system to be in equilibrium? Neglect all friction, and state any other assumptions.



Given

$m_A = 50$ kg

$\Theta = 25^0$

To find

$m_B = ?$,

Solution (Assumption – All cables attached to m_B are vertical)

$\Sigma Ft = 0$

$2T = 50 \times 9.81 \times \sin 25 = 207.30$ N

$T = 207.3 / 2 = 103.65$ N

$\Sigma fy = 0$

$3T = m_B \times 9.81$

$m_B = 3 \times 103.65 / 9.81 = 31.69$ kg

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

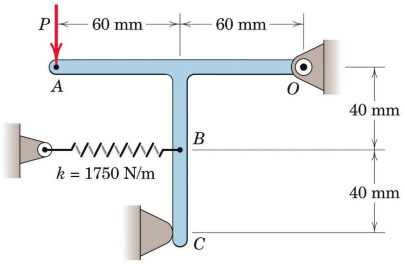
Figure G000021. UE26ME142A Unit 2 Class 16 Lecture PPT · source slide page 4.

EV0000307

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

3/19 When the 0.05-kg body is in the position shown, the linear spring is stretched 10 mm. Determine the force P required to break contact at C. Complete solutions for (a) including the effect of the weight and (b) neglecting the weight.



Given : m = 0.05 kg, Δ = 10 mm, Rc = 0

To find P = ?

(a) including the effect of the weight

$\Sigma Mo = 0$

$P(0.12) + 0.05\ (9.81)(0.06) – 17.5\ (0.04) = 0$

$P = 5.59$ N

(b) neglecting the weight

$\Sigma Mo = 0$

$P(0.12) - 17.5(0.04) = 0$

$P = 5.83$ N

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

Figure G000022. UE26ME142A Unit 2 Class 16 Lecture PPT · source slide page 5.

EV0000310

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

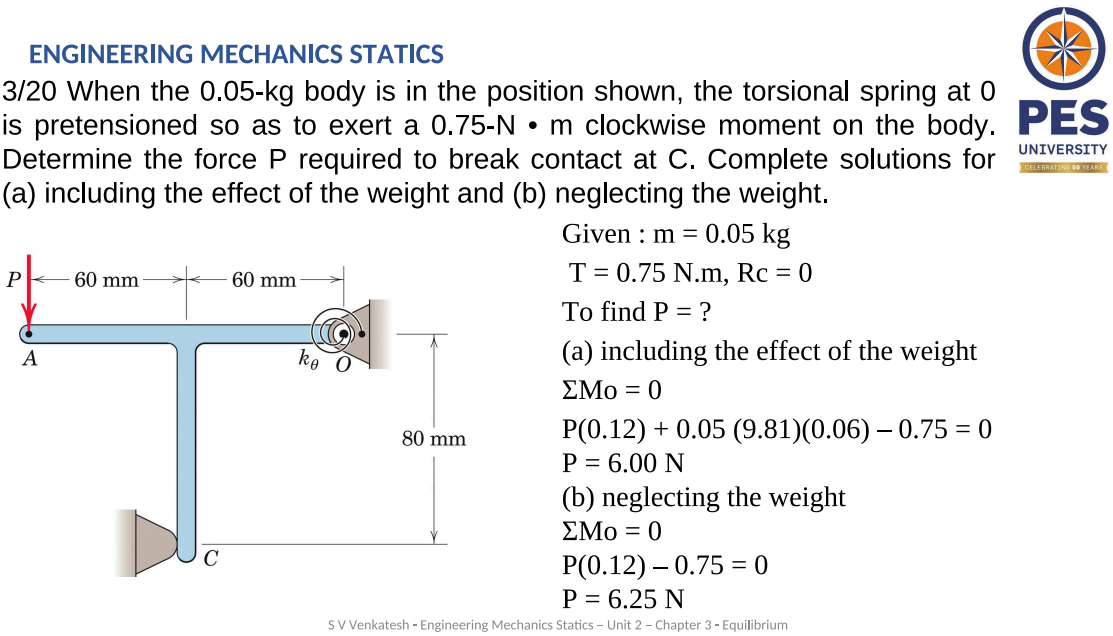


Figure G000023. UE26ME142A Unit 2 Class 16 Lecture PPT · source slide page 6.

EV0000312

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000024. UE26ME142A Unit 2 Class 16 Lecture PPT · source slide page 7.

EV0000313

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000297 · EV0000298 · EV0000299
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 3/12 The 500-kg uniform beam is subjected to the three external loads shown. Compute the reactions at the support point 0. The x-y plane is vertical. ΣMo = 0 M+1.4x1.2+15– 3xcos30x(1.2+1.8+1.8)– (500x9.81x4.8)/(1000x2) = 0 M = 7.56 kN - m Given ΣFx = 0 Self weight = Ho = 3 x sin 30 = 1.5 kN 500kg ΣFy = 0 To find Vo = M = ? -1.4+3xcos30+[(500x9.81/(1000) Ho = ? S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium = 6.1 kN V ? EV0000300 · EV0000301 · EV0000302
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS 3/13 A former student of mechanics wishes to weigh himself but has access only to a scale A with capacity limited to 400 N and a small 80-N spring dynamometer B. With the rig shown he discovers that when he exerts a pull on the rope so that B registers 76 N, the scale A reads 268 N. What are his correct weight W and mass m? Given Fb = 76 N Fa = 268 To find W = ? , m = ? S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000303 · EV0000304 · EV0000305
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 3/15 What mass mB will cause the system to be in equilibrium? Neglect all friction, and state any other assumptions. Given mA = 50 kg Θ = 250 To find mB = ? , Solution (Assumption – All cables attached to mB are vertical) ΣFt = 0 2T = 50 x 9.81 x sin 25 = 207.30 N T = 207.3 / 2 = 103.65 N Σfy = 0 3T = mB x 9.81 S V Venkatesh - Engineering MechanicsmB Statics= 3– Unitx 2103.65– Chapter 3 - Equilibrium/ 9.81 = 31.69 kg EV0000306 · EV0000307 · EV0000308
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS 3/19 When the 0.05-kg body is in the position shown, the linear spring is stretched 10 mm. Determine the force P required to break contact at C. Complete solutions for (a) including the effect of the weight and (b) neglecting the weight. (a) including the effect of the weight ΣMo = 0 P(0.12) + 0.05 (9.81)(0.06) – 17.5 (0.04) = 0 P = 5.59 N (b) neglecting the weight ΣMo = 0 P(0.12) - 17.5(0.04) = 0Given : m = 0.05 kg, Δ = 10 mm, P = 5.83 N Rc = 0 To find P = ? S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000309 · EV0000310 · EV0000311
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS 3/20 When the 0.05-kg body is in the position shown, the torsional spring at 0 is pretensioned so as to exert a 0.75-N • m clockwise moment on the body. Determine the force P required to break contact at C. Complete solutions for (a) including the effect of the weight and (b) neglecting the weight. Given : m = 0.05 kg T = 0.75 N.m, Rc = 0 To find P = ? (a) including the effect of the weight ΣMo = 0 P(0.12) + 0.05 (9.81)(0.06) – 0.75 = 0 P = 6.00 N (b) neglecting the weight ΣMo = 0 P(0.12) – 0.75 = 0 P = 6.25 N S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 7 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000312
7. Source slide page 7: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000313

Short-note recap

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.804] Dear students, welcome to Engineer Mechanics Statics classes. My name is Dr. Svankitish, Civil Engineering, Department of Science and Humanities, PS University. This is class number twenty. [00:00:17.764] Uh in class number nineteen, that is previous class, we had discussed how do we go about writing the free body diagram. [00:00:28.548] And then we have looked at what are the categories of equilibrium equations that we need to use for a particular type of force system. And also we discussed what are the alternate equations of equilibrium that we can use. [00:00:47.236] to go out and solve different problems. Earlier to this, we had also looked at what are the different types of support that we can get and how do we convert this support or replace the support by their respective support reactions. [00:01:04.452] Today we will be solving a few problems from the unsolved problems from the prescribed textbook that is Engineering Mechanics Statics authored by Miriam Craig and Bolton and published by Wiley Publications. Okay. [00:01:23.492] So I have chosen this problem. That is 3 bar 12. The statement of the problem goes something like this. The 500 kg uniform beam is subjected to the three external loads shown. Oh. [00:01:43.044] The XY plane is vertical. Okay. So what is given here from the statement is that there is a uniform beam. The self weight of the beam is 500 kgs, or I should say it is the mass of the beam is 500 kgs. [00:02:00.836] Okay, and we are supposed to determine the support reactions. And he says that the x-ray plane is vertical. So this is what we can know from the statement. Now look at the diagram. Okay, this is also given. So from the diagram we can make out that [00:02:20.740] You have a beam which is indicated by O to C and there is a support at O. If you recall this is a fixed support, then there are three forces, one is acting at point A, which is acting upwards, it is a point low. [00:02:39.652] And the magnitude of that is 1.4 kilonewtons. Then we have a movement here which is in the clockwise or no sorry, anti-clockwise direction of magnitude 15 kilonewton meters acting at B. [00:02:54.084] The distance of point A and B is also given, the horizontal distance from O to A is 1.2 meters, from A to B is 1.8 meters, and there is also an inclined point load of magnitude 3 kilonottons, acting at C [00:03:13.700] And the direction of this force is at thirty degrees to the vertical. They have also given you the x axis and y axis. So here we have to determine the support reactions. The first thing that we need to do is to draw a free body diagram. Of course the free body diagram is not indicated here. [00:03:32.484] But however I'm not be explaining how to draw the free body line. So what needs to be done is simply replace the given beam by a horizontal line. [00:03:43.044] Replace the six set support by two reactions. So the reaction that's going to be acting at O will be vertical. Either I can take the sense to be acting upwards or downwards. Then you will have the horizontal reaction acting to the left or to the right. [00:04:02.212] And then you will have a movement acting at war, which can be clockwise or indicwise. [00:04:07.236] Okay, you can take however you want the sense to be. If you get a positive answer, the sense that you have adopted is correct. If you get a negative sign, that indicates that the sense that you have adopted should be opposite to what you have assumed. Okay. [00:04:23.876] So you will get that. Then apply the force of 1.4 kilonot at A, apply the moment of 15 kilon meters at B, then you can resolve this force along vertical and horizontal. So vertical component of this force will be 3 cos 30 degrees acting downwards. [00:04:43.588] Then the horizontal component of this force will be three sin thirty degrees acting to the left. So that would be the free body diagram. So once you draw the free body diagram, next use the equations of equilibrium. In equilibrium chapter, we have to draw the free body diagram.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.956] Once you draw the free bad item, fifty percent of the problem will be solved. Then use the equations of equilibrium. When you are using the equations of equilibrium, remember that if you can use the movement equation first. [00:05:16.900] And you choose the point where you want to take the moment as a point where there are unknown that needs to be determined. If there are two points where the unknown reactions are acting, then choose the point where the maximum unknown forces are acting as a point where you have to take the moment. [00:05:36.676] By doing so, you will be able to get one of the unknown from this equight away. Okay? Then you can use the other two equation that is sigma f x is equal to 0 and sigma f y is equal to 0. [00:05:49.924] So here first I have used sigma m not to be equal to 0 because that is the point where the support reactions are acting. So when I take sigma m not equal to 0, I have assumed m to be acting at O, which is in the anti-clockwise direction. [00:06:07.108] So I am taking anti-clockwise moment as positive and clockwise moment as negative. So I take m positive plus this 1.4 into 1.2 the distance which will produce an anti clockwise moment which will be positive. So 1.4 into 1.2 will be the moment. [00:06:26.468] Next, 15 kilonewton meters acting in the antipower direction, that would also be positive straight away, add that without multiplying any distance because that is the moment that is there. Next. [00:06:38.372] Or you can take the force vertical force that is acting because of this or the component of the 3 cannon force acting around the vertical, which will be 3 cos 30 into the total distance from O to C, which would be 1.2 plus 1.8 plus 1.8. [00:06:57.188] Now this would produce a clockwise moment, so it is negative. Then the self weight of the beam, I forgot to tell you in the when I explained about the free body diagram. The self weight also has to be taken. The mass of the beam is 500 kgs. So mass into the acceleration. [00:07:16.292] due to graph D which is G would be the force and that would be acting at the center of the beam which would be 1.2 plus 1.8 plus 1.8 divided by 2, which would be equal to 4.8 divided by 2, which will be 2.4 [00:07:32.580] So this is the force that is caused because of the selfate which will be happening the center, that would create a clockwise movement, so it is negative into 4.8 is the total distance. So divided by thousand2. Okay, two is four. [00:07:52.580] The distance one 4.8 by 2. Now thousand is to convert the Newton, thus you will get a Newton because Kg into acceleration you to c you will get as 1 newton. So convert it into Knoton, so multi divide it by 1000. [00:08:08.196] So you have to have consistency of unit, that is very important. And equate that to zero from this equation, we get the moment to be 7.56 kilon meters. You we have got as positive. So I assumed that the moment at O is anti-clockwise. So we got positive, so that is correct. [00:08:26.532] Next, you take sigma f x equals 0, you have a reaction here. I'll assume the reaction at O to be acting to the right, but here the component of the three country force will be acting towards the left. So when I take fx is equal to 0, H0 is positive. [00:08:44.132] And the component is negative, or I can write H0 to be equal to 3 into sine 30 degrees. So I get it as 1.5 kilometers. Then we take all the algebraic sum of all the forces acting along the vertical. So 1 is here the reaction at O which will be acting upwards. So V0. [00:09:02.020] Then we have 1.4 is also acting upwards. So if I take it on the other side of the equation it becomes negative. Then the component of this, since it is acting downwards, on the other side of the equation it becomes uh 3 into cos 30 degrees. [00:09:18.724] Plus we should also take the sulfate of the beam, which is 500 into 9.81. Okay, divided by thousand because I have to convert it into kilonewtons. So by doing so, I get data as 6.1 kilonuttons. Now we have formed out all the three support reaction that is going to be acting at point O. [00:09:38.148] With the values by using the equations of equivalent. Hope all of you have understood this. The next problem that

I have chosen is 3 bar 30. The statement of the problem is given here, which goes something like this. A former student of mechanics wishes to weigh himself, but as [00:09:58.692] Only to a scale A with capacity limited to four hundred newtons and a small eighteen newton spring dynamometer B with the rig shown

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.724] He discovers that when he exerts a pull on the rope, so that B registers 76 Newtons. The scale A reads 268 Newtons. What are his correct weight W and mass M? [00:10:34.756] So from the statement, we can make out that the maximum capacity of the scale A is 400. We cannot measure anything more than 400. And also the maximum capacity of the spring dynamometer is 80 newton, so we cannot measure anything more than 80 newton. [00:10:53.060] But when he is doing this experiment, he finds that the scale will read two sixty eight. The maximum is four hundred, the scale is reading two sixty eight, it is still possible. It is not more than four hundred. Then [00:11:11.780] The rope dynamometer will indicate seventy six newtons, which is also less than eighty, so that is also possible. So what is given? The force FP is seventy six newtons. The F of force that is the scale rating is two sixty-eight. What we are supposed to find is the weight of the person and the mass of the person. [00:11:30.948] Now looking at the diagram, the weight of the person depends on what is the scale reading. And since he is holding this, a part of his weight will go here, which is seventy six, that is also given, that is attention. [00:11:48.868] Now this is connected to a system of police, which is in turn connected to this, and that is connected to the person. So whatever is the scale reading plus whatever is the tension in this FB, then whatever is going to be the tension here would be the weight of the person. [00:12:09.060] So, you can see the free body diagram that is written here. The tension that is applying here is 76 newton. The weight of the person, which will be W, that is acting at the center of mass of the person. [00:12:24.708] W is equal to M into G, then the scale reading which is at the bottom of S uh shoes, that is 268. Then this tension, if you observe, there is tension here which goes here, that this would also be D. [00:12:39.268] The same T is continuing here, so there will be T plus T, the same T is continuing here, and thus this is same. So you have T plus T plus T plus T which is connected to this two. So this the tension here will be half of 4 T which is 2t here, 2 T here. [00:12:57.540] Now this is connected to this triangle and there will be tension here. This would be equal to 40. And that is what is given written here as 4 into 76 newtons. So if I take some of all the forces, F is equal to 0 because there is only vertical forces that is acting. [00:13:16.900] So seventy six acts acting upwards, four of seventy six is also acting upwards, then two sixty eight is also acting upwards, only the weight of the person is acting downwards, that is W equals zero. So we get the weight of the person as six forty eight newtons. [00:13:33.604] So from this we can find out the mass which is equal to W by G. So we get the mass as 66.1 kg. Hope this is clear. Okay, let's move on to the next problem. [00:13:46.372] 3 bar 15, the statement of the problem is what mass mb will cause the system to be in equilibrium. Neglect all friction and state any other assumptions. [00:14:00.676] Now looking at the diagram, what are we supposed to find as mass mv will cause the system to be in equilibrium. So I have to find out this mass m b such that this is in equilibrium. [00:14:18.628] Now if this mass M B is not there, if you identify this as got roller, so the sulfate of the block, the component of that acting along the surface would make it to roll down. Okay? Now he also says that neglect all friction, the pulley is there. [00:14:38.148] There could be some small fraction that's going to be happening. So we have to neglect that. Okay. So this weight is also given the mass, which is 50 kgs. So I can find out the

weight, which is 15 to 9.81, which will be acting vertically down. I can resolve this along. [00:14:58.244] uh the plane that is this one and perpendicular to the plane. Now they have also given you the inclination of the surface as 25. So this and the what uh normal to the surface would also be inclined at 25 degrees.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:13.348] So I can find out the component acting normal to the surface as 15 to 9.81, cos 25. The component of the sulphate acting parallel to the surface will be 15 to 9.81 into [00:15:31.364] Sin 25. Now that whatever is going to be acting if I take the tension here, the same tension is going to be acting here. So if I cut it here, this would be 2T, which should support the component of the sulfate acting along the surface. [00:15:49.540] Now this T is connected to this, so the same T is continuing. So the same T there is a T here, it continues here, so there will be T here. Then this would also come. [00:16:00.900] Now if you observe this is slightly inclined, but they have not given any angle or anything. So I assume that this is also vertical. That is what is the assumption that I am making. So this would also have T. So this there will be 3D here. So 3D will be equal to your MB into [00:16:20.836] G. Okay. So if I could get T here from this, then I can get M B. So what is given? MA, T da, what is to be find? M B solution? Assumption all cables attach. [00:16:34.916] 2 L B or vertical as I told you, I am assuming that to be vertical. Now we are taking uh algebraic sum of all the forces along the T axis. That is this tangential axis. This is normal axis, T axis is equal to 0, so 2t. [00:16:52.964] is equal to 15 to 9.81 sine of this angle 25 I'll get the force to of this the component of the sulphate. So from that I get 2t to be this and t to be equal to 103.65 nucleus. [00:17:09.380] Now I take this and take allife sum of all the vertical forces to be equal to 0. So we have 3 T here which will be equal to MB29.81. So I get MB which is equal to 31.69 kgs. Hope this is clear. Next. [00:17:29.412] We will go to the next problem which is three bar nineteen. The statements goes like this. When the point zero five kg body is in position shown, the linear spring is stretched 10 mm. Determine the force P required to break contact at C. [00:17:47.364] Complete solutions from A, including the effect of the weight, B neglecting the weight. So the sulfate is given as 0.05 kg. Okay, which will be acting at the centroid of this. The centroid will be passing through this because it is symmetrical. [00:18:07.716] And somewhere along this line, some point it will be acting. And that is 0.05 is the mass that is given, so I multiply with 9.81 and [00:18:19.812] That would be the weight of the cell fate of the body that is given, which is 0.05 to 9.81, which will be passing through this line. Next he says the linear spring is structured by 10 mm. So this is structured by 10 mm. [00:18:37.892] Now you can see that they have also given you the stiffness of the spring k here, which is equal to 1750 newton per meter. So if I multiply 10 divided by [00:18:49.860] because it is in M I have to convert it into meters so divided by thousand so we get it as 1.7 into 10 which will be 17.5 into 10 would be the force that is going to be acting which will be pulling B to the left [00:19:04.900] Now what is asked is to determine this force P which is acting at A, such that there is no contact here, so the reaction here would be zero. So for this condition, we have to determine what would be this force B. So once including the effect of the weight, the weight is going to be acting here. [00:19:23.780] So I'll take a moment about. So we get this P into this distance, which is 120, so converting it into meters it is 0.12, which will be antichlocks so positive. Then the self rate, which will be 0.05 into 9.81. [00:19:42.116] Into the distance is 0.06, which is in meters, so we get it in Newton meters. Then the force here in this

spring, which will be 70.5, into this distance is 0.04, which will create a clockwise movement, so that is why it is negative. So we get P from this.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:02.180] Then neglecting the weight, that means we have to remove this. So we get only P into 0.12 minus 17.5 to 0.04. So from that we get this as the P value. [00:20:16.356] Hope everybody has understood this. The next one is 3 bar 20. This is similar to the previous problem except that instead of a spring that was attached here, linear spring, now they have given you the torsional spring attached at O. [00:20:32.836] And they have given that the torsional spring at goes pre-tention so that the so that it exerts a 0.75 N meter clockwise moment. [00:20:45.668] So there is a clockwise movement here which is going to be acting because of the spring, which will be equal to 0.7 newton meters. We are again supposed to find out what is the value of P when there is no contact here. So taking movement about this point. [00:21:02.116] Okay, is equal to 0, so P into 0.12, this we know. The self weight also we know, but there's a clockwise movement here, so minus equals 0, so we get P. Neglecting the weight, we have to neglect this. [00:21:16.932] And take it so we get P as 6.25 newtons. Okay. So we will stop today's class here. We will be continuing in the next class. We will solve some more problems in the coming classes. Okay. So thank you very much. My name is Dr. Sviving Tesh from PS University. [00:21:35.460] We'll see you in the next

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Full transcript: 3355 normalized words, 77 timestamped segments, 1297.152 seconds duration. Low-confidence segments flagged: 0.

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5 UE26ME142A_Unit 2_Class 17_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 17_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000058 · EV0000059

Graphs and diagrams



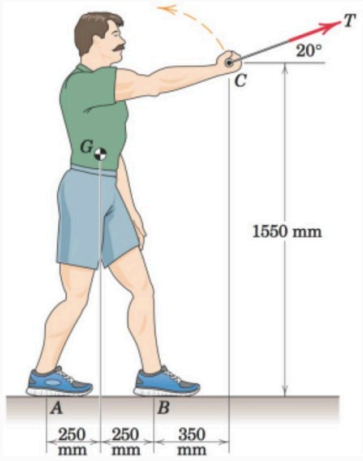
Figure G000025. UE26ME142A Unit 2 Class 17 Lecture PPT · source slide page 1.

EV0000315

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

3/23 The 80-kg exerciser is beginning to execute some slow, steady bicep curls. As the tension T = 65 N is developed against an exercise machine (not shown), determine the normal reaction forces at the feet A and B. Friction is sufficient to prevent slipping, and the exerciser maintains the position shown with center of mass at G.



Problem 3/23

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

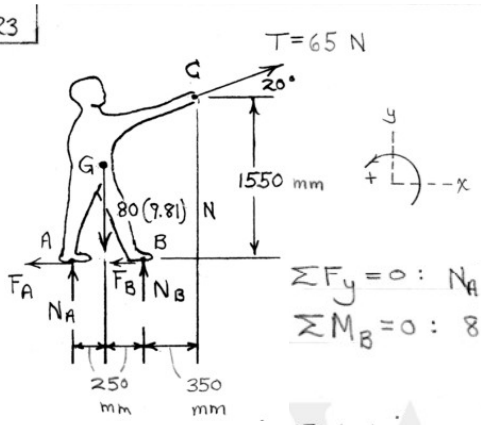
Figure G000026. UE26ME142A Unit 2 Class 17 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS

3/23



$\Sigma F_y = 0: \quad N_A + N_B - 80(9.81) + 65 \sin 20° = 0$

$\Sigma M_B = 0: \quad 80(9.81)(250) - N_A(500) - 65[1550 \cos 20° - 350 \sin 20°] = 0$

Solution: $N_A = 219$ N, $N_B = 544$ N

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Figure G000027. UE26ME142A Unit 2 Class 17 Lecture PPT · source slide page 3.

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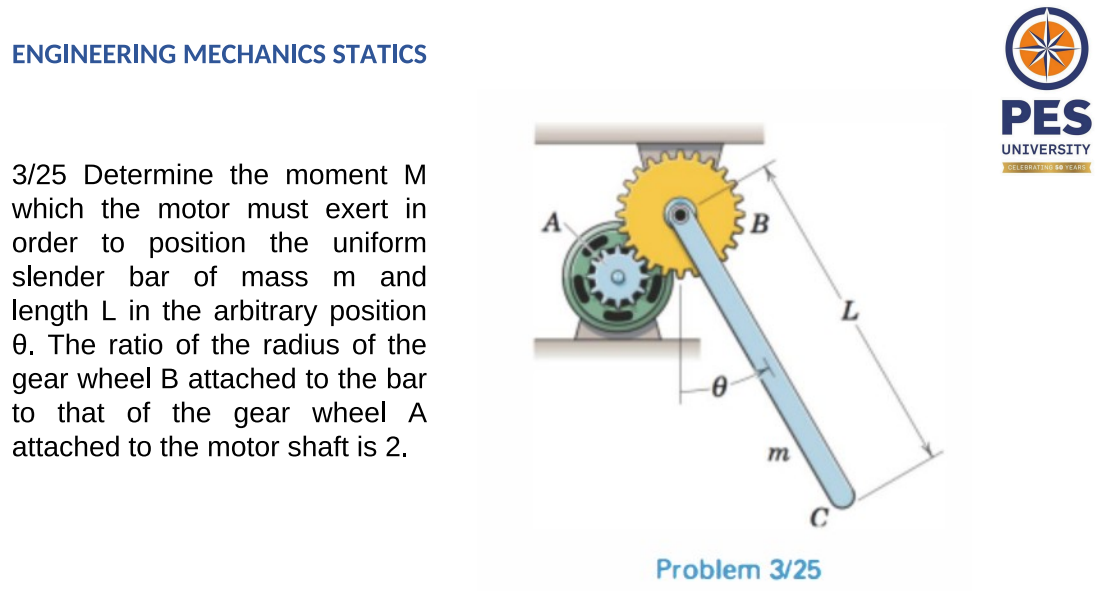


Figure G000028. UE26ME142A Unit 2 Class 17 Lecture PPT · source slide page 4.

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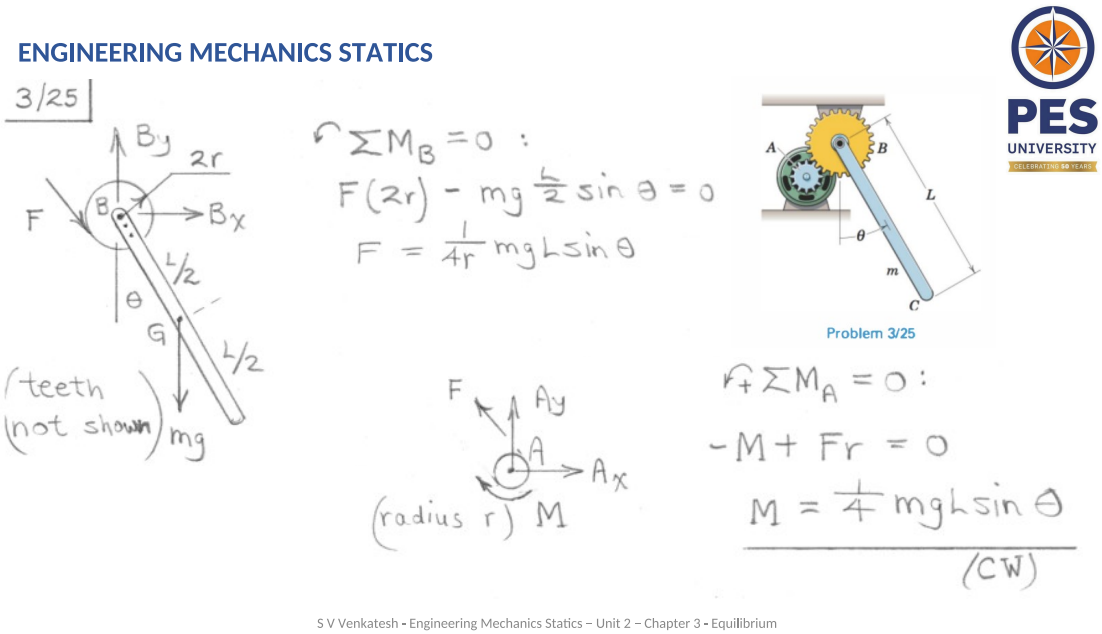


Figure G000029. UE26ME142A Unit 2 Class 17 Lecture PPT · source slide page 5.

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THANK YOU

Dr. S V Venkatesh
Civil Engineering
Department of S & H

Figure G000030. UE26ME142A Unit 2 Class 17 Lecture PPT · source slide page 6.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000314 · EV0000315 · EV0000316
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 3/23 The 80-kg exerciser is beginning to execute some slow, steady bicep curls. As the tension T = 65 N is developed against an exercise machine (not shown), determine the normal reaction forces at the feet A and B. Friction is sufficient to prevent slipping, and the exerciser maintains the position shown with center of mass at G. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 3 (OCR) ENGINEERINGMECHANICSSTATICS 3/23 T=65N PES UNIVERSITY CELEBRATING50YEARS 1550 mm 80 (9.81) N B ZF =0:NA+NB-80(9.81)+65sin20= M=0:80 (9.81)(250)-NA (500)-651550cos20 -350sin20° 250 350 mm mm Solutlon 219N 544N S VVenkatesh-Engineering Mechanics Statics-Unit 2- Chapter 3-Equilibrium EV0000317 · EV0000318 · EV0000319
3. Source slide page 3: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000320

4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 3/25 Determine the moment M which the motor must exert in order to position the uniform slender bar of mass m and length L in the arbitrary position ϑ. The ratio of the radius of the gear wheel B attached to the bar to that of the gear wheel A attached to the motor shaft is 2. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 5 (OCR) ENGINEERINGMECHANICSSTATICS 3/25 PES 2r UNIVERSITY CELEBRATING50YEARS F(2r)-mgsin日=0 日 Bx F 4mgLsinϑ 川 m G Problem3/25 (teeth EMA EO notshoun mg -M+Fr= AK M=本mghsin radius M (CW) S VVenkatesh-Engineering Mechanics Statics - Unit 2 - Chapter 3 -Equilibrium ## Page / slide 6 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000321
5. Source slide page 5: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000322
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000323

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.132] Dear students, welcome to Engineering Mechanics Statics classes. My name is Dr. Svink Tesh from Civil Engineering under the Department of Science and Humanities PS University. This is class number 21. In the previous class, that is class number 20, we had discussed a few. [00:00:17.792] uh problems on equations of equilibrium. Now today in our class we will be discussing more problems. Okay. So I have taken a problem three by twenty-three. [00:00:31.684] Which states that the ATKG exerciser is beginning to execute some slow, steady bicep currents. As tension T is equal to 65 newton is developed against an exercise machine not shown. Determine the normal reaction forces. [00:00:50.756] feet A and B friction is sufficient to prevent slipping and the exerciser maintains the position shown with the center of mass at G. Okay. Now if you [00:01:05.508] Uh look at what is given in the statement. They have given you the mass of the person, which is 80 kgs. They have also given you what is the tension in the cable, which is T that is equal to 65 newtons, and what is asked is to determine what is the normal reaction. [00:01:24.260] Under the feet A and B, and he also says that the friction is sufficient to prevent slippage. Now look at the diagram that is given. There's a tension T, there's a machine here which is not indicated. So there's a tension here which is T [00:01:41.828] The magnitude of tension is 65 newtons, which is given in the statement. The direction of this tension T with respect to the horizontal is 20 degrees. The centre of mass of the person. [00:01:57.252] is G where the 80 kg mass is going to be acting and of course if you convert it into a weight it will be 80 into 9.81 newtons which will be acting at G. Then you have two points are the under the feet of the person [00:02:13.348] Point A and point B, you will have a normal reaction exerted at A, the normal reaction exerted at B and there will be a friction force. Now here when he is trying to pull the exercise machine, the force in the tensile force will try to push this push this person. [00:02:33.380] Forward. So the friction force that is going to be developing here will be to the left of the screen. And in addition to the other details, they have also given you what is the vertical distance of this point C from the ground level. [00:02:50.052] They have also given you what is the distance of A from G, the horizontal distance that is two fifty. Similarly, what is the horizontal distance of G from B, which is two fifty. And also they have given what is the distance between B and C, the horizontal distance between B and C, which is three fifty. [00:03:05.860] So if

we draw a free body diagram for this, what all the things that is going to be there? One is the tensile force acting to the right of the screen, which is 65 newtons. Then the sulphate of the person which is 80 kgs into 9.81, which will be acting at point G [00:03:22.084] You will have a normal reaction N acting upwards at A, normal reaction at B acting upwards, frictional force at A which is acting to the left of the screen, frictional force at B acting to the left of the screen. So as I have indicated, you can see the free body diagram that is written here. [00:03:38.724] All the forces that I have discussed is indicated here along with the dimensions. Then the sign convention is also indicated here. X is positive, y is positive up third, x is positive to the right. Then movement, anti-clockwise movement is taken as positive. [00:03:56.196] Now here we are supposed to find NA and NB. Okay, so what are the equation that we can use? Now if you observe the tensile force is acting at an angle of 20 degrees, so we can resolve this along horizontal and vertical. [00:04:14.052] The horizontal component of the tension will be equal to the frictional force that is going to be acting F A and F B, but we are not interested in that. What we are interested is in N A and N B. [00:04:25.732] There are two unknowns. The horizontal summation of all the forces along horizontal cannot be taken, so we can take summation of all the forces along vertical algebraic sum. Then we can take movement about any point to be equal to zero. [00:04:39.012] So first equation summation of all the forces acting along Y to be equal to zero. So we get NE plus N B equals this eighteen to nine point eight one [00:04:49.092] Plus or this is negative, NEA is positive, N B is also positive, this is negative, then the component of this T will be acting upwards, so it is positive. That is 65 sin 20. That should be equal to 0 and that is what is indicated here.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.340] So from this I can express N A in terms of N B or N B in terms of N A. [00:05:12.324] Next taking moment about any point, it is better to take the moment at the point where you have the support reactions. So you can take it at B or at A. So here we have taken sigma m b is equal to 0. So moment about this point is 0. So what do we get? We get NA. [00:05:30.276] into the distance between A and B which is 500 millimeters. Okay. So that is what is indicated here and it produces a clockwise movement. So it's negative. [00:05:44.164] Then you have this sulfate of the person is 18 to 9.81 into the horizontal distance between G and B is 250. So that is what is indicated here, and that would be anti-clockwise, so minus. Then we have [00:06:01.092] The component of T along the horizontal, which is 65 into cos 20 into the vertical distance, that's what is indicated here. [00:06:15.588] Then we have a component of T which is acting along the vertical, which is 65 sin 20, into the distance from B to C, which is 350 mm, and that's what is indicated here. So by considering this and substituting the value. [00:06:33.604] We will get NA equals 219 from this. Substituting the value of NA here, we get the value of N B which is equal to 544 m. I hope you have understood here. So here it is very clear, what is asked is only to determine NA and N B. [00:06:53.476] So we are using only two equations of equilibrium. Of course the algebraic sum of all the forces along the horizontal if you take, you will have FA plus F B to be equal to 65 cos 20 degrees. [00:07:10.916] Right? I hope all of you have understood this. Now continuing further, I've taken this uh problem which is three bar twenty five. The statement of the problem is determine the moment which the motor e must exert. [00:07:28.484] In order to position the uniform slender bar of mass m and length L in the arbitrary position theta, the ratio of the radius of the gear whe attached to the bar. [00:07:45.476] To that of the gear wheel A, attach it to the motor shaft is 2. Okay. Now from the statement, what is required is to determine what should be the moment M, that should be applied

to the motor at A, that is this motor, this is a motor. [00:08:02.244] What am I supposed to apply the movement here such that this bar of length L would be in the position in an arbitrary position [00:08:15.972] That is at an angle theta is what is asked. No, if there is no movement say consider at this point, then this arm would simply rotate in the clockwise direction and this will come to a vertical position. [00:08:34.980] Now if this has to move by an angle theta, then this wheel needs to be rotated in the anti-clockwise direction. [00:08:45.956] If this has to rotate in the anti-clockwise direction, then this arm will also rotate along with that and it will come to the or uh in this arbitrary position at an angle T. If this has to rotate in this way, [00:09:01.700] Then the movement that needs to be applied here, so here will be in the clockwise direction. So you can see here how it is connected. So if this rotate in the clockwise direction, this would rotate in the anti-clockwise direction. [00:09:16.196] No, they have also given you the ratio of the radius of the shaft that is this and this or in the ratio of 2, or rather if I take this as R, then this would be 2 R. Okay? [00:09:35.428] Now what are the free body diagram and what are the forces that's going to be acting here? When I apply a movement here, so when I apply this movement, you will have a force that is going to be acting along this tangent line. [00:09:52.644] Now this needs to get rotated in the uh anti-clockwise direction. The weight of this would be acting at the center.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.692] So to balance this weight, if you take a moment here, this weight would produce a clockwise movement. So there should be a force here which will prevent it. So this force will act here, which will be in the [00:10:15.652] uh the clockwise direction, this would also be in the clockwise direction, so the force should act in the anti-clockwise direction. So that it will prevent this rotation. Now this force into this radius should be equal to this weight into this the horizontal distance. [00:10:33.348] Now that whatever force that we get, that will be acting here. So the movement that is going to be produced at this will be this force into this radius R. So if I look at the free void diagram of the upper case, that is this arm only with this gear at B. [00:10:52.100] So we will get it like this. The sulfate of the of this the O C would be acting vertically down which is equal to Mg. [00:11:02.244] The force that is acting on the gear would be like this, which is F, because this would produce a clockwise movement to prevent it from rotation, I should apply a force F here, which will act in the anti-clockwise direction. So the radius of this is 2 R. So if I take movement about this point. [00:11:20.420] sigma m or b is equal to 0, f into 2 r minus mg into this horizontal distance would be this is L by 2 sine of theta that's what is indicated here. So you get F in terms of R [00:11:37.700] M, G, and L as well as T. Now this force F which is acting here would be generated because of the gear system at A. So this force whatever is acting. [00:11:53.444] Here, the fourth that is going to be acting here will be in the opposite direction. So this is F here. So the movement that is going to be created here as I have explained it, it's going to be clockwise. So if I take movement about A, I will get M which is in the clockwise direction minus M. [00:12:10.340] Whereas this force would act in the anti-clockwise direction. So F into R is equal to 0. So we will get M to be equal to F R. Now substituting the value of F from this. [00:12:24.004] We will get this as R and R will get cancelled. So we get it as 1 by 4 mg L sin theta. That should be the moment that should be produced in the motor that is present at the point A, that is this one. So here [00:12:43.940] You have the motor, the movement that should be produced in this motor, which should rotate in the clockwise direction, should be equal to one fourth of your MGL sin theta. So the movement depends upon the mass of the [00:13:04.036] uh rod or the bar which is from B to C, then it will depend upon the

length of the bar from B to C. Then it is also depends upon at what angle this bar is to be [00:13:22.756] uh rotated. So these are the factors which affect the movement that is to be generated by the motor which is at A. I hope you have understood. Now if you look at all this what I have used in all this [00:13:39.780] uh problem so far comparing to today's class and the previous class that is class number twenty uh and twenty-one I have used [00:13:51.684] The equations of equilibrium that is summation of the forces along X, algebraic sum of all the forces along X, algebraic sum of all the forces along Y, then movement about any point should be equal to 0. But before using these three equations, [00:14:09.604] I am able to draw a clear free body diagram, right? Isolating what is that I want. [00:14:20.036] Okay, so that I can determine the unknown. Okay. As you can see here, I have taken the pulley at B and the rod that is acting at B separately along with the forces that is acting on this body. That is the force of [00:14:38.308] The sulfate of the rod okay is what I've taken. Now you can also see that there is BY and BX. What are this B will now act as an hinge. [00:14:52.196] So we know that for an hinge there will be two reactions, one vertical and one horizontal, and that is what is indicated here. Okay then coming to A, now the motor A is resting on this surface, then you will have a reaction here.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:11.140] Okay, that is what you have as a y and ax. Okay, so this also you can see that it is like an hinge, and because of that, you have these forces a y and ax. Of course, nowhere I have taken. [00:15:27.876] By, Bx as well as AY and AX, because in both the cases what I have considered is summation of all the movement about B is equal to 0, summation of all the movements about A is equal to 0. [00:15:41.860] I have not used the vertical force, algebraic sum of vertical force, or algebraic sum of the horizontal forces because that is not what I intend to determine. They have not asked us to determine Bx or BY as well as Ax or AY. [00:15:58.276] If they would have asked, I can easily use summation of all the forces along vertical to be 0, summation of all the forces along horizontal to be equal to 0, and then I could have determined what is Bx and BY. Okay. Since I have got F here, okay, this F. [00:16:15.236] And I can also determine what is the angle made by this force with horizontal or vertical. I can resolve this along vertical and horizontal, and I will get what is Bx in terms of F and also BY in terms of F and the sulfate of the rod. [00:16:33.156] Similarly here again I have F if I resolve this F I will be able to get AY and A X. [00:16:39.332] But in the problem, since they have not asked us to determine the reactions acting at B or A, we are not finding out that. I hope this is clear to all the students. Okay. So still now we have solved several problems. [00:16:58.884] Okay, in class number twenty and twenty one. Okay, and all the problems that are solved are example problems from the prescribed textbook, that is engineering mechanics. [00:17:12.804] A statics by Miriam and Craig and Bolton published by Wiley Publication and it is 8th edition. Okay Don't refer to any other edition other than the 8th edition and it is [00:17:27.684] Indian edition itself. Okay, you have foreign editions also, but you can refer to Indian edition. Okay. So with this we will end today's class. We'll be continuing solving of few more problems in the next class. [00:17:44.388] Okay so thank you very much. My name is Dr. Sri Vengtesh from Department of Civil Engineering under Department of Science and Humanities, PES University. Thank you very much. We will meet in the next class.

EV0000070 · EV0000071 · EV0000072 · EV0000073

Full transcript: 2748 normalized words, 68 timestamped segments, 1081.984 seconds duration. Low-confidence segments flagged: 0.

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6 UE26ME142A_Unit 2_Class 18_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 18_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000074 · EV0000075

Graphs and diagrams



Figure G000031. UE26ME142A Unit 2 Class 18 Lecture PPT · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

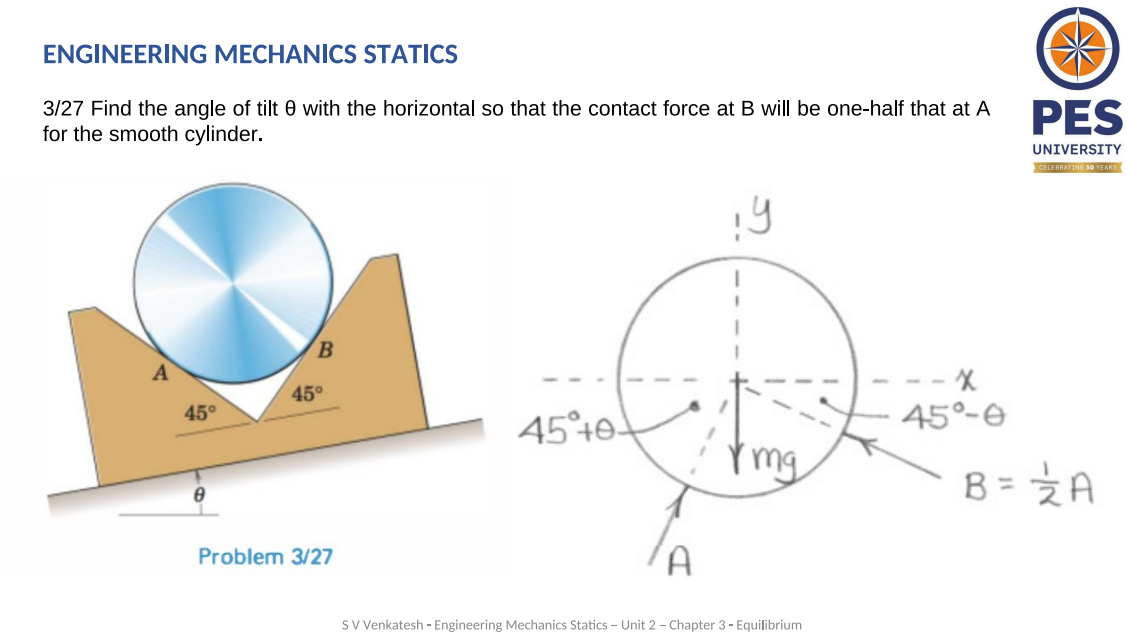


Figure G000032. UE26ME142A Unit 2 Class 18 Lecture PPT · source slide page 2.

EV0000328

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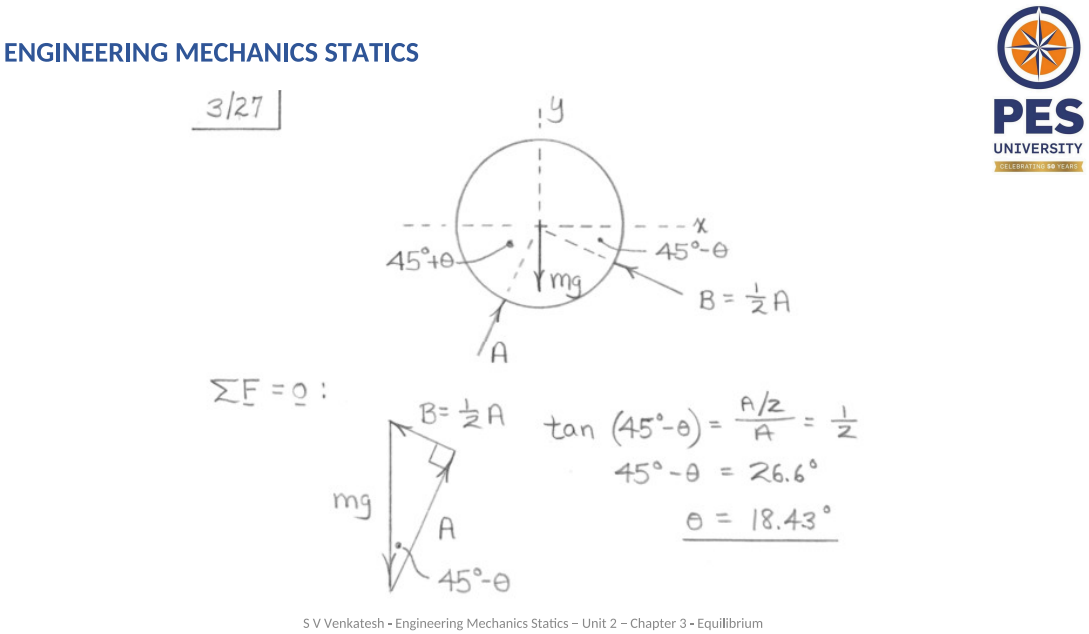


Figure G000033. UE26ME142A Unit 2 Class 18 Lecture PPT · source slide page 3.

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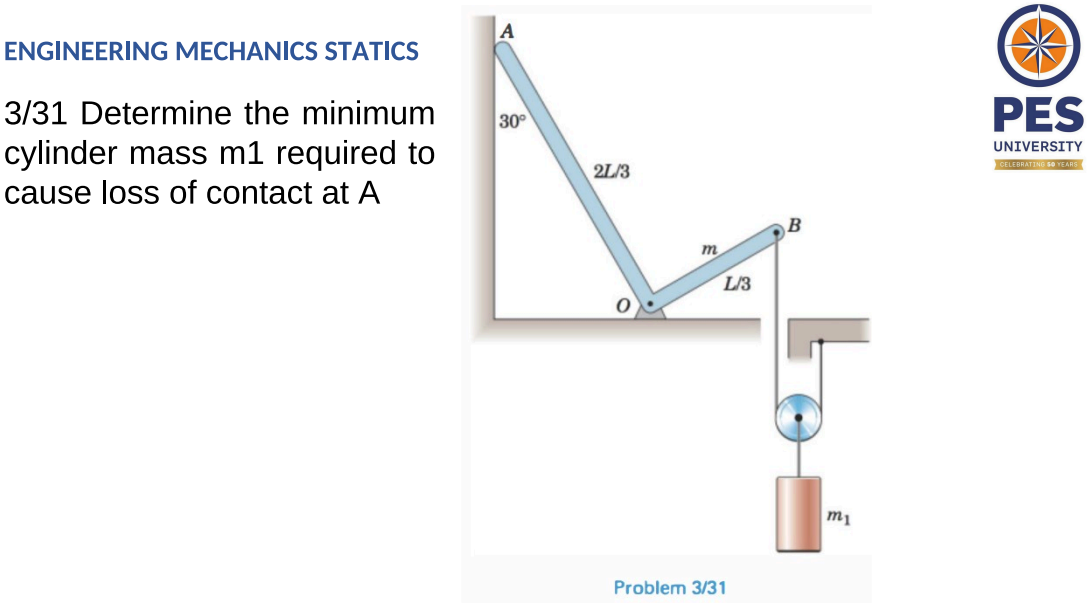


Figure G000034. UE26ME142A Unit 2 Class 18 Lecture PPT · source slide page 4.

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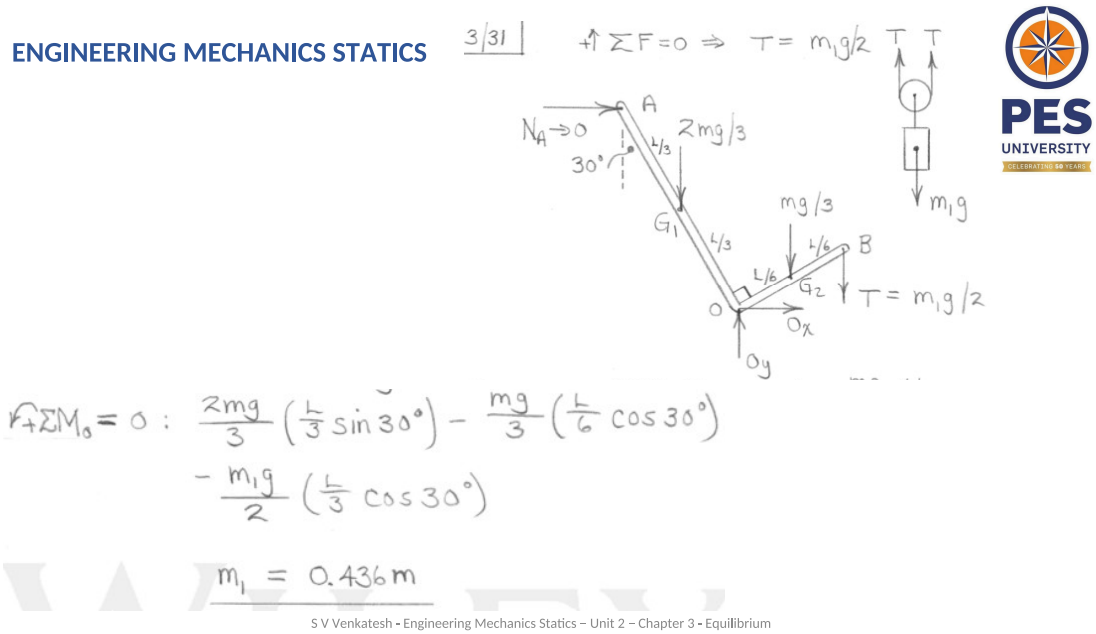


Figure G000035. UE26ME142A Unit 2 Class 18 Lecture PPT · source slide page 5.

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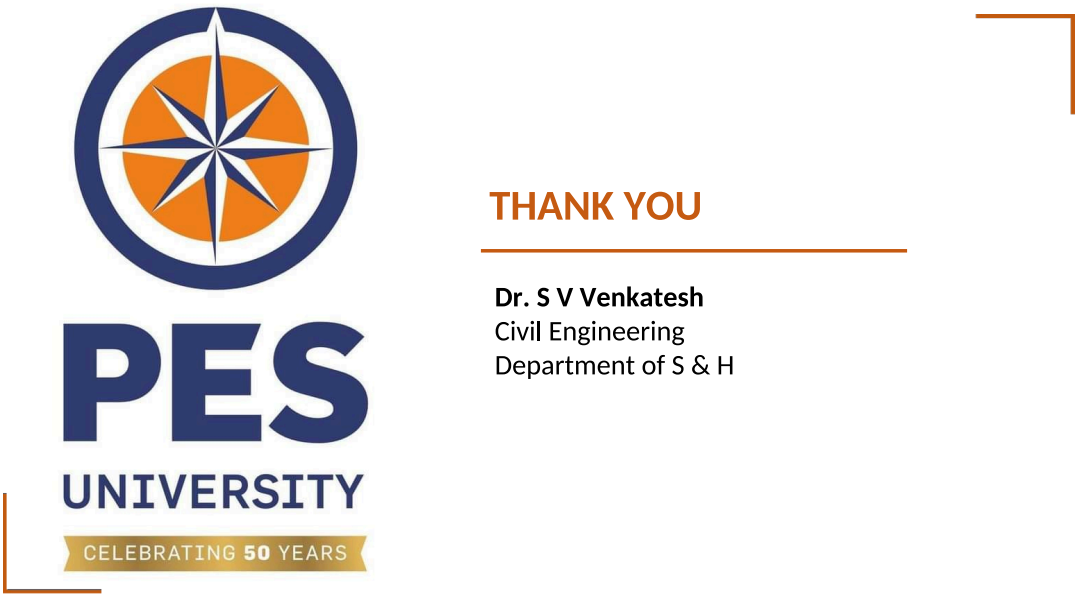


Figure G000036. UE26ME142A Unit 2 Class 18 Lecture PPT · source slide page 6.

EV0000335

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000324 · EV0000325 · EV0000326
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 3/27 Find the angle of tilt ϑ with the horizontal so that the contact force at B will be one- half that at A for the smooth cylinder. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000327 · EV0000328 · EV0000329
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000330 · EV0000331 · EV0000332
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 3/31 Determine the minimum cylinder mass m1 required to cause loss of contact at A S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 5 (OCR) 31 +∑F=0T=mg/2 ENGINEERINGMECHANICSS? A PES 2mg/3 UNIVERSITY 30° CELEBRATING50YEARS mg m,g T=mg/2 Rmg m9 EM=0 CoS30 Sin30 3 3 m9 Cos30 2 =0.436m S V Venkatesh - Engineering Mechanics Statics - Unit 2 - Chapter 3 - Equilibrium ## Page / slide 6 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000333
5. Source slide page 5: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000334

6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000335

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.516] Dear students, welcome to Engineering Mechanics Statics classes. My name is Dr. Isviving Tish from Civil Engineering under the Department of Science and Humanities, PS University. [00:00:13.188] In the last class, that is class number twenty one, we had discussed two problems on uh equilibrium. Today we will be discussing some more problems. So first one that I have taken as 3 bar 27. [00:00:30.916] The statement of the problem is find the angle of tilt theta with the horizontal so that the contact force at B will be one half that at A for the smooth cylinder. [00:00:47.332] The statement what we could identify as that we need to find out the angle theta for the condition that the contact force acting at B is equal to one half of that acting at A. [00:01:05.956] Now looking at the diagram you can make out that the angle theta is the angle made by this surface with the horizontal. Now they have given you a V block here. On this V block, they have kept a [00:01:23.620] Smooth cylinder. So that you can make out from the statement that this is a smooth cylinder. Now the cell fade of this cylinder will be acting at the center of the cylinder. [00:01:40.484] If the mass of the cylinder is M, then the sulphate of the cylinder would be mass times the accuration due to gravity, which will be acting at the center. Now this cylinder is placed on this V group. [00:01:57.028] The contact point of the cylinder on the V groove is at two point, that is A, that is this point, and B that is this point. Now the inclination of the groove with respect to the surface. [00:02:13.348] That is this one, this line, is also given and that angle is forty-five degrees, which you have to note down. So wherever there is contact between the cylinder and the V group, we will have a reaction. The self weight will be acting downwards. [00:02:31.780] That has to be balanced by the reaction. So there will be reaction at A, which will always be perpendicular to this line, the group, the V group at A. So this reaction at A will be perpendicular to this line, and also [00:02:50.020] You can know that this line is the tangent to the cylinder. Right? Because that would be only at one contact point, and that would be always going to be tangent to the cylinder. So the reaction, if it is normal to this, then that means it would be passing through the radial line. [00:03:09.316] Which will also pass through the center of the cylinder. Similarly, there will be reaction at B. This reaction at B is going to be normal to this line or this groove, which is also tangential to the cylinder. [00:03:27.300] So that would also pass through the radial line. [00:03:32.644] Which will also pass through the center of the cylinder. So you will have three different forces acting because there are only two contact points where the cylinder is touching the V group, that is at A and B. So you will have two V actions, plus there is a [00:03:47.684] Sulfate of the cylinder that is going to be acting. Now all these three would pass through the center of the cylinder. Okay. Next, to know the angle at which the reaction at A and B is acting, we have to identify this, which is [00:04:06.756] 45 degrees. Now if I draw in a horizontal line here, passing through this point, if I draw in a horizontal line, this angle theta will be the same angle made by this line with the horizontal, which will be teta, whereas here also it is going to be teta itself. So [00:04:26.148] The angle made by the group, V Group, which is passing through A [00:04:34.372] With respect to the horizontal, we'll be at an angle of 45 degrees minus this angle theta, which is 45 minus theta. Whereas here, if I you uh find out the angle made by the group passing through B with the horizontal, that would be equal to 45

plus theta. [00:04:53.636] So if I draw a line at A, an horizontal line passing through A, this line, these two are going to be corresponding angle. So this angle will be 45 minus theta. If I draw a

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.956] The reaction, if I draw the reaction here, this total angle will be 90 degrees. Since this would be 45 minus theta, the angle made by the reaction at A with the horizontal will be 45 plus theta. Similarly, if I draw a reaction at B [00:05:32.996] And this angle is 45 plus theta, and this reaction will be at 90 degrees to this line. The angle made by the reaction with the horizontal will be 45 minus theta. Now the free body diagram is written here. We have removed the V group. [00:05:50.820] The reactions at A, the reactions at B is indicated. The X and Y axis are indicated. You can see the sulphate of the cylinder is indicated as Mg, that is reacting vertically down. So this angle which the reaction at A [00:06:09.060] Next with the horizontal is going to be 45 plus theta. I have explained how to get that. Similarly, the reaction at B, the angle made by that with the x-axis would be 45 minus theta. I have explained that also. I hope it is clear till now. Let's continue. [00:06:27.588] So the same diagram, the free ball diagram is indicated here. You have to identify that the reaction at A is considered as A itself, the reaction at B is considered as B. [00:06:40.676] Which is equal to one half of that A. That is what is given in the statement of the problem. Reaction at B has to be one half of A. For that condition, we have to find out what is this T. Now there are three forces acting at this point. [00:06:57.508] At the center of the cylinder. So I can represent all these three forces on the sides of a triangle like this. Okay. By considering the reaction at B, this is reaction at B which is equal to one half of K. This is the reaction at A, which is A itself. [00:07:16.004] This is the cell fate of the cylinder that is Mg. So this angle, the angle made by the reaction A with the vertical, that is this angle. [00:07:35.012] would be 45 plus theta is this minus 90 minus 45 plus theta so this would be 45 minus theta that is what is indicated here. Now uh the angle made by the reaction B [00:07:51.204] with the vertical would be 90 minus 45 minus theta which will be 45 plus theta that will be this however if I take this triangle I can use tan theta minus tan of t 45 minus theta that is this one. [00:08:10.532] Tan of 45 minus theta, which will be equal to the opposite side by adjacent side, which is B by A, and we know that B is one half of A, so it is A by 2 divided by A. So I get tan, 45 minus theta as half. [00:08:31.108] So 45 minus theta will be tan inverse of off, which we will get as 26.6. So theta will be equal to 45 minus 26.6, which we will get as 18.43. Now this one we have solved it by using [00:08:51.236] the law of triangle of forces. But here instead of taking the distances, we have taken the reactions as the size [00:09:01.220] Or the force at the sides of the triangle, which will still work. You can represent the distances by forces or force by distances. So it will still work. However, if you want to solve it by using the equation of equilibrium, you can also solve it by taking [00:09:20.932] Sum of all the forces along X to be equal to 0. That means your force F cause of 45 minus Teta, which would be this, would should be equal to the reaction V cause of this. [00:09:36.324] Cause of forty-five minus theta which is this angle. So this force and this force has to balance with each other. So using that and using the uh value of B as half of A, we can find out. [00:09:54.468] The value of theta. Along with that, we can also use some of all the forces along acting along y. So you get mg to be equal to a sine 45 plus theta. Plus

EV0000078 · EV0000079 · EV0000080 · EV0000081

Transcript block 3 (source interval 000600-000899 seconds): [00:10:12.804] B sin 45 minus theta. So these two minus mg should also be equal to 0. You can use that also to solve this. Okay? Hope you have understood. Let's go to the next problem. I've considered problem 3 bar 31. [00:10:31.172]

The statement of the problem states determine the minimum cylinder mass m1 required to cause loss of contact at B. So what is to be determined is mass m1. So the condition that is given is there should not be any contact at A. [00:10:51.428] Now looking at the diagram, the mass m1 is given here. This mass m1 is attached to this cylinder. The cylinder is [00:11:05.348] Hanging by this row or wire. Okay. The wire is attached to this beam. One end of the wire is attached to this point beam. You can also see that this is an R. Okay. [00:11:25.316] They have given the inclination of this arm AO with respect to vertical as 30 degrees. The length of AO is given as 2L by 3. The length of OB is given as L by 3. The mass of this entire [00:11:45.156] AOL the L angle or L member is to be considered as M. So the mass distribution of OA will be equal to 2 M time 2 M by 2 by 3 M. [00:12:03.428] Whereas the mass of OB will be equal to M by 3. So that would be the distribution. So that would be acting at the center of OM. The force will always be acting vertical. So M times G [00:12:19.236] That is 2 by 3 m times g will be acting at the center of AO, which would be at a distance of 2 by 3 L divided by 2, that would be L by 3, whereas the mass m by 3 [00:12:38.500] Would M G by three will be acting at this point center and this distance will be equal to L by three into half or into half, which will be L by six, that distance where the force is going to be acting. Then you can see that here. [00:12:58.020] The point O, there is an hinge, so there will be two reactions, one vertical and one horizontal. Now at this point, there will be a reaction, but according to the statement, the force M1 or the mass M1 should [00:13:15.844] Make the contact here. There should not be any contact. So that means the reaction at A should be equal to 0. The free body diagram for this M1, if you take if I take the tension in this row as T. [00:13:31.396] The same rope is continuing here, so you will have T here also. So if I cut this at this point and control this separately, the weight of M1 into G which will be acting downwards is balanced by these two T. So T will T will be equal to [00:13:48.580] M1 by M1 G by 2, that is what is the force that's going to be acting here. This force will be acting downwards, which will make this two rotate in such a way that there is no contact here. So that is what we have to determine. So when there is no contact here, this force, the sulfate, [00:14:08.868] into this distance, should balance with this weight into this distance, as well as this force into this horizontal distance. Now let's look at the solution, the free body diagram. [00:14:24.068] So the sign convention is taken like this, the anti clockwise movement is considered as positive. [00:14:30.212] So if I take some of all the vertical forces acting on the mass m1g which I have explained it just now, so I get t to be equal to m1g by 2 because you have 2t here just to be balanced by this m1g. So you get the value of t. [00:14:48.708] Next let us take all the forces acting on the [00:14:52.484] uh L bar which is AOB. So as I have explained you have a reaction here NA. So this when I apply this tension P I should apply it or mass M1 should be applied in such a way that there is no contact here. So your NA has to be 0.

EV0000082 · EV0000083 · EV0000084 · EV0000085

Transcript block 4 (source interval 000900-001199 seconds): [00:15:10.020] Then you have the sulfate of this R which is 2 mg by 3. Then the sulphate of OB which is mg by 3 because this length and this length are 2 L by [00:15:24.868] 3 and L by 3. So with that we will get this. Then we also know that this length the inclined length of OA is L by 6. So half of that will be L by 3. Similarly the inclined length of OB is L by [00:15:43.364] 3, so we get this as L by 6, I am sorry, this is 2 L by 3, so I get this distance as L by 3 and L by 3. This distance is L by 6. So I get uh I are sorry uh L by 3. So half of that I will get as L by 6. [00:16:00.676] This inclination is 30 degrees, so obviously this will be 60 degrees. This if you observe AOB is at an angle of 90 degrees. So this is

this angle with the horizontal [00:16:18.852] We know that this is 30, so if I draw a vertical line here, this would be 30, this would be 60, so this obviously has to be 30 degrees. So if I take movement about O, why should I take movement about O? Because we know that there are two reaction that is acting here, which are unknown. [00:16:36.580] So I have to use the first equation of the equilibrium, which is the moment equation, at the point where you have the maximum or not. So I will apply moment about O to be equal to 0 by considering anti-clockwise moment as positive. [00:16:51.364] So if I take this force, it is 2 mg by 3 into this horizontal distance, which will be L by 2 into this angle, which is 60, or I can also take this distance and this angle. So the horizontal distance will be L by 3. [00:17:07.396] Sine 30 degrees, and that's what is mentioned here, which will produce an anti-clockwise moment, so positive. This would produce a clockwise moment, so negative. This force is mg by 3 into the horizontal distance of this force from O will be L by 3. [00:17:26.756] Cost 30 degrees because this angle is 30 degrees. Then we also have this tension T which will also produce a clockwise movement. So negative the T value is M1 G by 2 and that's what is taken here. The horizontal distance of B from O. [00:17:44.100] will be equal to L by 3 into cos thirty degrees and that's what is mentioned here. So if I simplify this I will get the value of M1 because M will in terms of M, M1 we will get it in terms of M [00:18:00.932] So if I simplify this, I'll get M1 as 0.436. Yeah. Hope everybody has understood this. Now what is asked is only to determine what should be the mass m1 such that the reaction at this point is 0, or there is no contact at A. [00:18:21.412] That's why we are using only the moment equation here and we are stopping it. But the extension of this problem can also be if they ask you to determine the reactions are to O. [00:18:34.340] So there will be an horizontal reaction O X and there's a vertical reaction OY. So if I want to find out these values, I can simply use sigma fx equals 0. So I'll get the value of O X to be equal to 0 in this case. [00:18:52.356] Then if I use summation of all the forces along y to be equal to 0, so I'll get the value of OY which will be equal to m into g, summation of this, plus this T would be the reaction that's going to be acting at OY. Now O X is 0 because there is no [00:19:12.004] Horizontal force that's going to be acting here because this was anyhow zero. Hope everybody has understood this two problems. Whatever I have discussed. So in equations of equilibrium. [00:19:27.300] What we need to do is only use the equations of equilibrium, the three equations of equilibrium, uh algebraic sum of all the moments to be equal to zero at any point. So use that point where the maximum unknown is acting. Then summation of all the forces along y should be equal to zero. [00:19:44.388] The algebraic sum of all the forces acting along y direction to be equal to zero, the algebraic sum of all the forces acting along x direction to be equal to zero. And that's what we have done now. But you can use only the earth equation, whatever is required for you to determine the unknown value, okay? [00:19:58.916] So we'll stop today's class here, we'll continue in the next class. My name is Dr. Svangesh. See you in the next class. Thank you very much.

EV0000086 · EV0000087 · EV0000088 · EV0000089

Full transcript: 2950 normalized words, 72 timestamped segments, 1206.827 seconds duration. Low-confidence segments flagged: 0.

EV0000074 · EV0000075 · EV0000076 · EV0000077

7 UE26ME142A_Unit 2_Class 19.0_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 19.0_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000090 · EV0000091

Graphs and diagrams



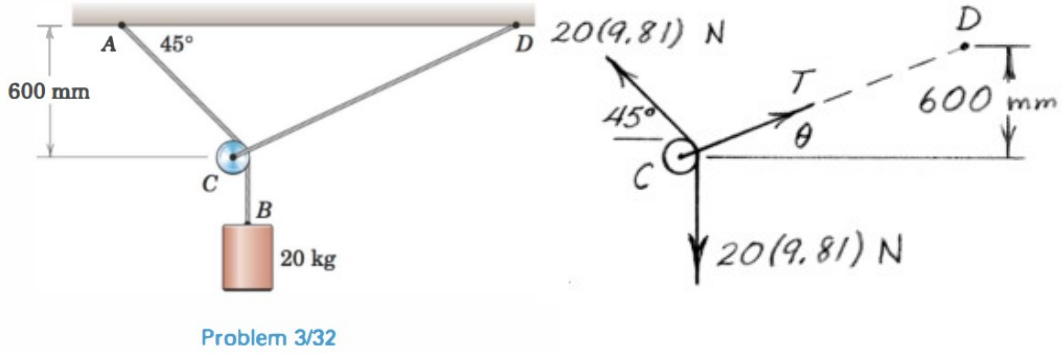
Figure G000037. UE26ME142A Unit 2 Class 19.0 Lecture PPT · source slide page 1.

EV0000337

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

3/32 Cable AB passes over the small ideal pulley C without a change in its tension. What length of cable CD is required for static equilibrium in the position shown? What is the tension T in cable CD?



Problem 3/32

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

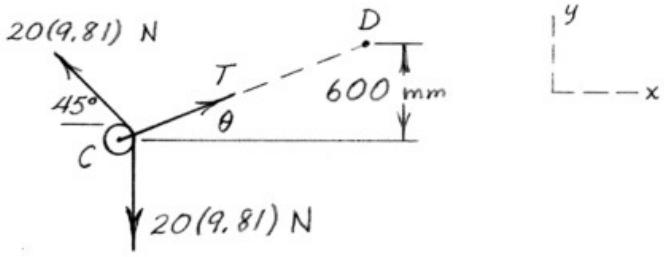
Figure G000038. UE26ME142A Unit 2 Class 19.0 Lecture PPT · source slide page 2.

EV0000340

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

3/32



$\Sigma F_x = 0$: $T\cos\theta - 20(9.81)\cos 45° = 0$

$\Sigma F_y = 0$: $T\sin\theta + 20(9.81)\sin 45° - 20(9.81) = 0$

Solve to obtain $\theta = 22.5°$, $T = 150.2$ N

$\frac{600}{\overline{CD}} = \sin\theta = \sin 22.5°$, $\overline{CD} = 1568$ mm

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

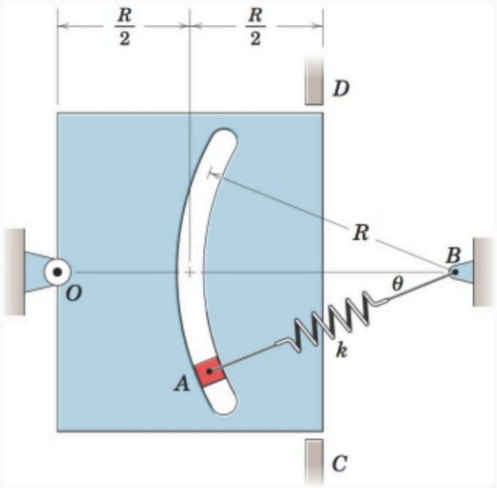
Figure G000039. UE26ME142A Unit 2 Class 19.0 Lecture PPT · source slide page 3.

EV0000343

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

3/34 The small slider A is moved along the circular slot by a mechanism attached to the back side of the rectangular plate. For the slider position θ = 20⁰ shown, determine the normal forces exerted at the small stops C and D. The unstretched length of the spring of constant k = 1.6 kN/m is R/3. The value of R is 25 mm, and the plate lies in a horizontal plane. Neglect all friction.



Problem 3/34

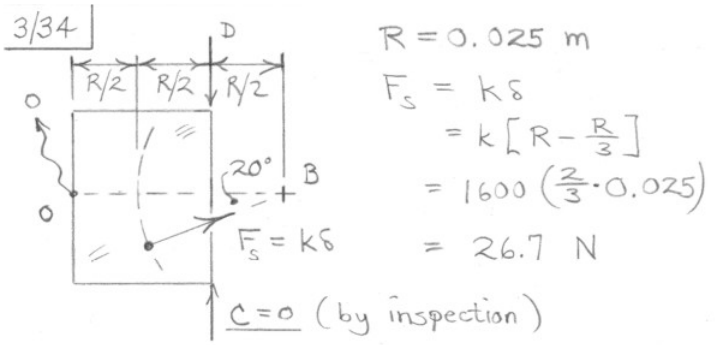
S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

Figure G000040. UE26ME142A Unit 2 Class 19.0 Lecture PPT · source slide page 4.

EV0000345

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS



$R = 0.025$ m

$F_s = k\delta$

$= k\left[R - \frac{R}{3}\right]$

$= 1600\left(\frac{2}{3} \cdot 0.025\right)$

$= 26.7$ N

$\curvearrowleft \sum M_O = 0: \quad 26.7 \sin 20°\left(\frac{3}{2} \cdot 0.025\right) - D(0.025) = 0$

$D = 13.68$ N

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

Figure G000041. UE26ME142A Unit 2 Class 19.0 Lecture PPT · source slide page 5.

EV0000346

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

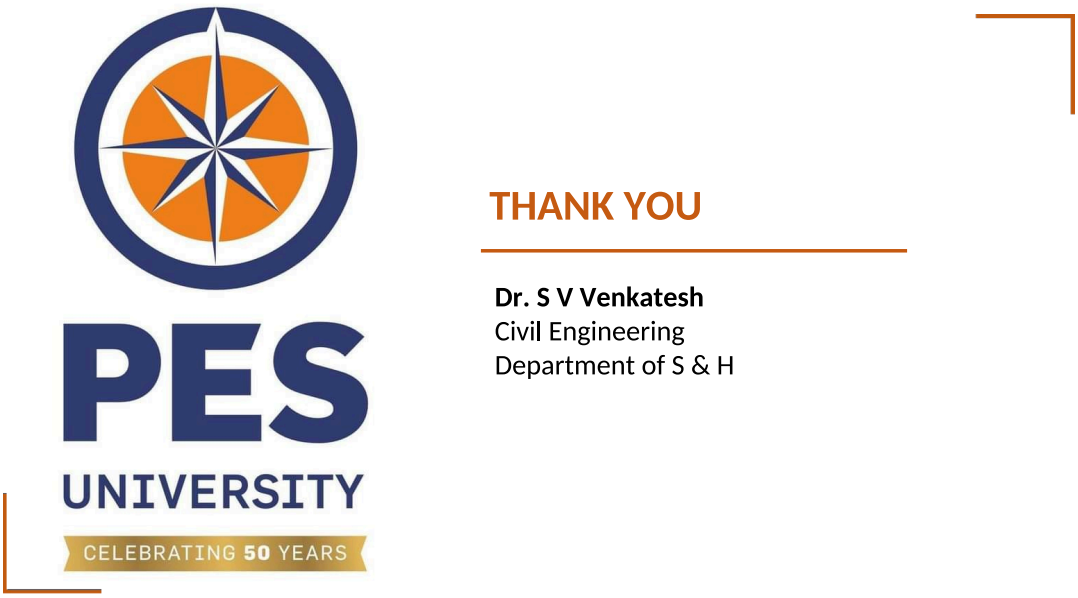


Figure G000042. UE26ME142A Unit 2 Class 19.0 Lecture PPT · source slide page 6.

EV0000347

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000336 · EV0000337 · EV0000338
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 3/32 Cable AB passes over the small ideal pulley C without a change in its tension. What length of cable CD is required for static equilibrium in the position shown? What is the tension T in cable CD? S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 3 (OCR) ENGINEERINGMECHANICSSTATICS 3/32 20(9.81)N PES UNIVERSITY 600 mm CELEBRATING50YEARS 20(9.81)N ∑F=0:Tc0sϑ-20(9.81)c0s45°=0 F=0:Tsinϑ+20(9.81)sin45°-20(9.81)=0 So1vetoob+ainϑ=22.5°T=150.2N 600 =sinϑ=sin22.5°CD=1568 mm CD S VVenkatesh-Engineering Mechanics Statics - Unit 2 - Chapter 3 -Equilibrium EV0000339 · EV0000340 · EV0000341
3. Source slide page 3: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000342 · EV0000343 · EV0000344
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 3/34 The small slider A is moved along the circular slot by a mechanism attached to the back side of the rectangular plate. For the slider position ϑ = 200 shown, determine the normal forces exerted at the small stops C and D. The unstretched length of the spring of constant k = 1.6 kN/m is R/3. The value of R is 25 mm, and the plate lies in a horizontal plane. Neglect all friction. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000345

5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 6 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000346
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000347

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.000] Dear student, welcome to the classes on engineering mechanics statics. This is class number thirty twenty-three, I'm sorry. This is class number twenty-three and I'm Dr. Sv Venkitesh, Civil Engineering. [00:00:17.028] Department of Science and Humanities under PS University. In the last class we had discussed about the problem three bar 31, I think. [00:00:31.140] So today we'll continue by solving some more problems. I have considered this problem, which is three bar thirty-two. The statement of the problem is cable A B passes over the small ideal pulley C without a change in its tension. [00:00:50.244] What length of cable C D is required for static equilibrium in the position shown. What is the tension T in cable C D? So if you read the statement and if you make out what is given here as you have a cable AB. [00:01:09.444] Which passes over a small pulley C without changing its tension. That means the tension in A C and the tension in C B would remain the same. That's what it means. Now with that. [00:01:27.364] Input, he has asked you to determine what should be the length of the cable CD, okay for static equilibrium for the given position, what is the tension T in the cable C D. So if you look at the diagram that is given. [00:01:45.700] Your ACB, this is the cable. So tension in this cable AC and C B would be the same, which would be equal to [00:01:56.868] The mass 20 kg into acceleration due to gravity. So that is 20 into 9.81, which will be the tension in this cable CB as well as the tension in the cable A C. Then he is also given in the diagram that this angle has to be 45. [00:02:16.100] For this condition, that means when this angle is 45, what would be the length of C D is what is asked. So this angle has to be 45, plus in addition to this angle, the vertical distance between AC should be equal to 600 mm. For this condition, [00:02:36.484] Mean you need to determine what should be the length of C D and also what is the tension in C D. So if you observe this, at this point there are three cabs, this A C B D, sorry, C D [00:02:56.324] And C B. So if I draw a free body diagram for this, this is how it's going to look. So since this is forty-five degrees, this would also be forty-five degrees because that is alternate angles. So that is what is indicated here. [00:03:10.852] The tension in this cable would be the same as the tension in cable C B, which is 20 into 9.81, that is how it is include indicated here. The tension in C D is indicated as T, the angle made by C D with respect to horizontal is considered as theta. [00:03:29.252] Okay. Then the tension in this cable is again twenty to nine point eight one. So this is what you can make out or you can draw as a free body diagram. And this distance is six hundred and now with this if you observe this there are three uh tension acting at a point. [00:03:47.140] So we can go for Lamis theorem or we can go for sine rule. Then what is the sine rule or Lamis theorem? Any force that you take where you have three forces acting at a point. [00:04:00.804] The force divided by the sign of the angle between the other two forces is a constant, or it is equal to the other force divided by the sign of the angle between the other two forces. Similarly, if you take this force, the sine of the angle between the other two, or they are all equal. So, by considering that [00:04:20.324] We know the tension in this. We know this is forty-

five, this angle would be 90 degrees. So this total angle would be 135. So when this is 135. [00:04:33.860] This would be equal to 90 plus theta, whereas this would be equal to 180 minus 45 minus theta. So I can take T divided by sine 135 to be equal to [00:04:50.468] This force divided by this angle, or this force divided by this angle. Okay. So using that I can solve the problem and then I can also find out.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:02.724] The distance C D once I get this angle theta, I know this distance. So using this angle I can find out what is the length that is required. Okay. So let's look at the [00:05:13.572] Solution. So here instead of using the Lamis theorem, of course we can also resolve the given forces along X and Y. So that is what is used here. So you resolve all the forces along X axis. So I get this as T cos teta, which is along X positive X. [00:05:32.388] Then 20 into 9.81, cos 45 will also be along X, but it is 9 minus. So that should be equal to 0. From this equation, I can easily find out what is T. Then you take all the sum of all the forces along y direction. So you get this 20 into 9.81. [00:05:50.724] Uh cost forty five. Uh [00:05:54.820] Or or cos 45 or you can also take it as sin 45 because since it is 45, this would also be 45. So 20 into 9.81 sin 45 is positive because it is acting upwards. Then t sine theta, then that would be the component along vertical, that would also be positive. [00:06:12.420] Then this one is negative, so that is twenty into nine point eight one equals zero. So from this [00:06:20.804] Since we have got T value from this equation, I can substitute here and find out what is angle theta. So by solving these two I can get the value of theta which is twenty two point five, as well as T which is one fifty point two newtons. [00:06:33.476] Using this angle theta and this uh triangle, right angle triangle, I can take tan theta to be equal to [00:06:41.252] Uh or rather I should take it as sine theta which is equal to 600 by Cd. So I can get the value of C D which is equal to 156 millimeters. Okay. So this problem. [00:06:57.348] You can solve it either using sign rule or Lamy's theorem, or you can resolve the given forces along vertical and horizontal, and then resolve. Here the solution, they have resolved the given forces along horizontal and vertical, then they have equated all the forces along vertical, and all the forces along horizontal, and then [00:07:17.232] Using these two equations the two unknowns which is T and Teta has been determined. Once you get the [00:07:23.268] Value of theta you can use the triangle C D C D and this triangle. Okay, if you take it as D dash and then you can use sine rule sine teta to find out what is the length C D. [00:07:38.308] Hope all of you have understood this. Right? So let's move on to the next problem. I have considered problem 3 bar 34 here. The statement of the problem goes like this. The small slider A is moved along the circular slot by a mechanism. [00:07:57.188] Attached to the back side of the rectangle or plate for the slider position theta equals 20 degrees, shown determine the normal forces exerted at the small stops C and D. [00:08:14.980] The unstretch length of the spring of constant k, which is equal to 1.6 kN per meter, is R by 3. The value of R is 25 mm, and the plate lies in the horizontal plane. [00:08:34.180] neglect all friction. So once you read the statement of the problem what you need to get from this is the angle theta is equal to 20 degrees and there are two stops at C and D. [00:08:50.884] The spring constant K is 1.6 kN meters. The unstretched length of the spring is R by 3 and R is equal to 25 mm. [00:09:05.188] Now if you read the diagram, what is that we can get? The radius R is this. Okay, that is AB is the radius, which is equal to R which is 25 mm. [00:09:18.852] The unstretched length of the spring is R by 2. That means if I don't stretch this, okay, if I don't elongate it, the length of the spring will be R by 3. So at any position that I keep this, [00:09:36.996] Okay, when I rotate it along this. So this distance

would be R, so the stretched length of the spring will be R minus R by 3. Okay. So unstretched length is R by 3, the stretched length is R here. [00:09:52.164] Okay, because it is A B the stretched length. So the the elongation of the spring will be equal to R minus R by 3. Then they have also given you this distance from this point to this point, it is R by 2. Similarly, from this point to this point is R by 2.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:10.436] The two stoppers is given here, which is C D. Okay. Now when I have the spring in this position, the tension that's going to be acting at A would be from A to B, the internal force would be acting from A to B along this. [00:10:28.324] Because of this, a component of the force will be acting upwards, a component of the force will be acting to the right. Because of this component at this position, which is an hinge here, so this would try to rotate this plate in the counterclockwise direction. [00:10:47.140] That would be prevented by the stopper here, so you will have a reaction at D. Whereas here there won't be any contact, so you will not have any reaction. But if this spring is moved above this horizontal in any position here, then you will have the [00:11:06.820] Tension in the spring acting like this from A to B. So because of which you will have a component acting downwards which will make the uh plate rectangular plate to rotate in the clockwise direction. At the top point you get the contact at C. So there will be reaction here, there won't be any reaction here. [00:11:25.796] Hope this is clear. So now if I draw the free body diagram of the plate, what do I get? I get an horizontal reaction here, a vertical reaction here, the sense can be anything. Then the force acting at A, okay, which should be at an angle of theta with respect to the [00:11:43.748] horizontal that is this angle theta so you will have two components here one will be whatever is the spring force f s this would be sine theta this would be cos theta then we will have a reaction at D which will be acting downwards okay. [00:12:00.740] So using this and the equations of equilibrium, I should be able to determine, what should I determine? I should determine the [00:12:11.812] Normal force exerted at C and D, that means at D, what is the normal reaction exerted? Now there is a mechanism which is at behind it, we need not to worry about it. So this won't come into picture because we are taking the forces acting on a plate only. So if I look at the free wire diagram, as I've told you, you can see here. [00:12:31.428] There's a spring force F S okay, which will be equal to K times delta. What is Delta R is equal to 25 mm which is 0.025 meters, F S is K delta [00:12:45.284] K into what is delta as I have explained it, it is R minus R by 3. So if I substitute the values here, I get the force Fs. By inspection here the reaction is 0, you get the reaction at D only. So I will take moment about O, sigma m at O equals 0. [00:13:05.572] The component of this force acting along the vertical is 26.7 sin 20 because this angle is given as 20 degrees. So this into the distance between this and this point, the [00:13:21.668] Horizontal distance, we need to calculate that. How do we calculate that? Now if you see this distance is R totally. Then they have given this distance as R by 2 and this also as R by 2. So if I want this distance, [00:13:40.644] Not this testance, we can calculate it as this is R. If I take the component of this along the horizontal, so I get this as R cos 20, which will be this. Okay. So totally this would this total is going to be [00:14:00.420] Uh 3 by 2 R because 1, 2, 3, 3 R by 2. So R is common. So what would be the distance from here to here? Would be this total R, this total is [00:14:19.012] R by 2, so that you have to take into account. Okay? I hope that is clear. What is clear? This total distance is 3 by 2 R. [00:14:35.108] That is 3 by 2 into R is 0.025. So we get that distance. Okay. Then minus R then we have this force D into this distance which is R which is

0.025. [00:14:53.476] Okay. So how do we get this distance? This is 1.5 R. So totally this is 3 by 2 R. So this distance from here to here would be equal to 1.5 R. Okay. So

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:12.932] That should be equal to 0. So only unknown in this equation is D. So if I calculate that D then we will get that to be equal to 13.68 newtons. I hope all of you have understood this. So this distance from here to here. [00:15:29.444] If there is a doubt, that would be equal to total distance, which is 3 by 2 r minus this distance, which is r cos 20 degrees. So I get this distance. Okay. So to calculate the moment. [00:15:47.268] 26.7 sine 20 is this into this distance which will be this component. There's one more component here. This is F S uh cos 20 degrees into this distance which will be equal to R [00:16:05.508] Cost 20 degrees. So that would give me this value. Okay? I hope that is clear to all of you. You can write down that and check it. [00:16:14.660] Okay. So or you can calculate what would be the perpendicular distance of this. Okay, from here, what would be the perpendicular distance? So to get that perpendicular distance, this is 3R by 2. Okay, if this is perpendicular, this would become the hypotenous. [00:16:34.692] Cause of that, okay, we will get this distance uh the sign of that we will get this distance. Okay, so this force into this distance will give me the moment here. So this is uh 26.7 sine 20, which is this. [00:16:51.908] You can do it in the other way, okay. If I directly want to take this force into this particular distance, that's one way of solving it, or I can resolve this force, okay, or I can take it to be acting at B itself. So this force, the component of that acting at B will be uh s. [00:17:12.452] Sine 20, which will be acting here, the total distance from here to here will be 3 by 2 r, and that's what is taken here. So there are three or four ways of calculating it. So here what they have done is they have taken the point of application of this Fs as being the component of this force acting along vertical. [00:17:31.876] Is Fs sin 20, which will be acting here. The distance from here to here will be 3 by 2 R. So that is what is taken here. Then they have taken this force into this distance. So you get the value of D here. So I have explained different ways of [00:17:51.088] Solving this problem by considering the force FS to be acting at B, or you can take the force to be acting at A, resolving that along vertical and horizontal, or you can take this force and find out what is the perpendicular distance. I have talked about all the three different ways of solving it. [00:18:09.476] But what is given here, the equation, is you are taking the force Fs to be acting at B, only this component will come into picture, whereas the horizontal component, since it is passing through O itself, it doesn't have the perpendicular distance, so that would not come into picture. [00:18:29.636] It is clear. If you have any doubts, you are always welcome to contact me and talk to me. I will explain it to you. Okay? Right? So we will end today's class here. We will continue this class in the next class. That is class number. [00:18:48.400] 24. Okay. So and we will try to solve some more problems. Okay. So thank you very much. And I am Dr. Sv Vintesh from Civil Engineering under the Department of Science and Humanities under PS University. Thank you. See you in the next class.

EV0000102 · EV0000103 · EV0000104 · EV0000105

Full transcript: 2882 normalized words, 69 timestamped segments, 1146.795 seconds duration. Low-confidence segments flagged: 0.

EV0000090 · EV0000091 · EV0000092 · EV0000093

8 UE26ME142A_Unit 2_Class 19.1_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 19.1_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000106 · EV0000107

Graphs and diagrams



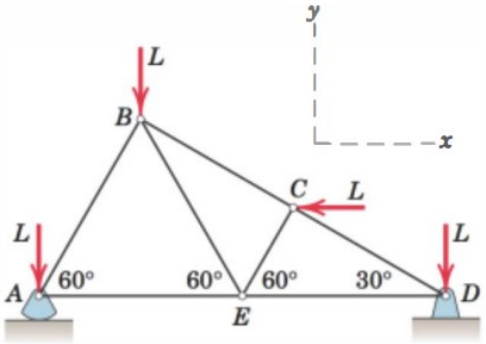
Figure G000043. UE26ME142A Unit 2 Class 19.1 Lecture PPT · source slide page 1.

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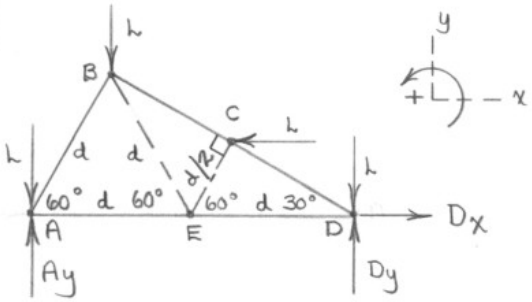
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS
STATICS

3/35 The asymmetric simple truss is loaded as shown. Determine the reactions at A and D. Neglect the weight of the structure compared with the applied loads. Is knowledge of the size of the structure necessary?



Problem 3/35



S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

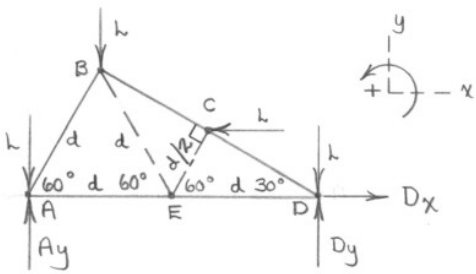
Figure G000044. UE26ME142A Unit 2 Class 19.1 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS
STATICS

3/35



$\Sigma F_x = 0: \; D_x - L = 0, \quad D_x = L$

$\Sigma F_y = 0: \; A_y + D_y - 3L = 0$

$\Sigma M_A = 0: \; D_y(2d) + L\left(\frac{d}{2}\frac{\sqrt{3}}{2}\right) - L\left(\frac{d}{2}\right) - L(2d) = 0$

Solving the last 2 equations : $A_y = \frac{L}{4}\left(7 + \frac{\sqrt{3}}{2}\right)$

$D_y = \frac{L}{4}\left(5 - \frac{\sqrt{3}}{2}\right)$

(or $A_y = 1.967L$, $D_y = 1.033L$)

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Figure G000045. UE26ME142A Unit 2 Class 19.1 Lecture PPT · source slide page 3.

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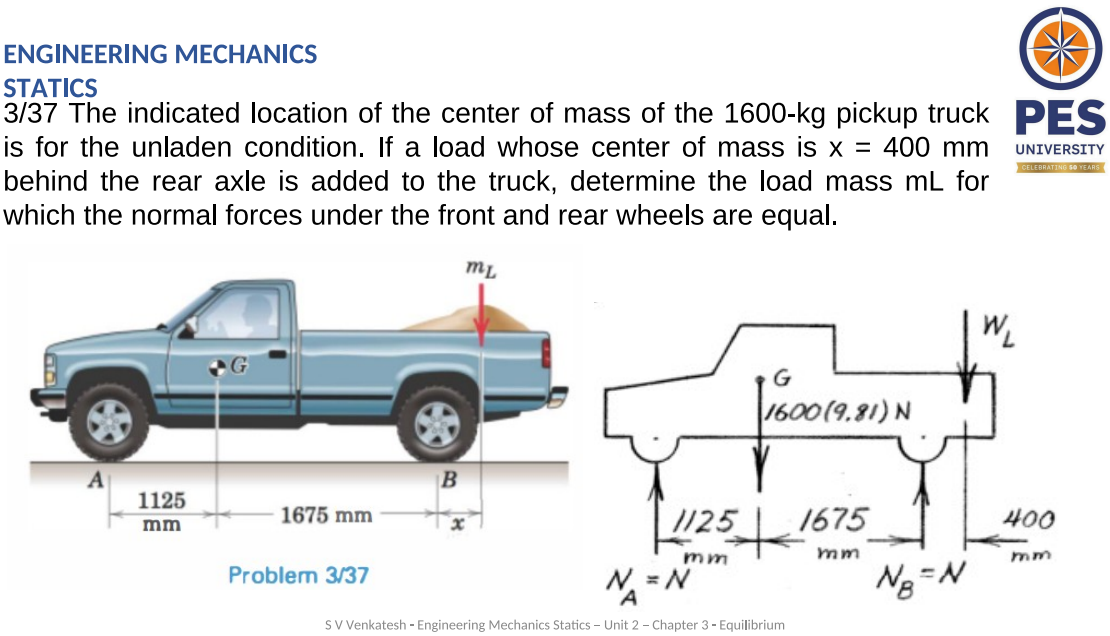


Figure G000046. UE26ME142A Unit 2 Class 19.1 Lecture PPT · source slide page 4.

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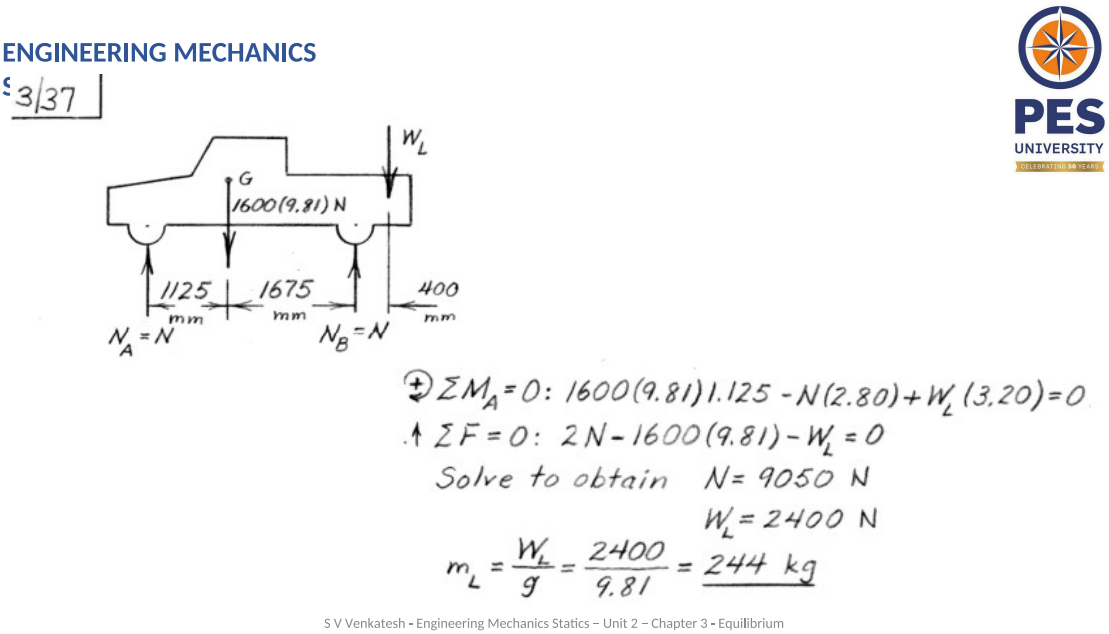


Figure G000047. UE26ME142A Unit 2 Class 19.1 Lecture PPT · source slide page 5.

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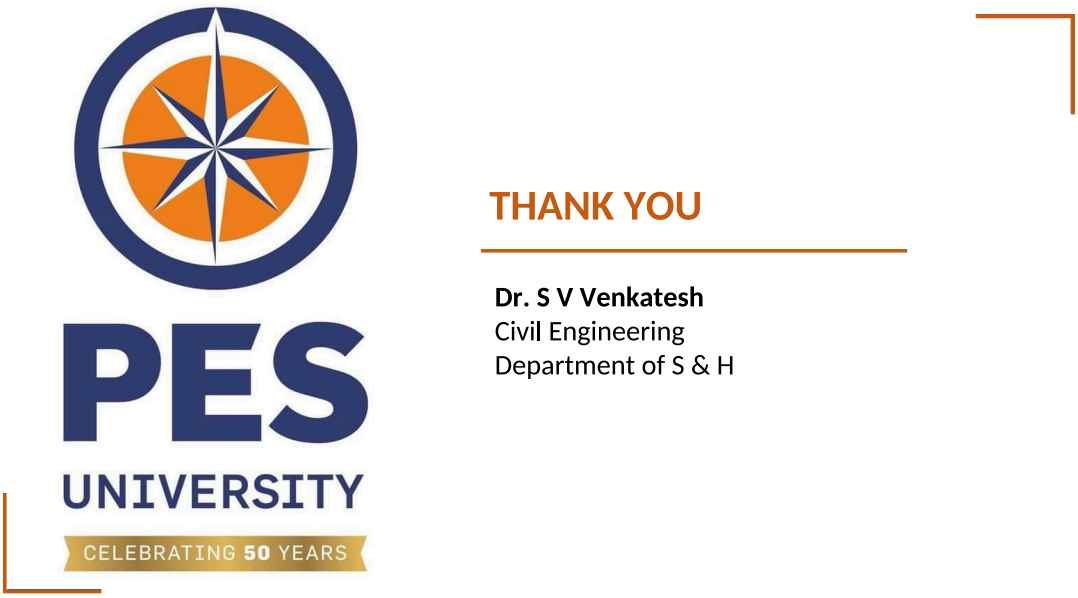


Figure G000048. UE26ME142A Unit 2 Class 19.1 Lecture PPT · source slide page 6.

EV0000361

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000348 · EV0000349 · EV0000350
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 3/35 The asymmetric simple truss is loaded as shown. Determine the reactions at A and D. Neglect the weight of the structure compared with the applied loads. Is knowledge of the size of the structure necessary? S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000351 · EV0000352 · EV0000353
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000354 · EV0000355 · EV0000356
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 3/37 The indicated location of the center of mass of the 1600-kg pickup truck is for the unladen condition. If a load whose center of mass is x = 400 mm behind the rear axle is added to the truck, determine the load mass mL for which the normal forces under the front and rear wheels are equal. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000357 · EV0000358 · EV0000359
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 6 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000360

6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000361

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.764] Hello, dear students. Welcome to Engineer Mechanics Statics classes. This is class number twenty four and I am Dr. Esvin Venkatesh from Civil Engineering Department under Department of Science and Humanities, PS University. In the last class that is class number twenty-three, we have [00:00:20.772] Worked out two problems on equilibrium. We'll continue to solve some more problems in today's class. That is class number twenty uh four. I've taken a problem three bar thirty-five, the statement of the problem goes like this. [00:00:38.692] The asymmetric simple truss is loaded as shown. Determine the reactions at A and B, neglect the weight of the structure compared with the applied loads is knowledge of the size of the structure necessary. [00:00:57.284] So if you go through the statement of the problem, you have to identify that what is given as an asymmetric truss. Truss is a structural member which is used for carrying out load from the roof to the columns that is supporting this truss. [00:01:17.060] We are supposed to determine the reactions at A and D. But to neglect the weight, sulfate of the structure itself, because that is going to be very, very small when compared to the load that is coming on the process. And they have also asked you to. [00:01:37.668] Let them know whether the size of the structure matters or not. And if you read the diagram that is given, this is the diagram that is given. So at A, there is a roller support. Okay? And there is a load of L acting vertically down. [00:01:57.828] Similarly, at B there is a load of L that is acting vertically down. At C there is a horizontal load which is acting to the left, which is negative of the [00:02:08.900] Axis that is given here, the positive axis X and Y. So this is this load is acting in the negative direction of X, that is horizontal direction. Then there's a load at D which is acting vertically down that is also negative. Then you need to identify that the support at D is an hinge support. [00:02:27.908] Then they have also given you the angle made by AB with respect to the horizontal which is EAE that is sixty degrees. This angle that is the angle made by the side B with EA is also sixty degrees. So this forms a triangle. [00:02:47.012] So when these two are 60 degrees, obviously this would also be 60 degrees. That makes this as an equilateral triangle. There were also even this angle as 60 degrees. So since this is 60, this is also 60, you can say that the side EC is parallel to AB. [00:03:06.852] Then they have also given this angle as thirty degrees, that is the angle made by the uh side B D with uh DA, which is thirty degrees. Now they have not given you the distances of any of the sites and that's what is asked. [00:03:25.764] Does the size of the structure matter or not? So they are not given. So how do we go about solving this? So to solve this, write the free body diagram which is given here. Replace the [00:03:41.444] Roller support at A by a reaction which is represented here as AY. Replace the reaction at D which is an in support by the reaction. There will be two reactions, one vertical and one horizontal. [00:03:57.764] Then apply the same forces whatever is there acting at different point at this. Okay. The sign convention, x axis positive to the right, y axis upwards positive, and anti-powered rotation as positive. Now I would [00:04:17.572] take AE distance orbitally as some distance D. So when I represent that as D, obviously your AB and BE will also be having the same distance because that forms an equilateral triangle. So that was it, that is what is mentioned here. [00:04:38.148] Next, if you observe ABD, this triangle. Okay. They have given this triangle as 60, this is 30, so obviously this has to be 90 degrees. Similarly, if you take the triangle, [00:04:56.420] C D E. This is sixty degrees, this is thirty

degrees, so this has to be ninety degrees. So this triangle AB and C D E. They are similar triangles. Okay, so

EV0000106 · EV0000107 · EV0000108 · EV0000109

Transcript block 2 (source interval 000300-000599 seconds): [00:05:16.100] If this is D and this triangle C D E and ABD are similar. So this angle also has to be 90 degrees. So this angle will also be 90 degrees. [00:05:33.604] So if this is ninety degrees, this is sixty, sixty, I can get this angle as sixty itself, because the total would be one eighty. So if this is sixty, this is ninety. What is remaining is? 180 minus 90 minus 60 is 30. So this would be 30 degrees. [00:05:51.332] So this forms a right angle triangle. So this distance BE is given as D. So using dichromatic relationship I can get the distance CE as D. [00:06:09.572] Cos 60 degrees, which will be D by 2, that is what is mentioned here. Okay. Next, using these similar triangles, since they are similar triangles, if this is D, this E point. [00:06:27.780] Okay, as to be the midpoint of this triangle because A B should be proportional to BE. AB should be proportional to E C. Similarly, your B should be proportional to [00:06:46.084] C D. So if that is the case then this entire length should be proportional to this also. So if I take this as D, it will be proportional only if E D is also D. So I can straight away write this as D or this point E is the midpoint of A D. [00:07:05.636] So if this is D then this also has to be D. And that's what is mentioned here. Okay. So with this uh findings, we can now go and apply the equations of equilibrium, the three equations of equilibrium. One is the moment about any point is zero. [00:07:24.068] So for equilibrium we have to take the point where we have the maximum unknowns. So probably we can take it as D. Then you can take sigma f [00:07:33.668] X to be equal to 0, that is sum of all the horizontal forces should be equal to 0. So by doing that I will get the value of the reaction, horizontal reaction at D, which should be equal to L. Then if I take some of all the vertical forces to be equal to 0, I get [00:07:53.572] A y and dy to be equal to L plus L plus L 3L. Or if I take the moment about D, I would have I would get the value of AY. So using that value, I can get the value of DY. So let's look at the solution here. [00:08:12.740] So the same free body diagram is mentioned here also. So first sigma f x equals zero sum of all the forces. So dy is the unknown which is acting in the positive x direction. So dy whereas there is only one horizontal reaction, horizontal force that is acting at C. [00:08:31.940] Which is in the negative direction, so dx minus l equals 0. So I'll get dx to be equal to l straight away. Then you take sigma f y equals 0. So I get f y plus dy. [00:08:45.412] To be equal to L plus L plus L as I have told you, so that is acting in the downward direction. So minus 3 L equals 0. Okay, that is one equation. Then taking moment either I can take D as the moment or A as the moment because here already we have got the value of DS. [00:09:03.684] So only unknown left is dy. So if I take the moment about A I get Dy into this distance which is 2D, that's what is mentioned here. That would be anti clockwise or positive plus L. [00:09:22.500] That is this L which will produce an anti-clockwise movement at A. So this L into what is the perpendicular distance. This would be the perpendicular distance. Now how do I get this distance? So they have written it as D by 2, which is this. [00:09:40.100] This angle is 60, so sine 60 I get this distance. Okay? So D by 2 sin 60 I get the vertical distance. That is root 3 by 2. So that is what is mentioned here. Next, y. [00:09:57.796] into D by 2 this L would produce a clockwise movement okay but this L into the perpendicular distance would be this which is definitely D by 2 because this is an equilateral triangle so this would be the midpoint of this base A

EV0000110 · EV0000111 · EV0000112 · EV0000113

Transcript block 3 (source interval 000600-000899 seconds): [00:10:15.204] So this distance I have considered as as D, so this would be D by 2, and that's what is mentioned here. Then we have this force also, that is that would also produce a clockwise movement, so minus L into this is D, this is D, so 2D should be equal to 0. So now solving these two equation. [00:10:34.948] We will get the value of AY and DY. So this is the value of AY and DY. Simplifying this, I would also get the value of AY and DY in terms of L. Because L is the load that is applied on the structure. Now looking at this diagram. [00:10:56.260] D can have any value. Now if you see the reaction values a y dy and dx, nowhere it is reflecting this d value. So whatever may be the value of d [00:11:12.708] This reactions would not change, it would remain same. So the knowledge of the size of the structure is not required to know or to get the reactions, provided you are not considering the cell fate of the truss. [00:11:29.764] Right, that should be the answer that you need to write here. The knowledge of the size of the structure is not influencing the support reactions. Okay? Hope you have understood this. Let's go to the next problem. The next problem is 3 bar 37. The statement of the problem goes like this. [00:11:50.756] The indicated location of the center of mass of the 1600 kg pickup truck is for the unladen condition. If a load whose center of mass is X, which is equal to 400 mm, [00:12:09.636] Behind the rear axis is added to the truck. Determine the lower mass M L for which the normal forces under the front and rear wheels are equal. So from the statement, what we can make out is [00:12:29.476] The centre of mass of 1600 pickup truck. 1600 is the total weight of the truck. And the centre of mass is for the unladen condition, means there is no load there. So they have given this G as the [00:12:49.348] Center of mass, then when there is a load that has been applied such that the center of mass of this load is at a distance of 400 mm from the rear axle of the truck, you need to determine [00:13:08.676] What would be the mass M L for which the normal reaction under the front and the rail wheel are equal. So if I look at the diagram here. [00:13:27.652] The G is the center of mass of the truck where your thousand six hundred kg low or mass will be acting. Then you have the mass of the load ml which is act at a distance of 400 mm from B. [00:13:44.996] There are two wheels if you are in contact with the surface, so you will have a reaction at this point, A and B. What is given in addition to that is what is the distance from A to the centre of mass of the truck, which is 1125. [00:14:01.892] What is the distance between central mass of the truck and the point B, which is 1675 mm? X distance is given as 400 mm. We are supposed to find out mm for the condition. [00:14:15.332] The reaction at A and reaction at B are equal. There are two reactions at A, one under this wheel what we are looking at. There is one more wheel on the earth. [00:14:26.180] Back side of A, so we can take it as 2 R A. Similarly here there is one wheel, there is one more wheel the back side. So we can take that as uh 2 R B. So 2 R A and 2 R B they have to be equal. 2 R A should be equal to 2 R B, R R A should be equal to R B. [00:14:43.364] And here he has taken that as NA, here he has taken it as N B which is equal to N. Both of them are equal. For that condition we have to find out ML. That is what is given in the problem. So if you look at the free body diagram, 1600 into 9.81 is the force that is acting at G.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:01.380] Then you have the weight ml or rather WL that is acting on the truck, but we have to find out what is the mass ml. So you divide this by 9.81, you will get it ml. Then you have the reaction at A, reaction at B, both of them are N and N itself. [00:15:20.132] So again use the equations of equilibrium, the three equations of equilibrium. Sigma Fx equals 0, that will not come into picture because there is no horizontal force, so no horizontal reaction. Then you take sigma f y equals 0, so you get 2n equals

[00:15:38.116] 1609.81 plus WL. Then you can also take moment about any of the supports, say I will take it at A or at B. Okay, or I can take it under the load W L also. So if I take it at [00:15:56.676] This point I will get this N into this distance, which is thousand one hundred and twenty-five plus thousand six hundred and seventy-five, which will be in the anti-clockwise direction. Whereas this force will produce a clockwise movement, this force into this distance. [00:16:12.420] Then this force into the total distance would also produce a proper moment. Using these two equations I can find out these two or not, which is N and WL. Same thing is what is given here. So first taking moment about point A. [00:16:30.564] Okay, 1600 into 9.81 into 1125. This is in the clockwise direction, so taken as positive. Then n which is this into this distance, you will get it as 2800 convert it into [00:16:47.300] uh meters which will be 2.8. This also converting it into meter it is going to be 1.125. Then we have this force which will produce a clockwise moment. [00:16:59.844] So that is positive WL into this total distance from here to here, which will be thousand one twenty-five plus thousand six seventy five plus four hundred, which you will get as three thousand two hundred. So converting it into meters it is three point two. [00:17:14.628] So that should be equal to zero. Solving this, you can see that there are two unknowns, N and W L. Okay, so you can express one in terms of the other. Then taking all the forces along the vertical to be equal to zero. [00:17:29.412] You get 2n because these two are the vertical forces acting upwards should be equal to this which is acting downwards, thousand six hundred into nine point eight one. Then we have this also which is acting downwards, so minus W L. So again here also there are two unknowns. By solving these two equations, [00:17:48.164] I will get the value of N and I will get the value of WL. So N value is 9050 nuclear and W L as two thousand four hundred by solving these two equations. So once I get the force that is W L acting [00:18:05.476] Because of the loaded or the load on the truck, I would divide it by acceleration due to the graphic to get the mass of the load that is added. So 2400 divided by 9.81, which will give me the mass of the load that is added on the truck. [00:18:25.188] such that the reactions at A and B are equal, which is 44 kgs, and the center of this mass should be at a distance of 400 mm from B. That is also given in the problem. [00:18:43.204] Hope you have understood this problem also. Okay? So today we have solved two more problems. This is class number twenty. [00:18:56.132] Or we will continue the same thing in our next class. We'll try to solve few more problems on equilibrium chapter and try to understand how we solve problems. [00:19:14.372] using the equations of equilibrium. Right? Thank you very much. I am Dr. SV Vengtish from Civil Engineering Department of Science and Humanities PS University. See you in the next class. Thank you

EV0000118 · EV0000119 · EV0000120 · EV0000121

Full transcript: 2813 normalized words, 66 timestamped segments, 1171.605 seconds duration. Low-confidence segments flagged: 0.

EV0000106 · EV0000107 · EV0000108 · EV0000109

9 UE26ME142A_Unit 2_Class 19.2_AVSummary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 19.2_AVSummary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or

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Graphs and diagrams

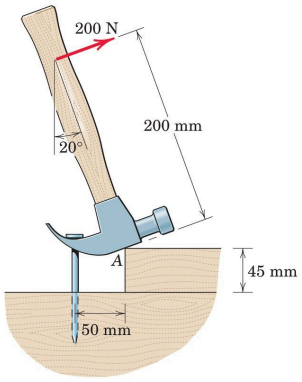


Figure G000049. UE26ME142A Unit 2 Class 19.2 Lecture PPT · source slide page 1. EV0000363

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

3/40) A block placed under the head of the claw hammer as shown greatly facilitates the extraction of the nail. If a 200-N pull on the handle is required to pull the nail, calculate the tension T in the nail and the magnitude A of the force exerted by the hammer head on the block. The contacting surfaces at A are sufficiently rough to pre



S V Venkatesh – Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

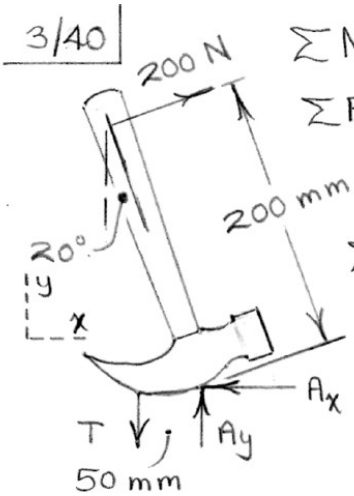
Figure G000050. UE26ME142A Unit 2 Class 19.2 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS

3/40



$\Sigma M_A = 0 : 200(200) - 50T = 0, \quad T = 800\ N$

$\Sigma F_x = 0 : 200 \cos 20° - A_x = 0$

$A_x = 187.9\ N$

$\Sigma F_y = 0 : A_y + 200 \sin 20° - 800 = 0$

$A_y = 732\ N$

$A = \sqrt{A_x^2 + A_y^2} = 755\ N$

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

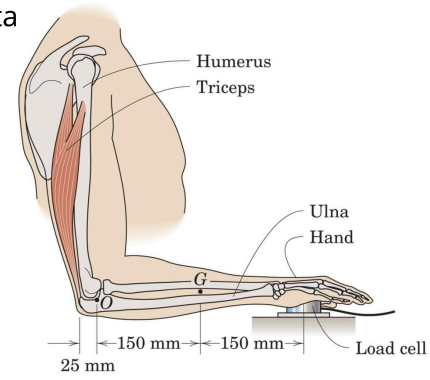
Figure G000051. UE26ME142A Unit 2 Class 19.2 Lecture PPT · source slide page 3.

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ENGINEERING MECHANICS STATICS

3/43) In a procedure to evaluate the strength of the triceps muscle, a person pushes down on a load cell with the palm of his hand as indicated in the figure. If the load-cell reading is 160 N, determine the vertical tensile force F generated by the triceps muscle. The mass of the lower arm is 1.5 kg with mass center at G. Sta



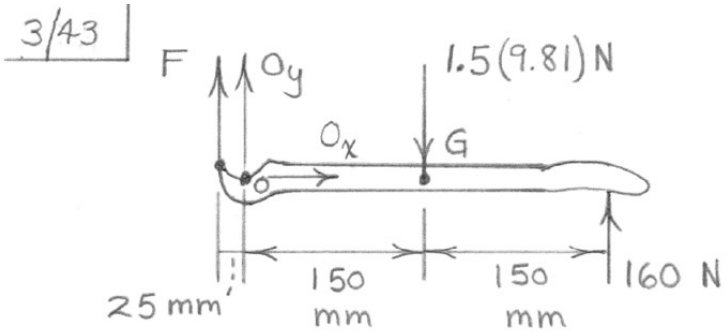
S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

Figure G000052. UE26ME142A Unit 2 Class 19.2 Lecture PPT · source slide page 4.

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ENGINEERING MECHANICS STATICS



$$\curvearrowleft \sum M_O = 0: \; -F(25) - 1.5(9.81)(150) + 160(300) = 0$$

$$F = 1832 \text{ N}$$

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium

Figure G000053. UE26ME142A Unit 2 Class 19.2 Lecture PPT · source slide page 5.

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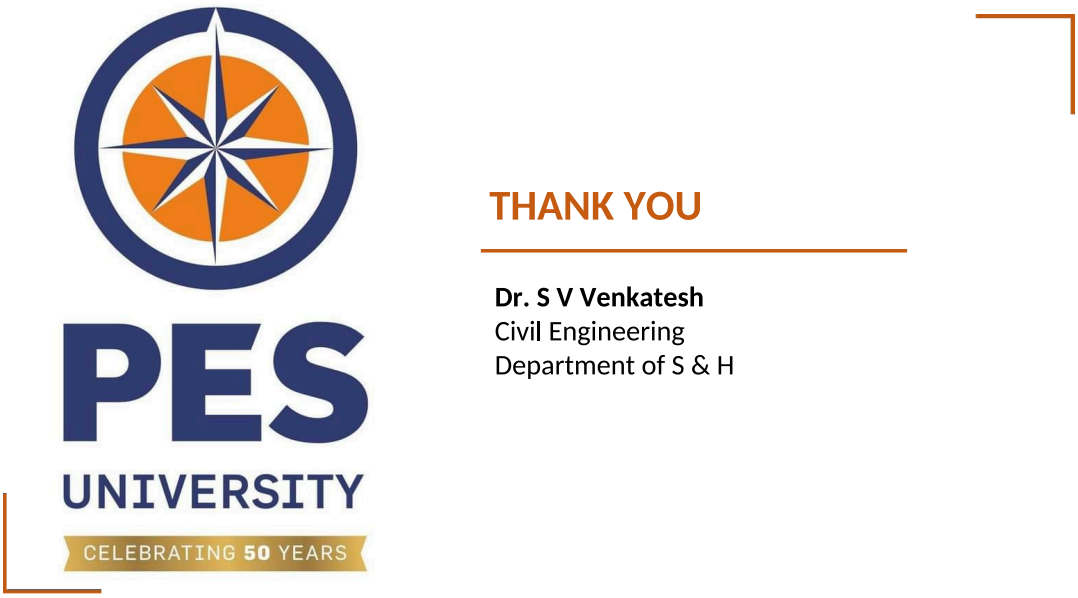


Figure G000054. UE26ME142A Unit 2 Class 19.2 Lecture PPT · source slide page 6.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000362 · EV0000363 · EV0000364
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 3/40) A block placed under the head of the claw hammer as shown greatly facilitates the extraction of the nail. If a 200-N pull on the handle is required to pull the nail, calculate the tension T in the nail and the magnitude A of the force exerted by the hammer head on the block. The contacting surfaces at A are sufficiently rough to prevent slipping. S V Venkatesh – Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 3 (OCR) ENGINEERINGMECHANICSSTATICS 3/40 MMA=0 :200(200)-50T=0,T=800 N PES 200 UNIVERSITY CELEBRATING50YEARS MFx=0:2o0Cos2o-Ax= Ax = 187. 9 N 2o0mm ZFy 20 =0:Ay+200sin20-800= 0 iy 732N Ax A= 755N T 50mm S V Venkatesh - Engineering Mechanics Statics - Unit 2 - Chapter 3 - Equilibrium EV0000365 · EV0000366 · EV0000367
3. Source slide page 3: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000368

4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 3/43) In a procedure to evaluate the strength of the triceps muscle, a person pushes down on a load cell with the palm of his hand as indicated in the figure. If the load-cell reading is 160 N, determine the vertical tensile force F generated by the triceps muscle. The mass of the lower arm is 1.5 kg with mass center at G. State any assumptions. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 5 (OCR) ENGINEERINGMECHANICSSTATICS 3/43 PES 1.5(9.81) N UNIVERSITY CELEBRATING50YEARS 150 150 1160N 25mm mm mm ∑M °=0 : -F(25)- 1. 5(9.81)(150) + 160(300) 二0 F=1832N S VVenkatesh-Engineering Mechanics Statics - Unit 2 - Chapter 3-Equilibrium ## Page / slide 6 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000369
5. Source slide page 5: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000370
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000371

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:02.308] Dear student, welcome to engineering mechanics statics classes. My name is Dr. Svingtish from Civil Engineering under the Department of Science and Humanities, PS University. This is class number [00:00:15.876] 25. We are solving problems related to equilibrium. We will continue to solve some more problems on this uh chapter equilibrium. So I have c taken this problem 3 bar 40. [00:00:32.644] The statement of the problem goes like this: a block placed under the head of the claw hammer as shown greatly facilitates the extraction of the nail. If a 200 newton pull on the handle is required to pull the nail. [00:00:51.844] Calculate the tension T in the nail and the magnitude A of the force exerted by the hand or head on the block. The contacting surfaces at A are sufficiently rough to prevent slippage. Now if you [00:01:11.364] Go through this statement, what is that we can understand as there is an hammer used to pull out a nail. A force of 200 newton is applied onto the handle. They have asked us to determine [00:01:31.140] The tension T in the nail and the magnitude A of the force exerted by the hamorate on the block. Okay. Now if you read the diagram, you can note down. [00:01:44.644] That the force applied onto the hammer, that is 200 newtons, is perpendicular to the axis of the hammer, which passes through A, and this axis of the hammer is at an angle of 20 degrees to the vertical. [00:02:03.652] The distance between point A and the point where the force is applied is 200 mm along the axis of the hammer. The nail is at a distance of 50 mm along the horizontal from point A, and at A, [00:02:21.924] What ES given is that it is the friction is quite sufficient the or the roughness is sufficient to take up the contacting uh surfaces at A. Okay. So the free body diagram has to be [00:02:39.652] The force applied on the hammer, 200 newtons. The distance between the force and the point A. Then the tension in the nail. The tension means the nail has been pulled. [00:02:56.580] Upwards. So tension in the nail will be acting downwards. So we have a downward force that is there. Then at point A we will have a normal reaction which is passed through the axis of the hammer and a friction which will prevent the [00:03:16.100] Slipping of the hammer. So this would be the free wire diagram, and of course the distance of the nail from A is also should be written. Okay. So if you look at the free wire diagram, this is what it is. The 200 newton force acting on the angle of the hammer. [00:03:35.268] the angle made by the axis of the hammer with respect to the vertical 200 twenty sorry twenty degrees. [00:03:43.556] Then the

distance between the force 200 newtons and point A, that is 200 mm, the tension in the nail which is acting downwards because it's the internal force P and the distance between A and this tension should be. As I told either you can take the [00:04:00.996] Normal reaction acting on the hammer along the axis of the shaft and the frictional force. Or we can also take the vertical reaction and the horizontal reaction that is acting at A, considering it to be a rough surface. [00:04:20.356] So these are the forces that's going to be acting on the ammo. Now taking moment about A, we can find out what is the tension T that is moment about A is equal to 0, so T into 50 should be equal to 200 into 200. [00:04:36.676] So straight away we get the tension in that which is equal to 800 newtons. Then take all the forces acting along the x direction and equally to 0. So we get Ax which is at which we have considered to be acting to the left. Then we have a force of 200. [00:04:54.212] Cost 20 degrees because this angle would be 20 degrees with respect to the horizontal. 200 cos 20 degrees. Only two forces are acting along the x axis. So we get a x there. Then taking all the forces along y, we have a by acting upwards minus t

EV0000122 · EV0000123 · EV0000124 · EV0000125

Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.636] Then a component of this which is also acting upwards, which is two hundred sine twenty degrees, that's what is given here. [00:05:22.660] Uh substituting the value of T as 800 newtons, we get AY which is equal to 725 newtons. Then we can find out what is the resultant of Ax and AY, which will be A which is equal to 755 newtons, which will be acting one force reaction at A. [00:05:42.148] Hope everybody has understood this problem. Now here again to reiterate what we are doing here is just using the equation of equilibrium for a two-dimensional body. There are three equations and what we are finding out is also three unknowns. [00:05:59.748] So three equation three numbers should be enough to solve this problem. The three equations of equilibrium is algebraic sum of all the movements about any point should be equal to zero if the body is in equilibrium. Then algebraic sum of all the forces along X direction or the component of all the forces along X direction should be equal to zero. [00:06:18.148] The algebraic sum of the components of all the forces acting along y direction should be equal to zero. And that's what we have used to solve this problem. Hope everybody has understood this problem. Okay. Next problem. [00:06:36.964] 3 bar 43. The statement of the problem goes something like this. In a procedure to evaluate the strength of the triceps muscle, a person pushes down on a lower cell with the palm of his hand. [00:06:57.924] in the figure. If the load cell reading is 160 newton, determine the vertical f tensile force generated by the triceps muscle. [00:07:15.236] The mass of the lower arm is 1.5 kg with mass center at the G. State any assumptions. So if you read the problem what we we have to take from the problem is that the reading in the load cell is 160 newtons. [00:07:33.604] The force is applied by the palm of the hand. Okay. So when you are pushing the load cell with your palm lower, the reaction that's gonna come would be I think about what you are applying a force lower. [00:07:53.604] No, if you push your hand in air, you won't feel any uh force in the tricept muscles. But if you have something which prevents then if you try to [00:08:05.444] Push it. You can try this. If you try to push it, you will you can feel that there is there is some strain or you can feel the uh force in the bicep muscles. Okay. So that force that is generated is 160 newtons. [00:08:24.836] He has asked you to determine the vertical tensile force in the tricep muscles because of the force that is exerted by the palm onto the low side. [00:08:36.804] Whatever is the force that you are applying by the palm, okay, there is that force is coming because of the tricep muscles. What would be the force experienced by the tricep muscles? Because of this force that you are generating. [00:08:50.884] At

the load cell. Okay. Now along with this, the mass of the lower arm is also given, which is 1.5 kgs, and the point of application of the mass of the lower arm is at G point. Then he has told to state any assumption. Okay. [00:09:10.468] Now looking at the diagram, they have given several things here, but what we need to take is that there is a load cell here that is below the palm. Then you have the point G which is at a distance of 150 mm from the load cell, horizontal distance. [00:09:28.484] Then we have the joint in the arm. The joint is at O. [00:09:33.988] The distance from G to that joint is 150 mm. This joint O will act as an hinge. Okay. Because you are able to move the arm. Because of which you have apply the force there. So that will act as an hinge. There will be two reactions, one vertical and one horizontal. Then they have given the small distance 25 mm. [00:09:54.500] which gives you where the tricep muscle is. Now here we will assume that this tricep muscle is vertical and the distance from O to that point is 25 minutes. That's the assumption. That's what he has told. State any assumption the assumption is that

EV0000126 · EV0000127 · EV0000128 · EV0000129

Transcript block 3 (source interval 000600-000899 seconds): [00:10:13.892] We are assuming that the tricep muscle is vertical. Okay. So what we need to draw is only this part with one sixty N force acting upwards. [00:10:25.988] Then G that is the way mass is acting downwards. Then you have the reaction at O, then tricep muscles which will be acting upwards. So the free body diagrams look like this. Very important thing is you have the joint at O which will act as an hinge. So you have two reactions, one vertical and one horizontal. [00:10:46.340] The mass of the arm is acting G, 1.5 kg into 9.81. The lower cell, the sense of the lower cell, when you apply this force, the reaction that you get it is acting upwards. That is why you get the [00:11:01.520] uh pain in the tricep muscles and that is acting up force. That sense of that force 160 un you have to clearly mention and or take it to be acting of course. Then only you will be able to get the force in the tricep muscles. So this is what is a free body diagram that you will get. Very simple free body diagram [00:11:21.220] Along with the forces, the distances of each and every point, that is from O to the force in the tricep muscle or the tricep muscle itself is 25 mm, O to the G point, the horizontal distance is 150 mm. [00:11:37.796] Then G2, the low cell point is also 150 mm. So again, you apply the equations of equilibrium here. They have asked you to determine the force in the tricep muscles only. So what we can do is you take the moment about point O. [00:11:55.428] Okay, when I take the moment about 0.1, what do I get? I get f into 25 which will produce a clockwise moment. I will consider that as negative. So minus f into 25. [00:12:08.452] Then we have the self rate of the R, which is 150 to 9.81, into the distance, perpendicular distance is 150, which will also produce a clockwise moment. So that is also negative, minus 1.5 into 9.81, into the distance is 150. [00:12:28.228] Now the force in the load cell which is acting upwards 16 untons that would produce an anti-clockwise moment. This would produce an anti-clockwise moment at point O. [00:12:43.236] So that 160 into the total distance is 150 plus 150, which is 300, and that should be equal to 0. Now from this equation there is only one unknown that is your force in the tricep muscles, that is F. So we can get F equals [00:13:01.188] 1832 Newton from this. Okay. Now you can ask, sir, you have considered only one equation of equilibrium. What about the other two? So here we have considered the [00:13:20.068] Moment at point O only. Right? And at point O you have the unknown as OY and O X. So when we consider at point O the moment, this unknown O X and OY will not produce any moment. [00:13:37.828] But in the problem since they have asked only the force in the tricep muscle, we are not bothered about OX and OY. [00:13:45.764] Further, if you would have asked you to determine the reaction that you get at O, then you should have taken

sigma f x equals 0, sigma f y also to be equal to 0, and then you should have calculated the value of Ox and OY. [00:14:03.908] Since that is not asked, we are not interested in the other equations. Okay. Now other than point O, if you would have taken the moment about any other point, say for example, if I would have taken the moment about G. [00:14:22.244] You would have got F into 150 plus 25, which is 175 as the clockwise moment. Then you would have got OY into 150 as again the clockwise moment. Then you would have got 160 into 150 as the anti-clockwise moment. [00:14:39.908] So in that equation there would have been two unknowns, one is the F, and then OY. Now to solve this, we should have generated one more equation, which would have been sigma f y equals 0, then we would have got F plus OI. [00:14:57.540] Minus 1.5 into 9.81 plus 160 to be equal to 0. By solving those two equations, we could have got the value of f as well as OY. However, the value of Ox would be 0.

EV0000130 · EV0000131 · EV0000132 · EV0000133

Transcript block 4 (source interval 000900-001199 seconds): [00:15:14.372] Because there is no horizontal force that is acting on the arm. So when there is no action, there won't be any reaction because all reactions are passive forces. Okay. There is no active force, so you may not have passive force, that is no reaction. When there is no action, [00:15:34.084] There won't be any reaction. Okay. So again this problem we have used only the equations of equilibrium. To use the equations of equilibrium, an appropriate free body diagram has to be written. [00:15:51.940] To write the appropriate free body diagram, you should identify what are the forces acting, what are the supports that you get, and how many reactions are going to be acting at that support. See, taking all this into consideration, you have to draw the free body diagram. [00:16:11.044] If you neglect that there is a joint at O, if you don't take these two, then there is chance of you making mistake. So kindly remember and note down that you have to consider appropriate [00:16:30.084] Support reactions at the supports. Okay, depending upon what is the type of support it is. So if it is a simple support or if it is a cable support, only one reaction. If it is an hinge, there will be two reactions, one vertical and one horizontal. [00:16:46.980] If it is a built-in support, there will be three reactions, one vertical, one horizontal. Along with that, there is a moment also. Okay, if it is a roller support, you will get only one reaction. That I have discussed very much in the introduction classes of [00:17:04.644] the equilibrium chapter okay so writing a free body diagram is very very important and fifty percent of the problem is solved if you write the correct and appropriate free body diagram [00:17:23.620] If you make a mistake in writing the free body diagram, the entire solution that you are going to get will be wrong and you will not get the appropriate or the correct answer. So if you do that, then you will lose marks. [00:17:39.780] in the examination as well as in the test. Okay. I hope all of you have understood these two problems clearly. Okay? We will stop today's class here. We will continue in the next class. That is in class number 26. [00:17:55.108] We will try to solve few more problems where we will try to understand few other concepts what needs to be applied to solve such problems. Okay. So we have come to the end of today's.

EV0000134 · EV0000135 · EV0000136 · EV0000137

Full transcript: 2599 normalized words, 64 timestamped segments, 1092.544 seconds duration. Low-confidence segments flagged: 0.

EV0000122 · EV0000123 · EV0000124 · EV0000125

10 UE26ME142A_Unit 2_Class 19.3_AV Summary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 19.3_AV Summary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000138 · EV0000139

Graphs and diagrams



Figure G000055. UE26ME142A Unit 2 Class 19.3 Lecture PPT · source slide page 1.

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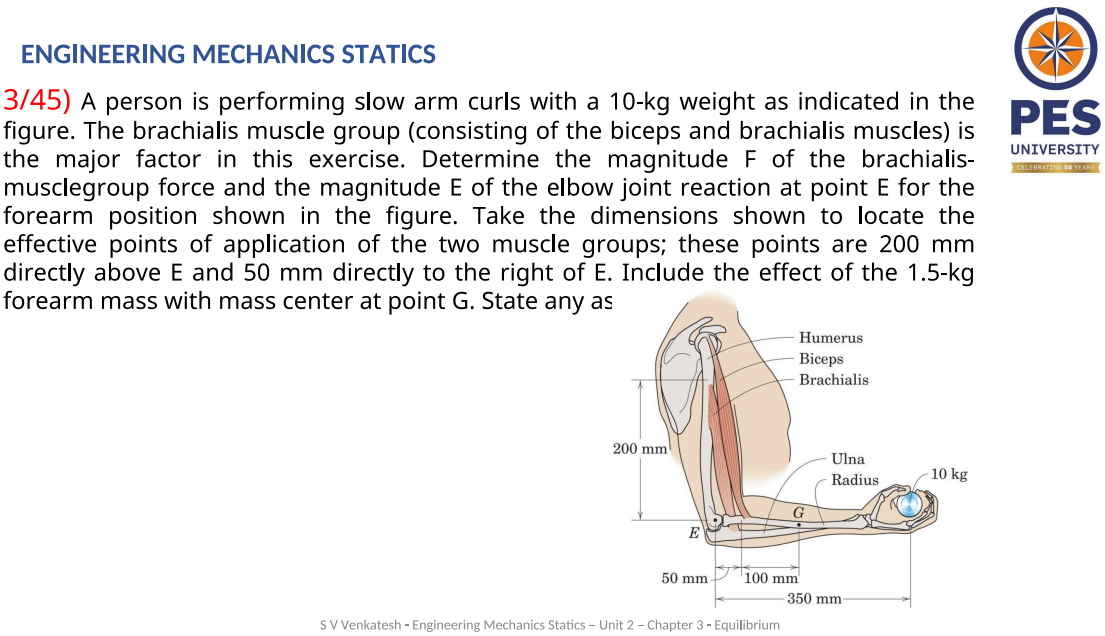


Figure G000056. UE26ME142A Unit 2 Class 19.3 Lecture PPT · source slide page 2.

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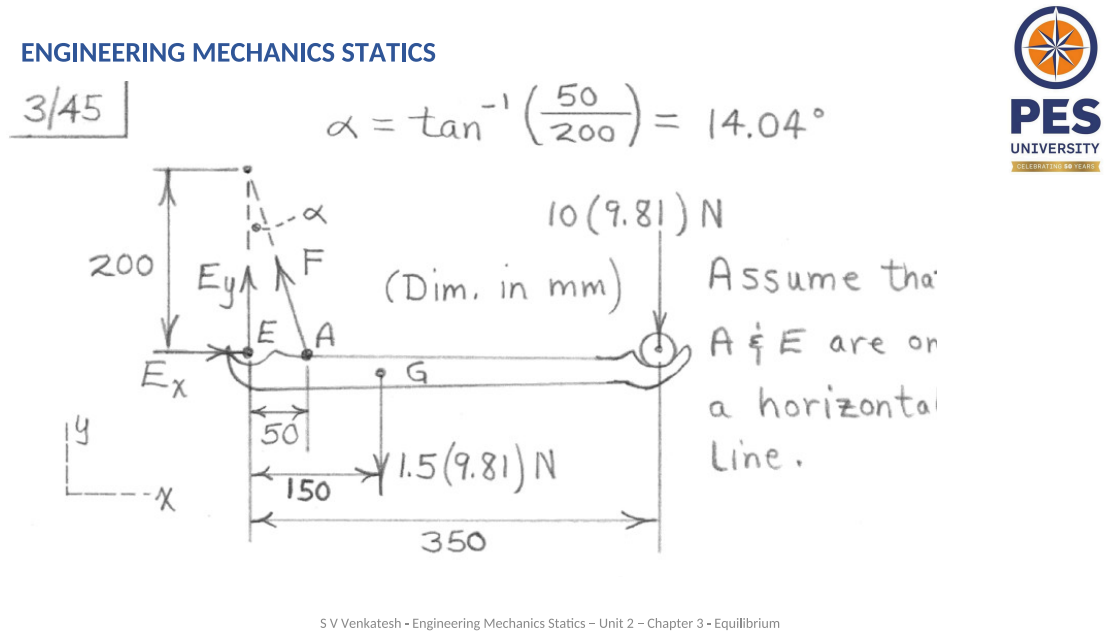


Figure G000057. UE26ME142A Unit 2 Class 19.3 Lecture PPT · source slide page 3.

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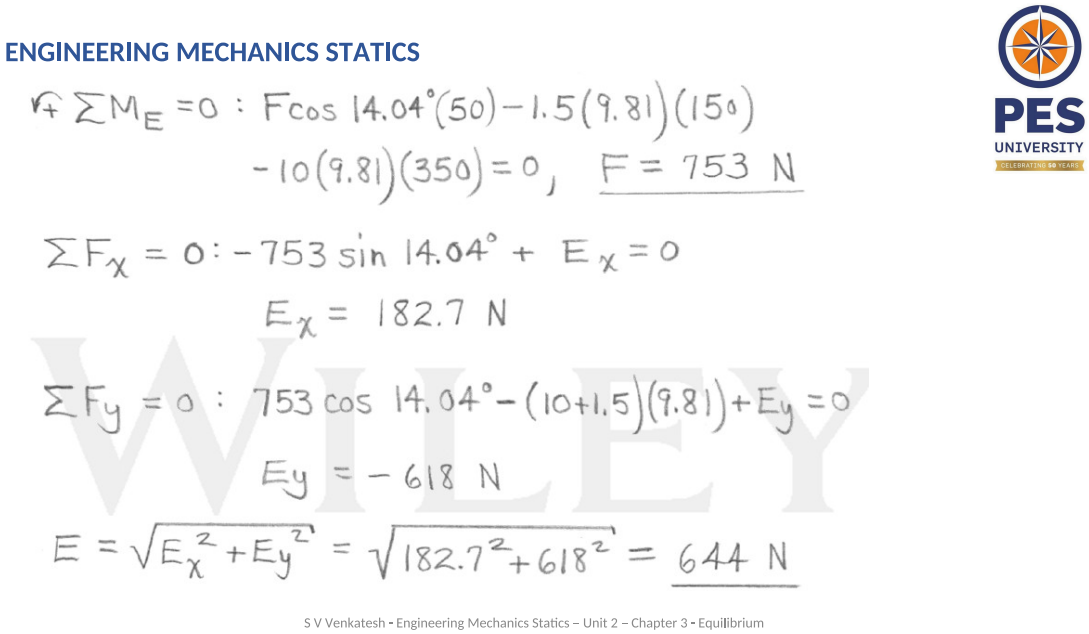


Figure G000058. UE26ME142A Unit 2 Class 19.3 Lecture PPT · source slide page 4.

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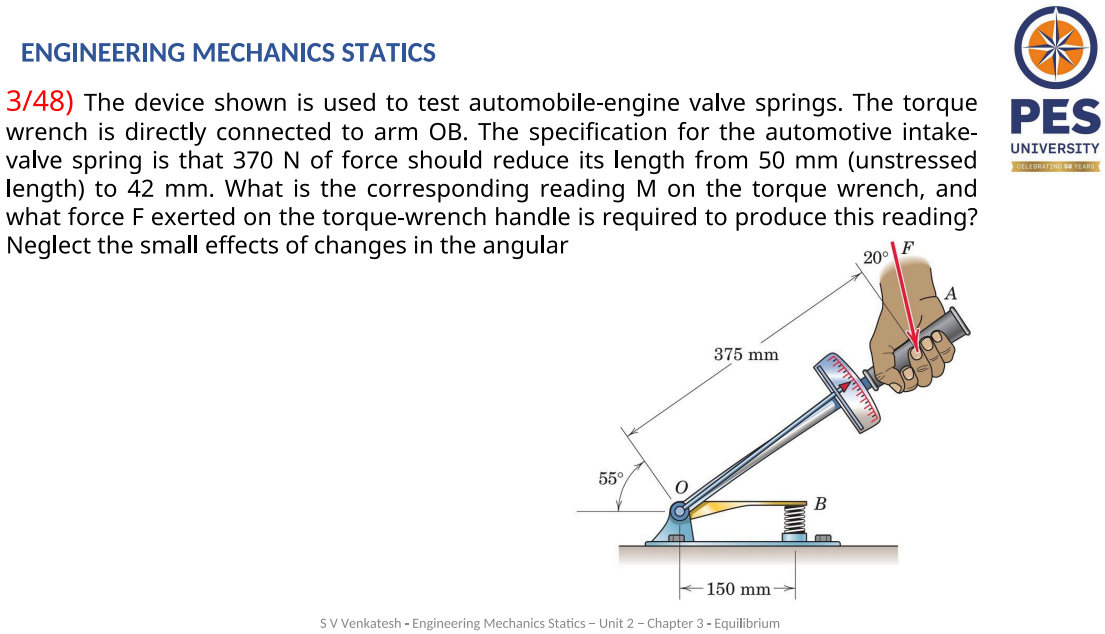


Figure G000059. UE26ME142A Unit 2 Class 19.3 Lecture PPT · source slide page 5.

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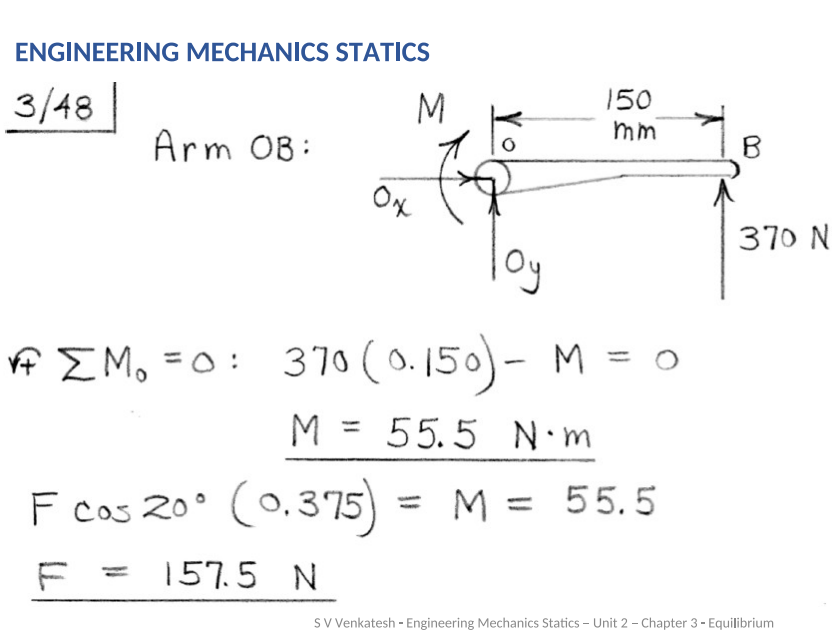


Figure G000060. UE26ME142A Unit 2 Class 19.3 Lecture PPT · source slide page 6.

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Figure G000061. UE26ME142A Unit 2 Class 19.3 Lecture PPT · source slide page 7.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000372 · EV0000373 · EV0000374
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 3/45) A person is performing slow arm curls with a 10-kg weight as indicated in the figure. The brachialis muscle group (consisting of the biceps and brachialis muscles) is the major factor in this exercise. Determine the magnitude F of the brachialis-musclegroup force and the magnitude E of the elbow joint reaction at point E for the forearm position shown in the figure. Take the dimensions shown to locate the effective points of application of the two muscle groups; these points are 200 mm directly above E and 50 mm directly to the right of E. Include the effect of the 1.5-kg forearm mass with mass center at point G. State any assumptions. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 3 (OCR) ENGINEERINGMECHANICSSTATICS 3/45 50 PES α=tan 14.04 200 UNIVERSITY CELEBRATING50YEARS 10(9.8!) N 2OO Assume tha (Dim,inmm E AE areon X ahorizonta 1.5(9.81) N Line. 150 350 S VVenkatesh-Engineering Mechanics Statics - Unit 2 - Chapter 3 -Equilibrium EV0000375 · EV0000376 · EV0000377
3. Source slide page 3: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000378 · EV0000379 · EV0000380
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000381 · EV0000382 · EV0000383
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS 3/48) The device shown is used to test automobile-engine valve springs. The torque wrench is directly connected to arm OB. The specification for the automotive intake-valve spring is that 370 N of force should reduce its length from 50 mm (unstressed length) to 42 mm. What is the corresponding reading M on the torque wrench, and what force F exerted on the torque-wrench handle is required to produce this reading? Neglect the small effects of changes in the angular position of arm OB. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000384
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 7 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000385
7. Source slide page 7: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000386

short-note recap

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.228] Dear students, welcome to Engineering Mechanics Statics classes. This is class number twenty six. My name is Dr. Sv. Venkatish from Civil Engineering under the Department of Science and Humanities, PS University. [00:00:18.052] In the last class we had discussed few problems from unit two on equilibrium. We'll be continuing to solve few more problems in today's class. This is problem number three bar forty five from [00:00:35.780] The prescribed textbook, Engineering Mechanics Statics by Miriam Craig and Bolton, Volume 1, 8th edition. [00:00:46.180] So the problem statement goes something like this. A person is performing slow arm curls with a 10 kg weight as indicated in the figure.

[00:00:59.108] The brachies muscle group consisting of the biceps and brachialis muscles is the major factor in this exercise. Determine the magnitude F of the brachialis muscle group [00:01:18.660] force under the magnitude E of the elbow joint reaction at point E for the forearm position shown in the figure. [00:01:30.372] Take the dimensions shown to locate the effective point of application of the two muscle groups. These points are 200 mm directly above E and 50 mm directly to the right of E. [00:01:49.028] Include the effect of the 1.5 kg for arm mass with mass center at point G. State any assumptions. Now if you look at the entire statement. [00:02:06.596] What you need to take from this statement is that the Chenkeji weight which is being held by the person in his hand, which will definitely be acting downwards, that distance is also given here. [00:02:25.348] which is 350 mm. If I'm not wrong. Then he also says that you take the weight of the forearm as 1.5 kg, which will be acting at point G, which is given here. The distance of G from [00:02:43.492] E can be taken at as 50 plus 100. At this point the force is going to be acting downward because it's the self fate of the forearm, 1.5 kgs into 9.81. Then he has asked you to determine the uh [00:03:02.948] Magnitude F and the magnitude E of the elbow joint. So this is the elbow joint E. What is the magnitude of the resultant force acting at elbow joint E? And also what is the force acting in the muscle group. The muscle group [00:03:21.636] could be taken to be acting along a line of action which is oriented in such a way that it is 200 mm above e point vertically, that is this one, and then it is 50 mm along the line EG. [00:03:38.468] Okay. So the uh muzzle forces will be acting along this line, which will be joining the point to the right of E by distance of 50 mm. [00:03:51.908] above E vertically by distance 200 mm. That line will be the line along which the muscle group force is going to be acting. Now whenever you take the problem as I have told you earlier also [00:04:09.412] That whatever may be the body that is acting, we are not interested in all that. What we are interested in is what is the magnitude of the force, what is the sense of the force, what is the line of action of the force, what is the direction of the force. What are the forces acting, what are the forces that we need to determine. [00:04:28.580] So that's what we are interested in. Let's not worry about which muscle it is, why is that muscle coming, what for is that muscle, all that is not required for us. What body is on which these forces are acting is not our concern. Our concern is only what are the characteristics of forces, where it is acting. [00:04:48.068] What we need to find. That's all is what we are interested in. So if you look at the free body diagram, so we need to take the [00:04:56.004] 10 kg weight that is acting because of the spear that the person is holding. Then 1.5 kg force, which is the sulfate of the forearm that is acting, 1.5 kg into 9.81 at point G.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.028] Then the joint at the elbow, which will act as an hinge point. So there will be two reactions, one vertical and one horizontal. [00:05:17.572] Then the force in the muzzle, as I have earlier explained, it is to the right of E at a point A, which is mentioned here, which is 50 mm from E horizontally, then above E vertically by a distance of 200 mm. [00:05:34.576] The line joining this, that is from A point to this point, is the force, that is the sense of the force that's acting. The muscle is has to resist, so that's why [00:05:46.468] The arm will try to go down, the arm will try to go down, but that is been held in position because of that muscle and that is a line of action of the force if that is going to be acting. [00:05:56.356] So we also have the distances from E to G, which is fifty plus hundred, that is one fifty, then we also have the distance from E to the point where the spear is uh weight of the spear is acting, that is about three fifteen. [00:06:10.852] So first thing is find out what is the angle

alpha made by the force F with the vertical as indicated here. So alpha is going to be equal to 10 inverse of this distance which is 50 divided by this distance which is 200, that would be equal to 14.04 degrees. [00:06:27.556] Then as I told you, we are assuming that the force is going to be acting along this line, that is the assumption that we are making. Okay. They have also given you the x axis and y axis positive x and positive y. So first thing is we need to find out what is the force acting at point E. So [00:06:46.916] To find out the forces acting at E, we need to find out what is the force F that is acting. So to find out the force that is acting, let us take moment about E. So what we get is we will resolve this force. So along the vertical, it is going to be F. [00:07:03.268] Cause alpha and along the horizontal passing through E it will be F sin alpha, which will not produce any movement. So movement about E will be equal to 10 into 9.81 into 350, which will be clockwise, so that is negative. Then 1.5 into 9.81 into 150, which will also be clockwise. [00:07:22.980] That should be equal to the component of the fourth that is F, cos alpha vertically upwards, which will cross an anti-clockwise movement. That F cos alpha into 150 which will be anti-clockwise. So equal to 0, so you get the value of F. [00:07:39.044] Once you get the value of F, you can resolve it along vertical and horizontal. The horizontal component of the force is the only horizontal force that is acting on the body, but that has to be resisted by the reaction at E. So your EX will be equal to F sine alpha. [00:07:58.148] Because your sigma f x is equal to 0. Then you take sigma f y is equal to 0, so you get e y to be equal to 1.5 into 9.81 plus 10 into 9.81 minus the component acting along the vertical because of the force f which will be f cos alpha. [00:08:17.156] So you get the vertical reaction at E. Now that's what is indicated here. So take movement about point E, which is equal to 0. So F cos 14.04 into 50 B. That's a component of the force acting in the muzzle, which will cause an anti-clockwise movement. So positive. [00:08:37.028] Then four arm weight, which is 1.5 to 9.81 into the distance, the horizontal distance between the G point and E point, which is 150, and that would produce a clockwise movement, so negative. Then the self weight of the spear. [00:08:54.340] Which is 10 into 9.81, the distance from E to the point where the weight of the spar is going to act is 350, that is it. That is what is indicated here. The sum of all this should be equal to 0. So you get the force F which is equal to 753 newtons. Next take sigma f x equals 0. [00:09:14.532] So minus seven fifty three which is the value of F. Why is it negative? Because when I resolve the force F along the X it will go to the left. [00:09:24.548] But uh right is the positive sense of x, so it is negative. So when fifty-three is sin 14.04, that's a component. Plus we have taken the force of EX to be acting to the right, so it is positive. So we get the value of E x which is equal to 182.7 newtons. [00:09:43.588] Next takes summation of all the forces along vertical that is sigma f y equals zero, so component of the force F seven fifty three cost fourteen point zero four, acting vertically upwards, so positive. You have the force [00:09:56.996] of the sphere acting downwards, the sulphate of the arm acting downwards. So that will be 10 plus 1.5 kg multiplying to 9.81 to convert it into mutants. Plus we have a force acting upwards which is EY. So plus EY equals 0. So we get the value of E1.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:15.044] Now in the problem he is asking you to find out the resultant of the force acting at the joint E. So resultant E will be equal to the [00:10:24.356] root of E square plus EY squared ph so you find that you get the value of E to be six forty four Newtons. I hope you have all of you have understood this problem. Okay. Next let's go to the next problem number three bar forty eight. [00:10:42.436] So the problem statement goes like this. The device shown is used to test automobile engine valve springs. The dark wrench is

directly connected to ROB. The specification for the automotive intake valve. [00:11:02.180] Spring is that 370 newton of force should reduce its length from 50 mm, unstretched length to 48 mm. What is the corresponding reading here on the dark range? [00:11:18.148] And what force F exerted on the torque range handle is required to produce this reading? Neglect the small effects of changes in the angular position of R O B. Now if you read the statement of the problem. [00:11:39.300] Clearly you can see that the spring will reduce its length from 50 mm to 42 mm. That's the difference in 8 mm. It will reduce or decrease in its length by 8 mm. [00:11:56.004] When a force of 370 Newton is applied. Now corresponding to this, what is the corresponding reading M on the dark orange, that is what will be the [00:12:13.028] Moment that is produced when you have this force and this much reduction is what is given. Then he has also asked you to find out what is the corresponding force that needs to be applied. So that this [00:12:29.284] Movement is indicated in the uh ridge. Okay. Then when you apply this force here, this angle that is given here as 55 will also slightly change because this would come down a little bit. [00:12:44.260] So what would be or that angular change can be neglected is what is given. Now you can if you observe here when I'm trying to apply this force, this this would try to rotate because of which this would get compressed. [00:13:04.676] When this has to get compressed, the force that has to be exerted on this is 370 newtons, and when I apply that 370 newtons, this would reduce the length by 8 mm. [00:13:20.708] Okay, that is what is given in the problem, and we need to find out what would be the moment that is going to be acting at O because of this, and what is the corresponding force that needs to be applied. So movement at O. [00:13:38.500] will be equal to whatever is the force acting at B into the distance. So since the spring is vertical, the force acting on the spring also has to be vertical. [00:13:49.540] And it is going to be downwards. So this force into this distance will produce a moment here, and that is what is going to be indicated here. Okay. So moment at O will be 370 into 150 mm. Okay. So the same moment. [00:14:07.460] should be produced when we apply this force F. [00:14:11.204] So if I resolve this force F along this axis, which is normal to OA, then I get it as F cos 20 and parallel to OA, I got it as F sine 20. This would not produce any moment, that is the component acting along OA. [00:14:28.240] which is F sine twenty will not produce any moment at O. Only the component F cos twenty which is acting normal to OA will produce a moment at O. [00:14:38.180] Which will be equal to F cos 20 into 375. So this should be equated to whatever moment that we have gotten. Oh, because of this 370 into 150 mm. So you will get the value of F there. So I've explained. So moment produced. [00:14:55.716] At O is equal to 370. I told you that the force that's going to be acting on the spring is 370, but here the reaction force is taken, both of them are same. So it's 370 into 150 will be the moment that is produced at this point. Now this moment will be the moment

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:16.132] That should be produced by the F force that is acting. So movement that we have got is 55.5. The movement produced by the force applied is F cos 20 degrees. [00:15:30.084] into the distance between uh along OA is 375 mm converting it into meters. So that should be equal to y so equating that we get it as 55.55 which we have calculated earlier. [00:15:44.612] So equating that we get F to be equal to 157.5 kilon. So when I apply a force F at an angle of twenty degrees to the normal axis of OA. [00:16:01.732] With a magnitude of 157.5, when I apply that, the moment produced at O will be equal to 55.5. Because of this, the spring would get [00:16:16.612] shortened by a length of 8 mm, which is earlier was 50 mm. It will reduce to a length of 42 mm, that is it will get compressed by a

length of 8 mm. Okay. I hope all of you have understood these two problems. These two problems that I have chosen [00:16:36.324] Are slightly difficult when compared to other problems. So I have chosen only a few problems which I felt are difficult and need some explanation from the teacher so that the students can understand it and they can [00:16:55.300] Get to know how to solve the other problems which are not solved, which are relatively easier than the problems that I have chosen so far. Those problems which are not solved, kindly solve it. And if you have any doubts in this, [00:17:12.004] In those problems, you can refer to the notes which is uploaded in Peso Academy, or you can approach me or any other faculty members in the Department of Civil Engineering. [00:17:24.868] They will certainly be helping you out to understand how to solve those problems which we have not solved. Okay. So with this, we have come to the end of this. [00:17:38.084] uh class. That is class number 25. We'll be continuing to solve few more problems under equilibrium in the coming classes. Okay. So with this, thank you very much. I am Dr. Svangitesh from [00:17:54.884] the Civil Engineering Department, under the Department of Science and Humanities, PS University. See you in my next class. Thank you very much.

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Full transcript: 2677 normalized words, 68 timestamped segments, 1085.3843 seconds duration. Low-confidence segments flagged: 0.

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11 UE26ME142A_Unit 2_Class 19.4_AV Summary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 19.4_AV Summary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000154 · EV0000155

Graphs and diagrams



Figure G000062. UE26ME142A Unit 2 Class 19.4 Lecture PPT · source slide page 1.

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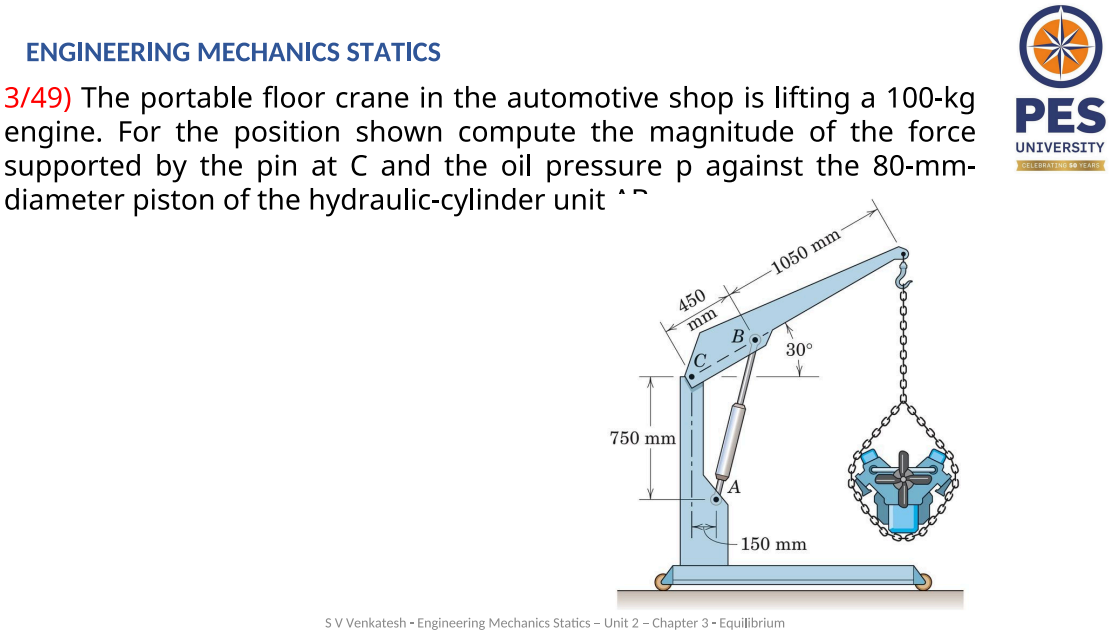


Figure G000063. UE26ME142A Unit 2 Class 19.4 Lecture PPT · source slide page 2.

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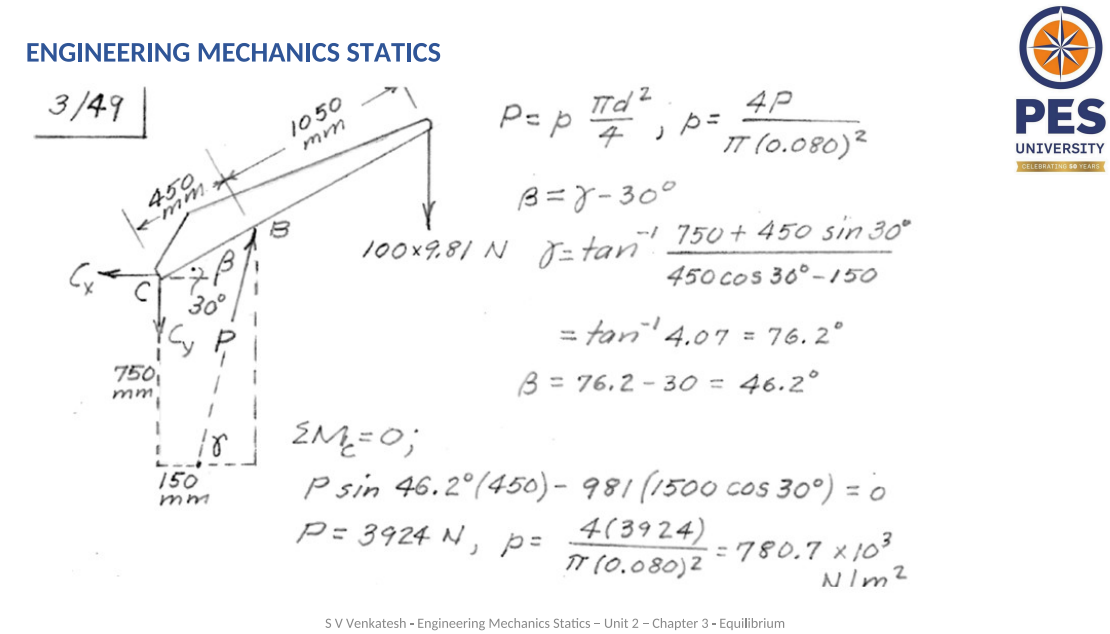


Figure G000064. UE26ME142A Unit 2 Class 19.4 Lecture PPT · source slide page 3.

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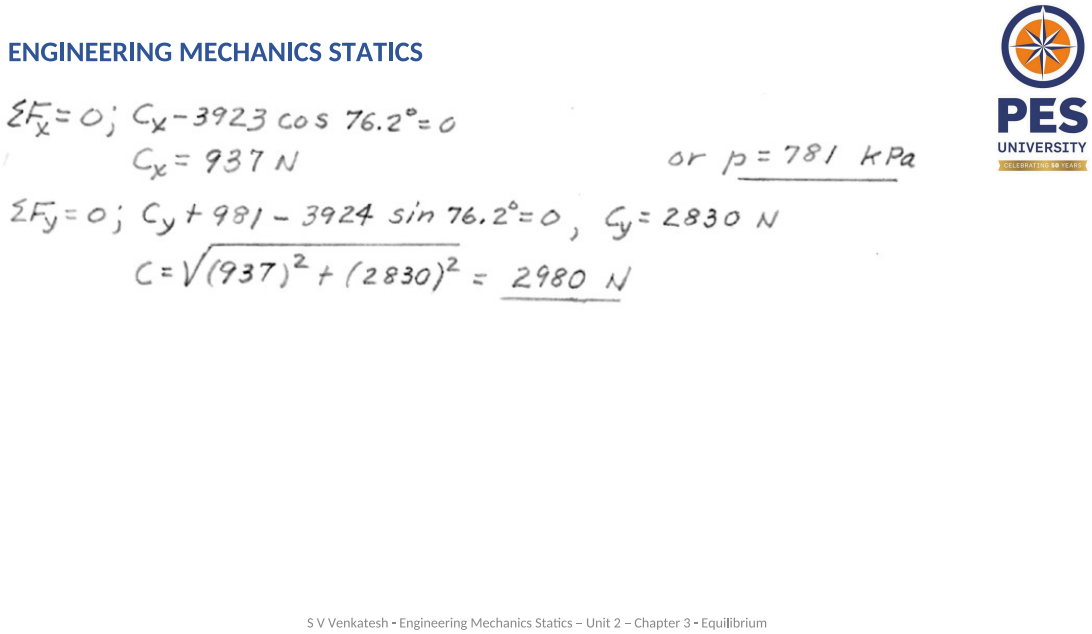


Figure G000065. UE26ME142A Unit 2 Class 19.4 Lecture PPT · source slide page 4.

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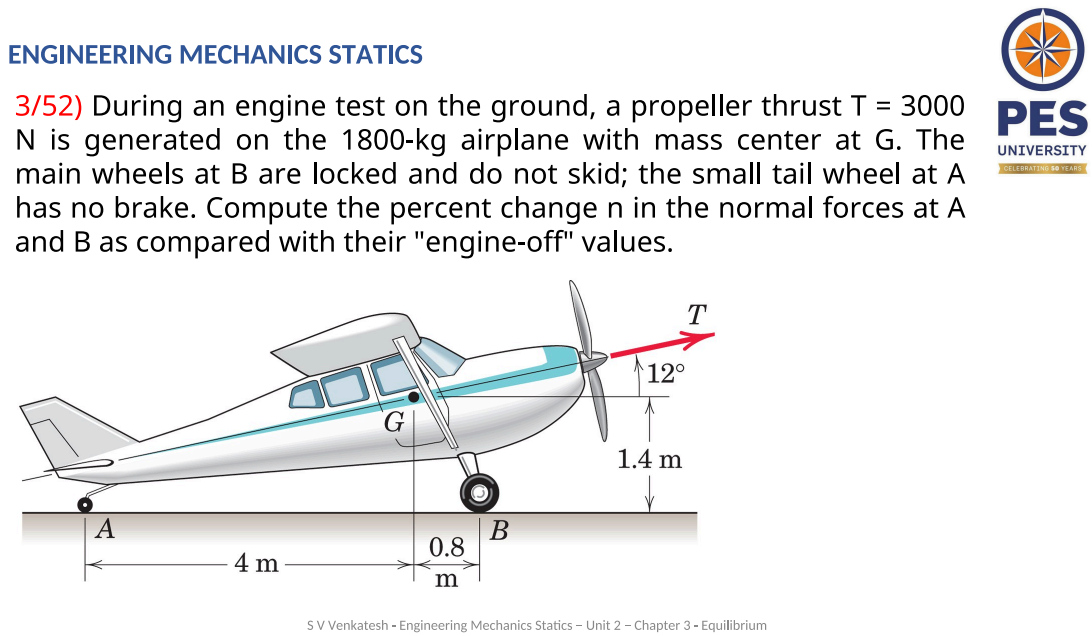


Figure G000066. UE26ME142A Unit 2 Class 19.4 Lecture PPT · source slide page 5.

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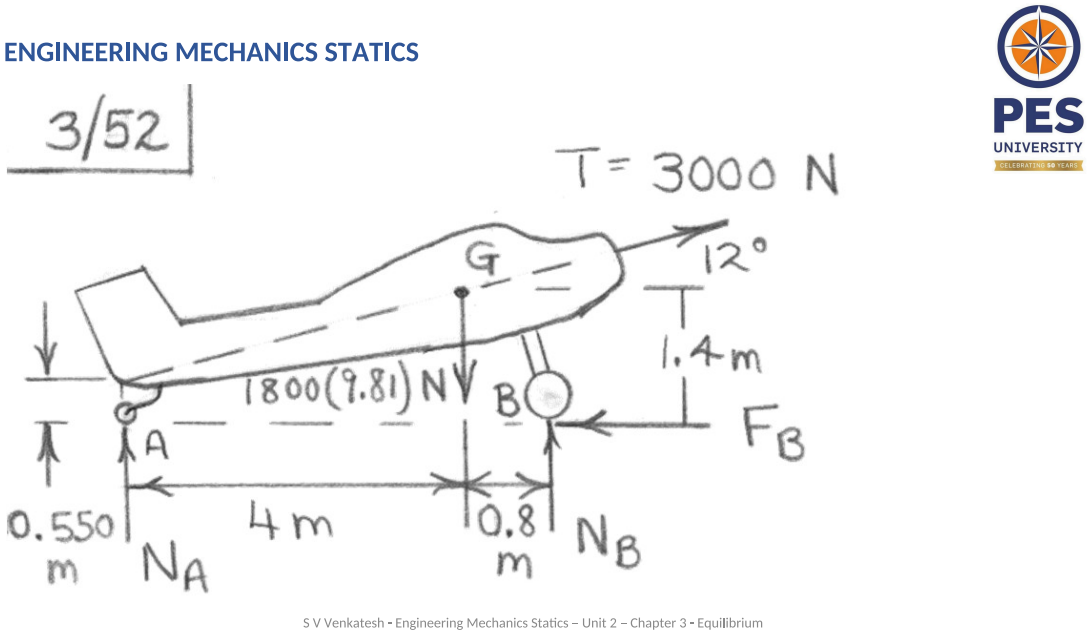


Figure G000067. UE26ME142A Unit 2 Class 19.4 Lecture PPT · source slide page 6.

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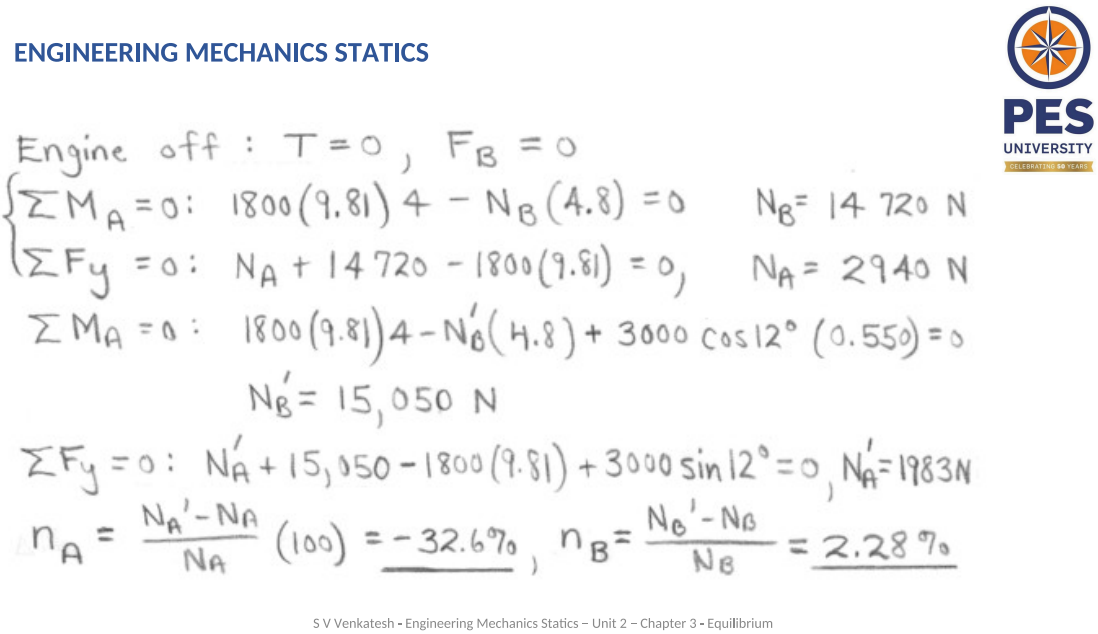


Figure G000068. UE26ME142A Unit 2 Class 19.4 Lecture PPT · source slide page 7.

EV0000405

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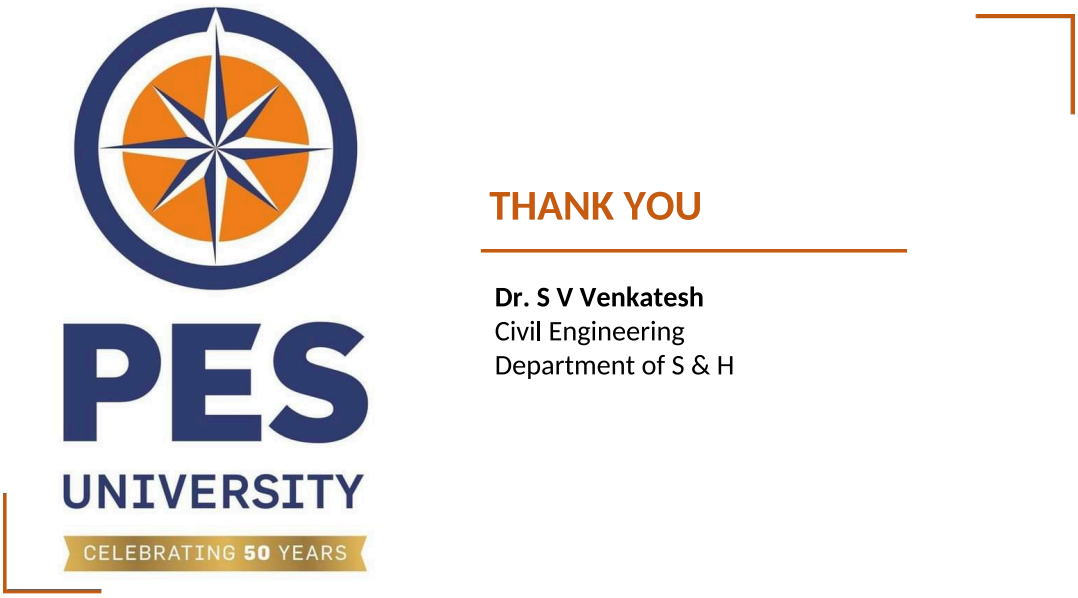


Figure G000069. UE26ME142A Unit 2 Class 19.4 Lecture PPT · source slide page 8.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000387 · EV0000388 · EV0000389
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 3/49) The portable floor crane in the automotive shop is lifting a 100-kg engine. For the position shown compute the magnitude of the force supported by the pin at C and the oil pressure p against the 80-mm-diameter piston of the hydraulic-cylinder unit AB. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000390 · EV0000391 · EV0000392
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000393 · EV0000394 · EV0000395
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000396 · EV0000397 · EV0000398
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS 3/52) During an engine test on the ground, a propeller thrust T = 3000 N is generated on the 1800-kg airplane with mass center at G. The main wheels at B are locked and do not skid; the small tail wheel at A has no brake. Compute the percent change n in the normal forces at A and B as compared with their "engine-off" values. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000399 · EV0000400 · EV0000401
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000402 · EV0000403 · EV0000404

7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 8 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000405
8. Source slide page 8: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000406

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.924] Dear students, welcome to Engineering Mechanics Statics classes. My name is Dr. Sv Venkatish from Civil Engineering, Department of Science and Humanities, PS University. This is class number twenty seven. [00:00:20.164] In the last class we had discussed two problems on equilibrium and also I had informed that we'll be continuing in solving a few more problems before we go to the next topic in unit two. [00:00:37.540] So let's look at the next problem. I've considered problem number three bar 49 from the prescribed textbook, which is Engineer Mechanics Statics by Miriam Craig and Borgan, Volume 1, 8th edition. [00:00:57.572] So the problem statements is something like this. The portable floor crane in the automotive shop is lifting a 100 kg engine for the position shown. [00:01:15.684] Compute the magnitude of the force supported by the pin at C and the oil pressure P against the ATM diameter question of the cylindric of the sorry of the hydraulic cylinder. [00:01:35.364] unit A B. So when you read this statement, what you should be able to make out is that there is an hundred kg engine that has been lifted by a automotive or crane in the automotive shop. [00:01:55.748] For the angle that is given in the diagram, we need to find out what is the support at the ping C and also what is the pressure on the cylinder, hydraulic cylinder, the pressure on the [00:02:12.356] oil pressure on the hydraulic cylinder is what we have to look at. Now look at the diagram. They have clearly given that there is a crane here and you have wheels also which can be moved. Okay? Then you can see that there is a hook here to which there is some load that is there. [00:02:32.228] And he says that it is hundred kg. So there's an hundred kg acting here, mass. So into nine point eight one I get the force that is to be acting here. [00:02:42.404] For the position shown means this angle is 30 degrees, that is very clearly indicated along the line B C which goes and touches here. So that angle is 30 degrees. And there is a pin at C, so you will have two reactions here, one vertical and one horizontal. [00:03:02.340] Now we need to find out what are these values. That is what is the support at Pen P, that is first thing. And also you can see that there is a cylinder here, which is the piston. Piston is there, then there is a oil filled in this this cylinder. [00:03:21.444] So when I apply this force that has been resisted by the by this piston which would try to compromise because of the oil present there, it is resisted by that, along with the support at C. Okay. So and also they have given you [00:03:40.612] What is the distance between B C along B C, and then what is the distance between B to the point where the load is applied along B C again? Okay, that is this distance. And they've also given you what is the vertical distance between A to C, which is 750, what is the horizontal distance between C to A, which is 150 and [00:04:00.580] Okay. Now with the data that you can look at, what are the force that is acting? Now there is a force here which is acting that is 100 into 9.81. [00:04:11.588] Then this piston is getting compressed, so internal forces will be moving away from each other. So a force is going to be moving from A to B and B to A. At B the force will be acting towards B from A. Okay. So you will have a force here. [00:04:29.284] Then you will have two reactions here. So if you

draw the free body diagram, you will have one vertical reaction at A, one horizontal reaction at sorry, one vertical reaction at C, one horizontal reaction at C. If I take only this part separated out, then I have the pressure in the cylinder. [00:04:49.892] Which will be acting towards B and if I take the force as F B, the line of action of that force which is acting at B has to be determined. Then we have this force hundred. So if I take the moment about C

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.628] After determining what is the angle made by B C with the vertical or horizontal, preferably with the vertical, how can I determine this angle? So if I draw a vertical line from B till the horizontal line extended from A, I get this to be a right angle triangle. [00:05:24.900] Okay. So I can find out what is the horizontal distance between B to C, which will be four fifty cos thirty, which will be this distance. Okay. That minus one fifty, I get the horizontal distance between A B. Then [00:05:42.884] We already know what is the vertical distance between A C, I can also find out what is the vertical distance between C B, which would be four fifty sin thirty, which I'll get this one. [00:05:53.380] So this plus seven fifty will be the vertical distance between A B. So we know the vertical distance and the horizontal distance, so I can use tan inverse of the horizontal distance between A B divided by the vertical distance between A B. So I get that angle. [00:06:10.436] So once I get that angle I can resolve the force at B, F B cos of if I take this as alpha, F B cos alpha will be acting here. Okay. So definitely I can resolve this force along horizontal and vertical or along B C and normal to B C. [00:06:29.892] So if I take along B C and normal to B C, the force normal to B C into four fifty will be the moment produced at C. [00:06:37.828] Then I have this 100 kg force into the horizontal distance between them will be equal to 150, 1500, which is 1050 plus 450, which is 1500. [00:06:50.532] Cross thirty degrees will be the horizontal distance. So that will be the moment which will be acting in the clockwise direction. So I can easily get the force in F B. So once I get F B, [00:07:03.076] Then I can take algebraism of all the forces acting along the horizontal, so I get the horizontal reaction at C because the component of F P acting in the [00:07:17.892] horizontal direction which will also be towards the right. So reaction at C has to be towards the left. Then the force acting because of F will be acting upwards. So if I take all the forces acting along the vertical, so I get the reaction at C to be acts. [00:07:37.284] Then we have the force or the component of FP acting upwards, then we have this hundred kg which is acting downwards. So summation of this I get the reaction at C. Then once I have got the force in acting at B, F [00:07:54.116] That force divided by the area of the piston, I get the pressure acting in the hydraulic cylinder. The area I can always determine because they have given the diameter of the piston. So pi into 80 square divided by 4. [00:08:10.916] I get the area. So I get the pressure there. Okay. I hope all of you have understood this explanation. So let's look at the solution here. So as I told you, find out the area. [00:08:24.996] That is pi d square by 4. Multiply that with the force that is acting, so you get the pressure. Or you get the pressure into the force, we get the force, pressure into the area, we get the force. Okay. Or pressure is equal to [00:08:43.332] Force divided by area, that's what is indicated here. So to find out the force, here this angle is taken as beta. This angle. So beta is equal to alpha minus thirty. What is that? [00:09:01.060] So if I extend this line, okay, so this is beta, this is thirty degrees. So they have given this as uh or this as this is considered as alpha. Okay? So if I draw an horizontal line here, [00:09:19.940] So this horizontal line and this line will be at thirty degrees because this is alternate angles. So uh beta which is the angle made by the line, this line with C V, this angle. So this is [00:09:39.396] Uh alpha, this total is alpha. So alpha minus 30 I'll get beta.

Okay. So he has found out alpha as I have explained earlier. I had considered this as alpha, but they have considered this as alpha, so time inverse of this distance by this, I have explained that how to get it. [00:09:59.748] So that is what is indicated here, which is this seven fifty plus this four fifty into this is four fifty sign of this angle will give me the distance B to this point. That is the vertical distance. The horizontal distance between this point to B

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:20.164] Will be 450 which is this line. Cos of thirty which I get this distance total minus this one fifty, so I get the horizontal distance. So by which I will get the angle alpha, then I can find out the angle beta. [00:10:37.540] So once I get angle beta I can get the component of this force along C V and perpendicular to it. So you take the movement about C point. So you get P is a force this force. [00:10:55.172] And sine of this angle beta which will be the component perpendicular to C B into the distance 450 will give you the moment which is in the anticlocal direction. So it's positive. Next this force is 981. [00:11:13.124] into what is the horizontal distance between C to this force, which will be one 150 plus 408 is 1500. Cos of 30, which I get this as the horizontal distance. That's what is indicated here. To be equal to zero. So from this equation [00:11:32.164] I get the force P which is acting in the piston, which we get it as three thousand nine hundred and twenty four newtons. So using that I can find out the pressure which we have the formula here already. So pressure is equal to this. [00:11:46.788] So I get the pressure to be equal to seven eighty point seven into ten to the power of three Newton per meter square. Okay, that is possible. Okay. Next find out the [00:12:05.796] Reaction at C, take all the forces acting along X axis. So Cx uh minus [00:12:17.412] The force in the piston, cause of the angle, will give you CX. Because force in the piston is seven eighty one. [00:12:30.180] Measure is seven eighty one. Then all the forces acting along the vertical, C Y acting upwards, plus nine eighty one which is acting downwards. If I take C Y to be acting downwards itself. [00:12:46.372] That is positive. 981 acting downwards is also positive. The component of the force in or the piston is 3924. [00:12:58.756] uh which is the force into sine of this angle. So I get the value of C y. Then to find out the magnitude of C I will take the root of 937 square plus 2830 square. So we get the [00:13:17.764] Resultant of the support reactions acting at CS two nine eight zero newtons. Hope everybody has understood this. Let's move on to the next problem. This is a very interesting problem, slightly different than what we have solved so far. [00:13:35.108] So the problem number is three bar fifty two. The statement of the problem was like this. During an engine test on the ground, the trust T is equal to three thousand newton is generated on the thousand eight hundred kg airplane. [00:13:52.868] With masks center at G, the main wheels at B are locked and do not skip. The small tail wheel at A has no brake compute. [00:14:08.004] the percentage change in N in the normal forces at A and B as compared with their engine of values. So in this problem they have given you the trust T which is equal to 3000 newtons. [00:14:27.428] The self weight of the airplane is 1800 kgs and that is located or at B at point G. Then the main wheels at B are locked. Lock means the B wheels are not skidding, it won't skid. [00:14:44.932] The small tailwheel at A has no brake, so this does not have brake. Okay, so this would act uh free. Okay, there will be only one reaction that is vertical. Whereas at this point, since there is uh

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:02.788] Locked locked means it won't rotate. So you will also have an horizontal reaction there. So what is asked is to compute the percentage change in normal in the uh on A and B. Okay? When the engine is switched on, when

the engine is switched off. [00:15:21.956] So when the engine is switched off, you won't get this trust into picture. So what you will have is only the sulfate of the aeroplane, which is 1800 kg is acting at G, so 1800 into 9.81. So simple take moment about B. So you get RA. [00:15:41.604] into the distance for point eight should be equal to 1800 into nine point eight one into this distance 0.8 so you get r n then take algebraic sum of all the forces along the vertical so you get r b here. So that is when the engine is switched off. [00:15:58.916] When the engine is switched on, you will have in addition you have this thrust and also a reaction at B, horizontal reaction. So this could be resolved because they have given this as 12 degrees. So I get T cos 12. [00:16:12.740] acting along the horizontal. T is sine 12 acting along the vertical. So again I will take moment about B. So I get the reaction at A. Then I take some of all the forces along vertical. So I get the vertical reaction at B, then sum of all the forces along horizontal, so I get the [00:16:32.868] Reaction, horizontal reaction at B. Okay? Then of course I can find out what is the uh percentage variation acting at A and acting at B because of engine off and engine on position. So uh [00:16:54.052] The free body diagram, as I have told you, this is what it is. A point, then B point, you will have the normal reaction, NA, NB. Since this is locked, you will also have a horizontal reaction. [00:17:08.164] Then the force acting at G and the engine when it is switched on you will have T equals three thousand metals. This is acting at twelve degrees. [00:17:18.564] Next one engine of T equals 0, F B is equal to 0, that is the horizontal reaction of B will be 0. So taking sigma M A equals 0, we get N B. Taking sigma F is equal to 0, we get N A. Taking sigma MA equals 0. [00:17:38.948] You have the engine switched on. So you can see the variation in this equation and this equation. There is this three thousand cross twelve degrees into the corresponding perpendicular distance should be equal to zero. So you get [00:17:57.188] N B dash that is the normal reaction acting at P. Then you take sigma F equals 0 normal reaction acting at E you can find out. So percentage variation at in the reactions at A. So it is N A dash when the switched on condition, NG switched on, minus [00:18:17.220] Switched off condition, you have a switched off condition in 200 because percentage. So you get as thirty two point six and it is a negative because the reaction when the engine is switched on is lesser than the reaction when the engine is switched off. [00:18:35.748] Similarly the percentage change at the reaction at B will be equal to N B dash minus N B divided by N B. So that you get us into hundred of course. That you will get us or divided by hundred I should say, I'm sorry. [00:18:52.900] So you get it as two point two eight percent. Okay so uh I hope all of you have understood. This is a slightly a different problem than what we have solved earlier, but the procedure is the same except that you need to find out the percentage variation when the engine is switched on and switched off. [00:19:13.156] Okay so I hope all of you have understood this. So we'll be continuing to solve few more problems in the next class on equilibrium. So thank you very much. My name is Dr. Sv Vengadesh from Civil Engineering under the Department of Science and Humanities, PS University. [00:19:31.172] Let's meet in the next class. Kindly solve problems which I have not done. If you have any doubts, you can always contact me in the Department of Civil Engineering PS University. Thank you very much. Meet you in the next class.

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Full transcript: 2840 normalized words, 69 timestamped segments, 1186.965 seconds duration. Low-confidence segments flagged: 0.

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12 UE26ME142A_Unit 2_Class 19.5_AV Summary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 19.5_AV Summary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000170 · EV0000171

Graphs and diagrams



Figure G000070. UE26ME142A Unit 2 Class 19.5 Lecture PPT · source slide page 1.

EV0000408

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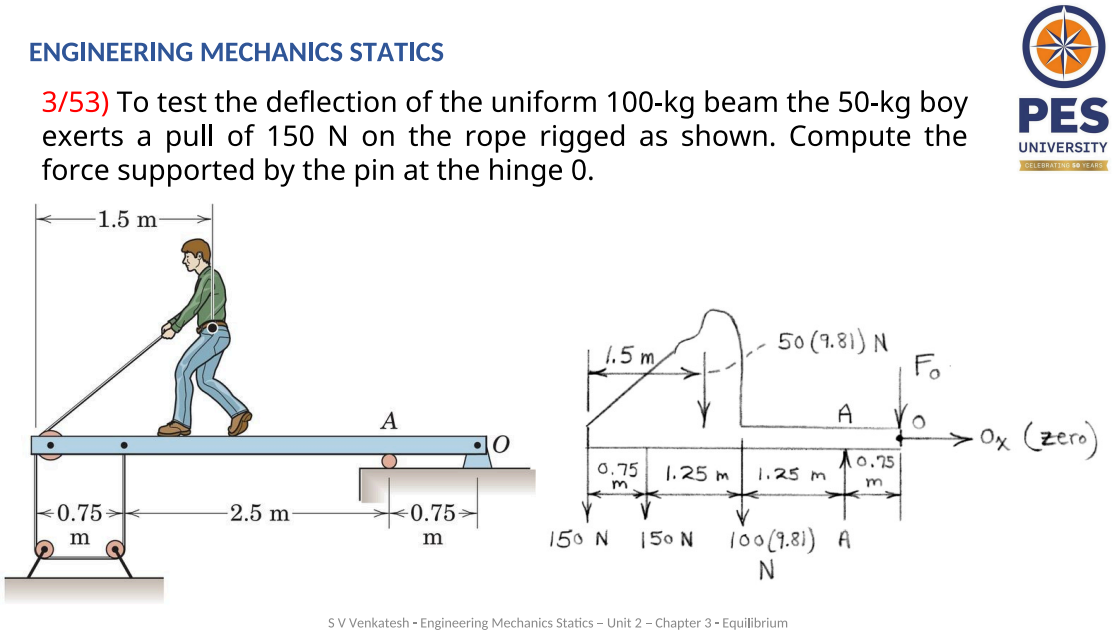


Figure G000071. UE26ME142A Unit 2 Class 19.5 Lecture PPT · source slide page 2.

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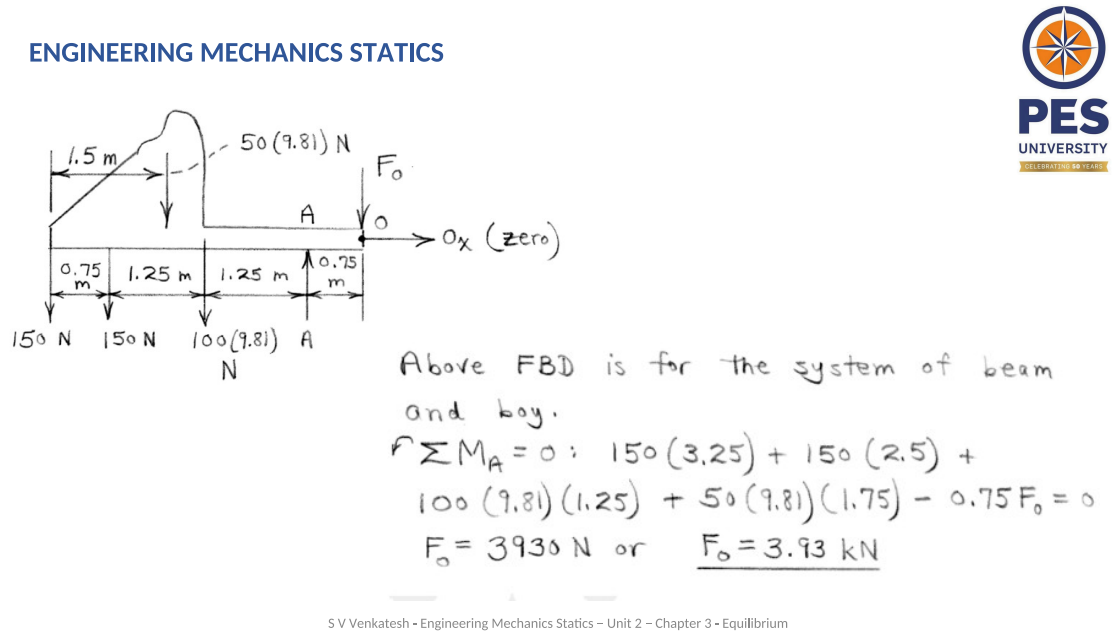


Figure G000072. UE26ME142A Unit 2 Class 19.5 Lecture PPT · source slide page 3.

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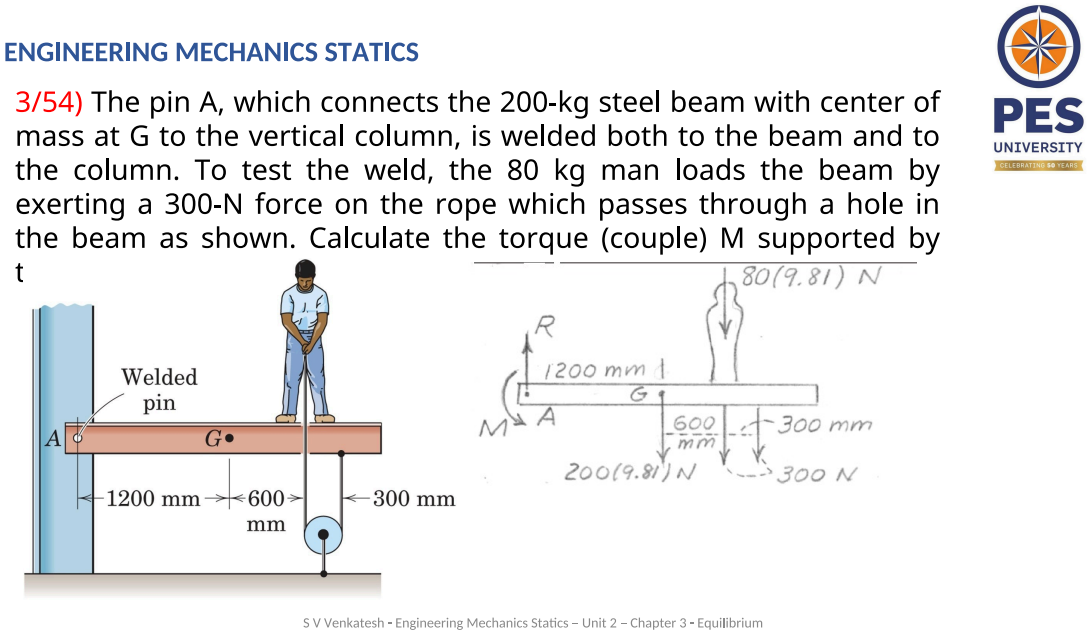


Figure G000073. UE26ME142A Unit 2 Class 19.5 Lecture PPT · source slide page 4.

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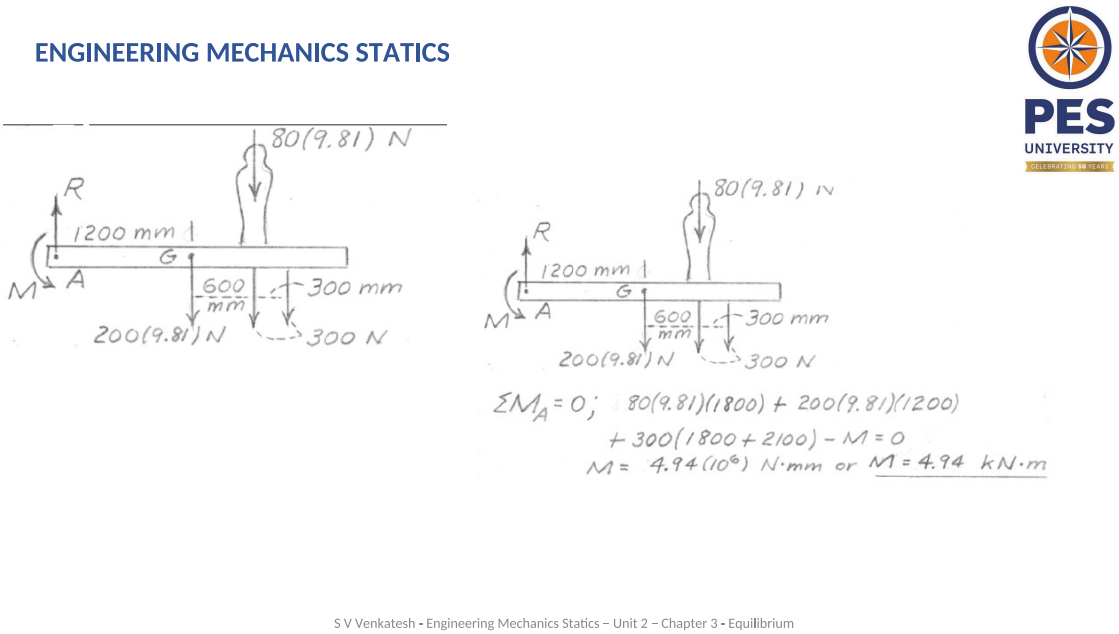


Figure G000074. UE26ME142A Unit 2 Class 19.5 Lecture PPT · source slide page 5.

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Figure G000075. UE26ME142A Unit 2 Class 19.5 Lecture PPT · source slide page 6.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000407 · EV0000408 · EV0000409
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 3/53) To test the deflection of the uniform 100-kg beam the 50- kg boy exerts a pull of 150 N on the rope rigged as shown. Compute the force supported by the pin at the hinge 0. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000410 · EV0000411 · EV0000412
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000413 · EV0000414 · EV0000415
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 3/54) The pin A, which connects the 200-kg steel beam with center of mass at G to the vertical column, is welded both to the beam and to the column. To test the weld, the 80 kg man loads the beam by exerting a 300-N force on the rope which passes through a hole in the beam as shown. Calculate the torque (couple) M supported by the pin. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium EV0000416 · EV0000417 · EV0000418
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 3 - Equilibrium ## Page / slide 6 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000419

6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000420

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.548] Dear students, welcome to Engineering Mechanics Statics classes. My name is Dr. Svinkitish from the Civil Engineering under the Department of Science and Humanities, PS University. This is class number twenty eight. [00:00:21.028] We are in the process of solving few problems. Under chapter three, there is equilibrium chapter. And probably this would be the last class in this chapter. [00:00:36.772] In the coming classes I will be starting with the new chapter, that is chapter number four, that is process. Okay. In the previous class, that is class number twenty seven we had discussed few problems, how to solve them. [00:00:53.668] under equilibrium chapter. So we will continue to solve few more problems in this class as well. So for this class I have chosen this problem that is 3 bar 53 from the prescribed textbook that is engineering mechanics statics 8 edition [00:01:15.140] The authors are Miriam, Craig and Bulton. The statement of the problem goes something like this. The 50 kg point exerts a pull of 150 newton on the row. [00:01:35.268] Rig the ashton. Compute the force supported by the pin at the hinge O. So what this statement indicates is that the self weight of the beam. [00:01:53.188] is hundred kg and we all know that the sulphate is always going to be acting at the center of the total span or total distance of the beam. [00:02:03.716] So what is the total distance of the beam in this case if you look at the diagram that is given, it is equal to 0.75 plus 2.5 plus 0.75 which will come to 4 meters. So from each end, that is extreme left end as well as extreme right end. [00:02:22.596] The distance at which this 100 kg into 9.81 newtons will be acting is at a distance of 4 by 2, which is equal to 2 meters. And also he says that the weight of the boy who is standing on the beam is 50 kgs. [00:02:39.908] Now where will this weight act? It will go to act at the center of gravity of the boil, which is indicated in the diagram as 1.5 meters from the left support, that is this distance where the [00:02:54.928] weight of this person or the boy is gonna act that is fifteen to nine point eight one kgs will be acting at this point. Okay. [00:03:03.076] And it also says that the boy exerts a pull of 150 newtons. So here is pulling this rope by 150. So the internal force or we should say that this rope will be under tension when he's got a pull it. [00:03:21.732] Okay. So internal force that is going to be acting in the uh row here will be from he is pulling it this way, that is upwards, whereas internal forces will be acting downwards at this point. Whereas at this point it will be acting upwards. So [00:03:40.036] Since the boy is standing on the beam, the weight of the boy is going to be acting on the beam. Whereas when he tries to pull this rope, the tensile force always going to be in pair, acting towards each other. [00:03:59.908] So this force and this force will get cancelled. However there will be force in this cable also. Again here it will be opposed to this which will be acting. [00:04:10.788] Like this, from moving away from here, here it will be moving away. Similarly, here it will be moving away, here it will be moving away, this also will be acting upwards, here it will be acting upwards. Now this force which is attached to this police is going to be taken by this, but however, if I cut it here, [00:04:29.316] We will have a force here which will be acting downwards. We will have a force here which will be acting downwards. In addition to this, we will also have a roller here at point A. So we will have a reaction at A acting upwards. [00:04:43.460] Then we have a pin or an inch connection at O. So we will have a reaction acting upwards or vertical reaction and then we will also have a horizontal reaction at O. So [00:04:56.324] What are the forces that's going to be acting on this beam now as

two reaction at O, one vertical, one horizontal, one reaction at A which will be acting up vertically upwards. We have the

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.036] Tension force in this cable which is acting downwards of magnitude 150 newtons. Here also there will be one more cable which will be acting downwards of magnitude 150 newtons, and we will have the weight of the [00:05:23.268] which is fifteen to nine point eight one. In addition we will have also hundred into nine point eight one will be active in the center of the beam. [00:05:32.004] The forces in this cable will balance with each other, remember this carefully. So how do we go about solving this problem? Or what is that we have to find out? We have to compute the force supported by the pin at the end. So we have to find out what is the force supported at this. Okay. [00:05:49.348] Uh that is O. So how do we solve this? You take the momentum out EA. [00:05:54.948] Or movement about O because that's where you have the unknown support reaction that is acting. If you take the moment about A, you get the reaction acting at O, vertical reaction into 0.75, I assume it to be acting upwards, it will be anti-clockwise. Then the sulfate of the beam which will be acting [00:06:14.660] But the center of magnitude hundred into nine point eight one. [00:06:18.628] But from A it is going to be at a distance of 1.05. So that would produce a clockwise moment. Then the force in this will also act in the anticlockwise moment, which will be equal to 115 to 2.5. Then we will have the uh [00:06:37.860] force that is acting in this cable which is also equal to 150 newton into force is 3.25. In addition we will also have the weight of the boil which is 15 to 9.81. From this end if I want to find out it will be equal to [00:06:54.224] 3.25 minus 1.5, which is going to be 1.75. So if I calculate the only unknown in that case will be the vertical reaction at O. And that vertical reaction we will find out. Okay. So uh going forward, uh [00:07:14.436] Yeah. Uh free body diagram uh I'm not sure it is not visible here, however I have told you. So taking movement about A. That is at the point where the roller is acting, as I have told you, and one fifty is the uh [00:07:30.436] force in the cable into 3.25 that is the extreme end left left cable, then one more cable which is acting 1.15 into 2.5 plus the sulphate of the beam. [00:07:42.884] Should be acting at the center from A it is 1.25 plus the weight of the boy which is 15 to 9.8 one acting at it. So 1.75. All this are clockwise, so it is taken as positive minus the reaction acting at the uh hinge into the distance which is uh [00:08:02.820] 3.25. So by that we will get the value of the reaction acting at the point at D, which we will get as 3930 newton, which is equal to 3.93 kilonuton. I hope this is clear. Let's move on. Okay, the free body diagram has come now. [00:08:22.948] So uh the force in the or the reaction at O is taken to be acting downwards, which will produce an anticlockwise movement or clockwise movement here, whereas all this will produce an anticlockwise movement. [00:08:37.124] However, if you take clockwise as positive and anti-clockwise as negative, or clockwise as negative and anti-clockwise as positive, you get the same equation. And that's what I've told you. [00:08:53.636] Force in the cable is not considered because that will be acting opposed to each other, that will balance with each other. Okay. And also there is no other horizontal reaction that is acting, that's why the reaction at [00:09:08.708] O along the horizontal direction is a zero. Okay? Right. So we move on to the next problem. Next problem is 3 bar 54. So the problem statement goes something like this. [00:09:25.604] The pin A which connects the 200 kg steel beam with center of mass at the G to the vertical column is welded both to the beam and to the column. To test the weld [00:09:42.884] The eighty kg man loads the beam by exerting a

eighty newton force on the row, which passes through a hole in the beam as shown, calculate the chart. [00:09:59.524] or couple yum supported by the pin. So what is given here is

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:10.340] There is a pin at here, that is this one. So when you have a pin here you have two reaction, one vertical and one uh uh horizontal force. [00:10:22.980] The sulphate of the beam is given as two hundred kgs, which will be acting at the center, that is given as G here, which is at a distance of thousand two hundred mm from the left hand. Then he says that the beam is welded. [00:10:46.948] To the column. You can see here, they have given that it is welded here. Welded pin. Okay. To test this weld, because without the weld, the beam would try to rotate in the clockwise direction. Since there is a weld there. [00:11:04.996] This would resist the rotation. Okay. And they have also given you the weight of the person which is ADKG, which will be acting along the center line of the person, which is at a distance from A by 1200 plus 600, which is 1800. [00:11:25.796] And he also says that the person exerts a 800 newton force on the rope, which is passing through the beam by a hole, which is drilled. So when he exerts a force, he is pulling it. So when he exerts that [00:11:45.316] The same rope is continued here. So you will have the same tension in this also, which is equal to 800 newtons. Here also you will have the 800 newton force. And that is taken by this pulley. So here in this case, it will be 800 plus 800 will be the force that is acting. [00:12:04.260] But however if I cut it here, there will be 1804 newton which will be acting downwards, plus here also. Okay. Because of this is asking calculate the target M or the couple M. [00:12:19.012] Supported by the pin that is at this point what is the moment that is being supported or the target that is being supported. So what are the forces acting? The free body diagram is given here. You have the reaction R here. The horizontal reaction is not indicated because there is no horizontal force here. [00:12:37.892] Then you have a momentum that is indicated in the anticous direction. Cell weight of the beam, cell feed of the person, then the cable weight. [00:12:49.572] Which will be acting on the person itself, even though there is a hole there, it is not going to be directly acting on the beam. It is going to be transferred from the [00:12:58.404] So that's why it is acting along that. Plus, in addition to that, you will have also the cable which is attached to this point, so you will also have that force. So totally these are the forces that's going to be acting. The force acting is uh [00:13:20.580] The sulfate of the bee? Self rate of the person. So that is gonna be acting, right? As I've explained. Okay. Now uh taking moment about A [00:13:39.172] Okay, that is this point. So what do we get? We get the s first sulfate of the person eighteen to nine point eight one. The distance at which it is acting is thousand eight hundred from the left end, that is A. Plus we have one hundred. [00:13:55.268] Uh Kg that is 29.81 into thousand two hundred is the sulfate of the beam that is taken here plus the cable whatever is there that has been taken here. Only one unknown is M and this is acting in the [00:14:13.508] Uh anti-clockwise direction, all the clockwise forces or the moments are taken positive, anti-clockwise moment is taken as negative. So from this we will directly get the value of yeah. Okay. I hope all of you have got this. I have a doubt of whether [00:14:31.876] the force that is acting [00:14:36.260] Or the force with which the person is pulling is not 800, I think it is 300. Make that correction, please. It is 300 newtons. So with that we get the moment acting at A, that is A equals 4.94. [00:14:54.532] Uh Newton. No, sorry, it is kilo Newton meetings. Okay. Right. So I hope uh all of you have understood the these problems.

EV0000178 · EV0000179 · EV0000180 · EV0000181

Transcript block 4 (source interval 000900-001199 seconds): [00:15:12.868] We have come to the end of equilibrium chapter. So basically what we have to understand in equilibrium chapter is first thing is the equations of equilibrium. For a two dimensional there are three equations of equilibrium. One is sigma [00:15:30.916] F along the vertical, algebraic sum of all the forces acting along the vertical is zero. Algebraic sum of all the forces acting along the horizontal is zero, then algebraic sum of the moment at any point should be equal to zero. So when we have these three equations, we can solve [00:15:49.988] Three unknowns. So if you have a structure which has got only three constraints or three unknowns, we can use the equations of equipment solve it. They are called as determinate structure. If you have more than three constraints, [00:16:10.180] Then the number of constraints more than three is called as a redundant reaction or constraint. And we need to have more equations to solve such problems other than the three equations of economy. Such structures are called as indeterminate structure. [00:16:29.732] Okay. Then we should also understand what are the different types of uh supports that we have. I have explained this, which is given also in the textbook on page number one hundred and eleven if I'm not wrong, and as well as hundred and twelve, where we have discussed about cable. [00:16:48.452] Smooth surface, rough surface, roller surface, or roller support, then sliding guide, then pin connection, building connection, gravitational force, linear, spring, [00:17:07.044] And torsional spring. Okay. So this we have discussed, so you should know about this. Then how to draw the free body diagram. Then what are the different types of supports that we have? How do we replace these supports by reaction. [00:17:23.876] Okay, and then we have discussed about what are the different types of structures that is determined and indeterminate. Okay. With this we will be able to solve the problems in case of uh equilibrium chapter. Okay. So [00:17:42.308] In the next class, I'll be starting with chapter four, which is on trusses. Again, in trusses, we will be continuing to use these equations of equilibrium along with what we have learnt in unit one, that is resolution of forces. [00:18:00.996] Moment topic, couple as well as the resultant. So your unit one and unit two, chapter three. Knowledge is very much required to understand chapter four, that is process chapter. [00:18:19.332] Okay, along with that you should also understand other things related to trusses which I will be discussing more in depth when I go to take it in the next class. Okay. So [00:18:33.060] Whatever problems I have discussed here, there are other problems also which I would request you to go through and solve them. If you have any doubts, you are always welcome to discuss with me. I'll be very happy to help you. So with this [00:18:49.124] We have come to the end of today's class as well as end of chapter three, that is equilibrium. Thank you very much. My name is Dr. Sankitesh from Civil Engineering under the Department of Science and Humanities. [00:19:04.292] PS University Thank you, see you in the next class. That is class number thirty. However, we have one more class that is 29, which is kept for experiential learning. Thank you and see you in the next class.

EV0000182 · EV0000183 · EV0000184 · EV0000185

Full transcript: 2734 normalized words, 71 timestamped segments, 1158.336 seconds duration. Low-confidence segments flagged: 0.

EV0000170 · EV0000171 · EV0000172 · EV0000173

13 UE26ME142A_Unit 2_Class 21_AV Summary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 21_AV Summary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000186 · EV0000187

Graphs and diagrams



Figure G000076. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 1.

EV0000422

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ENGINEERING MECHANICS STATICS

Engineering Mechanics

Statics

Eighth Edition
SI Version

J. L. Meriam and L. G. Kraige

Unit 2 - Chapter 4

Structures - Trusses

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses

Figure G000077. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 2.

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ENGINEERING MECHANICS STATICS

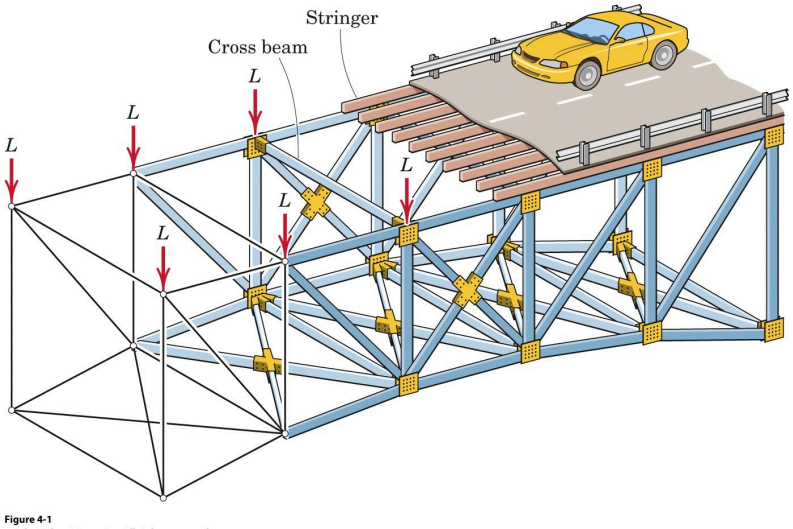


Figure 4-1
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S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses

Figure G000078. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 3.

EV0000428

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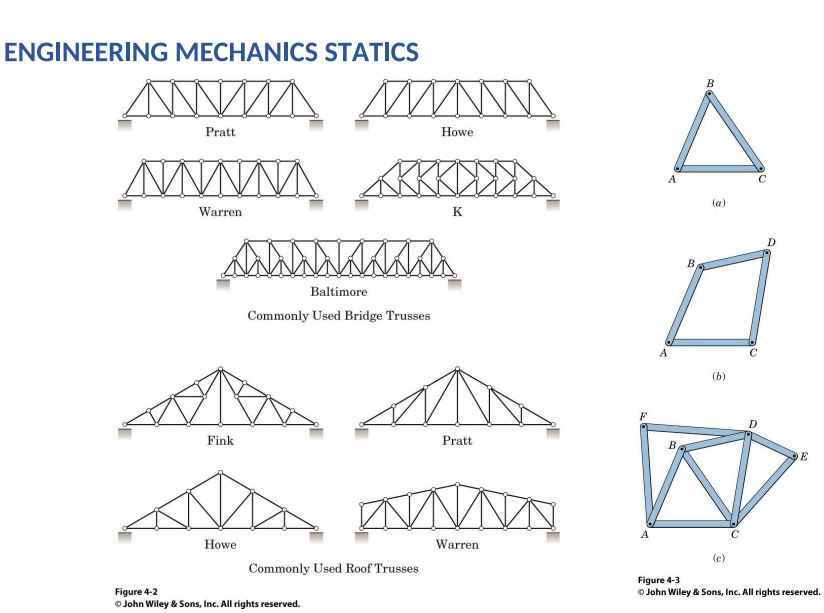


Figure G000079. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 4.

EV0000431

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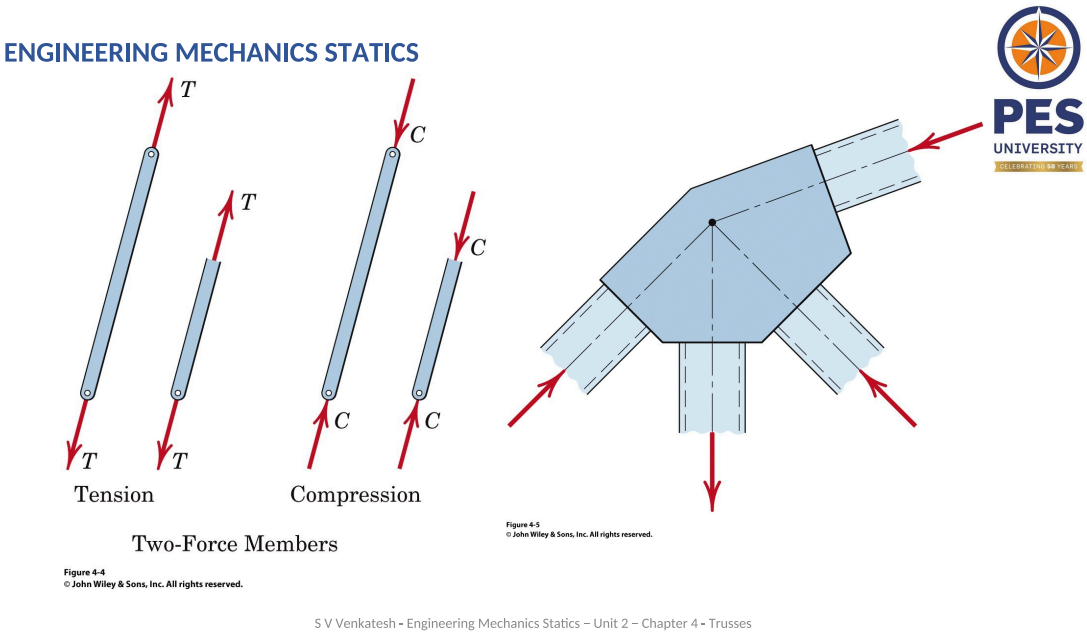


Figure G000080. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 5.

EV0000434

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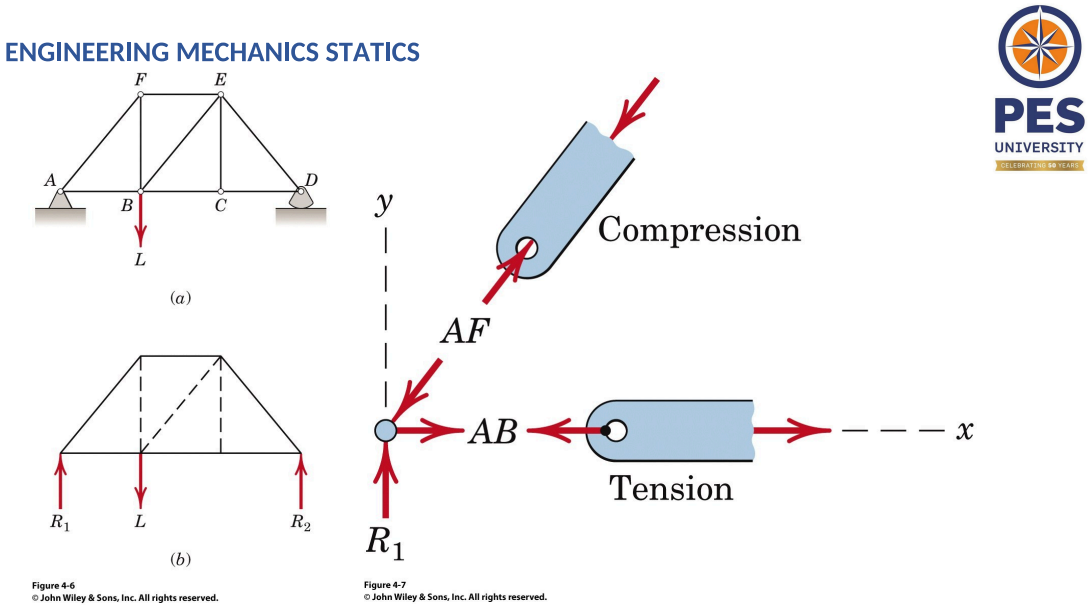


Figure G000081. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 6.

EV0000437

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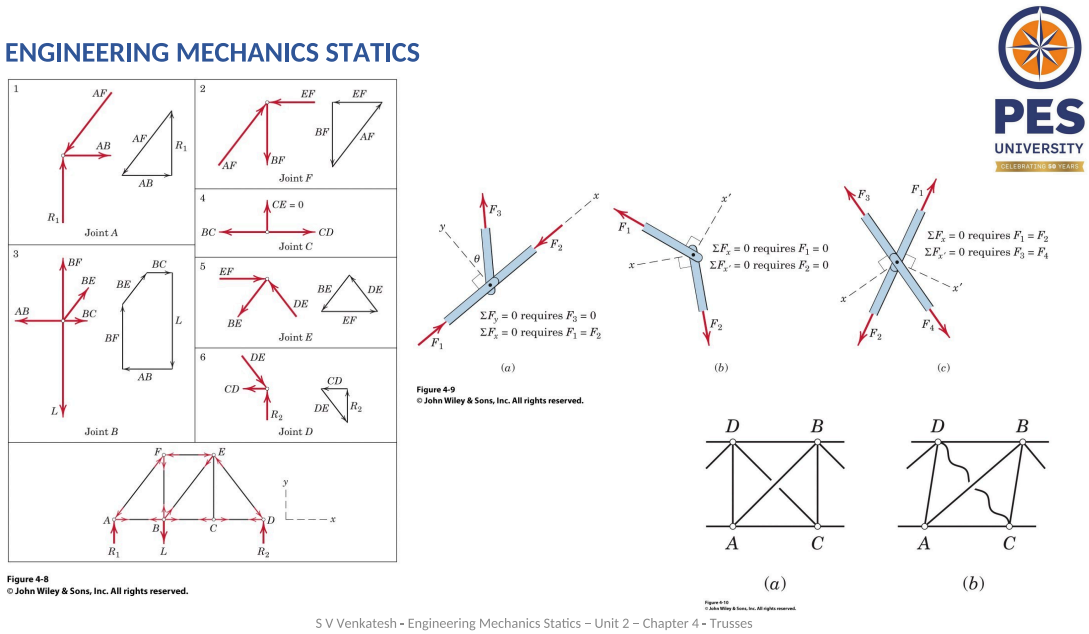


Figure G000082. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 7.

EV0000440

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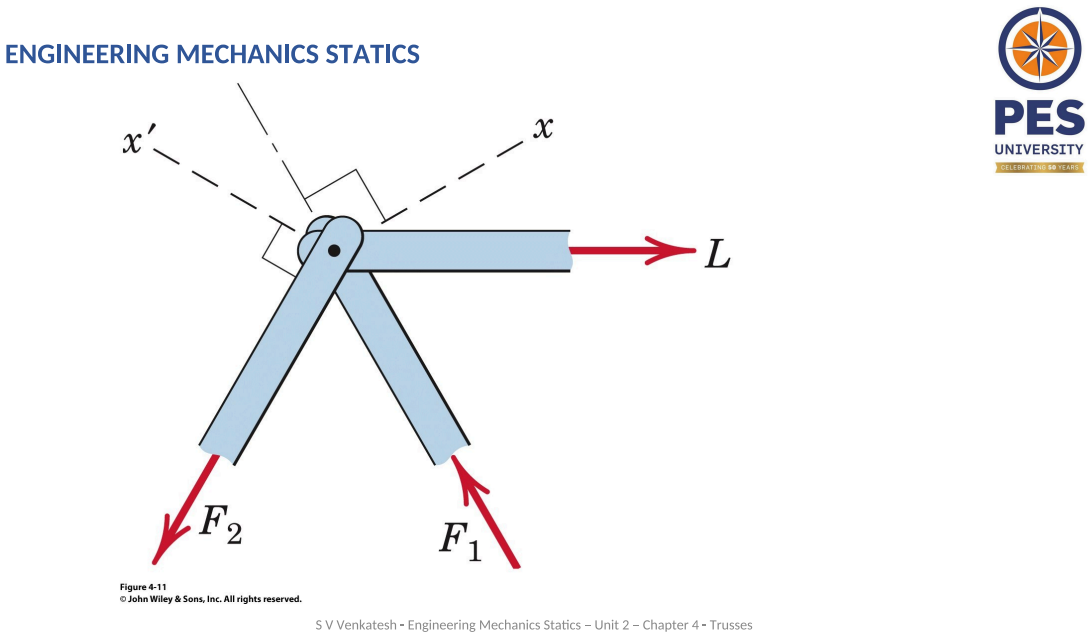


Figure G000083. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 8.

EV0000443

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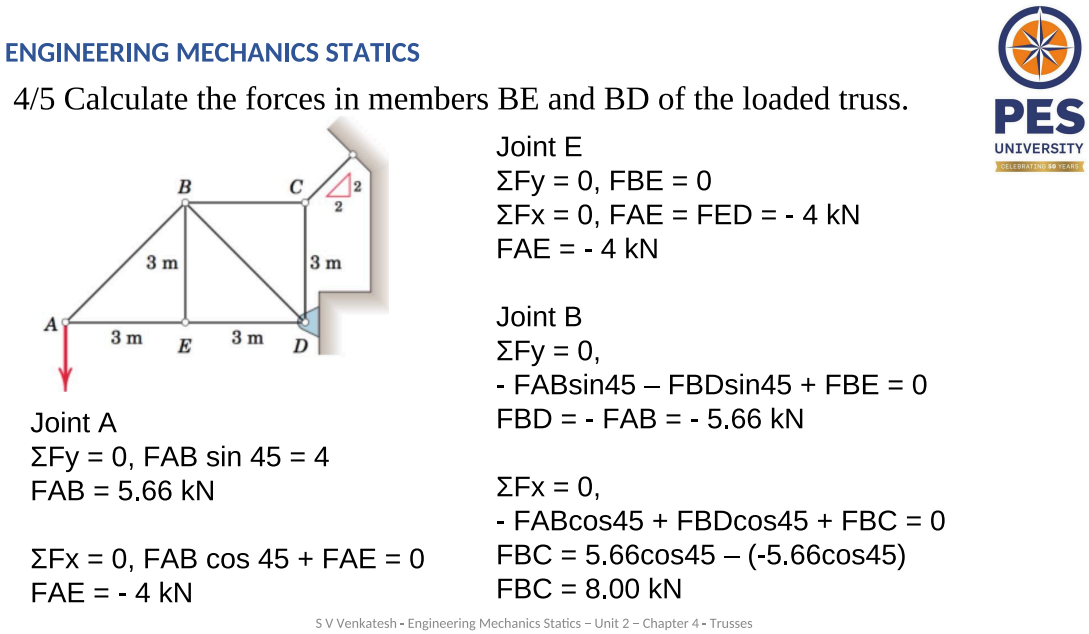


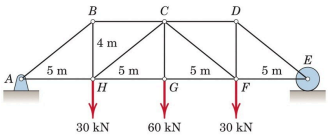
Figure G000084. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 9.

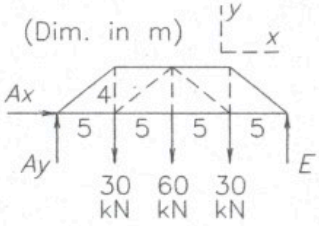
EV0000446

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ENGINEERING MECHANICS STATICS

4/10 Determine the force in each member of the loaded truss. Make use of the symmetry of the truss and of the loading.





As a whole: $\Sigma F_x = 0 \Rightarrow A_x = 0$

$A_y = E = 60$ kN by

$\Sigma F_y = 0$ and symmetry.

Joint A: $\left(\theta = \tan^{-1}(4/5) = 38.7°\right)$



$\Sigma F_y = 0 : 60 - AB\sin\theta = 0,\ \underline{AB = 96.0 \text{ kN } C}$

$\Sigma F_x = 0 : AH - 96.0\cos\theta,\ \underline{AH = 75 \text{ kN } T}$

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses

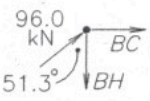
Figure G000085. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 10.

EV0000449

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ENGINEERING MECHANICS STATICS

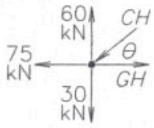
Joint B:



$\Sigma F_x = 0 : BC + 96.0\sin 51.3° = 0,\ \underline{BC = -75 \text{ kN } (C)}$

$\Sigma F_y = 0 : -BH + 96.0\cos 51.3° = 0,\ \underline{BH = 60 \text{ kN } T}$

Joint H:



$\Sigma F_y = 0 : -CH\sin\theta + 30 = 0,\ \underline{CH = 48.0 \text{ kN } C}$

$\Sigma F_x = 0 : 48.0\cos\theta + GH - 75 = 0,\ \underline{GH = 112.5 \text{ kN } T}$

Joint G:



$\Sigma F_y = 0 \Rightarrow \underline{CG = 60 \text{ kN } T}$

By symmetry:

$FG = 112.5$ kN T, $CF = 48.0$ kN C

$CD = 75$ kN C, $DF = 60$ kN T

$\underline{EF = 75 \text{ kN } T,\ DE = 96.0 \text{ kN } C}$

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses

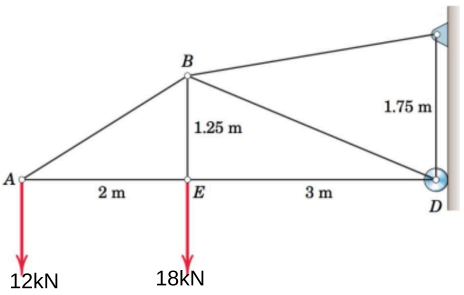
Figure G000086. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 11.

EV0000452

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ENGINEERING MECHANICS STATICS

4/8 Determine the force in each member of the loaded truss.



Tanθ1 = 1.25/2,
θ1 = 32
Tanθ2 = 1.25/3,
θ1 = 22.61
Tanθ3 = (1.75 - 1.25) / 3,
θ3 = 9.46

Joint A
ΣFy = 0, FABSin32 = 12
FAB = 22.65 kN

ΣFx = 0, FABcos32+FAE = 0
FAE = - 19.20 kN

Joint E
ΣFy = 0, FEB = 0
ΣFx = 0, FED = FAE = -19.20 kN

Figure G000087. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 12.

EV0000455

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ENGINEERING MECHANICS STATICS

•4/8 Conti…..

Joint B
ΣFy = 0
- FABsin32 - FBE - FBDsin22.61 + FBCsin9.46 = 0
FBD = (FBCsin9.46 - 22.65sin32) / sin22.61
FBD = 0.43FBC – 31.22

ΣFx = 0
- 22.65cos32 + FBDcos22.61 + FBCcos9.46 = 0
19.20 = (0.43FBC – 31.22)0.92 + 0.99FBC
FBC = (19.20 + 28.82) / 1.38
FBC = 34.79 kN

FBD = - 24.54 kN

Figure G000088. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 13.

EV0000458

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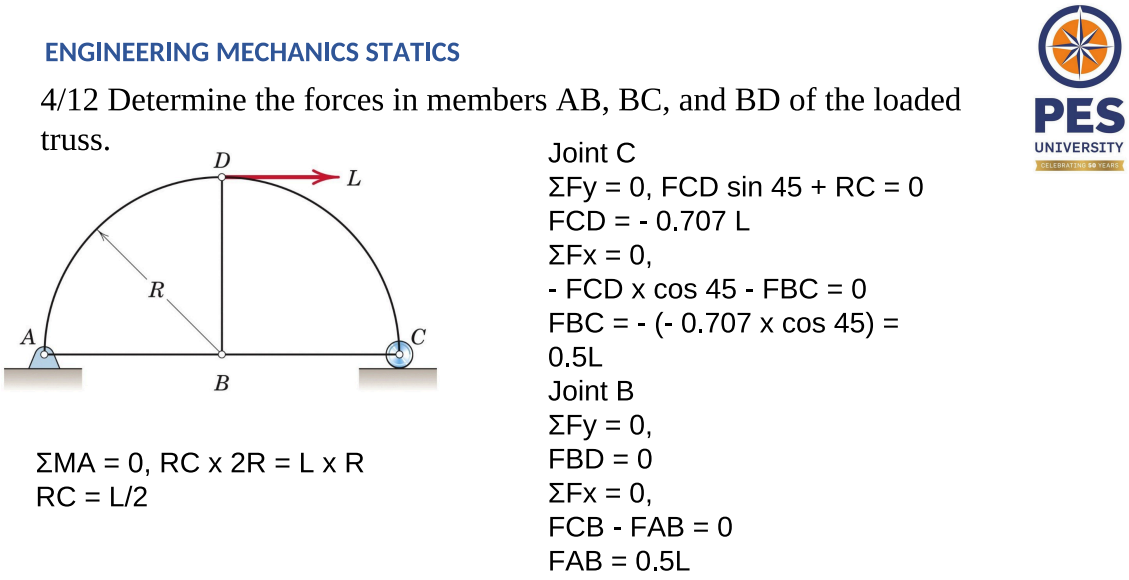


Figure G000089. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 14.

EV0000461

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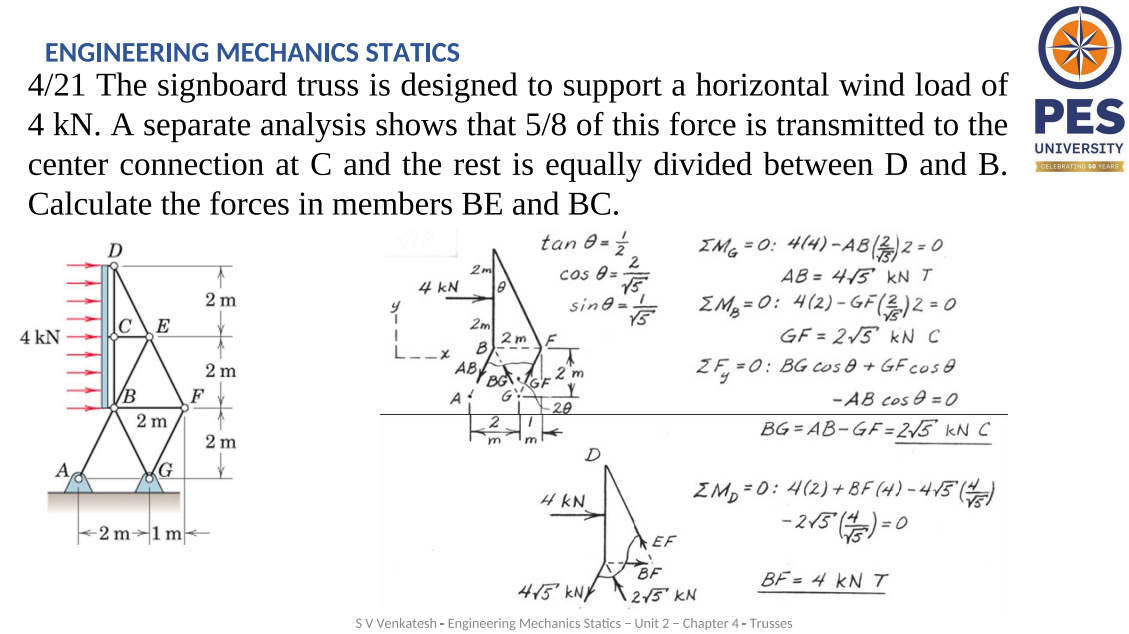


Figure G000090. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 15.

EV0000464

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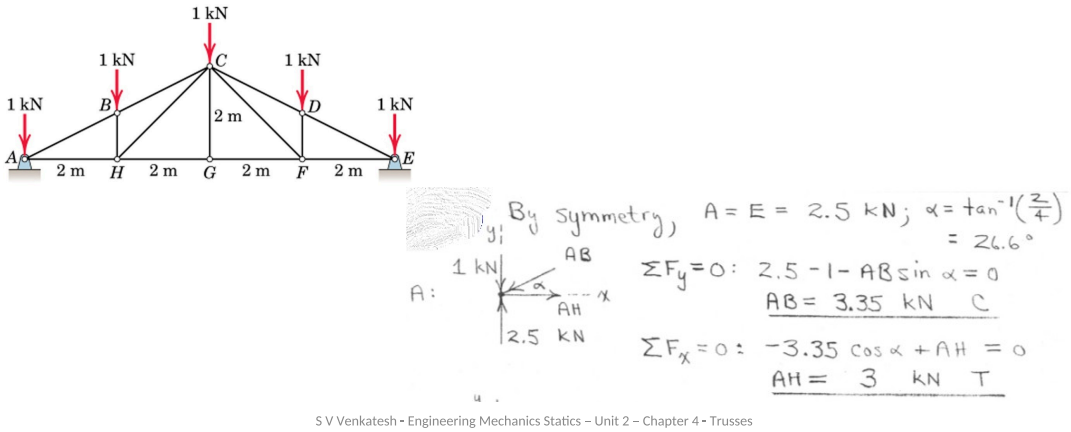


Figure G000091. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 16.

EV0000466

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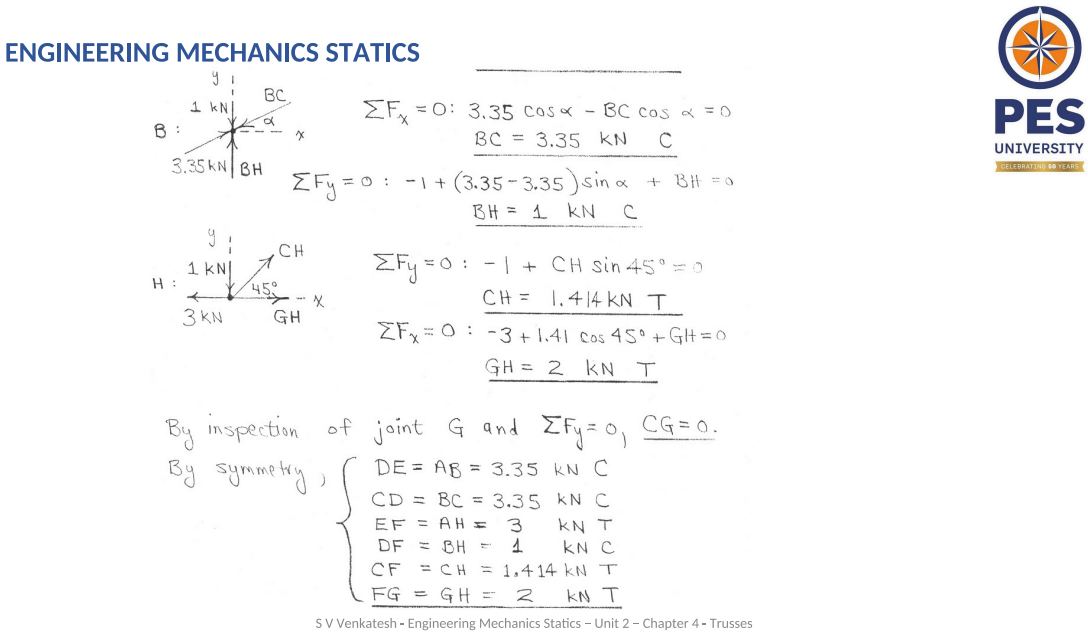


Figure G000092. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 17.

EV0000467

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Figure G000093. UE26ME142A Unit 2 Class 21 Lecture PPT · source slide page 18.

EV0000468

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000421 · EV0000422 · EV0000423
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Engineering Mechanics Statics Eighth Edition SI Version J. L. Meriam and L. G. Kraige Unit 2 - Chapter 4 Structures - Trusses S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000424 · EV0000425 · EV0000426
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000427 · EV0000428 · EV0000429
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000430 · EV0000431 · EV0000432
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000433 · EV0000434 · EV0000435
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000436 · EV0000437 · EV0000438
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000439 · EV0000440 · EV0000441
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000442 · EV0000443 · EV0000444

9. Source slide page 9 (extracted text): ENGINEERING MECHANICS STATICS 4/5 Calculate the forces in members BE and BD of the loaded truss. Joint E ΣFy = 0, FBE = 0 ΣFx = 0, FAE = FED = - 4 kN FAE = - 4 kN Joint B ΣFy = 0, - FABsin45 – FBDsin45 + FBE = 0 FBD = - FAB = - 5.66 kN Joint A ΣFy = 0, FAB sin 45 = 4 ΣFx = 0, FAB = 5.66 kN - FABcos45 + FBDcos45 + FBC = 0 FBC = 5.66cos45 – (-5.66cos45) ΣFx = 0, FAB cos 45 + FAE FBC = 8.00 kN = 0 S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses FAE = - 4 kN EV0000445 · EV0000446 · EV0000447

10. Source slide page 10 (extracted text): ENGINEERING MECHANICS STATICS 4/10 Determine the force in each member of the loaded truss. Make use of the symmetry of the truss and of the loading. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses ## Page / slide 11 (OCR) ENGINEERINGMECHANICS STATICS Joint B: PES 96.0 EF.=0:BC+96.0sin51.3°=0,BC=-75kN (C) UNIVERSITY kN BC CELEBRATING50YEARS BH ∑F=0:-BH+96.0c0s51.3°=0,BH=60kNT 51.3 Joint H: 60 kN EF=0:-CHsinϑ+30=0.CH=48.0kNC 75 kN GH ∑F.=0:48.0c0sϑ+GH-75=0,GH=112.5kNT 30 kNI Joint G: CGA EF=0=CG=60kNT FG By symmetry: 112.5 FG=I12.5kNT,CF=48.0kNC kN 60 CD=75kNC,DF=60KNT kN EF=75kNT.DE=96.0KNC S V Venkatesh - Engineering Mechanics Statics - Unit 2 - Chapter 4 - Trusses EV0000448 · EV0000449 · EV0000450

11. Source slide page 11: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000451 · EV0000452 · EV0000453

12. Source slide page 12 (extracted text): ENGINEERING MECHANICS STATICS 4/8 Determine the force in each member of the loaded truss. Joint A ΣFy = 0, FABSin32 = 12 FAB = 22.65 kN ΣFx = 0, FABcos32+FAE = 0 FAE = - 19.20 kN 12kN 18kN Tanϑ1 = 1.25/2, Joint E ϑ1 = 32 ΣFy = 0, FEB = 0 Tanϑ2 = 1.25/3, ΣFx = 0, FED = FAE = ϑ1 = 22.61 -19.20 kN Tanϑ3 = (1.75 - 1.25) / 3, S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses ϑ3 = 9.46 EV0000454 · EV0000455 · EV0000456

13. Source slide page 13 (extracted text): ENGINEERING MECHANICS STATICS •4/8 Conti..... Joint B ΣFy = 0 - FABsin32 - FBE - FBDsin22.61 + FBCsin9.46 = 0 FBD = (FBCsin9.46 - 22.65sin32) / sin22.61 FBD = 0.43FBC – 31.22 ΣFx = 0 - 22.65cos32 + FBDcos22.61 + FBCcos9.46 = 0 19.20 = (0.43FBC – 31.22)0.92 + 0.99FBC FBC = (19.20 + 28.82) / 1.38 FBC = 34.79 kN S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses FBD 24 54 kN EV0000457 · EV0000458 · EV0000459

14. Source slide page 14 (extracted text): ENGINEERING MECHANICS STATICS 4/12 Determine the forces in members AB, BC, and BD of the loaded truss. Joint C ΣFy = 0, FCD sin 45 + RC = 0 FCD = - 0.707 L ΣFx = 0, - FCD x cos 45 - FBC = 0 FBC = - (- 0.707 x cos 45) = 0.5L Joint B ΣFy = 0,ΣMA = 0, RC x 2R = L x FBD = 0R ΣFx = 0,RC = L/2 FCB - FAB = 0 FAB = 0.5L S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000460 · EV0000461 · EV0000462

15. Source slide page 15 (extracted text): ENGINEERING MECHANICS STATICS 4/21 The signboard truss is designed to support a horizontal wind load of 4 kN. A separate analysis shows that 5/8 of this force is transmitted to the center connection at C and the rest is equally divided between D and B. Calculate the forces in members BE and BC. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000463 · EV0000464 · EV0000465

16. Source slide page 16 (extracted text): ENGINEERING MECHANICS STATICS 4/23 A snow load transfers the forces shown to the upper joints of a Pratt roof truss. Neglect any horizontal reactions at the supports and solve for the forces in all members. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000466
17. Source slide page 17 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses ## Page / slide 18 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000467
18. Source slide page 18: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000468

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.540] Dear students, welcome to engineering mechanics statics classes. My name is Dr. SV Vengadesh from Civil Engineering under the Department of Science and Humanities, PS University. This is class number thirty-one. We are in the process of understanding. [00:00:21.988] Chapter 4, that is trusses under unit 2. In the last class I had explained about what is a truss, what is trust made up of, what are the different types of trusses that we have, what is plain truss, what is [00:00:36.644] The load that can be applied on trusses as well as what is tension forces acting on the members and what is compressive forces acting on the members. And also I talked about what are the different methods of determining the forces in the members. [00:00:53.312] which is either in going to be in the compression or at tension. So we can apply the forces only at the joint. Okay. So we'll continue from there. So in the last class I had also talked about this slide. [00:01:06.788] which explains what are the different methods of determining the forces in the members. The first method is method of joint. There's one more method which is called as method of section, but in the syllabus prescribed for you, we have only method of joint, so we'll concentrate only on method of joints. [00:01:24.964] Now to explain the method of joints I must take a simple example as plane trust ABCDEF At A we have a pin connection, at D we have a roller connection and there is a force applied at B which is here which is also vertically downwards. [00:01:43.556] Since there is no horizontal force acting on the truss, there is no horizontal reaction acting at A. So we will have only two reactions that is vertical reactions acting at A, which is considered as R1, and vertical reaction acting at D, which is considered as R2. Now these two reactions can be determined by using the equations of equilibrium. [00:02:04.644] Okay. Now coming to method of joints, the first step that is involved in method of joint is to identify a joint which has got only two unknowns. Okay. Now if you observe the joints from A to E. [00:02:19.268] None of the joints have got only two unknowns unless and until we determine the support reaction R1 and R2. If we determine the support reaction R1 and R2, there will be two joints that is joint A and joint D which will have only two unknowns. [00:02:34.928] That is in member AF and member AB as well as if you take joint D it will be member DC and DF. However, here we have considered a joint A, that is this this joint A. Okay, the free body diagram is represented here. So the force that is acting at joint A, if you observe here. [00:02:55.012] is the external force, which is the reaction force, that is R1, and then there are two members, AF and AB. [00:03:03.460] I am assuming the force in the member A B as tension. So earlier I discussed if the external forces are moving away from each other, the body would elongate, whereas the internal forces will be acting towards each other, the body will

still elongate and that member will be under tension. So here [00:03:20.528] This is external force, this is internal force. So since here R1 is external force, so we have to consider the internal force for this to be in equilibrium. So the force in the member AF is taken to be moving away from the joint. [00:03:35.940] Okay. Next the force in A uh the force in AB is considered to be moving away from the joint, which is under tension. Now coming to the other member that is AF it is assumed that the member is under compression, so in external forces will be acting towards each other. [00:03:53.028] Whereas the internal forces will be acting away from each other. This is the external force is acting towards there will be one more here. Internal forces is acting away from each other. So that means it will be towards this joint. Okay. So that is what is assumed. Now if I could find out what is the angle made by A, sorry, F [00:04:12.132] AB that is this angle. I can resolve the force in the member AF along vertical as well as along horizontal and then use the equations of equilibrium to determine the forces in the member AF and AB. There are two equations of equilibrium, so we can determine these two forces. [00:04:30.948] Okay. Next after taking this joint A, next we have to choose the other joint which has got only two unknowns. Now since we have already determined the force in Af, if I take joint of, there will be two unknowns, that is these two. [00:04:45.828] Again, we can resolve the force in the member AF along horizontal and vertical, use the equations of equilibrium to find out the forces in F E and F. [00:04:59.044] So, once I do this by using the equations of equilibrium, then I can choose joint B where I will be knowing the forces in these two members. So these two members can be determined. Once I determine these two members, now I can take joint C, I can determine the forces in these two members.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:18.724] Then I can take joint E where I will be knowing all these three members. There will be only one non-on which is E D which I can solve. Then if I take joint D, now we know the force in the member E D as well as the force in the member C D. [00:05:35.780] And if I had already determined R2, I would also be knowing R2. But what we could do here is we can take R2 as an unknown. [00:05:45.092] Determine R2 by using the forces that we have got from the member ED and C D, and check whether we get the same answer, whatever we would have got earlier. So this ensures that whatever procedure that we have adopted or whatever values that we have got, if they are equal, are correct. [00:06:03.972] in the members. Now in the entire engineering mechanics, whatever that is included in the syllabus, only for trusses we would be able to say whether the forces that we have got in the members are correct or not because we have a chuck in this case. [00:06:22.916] So whatever I have explained is what is given here. All the joints are given here, and then finally we will find out the forces in all the members, and once we get the forces in all the members, then we will represent those forces [00:06:38.672] on the truss like this. So this indicates the [00:06:43.588] uh type of forces that is acting in each of the member. For example, if you take AF, since the forces are acting away from each other, they are internal forces. So this is under compression. Now if you take these two the forces is acting towards each other that is AB [00:06:58.928] That is under tension. Then if you take the member F B, again the forces are acting towards each other, that is under compression. Then if you take FE, the forces are acting away from each other is under tension. So similarly we can represent all the [00:07:15.024] forces in the members and also that would give whether it is under compression or tension. This is the method that we should use to get the forces in the members. I hope all of you have understood. Now let's look at some special cases how we can [00:07:33.860] uh consider these special cases to find out the forces in the members easily. Okay.

The first thing is you have a member A B C D okay like this, it's a square one. And then we also have the member [00:07:51.972] B C and AB. Now here this is not a joint. This is just overlapping each other and each of them can move individually without the joint considering them. Okay. Now if because of the external forces, if the body is struck like this, [00:08:10.788] And also if you observe at each of the joint there are more than [00:08:16.644] three members. So we cannot take any of the joint unless and until we find out forcing one of the member. Right? Here these diagonal members are capable of supporting only tension. It cannot support compression. That means [00:08:35.556] This member AB and D are cables which can take tension which cannot take compression cable or row. Okay. So when you have an external force which makes the body to distract like this or deform like this. [00:08:55.460] You can easily identify that this diagonal T C is getting compressed, whereas this diagonal AB is getting elongated. So because of which I can say that [00:09:10.340] The force in the member A B will be under tension, whereas the force in the member D C will be under compression. [00:09:17.924] So since it is cable, it cannot support compression, the force in the member T C will be zero, whereas there will be some force in the member A B. Now I can take this joint D and then analyze for these two forces. [00:09:31.680] Then I can take this joint since I have no force in this, I can find out the force in these two, and similarly I can go out and join the other joints also. Okay that you have to remember if the members are in [00:09:46.020] capable of taking compression then we have to see whether the member is under compression or tension and take the suitable action according to that. Then now coming to other special cases if there are only three members of the joint and if two members are

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.132] Collinear, okay, the forces in these two members are going to be equal because of which the force in this member when I resolve it, if I take some of all the forces along this, the force in this member will be simply zero. Similarly, if you have [00:10:23.044] Only two members of the joint and they are uh acting like this, then if you take any axis x and x dash, the forces in those two members will be zero. [00:10:41.636] Okay. So if you remember this then it would be very easy. Then if you have a uh crossing of a members, since this member is collinear, the member in these two four. [00:10:55.664] forces in these two members will be same similarly the forces in these two members is going to be same it is not essential that they have to be at retire decrease only so if the members are collinear if they are in a straight line then the forces in those two members is going to be same okay [00:11:13.392] So I hope you have understood this. Now let's take a simple problem. Problem 4 bar 5 is what is taken here. Now in this problem they have asked you to calculate the forces in members BE and BD of the loaded trust. So this is the trust that is given. This is the trust that is given. [00:11:32.868] So in this truss uh Yeah in this truss there is a load acting at A [00:11:42.276] And you can also look at those joints. At joint A, there are only two members since we know the load at A already, so there are uh only two unknown forces that is acting. So I can straight away take joint A. [00:11:58.368] And the force acting in this member is 4 kilometers. I think that is missing here. Take that force acting at joint A as 4 kilometers. Okay. So we can determine the angle made by the [00:12:13.092] Angle BAE which is going to be this angle. So this is three meters, this is three meters. So this would be forty five degrees. Similarly, angle made by BDE is also forty-five degrees, and that's what is mentioned here. Now if I take this joint, I will assume [00:12:28.836] that the forces in all the members are under tension. Why am I doing this? Because it is easier to say initially that the forces are moving away from the joint. [00:12:43.108] And once I get the values of

this, if it if my assumption is correct, I get a positive answer. If my assumption is wrong, I'll get a negative answer, which indicates that the force in that member is a compressive force. So when I take joint A. [00:12:58.864] This one, this force in this member AF is acting away from A. Similarly, the force in the member A is also acting away from A. So I can resolve the force in the member AF along vertical and horizontal and take sigma f y equals 0. So what do I get? [00:13:16.004] I get the force in this member F A B. This is forty five, so sine forty five which is acting upwards. [00:13:22.308] And the force, the external force which is acting downwards is 4 kilonewtons. So I have taken it on the other side of the equation. So I get the force in the member F AB as 5.66 kilonewtons and it is positive, which indicates that that member is under tension. [00:13:40.612] Next, I'll take the other equation of the equilibrium that is summation of all the forces, algebraic sum of all the forces along X equals 0. So a component of this force is going to be acting along this, which is F. [00:13:53.284] Uh AB cos 45 which is acting towards the right so positive. Similarly we have the force in the member AE also which is acting towards the right that is also positive. So only two forces are acting, so I can find out what is FAE which we will hate as minus 40. [00:14:10.896] So since it is minus it is under compression. Now what we can do is we don't have to change the sign here itself. We will keep the sign positive or negative as it is. When we are substituting somewhere else for the value of this member forces A. [00:14:28.432] I will take the sign also into consideration as it is as negative. Or if it is positive, I will take it as positive. Because I am assuming the forces in all the members as tension itself that is moving away from each other, it will help me out very much. Okay. Next I will take joint E. [00:14:46.276] Okay. So when I take joint D I can take sigma f y equals zero. So by which I get the force in the member F BE to be equal to zero. Then if I take sigma f x is equal to zero. [00:14:58.564] Uh we get the force in the member E D to be same as the force in the member. Uh AE or EA, both of them are same, which is also going to be equal to when I substitute the value of minus 4, I get this to be minus 4. Okay. So that is also under compression. Next I will move on to joint B.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:17.604] So again here I'll take sigma f y equals zero. So these two forces when I equate it, since the force in this is zero, I will get the force in the member B D, which is equal to the force in the member. [00:15:31.012] FAB with a negative sign. So the force in the member is positive, so I get a negative value, which also indicates that this member B is under compression. Okay. [00:15:43.428] Next, I'll take summation of all the forces along X is equal to 0. So I get the component of this force, component of this force, and the force in B C, so I will get the value of the force in Bc as 8 kn, which is under tension. [00:16:00.388] Okay. So what is asked here is to find out the forces in the member B and P D. So if I solve this joint I will get A B and AE. If I solve the joint E I will get E B. Then if I solve for joint B I will get the force in member B D. [00:16:20.484] Okay now using the special conditions I can initially say that the force in the member BE is equal to zero because there is no force acting at E. [00:16:32.580] And these two member that is A and E D are collinear, the forces in those two members are going to be equal. So the force in the member B is equal to 0. So in this case I can solve only three joints to get the forces in the member B and B D. [00:16:52.868] Okay so we have already found out the forces in the member. We have got the force in the member BE as zero, okay because there is no force acting at E and B D as minus [00:17:11.108] uh five point six six kilometer which indicates that it is under compression. I hope all of you have understood this. Let's move on to the next problem. I've considered four

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Full transcript: 2873 normalized words, 66 timestamped segments, 1041.451 seconds duration. Low-confidence segments flagged: 0.

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14 UE26ME142A_Unit 2_Class 22_AV Summary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 22_AV Summary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000202 · EV0000203

Graphs and diagrams



Figure G000094. UE26ME142A Unit 2 Class 22 Lecture PPT · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

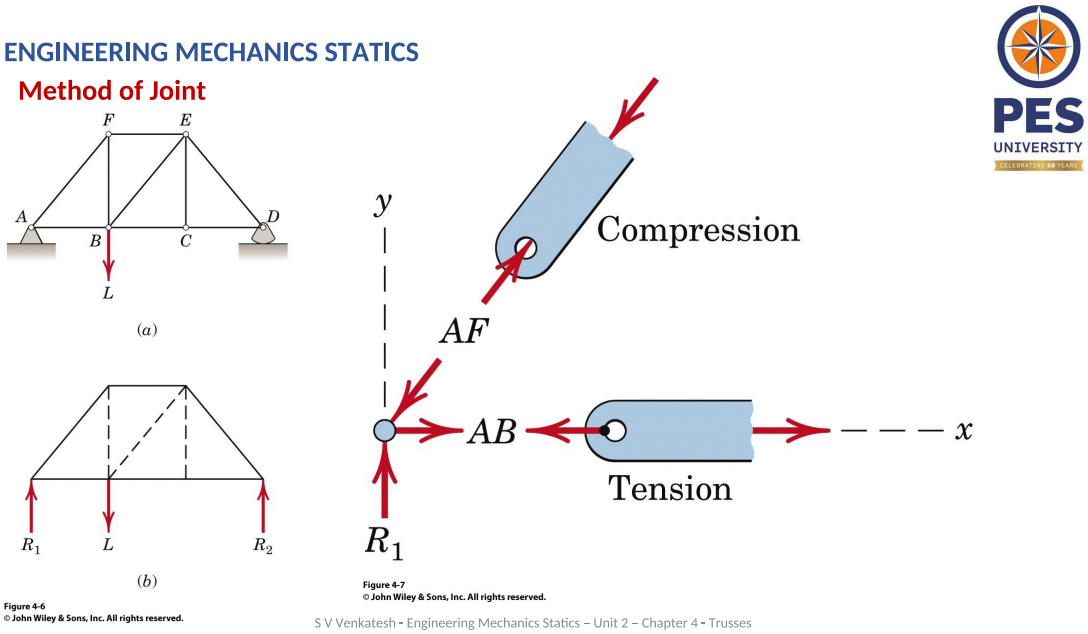


Figure G000095. UE26ME142A Unit 2 Class 22 Lecture PPT · source slide page 2.

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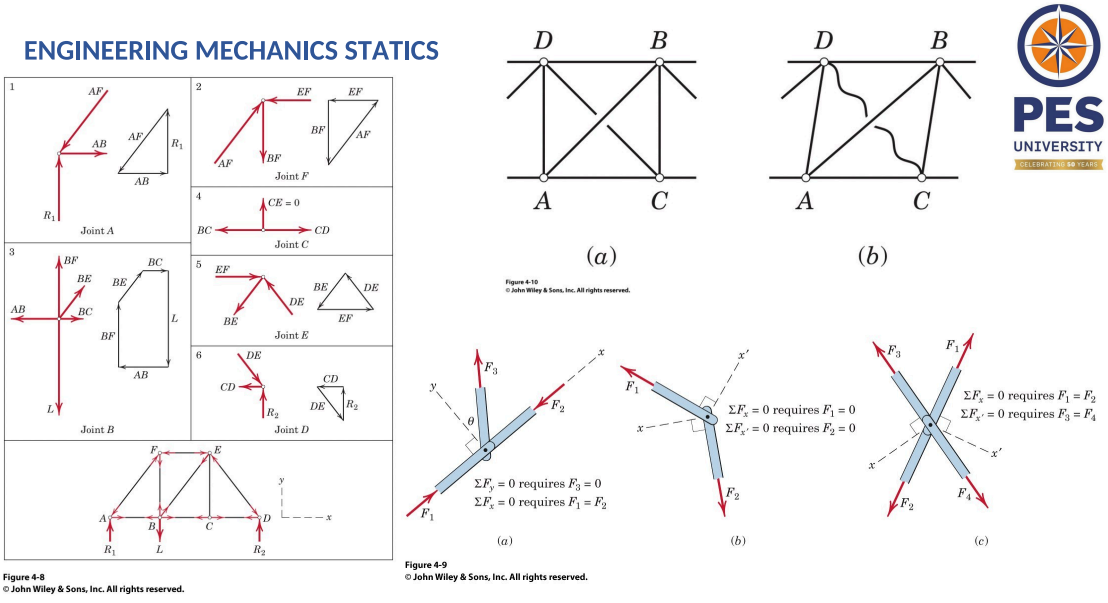


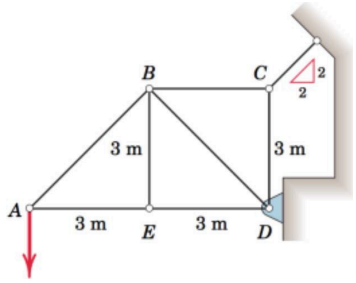
Figure G000096. UE26ME142A Unit 2 Class 22 Lecture PPT · source slide page 3.

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ENGINEERING MECHANICS STATICS

4/5 Calculate the forces in members BE and BD of the loaded truss.



Angle BAE & BDE = 45⁰
Joint A
ΣFy = 0, FAB sin 45 = 4
FAB = 5.66 kN
ΣFx = 0, FAB cos 45 + FAE = 0
FAE = - 4 kN

Joint E
ΣFy = 0, FBE = 0
ΣFx = 0, FAE = FED = - 4 kN
FAE = - 4 kN

Joint B
ΣFy = 0,
- FABsin45 – FBDsin45 + FBE = 0
FBD = - FAB = - 5.66 kN

ΣFx = 0,
- FABcos45 + FBDcos45 + FBC = 0
FBC = 5.66cos45 – (-5.66cos45)
FBC = 8.00 kN

S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses

Figure G000097. UE26ME142A Unit 2 Class 22 Lecture PPT · source slide page 4.

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THANK YOU

Dr. S V Venkatesh
Civil Engineering
Department of S & H

Figure G000098. UE26ME142A Unit 2 Class 22 Lecture PPT · source slide page 5.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000469 · EV0000470 · EV0000471
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS Method of Joint S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000472 · EV0000473 · EV0000474
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000475 · EV0000476 · EV0000477
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 4/5 Calculate the forces in members BE and BD of the loaded truss. Joint E ΣFy = 0, FBE = 0 ΣFx = 0, FAE = FED = - 4 kN FAE = - 4 kN Joint B ΣFy = 0, - FABsin45 – FBDsin45 + FBE = 0 Angle BAE & BDE = 450 FBD = - FAB = - 5.66 kN Joint A ΣFy = 0, FAB sin 45 = 4 ΣFx = 0, FAB = 5.66 kN - FABcos45 + FBDcos45 + FBC ΣFx = 0, FAB cos 45 + FAE = 0 = 0 S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses FBC = 5.66cos45 – (-5.66cos45) FAE = - 4 kN ## Page / slide 5 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000478
5. Source slide page 5: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000479

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.516] Dear students, welcome back to Engineer Mechanics Statics classes. My name is Dr Svinkitesh, a civil engineer from the Department of Science and Humanities, PS University. [00:00:15.876] Uh in the last class we had discussed about uh unit uh two equilibrium chapter and also we had discussed few problems on equilibrium chapter. Today we will be starting with the next chapter, that is chapter number four. [00:00:35.940] uh from the textbook which is on trusses under unit two as prescribed by the university PS University. So this chapter [00:00:51.684] uh four from unit two is taken from engine mechanics statics uh textbook eighth edition si version written by Miriam and Craig okay and you can find the content of this [00:01:08.772] uh chapter in chapter four of the same textbook under trusses structures okay so what is a truss [00:01:19.524] A truss is a structural member which actually carries the load from the roofing element or a slab or a deck increase of a bridge. [00:01:32.964] And transfers the load coming on this roofing elements to the column on either side and finally this load will get transferred to the foundation. From foundation it will get transferred to the ground below. [00:01:49.988] So here in this picture you can see a bridge, okay? So you can see a car that is there on the Dex Lab. This is the Dex Lab and you can see this, they have given it as strangers, okay [00:02:04.900] The stringers are supported by this rafters or cross beams as they are called. And this cross beams are attached to this structure, which you can see on either side of the bridge. There is one here and there's one more here. [00:02:23.844] Which are nothing but the actual truss. Okay. The load from the vehicle and the Dex Lab will transfer to this stranger. Then the string from stranger it will get transferred to the cross vehicle. [00:02:39.652] From cross beam it will get transferred to the truss on either side, and you can see they have given the load at the joint, which is given as L L. Wherever there are joints. Here also you will have a load L. So there could be several parallel trusses. [00:02:59.172] to each other and you will have the vehicle moving

here and in case if it is a roof truss, usually the roof trusses are going to be slope. Okay, and the slope [00:03:11.428] uh you will have uh the process uh perpendicular to this slope and the cross beams will be in between the two uh [00:03:23.012] truss and your roofing element will be placed on this cross beam directly. Okay. So the load from the roofing element will transfer to the cross beam from which are also called as purlins in case of trusses or roofing trusses. [00:03:39.748] Then from the purlin or the cross staff, it will get transferred to the trusses on either side. Now, why do we go for such structure? A truss is basically used when the span [00:03:55.780] between the supports. Now in this case you have the truss here, this is the truss and this is the truss and you will have a support here as well as a support only half of the bridges given here. There will be another half on the left side of the screen. [00:04:11.392] And you will have a support on either the side. Now when this support between the the the span between the two supports is quite large, okay [00:04:23.748] Then we go for such type of structure. Okay? And also when we are not intended to construct an upper floor above the truss, we go for truss. [00:04:39.088] And usually the truss will be uh constructed of either wood. Earlier we used to do it in wood, but nowadays since it is very difficult to get wood and also there is restriction from the government for the use of wood. Uh we have moved on to using uh [00:04:56.164] iron or steel. It can also be manufactured using aluminium or any other suitable metal okay for the purpose. Now usually in case of trusses the truss will be

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:11.812] Constructed of a lighter material, lighter weight material. So the self weight of the truss will be very very less when compared to the load that it is going to be taking. For example, if you take the weight of the [00:05:29.028] Vehicles that will be moving on this uh bridge. [00:05:33.956] That weight will be quite large when compared to the self weight of this truss itself. So usually the weight of the trust, the self weight of the trust will not be considered for analysis of the supports or the forces in the members, but when [00:05:52.644] The in the problem if it is mentioned to be considered the sulphate, we have to take the sulphate into account. Okay. With this background, let's now look at what are the different types of uh [00:06:08.580] Bridge trusses that are employed. So the first five which is given here are the trusses for bridges. Now you can see here it is given as Pratt and Hobby, Warner, K and Batti Moore Truss. [00:06:28.836] Now this you can see the difference between these two. This case you have a central member and of course it is symmetrical when you see the geometry of the truss, but you can see the inclined members which are moving from top to bottom. [00:06:47.428] from left to right. Okay except the uh end once and it is symmetrical with respect to this axis whereas the right side the members the implant members are moving from [00:07:00.420] bottom to top from left to right. So that is the difference between these two. And you have four members here. Okay. Now before getting into this, a truss is made up of what are called as members and what are called as joints. [00:07:17.988] Members are the structure which are in between two joints. Now you can see here this is a member, this is also a member, this is also a member, this is also a member. Likewise you have several member here. [00:07:32.004] Ends of the member will be will be connected to what is called as a joint. So a joint is one where two or more members joint. So unless you understand a member, you can't understand joint. Unless you understand a joint, you cannot understand member. [00:07:51.620] So I'll repeat what is a member? A member is a structure which is in between two joints. Okay so this are members. So you have a joint here, this is a joint, this is a joint. So this is a member, this is also a member, this is also a

member. [00:08:10.052] Now, this is a joint where you have one, two, three, four members meeting at this point. But this is a joint where you have two members meeting. Plus, of course, you have the support reactions also here. This is also a joint where you have three members meeting. [00:08:25.284] Okay. So likewise you have different ways of connecting the members to the joints. Okay. Now the difference between this trust that the Pratt and Howie, okay, you can see here you again have a symmetrical axis here. [00:08:42.116] If you see the left side, the inclined members are moving from left to right, okay, and it is upwards, it's moving upwards. Whereas here, except the end, the other was moving from right to left, from up to down, okay. [00:08:57.796] And the right side of the uh members, the income members are moving from uh top to bottom from right to left. So that is the difference between these two. Similarly, you have Warner where you have [00:09:11.460] Each member moving in a zigzag way like this, the inclined member. One member if it is from left to right, the other member is from right to left, top to bottom, and this is from bottom to top. [00:09:22.660] And this is a K member you can see that this looks like a K joint, okay? Here also it is inverted K joint. So everywhere it you look like a K1, and this is Baltimore. That is named after the person who has invented this. [00:09:39.172] It is based on how you connect the the members and how do you uh take the joints into account. One more thing that you should remember in case of processes. [00:09:49.988] The members in the trusses will be loaded at the joints only. None of the member will be loaded in between. That is because [00:09:59.876] The members of a truss can take either a compressive load or a tensile load only. Okay. Next, coming to what are the different types of uh roof uh trusses that you have, there are four different types that is mentioned here.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:17.536] One is Fink, then it is Pratt. Again Hovi and Warner. Now again you can see what is the difference between a Pratt and a Hovi. Here you have [00:10:28.580] Member from going from the incline members going from bottom to top, from left to right, whereas in this case it is moving from top to bottom from left to right, the left side. Whereas the right side, you can see here it is the incline members are moving from [00:10:44.708] Top to bottom from left to right, whereas here it is bottom to top from left to right. And of course they are symmetrical, all of them are symmetrical here. Now you have to remember these type of trusses because in the examination, ISA or ESA they can ask you to draw a [00:11:03.524] A Pratt trust or a hobby trust, or they can give you a trust and ask you to name the type of trust. Okay, so you should remember this. [00:11:13.028] Okay. Now what is a plane truss? A plane truss is a truss in which you have all the members and the joints in the same plane. For example, if you take here, you have three members here, all these three members [00:11:28.032] If you take it, it is in the horizontal plane only. Similarly, there is a four members here, all the four members are in the same horizontal plane, and of course here you have several members, all the members are in the same plane. So that is what is called as a [00:11:45.040] Plain truss. Okay. Moving on. As I have told you that the members of a truss is made up of either only compression or only tension. [00:12:00.164] So here is an example of what is tension and what is compression. So there is a member that has been taken here, and if the external forces, these are external forces, if there are external forces which are moving away from each other, always remember. [00:12:17.248] Tension and compression will have a pair of forces, either they will be moving away from each other or moving towards. If the external forces are moving away from each other, the body would tend to elongate. [00:12:32.388] Okay, and that is what is called as tensile member or tension or the

member is under tension is what we call. [00:12:39.812] Now if you cut this and look at the internal member, the internal force in this member is acting opposite to this. So the forces be acting like this. So if I if this is only this part of the uh [00:12:54.628] member that is taken if you take the upper part of the member you have the external force acting upward so internal force will be acting towards each other. So if you take the internal forces the internal forces will be moving away from each other. Here it will be moves towards this joint. [00:13:09.392] Here it will be moving towards this joint. So the internal forces if they are moving towards uh each other. Now if you take this it will be moving towards each other, and if you take this one it will be moving towards each other, they are moving towards each other. [00:13:23.744] So the external forces will be moving away, so the member will get elongated. Such forces are defined as tensile force. Comping coming to compressive force. [00:13:34.820] If the external forces are moving towards each other along the longitudinal direction of the member, the member will try to get compressed. Okay. Such members are called as compressive members. Now if I cut this. [00:13:48.964] The bottom half and take it separately, you can see here. This is external force, compressive force, C, the internal force will be acting towards the joint for this half of it. Whereas if you take the upper half, the external the internal force will be moving towards this. So if the forces are moving towards the joint, [00:14:08.356] The internal forces that is away from each other, the external forces will be moving towards each other, the body will be under compression. Okay? I hope this is clear. In fact, I am going to repeat this again, taking the tension into consideration. [00:14:23.376] If the external forces are moving away from each other, the body would elongate. That is what is called as tensor member. Now, because of the external forces, if I take the internal forces, the internal forces will be acting towards each other, that is towards the joint. The body would get elongated, so it is under tension. [00:14:42.500] Similarly, if I take the compressive forces, the compressive forces, external forces will be acting towards each other, the member would get compressed, whereas if you take the internal forces, the internal forces will be acting away from each other, the body would get compressed. So it is called as a compressive member.

EV0000210 · EV0000211 · EV0000212 · EV0000213

Transcript block 4 (source interval 000900-001199 seconds): [00:15:01.796] Now, if I take any joint in case of a trust, some members will be under compression, some members will be under tension. [00:15:11.844] Now, what because of what these forces are coming? Because of the external force that is acting at the joint. And as I told you, in case of trusses, the lore cannot be applied in between the members. It has to be applied at the joint only. [00:15:28.624] So at the joints you can have a vertical force or an horizontal force or an inclined force which can be towards the joint or away from the joint. These are external forces. Because of these external forces, you get the forces in the members which are internal. [00:15:47.620] Okay so if I take this member this is the internal force that is acting in this member which is acting towards a joint so as I told you here if the internal member is acting towards the joint it will be in compression so similarly this one also is under compression [00:16:03.520] This one also is under compression, whereas this one is moving away from the joint. So this is under tension. Kindly remember this because this is going to be helpful when we are going to be solving problems on trusses. And especially in case of trusses. [00:16:20.324] What we are going to be doing is to determine what is what are the forces that is acting in each of the member and what is the type of force that is acting, whether it is compressive or tensile. Okay. Now there's a simple example that is given here, a roof thrust. Okay. [00:16:39.108] And how do we determine the forces in the members? There are two methods. First

method is method of joints, the second method is method of section. [00:16:51.108] Now your syllabus you have only method of joints and that is what we are going to be concentrating on. So we have an example here which is a roofing uh truss that is taken. [00:17:04.292] The end A is supported with a hinge support or a pin support, so you have two reactions, one vertical and one horizontal, and end B is supported with a roller support, so you have one reaction which will be acting vertically upwards. [00:17:22.020] We have to first determine the support reactions. [00:17:26.500] In this case, to identify a joint which will have the number of unknowns or the number of members where the unknowns are to be determined, which are the member forces, which are only two members are there. [00:17:43.716] So then we have to solve for this joint. So in case of method of joints, what we do is we take each joint separately as indicated here. You can see here they have indicated the joint A here. [00:17:58.436] Okay, after determining the support reaction R1 and R2, since there is only vertical force that is acting at B. [00:18:07.104] No other forces are acting at any of the joint. The reactions are also going to be vertical. So they have indicated the reaction at A as R1 and reaction at D as R2. And they have written the diagram here. [00:18:24.208] Considering the joint A only separately here. This is joint A where you have a reaction at A acting which is written as R. Then you have the member AF as well as the member AB. Now in [00:18:40.324] The member EAF, it is considered to be under compression. That means forces will be acting towards the joint. You can see that is what is considered here. See here the member internal forces is acting towards the external forces will be acting away. So we have to take the internal member internal forces so it is acting towards the joint. [00:18:59.524] Whereas the member A B it is considered to be in tension, so that means the forces will be acting away from the joint. So that is why he has taken this as a way. This is the [00:19:10.820] Force that is reacting external force. So we have to consider here the internal force that is what is considered here. So now in this joint, if I have con found what is RA, then I can apply the equations of equilibrium for this. That is [00:19:27.300] algebraic sum of all the forces along X to be equal to 0 and algebraic force of all the forces along Y to be equal to 0. I can determine the forces in the member AF and AB. So we would have got the forces in these two members. Similarly [00:19:44.388] Once I find the forces in these two members, I can next go to the next joint, which will have only two unknowns. That is, the next joint is A is F after A [00:19:56.400] I am taking the next joint as F. The m force in the memory AF is already determined from this joint. So you can see here the sense of the force in this is moving towards the joint. The sense of the force in this is moving towards the joint again.

EV0000214 · EV0000215 · EV0000216 · EV0000217

Transcript block 5 (source interval 001200-001499 seconds): [00:20:11.588] This one AB will not come into picture when I take the joint Ef, you can see here this is F. So this is a force that we have taken and the force in EF is also taken to be acting towards the joint, whereas the force in the member Bf is taken to be acting away from the joint. [00:20:30.672] Then we will analyze this again by using the equations of equilibrium to find out the these two unknowns, okay, which we can determine by using sigma fx equals 0 and f y equals 0. Once I find this, so now we have found out these three members along with this. So now I can take this joint where I will know these two members already. [00:20:50.724] The external force that is applied is also known. So I will have these two as unknown. So that is what is written here. So again, using the equations of equilibrium, I can find out these two members. Then you can take the next joint as [00:21:04.836] Joint C here. This member is known. These two I can find out. The mem force in this member will be zero. We will look at that a little bit later. So I can find out the forces in this. Then I can take the force in

the in the joint E. [00:21:19.232] So this is joint D. We have already determined the force in this, that is zero member in this. So we will have three here. So we have determined this, these two we can find out. Then coming to this joint D, here we would have determined these two earlier itself, and also R2, but we can take R2 as an unknown. [00:21:39.268] from these two uh forces whatever we know, if I I can find out R two and I can compare what is the value that I have got earlier to what I will get here. So that would become a chuck for us to determine. [00:21:52.192] Okay so we will stop uh it uh here we'll continue in the next class as we are running short of time I'll explain the other things in the next class. Thank you very much. See you in the next

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Full transcript: 3478 normalized words, 87 timestamped segments, 1333.141 seconds duration. Low-confidence segments flagged: 0.

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15 UE26ME142A_Unit 2_Class 23_AV Summary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 23_AV Summary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000222 · EV0000223

Graphs and diagrams



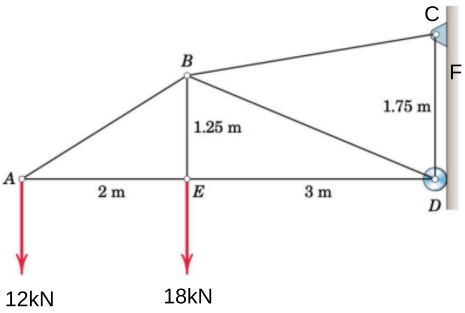
Figure G000099. UE26ME142A Unit 2 Class 23 Lecture PPT · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS STATICS

4/8 Determine the force in each member of the loaded truss.



Tanθ1(BAE) = 1.25/2,
θ1 = 32⁰
Tanθ2(BDE) = 1.25/3,
θ1 = 22.61⁰
Tanθ3(CBF) = (1.75 - 1.25) / 3,
θ3 = 9.46⁰

Joint A
ΣFy = 0, FABsin32 = 12
FAB = 22.65 kN

ΣFx = 0, FABcos32+FAE = 0
FAE = - 19.20 kN

Joint E
ΣFy = 0, FEB = 18 kN
ΣFx = 0, FED = FAE = -19.20 kN

Figure G000100. UE26ME142A Unit 2 Class 23 Lecture PPT · source slide page 2.

EV0000484

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ENGINEERING MECHANICS STATICS

•4/8 Conti…..

Joint B
ΣFy = 0
- FABsin32 - FBE - FBDsin22.61 + FBCsin9.46 = 0
FBD = (FBCsin9.46 - 18 - 22.65sin32) / sin22.61
FBD = 0.43FBC – 78.4

ΣFx = 0
- 22.65cos32 + FBDcos22.61 + FBCcos9.46 = 0
19.20 = (0.43FBC – 78.4)0.92 + 0.99FBC
FBC = (19.20 + 71.8) / 1.38
FBC = 65.94 kN

FBD = - 50.05 kN

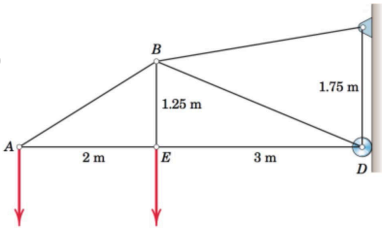


Figure G000101. UE26ME142A Unit 2 Class 23 Lecture PPT · source slide page 3.

EV0000487

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ENGINEERING MECHANICS STATICS

•4/8 Conti…..

Joint D
ΣFy = 0
FBDsin22.61 + FCD = 0
FCD = -(-50.05sin22.61)
FCD = 19.24 kN

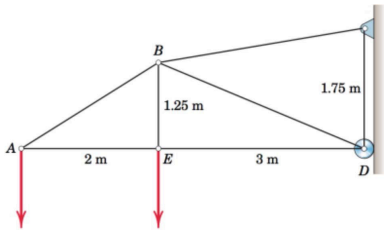


Figure G000102. UE26ME142A Unit 2 Class 23 Lecture PPT · source slide page 4.

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ENGINEERING MECHANICS STATICS

•4/8 Conti…..

Sl No.	Member	Force	Type
1	FAB	22.65 kN	Tension
2	FAE	- 19.20 kN	Compression
3	FEB	18.00 kN	Tension
4	FAE	-19.20 kN	Compression
5	FBC	65.94 kN	Tension
6	FBD	- 50.05 kN	Compression
7	FCD	19.24 kN	Tension

Figure G000103. UE26ME142A Unit 2 Class 23 Lecture PPT · source slide page 5.

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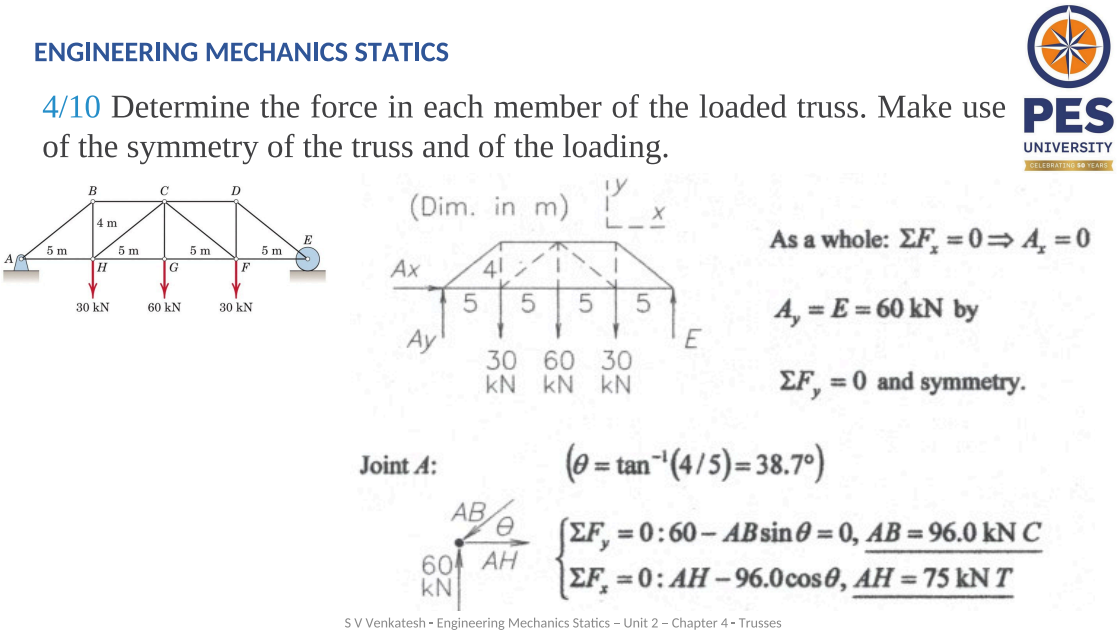


Figure G000104. UE26ME142A Unit 2 Class 23 Lecture PPT · source slide page 6.

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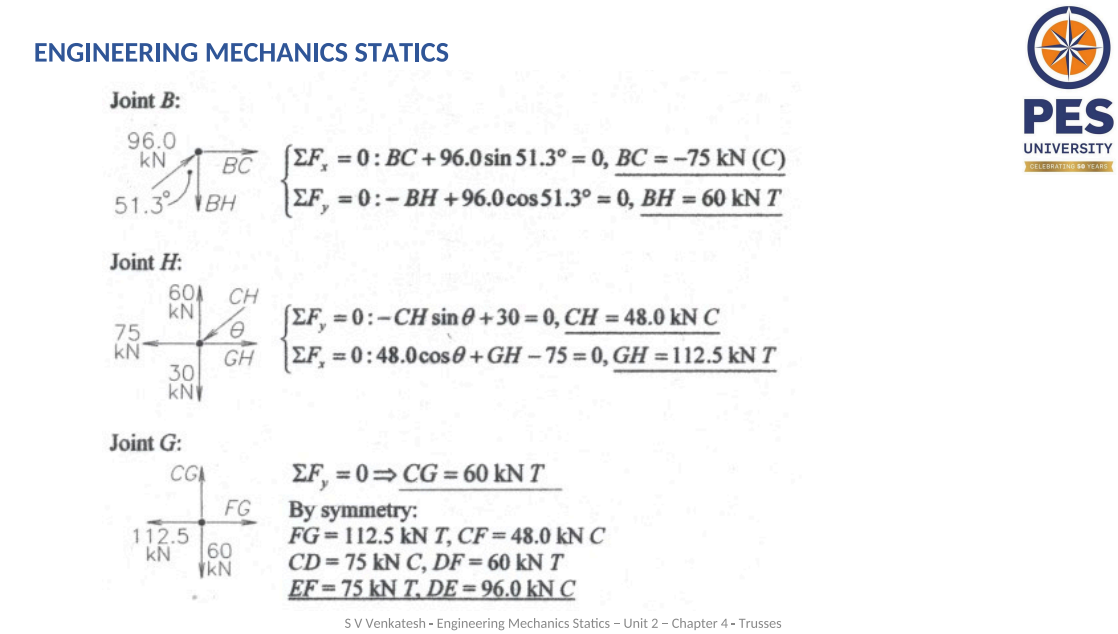


Figure G000105. UE26ME142A Unit 2 Class 23 Lecture PPT · source slide page 7.

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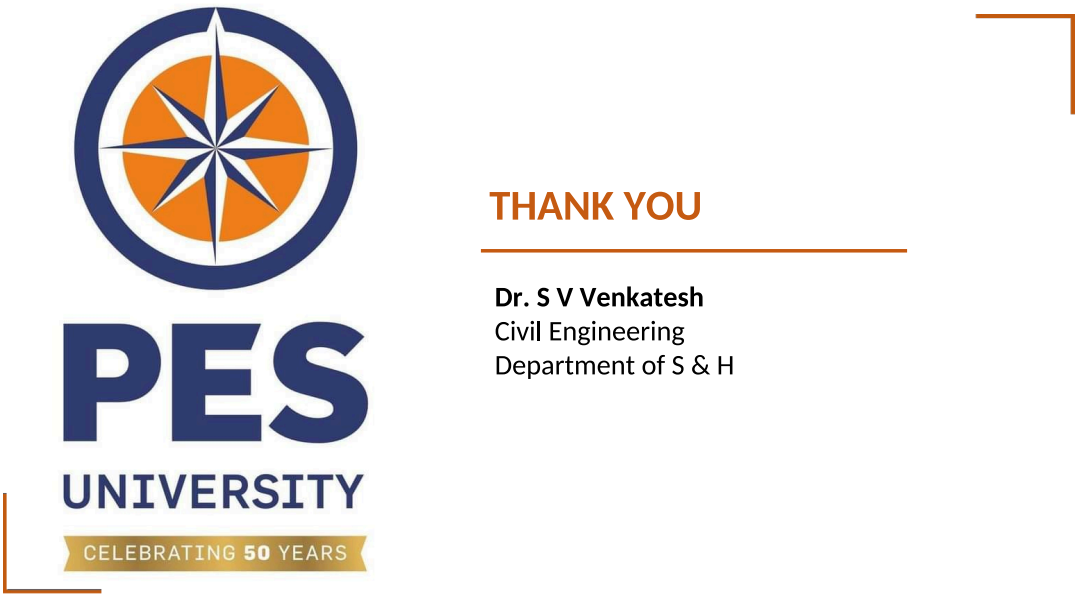


Figure G000106. UE26ME142A Unit 2 Class 23 Lecture PPT · source slide page 8.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000480 · EV0000481 · EV0000482
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 4/8 Determine the force in each member of the loaded truss. C F Joint A ΣFy = 0, FABsin32 = 12 FAB = 22.65 kN ΣFx = 0, FABcos32+FAE = 0 FAE = - 19.20 kN 12kN 18kN Joint E Tanϑ1(BAE) = 1.25/2, ΣFy = 0, FEB = 18 kN ϑ1 = 320 ΣFx = 0, FED = FAE = Tanϑ2(BDE) = 1.25/3, -19.20 kN ϑ1 = 22.610 Tanϑ3(CBF) = (1.75 - 1.25) / 3, S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses ϑ3 9 460 EV0000483 · EV0000484 · EV0000485
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS •4/8 Conti..... Joint B ΣFy = 0 - FABsin32 - FBE - FBDsin22.61 + FBCsin9.46 = 0 FBD = (FBCsin9.46 - 18 - 22.65sin32) / sin22.61 FBD = 0.43FBC – 78.4 ΣFx = 0 - 22.65cos32 + FBDcos22.61 + FBCcos9.46 = 0 19.20 = (0.43FBC – 78.4)0.92 + 0.99FBC FBC = (19.20 + 71.8) / 1.38 FBC = 65.94 kNS V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000486 · EV0000487 · EV0000488
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS •4/8 Conti..... Joint D ΣFy = 0 FBDsin22.61 + FCD = 0 FCD = - (-50.05sin22.61) FCD = 19.24 kN S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000489 · EV0000490 · EV0000491

5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS •4/8 Conti..... Sl Member Force Type No. 1 FAB 22.65 kN Tension 2 FAE - 19.20 kN Compression 3 FEB 18.00 kN Tension 4 FAE -19.20 kN Compression 5 FBC 65.94 kN Tension 6 FBD - 50.05 kN Compression 7 FCD 19.24 kN Tension S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000492 · EV0000493 · EV0000494
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS 4/10 Determine the force in each member of the loaded truss. Make use of the symmetry of the truss and of the loading. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses ## Page / slide 7 (OCR) ENGINEERINGMECHANICS STATICS Joint B: PES 96.0 EF.=0:BC+96.0sin51.3°=0,BC=-75kN (C) UNIVERSITY kN BC CELEBRATING50YEARS BH ∑F=0:-BH+96.0c0s51.3°=0,BH=60kNT 51.3 Joint H: 60 kN EF=0:-CHsinϑ+30=0.CH=48.0kNC 75 kN GH ∑F.=0:48.0c0sϑ+GH-75=0,GH=112.5kNT 30 kNI Joint G: CGA EF=0=CG=60kNT FG By symmetry: 112.5 FG=I12.5kNT,CF=48.0kNC kN 60 CD=75kNC,DF=60KNT kN EF=75kNT.DE=96.0KNC S V Venkatesh - Engineering Mechanics Statics - Unit 2 - Chapter 4 - Trusses ## Page / slide 8 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000495
7. Source slide page 7: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000496
8. Source slide page 8: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000497

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.996] Dear students, welcome to Engineering Mechanics Statics classes. My name is Dr. Sv Vength, Civil Engineering, Department of Science and Humanities, PS University. This is class number thirty-two. We are in the process of solving some problems. [00:00:20.324] In chapter four, unit two, that is on trusses. In the last class we had solved one problem. Today we will be continuing and solving some more problems. This is I've taken problem number four bar eight. [00:00:34.852] The statement of the problem goes like this. Determine the force in each member of the loaded truss. Now they have given you a truss where you have a pin support at point C and you have a roller support at point D. [00:00:53.412] You have load at A, which is twelve kilonewtons, and load at E which is eighteen kilonewtons. So we have to determine the forces in all these members that is A B, A, B E, E D, [00:01:10.180] B D and B C as well as C D. So to find out the forces in this member, first thing that we have to do is determine the angle made by the member AB with the horizontal that is angle BAE. [00:01:28.836] which we will get it using this triangle ABE. So this angle A is considered as T1. So we get theta 1 to be equal to 32 degrees, which will be equal to 10 inverse of 1.25 by 2 meters. Similarly we need to find out the angle made by B [00:01:48.740] uh D with uh D E that is this angle angle B De E which is equal to tan theta 2 as I have taken here. Sorry about this this should be tan theta 2. So that angle will be tan inverse of 1.2 by 3 which is 22.61 degrees. [00:02:07.812] Then if I draw a horizontal line from B till the line C D, which will meet the line C D at point F, then the angle made by [00:02:22.020] the line B C with the horizontal that is C B F, this angle, if I take it as theta three, that would be equal to tan inverse of one point seven five minus one point two, which is the distance between C F divided by [00:02:40.100] Three meters. Okay. So we get that as nine point four six degrees. Now we consider the joint A. [00:02:47.684]

Consider all the members to be under tension, that means we are taking the forces in each of the member moving away from the joint. So the force in AB will be from A to B as well as from B to A. [00:03:01.392] Because we are assuming it to be under tension, internal forces in case of tension will be acting towards each other. Similarly, A if I take the force will be moving from A towards E and from at E it is from E to A. [00:03:15.492] So if I take the forces in this, I can resolve the force in A B along vertical as well as horizontal, take the algebraic sum of all the forces along vertical, so we get it force in A B. [00:03:29.668] Sine of angle theta 1 should be equal to 12 kilonewtons, that is what is mentioned here. So we get the force in the member FAB which we will get as 22.65 kilonewtons and it is positive, so it is under tension. [00:03:45.092] Next we take some of all the forces along the x direction. So we get F A B [00:03:50.148] cos theta 1 which is thirty two degrees plus f a e to be equal to 0. So we get Fe to be equal to minus 19.2 kilovattan, which means that it is under compression. However we will keep the sign as it is as minus and take that force to [00:04:06.928] B in the member A E to be in tension only. Next coming to joint E. If I take joint E, the force in the member E B will be equal to 18 kilonewtons, that's what is mentioned here, which will be under [00:04:22.596] tension itself. Then you take summation of all the forces along X. So force in the member E D will be equal to force in the member AE. So [00:04:32.868] We have got uh AE as negative, so the force in the member E D will also be negative, which indicates that it is under compression, which will be of the magnitude nineteen point two itself here also. Okay. Next we will take the joint B [00:04:48.356] We have determined these two joints already, we know the the these two members, sorry, not the joint. The forces in these two members is known, so we can find out the force in these two members. So we will take joint B, resolve the force in member B C a long vertical, and

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.808] as well as along horizontal. Similarly force in the member B D along vertical and along horizontal and questions of equilibrium as you can see here. Sigma F y equals 0. So forces at this junction B all the forces along y2 [00:05:18.980] So we get the component of this, which should be positive. So that is F B C sin 9.46, that will be this component. Whereas this would be negative, so minus FP. Then [00:05:34.372] B D also will be negative minus F D sine twenty two point six one which is this angle. [00:05:41.988] Then we have the force in the member BA, which will also be negative, which is FAB sin 32. So solving this, we get a relationship between the force in the member B D and the force in the member B C. [00:05:59.524] Okay. So this is the relationship that we have got. Now take all the forces along the x axis. So when we take all the forces along the x axis we get the component of the force B C along X, component of the force B D along X. Okay. So uh [00:06:17.956] That is what is indicated here and also the component of the force B a along x. BA along x will be negative, so that is what is indicated here. Whereas this two is going to be positive because of these two forces. So equating to 0 and then substituting a value of [00:06:35.440] F B D in this equation, we get the force in the member B C which is positive sixty five point nine four kilonewton. So that means [00:06:47.780] The force in B C will be tensile force of magnitude 65.94. Substituting this value back in this equation, we will get the force in the member Bd as minus 50.05 kilonewton. So this member will be under compression. [00:07:06.596] So we have determined AB, AE, BE, BD and BC. Only one member that is left is this member that is C D. So we can take the joint D here. [00:07:20.740] And we can we know the force in this member, this member. So resolving this along the vertical, we get the force in the member D C and that would be under

tension which is equal to 19.24 kilonewton. So we have determined the forces in all the members. [00:07:39.940] some of the forces that we have got it as positive, some of the forces we have got it as negative. [00:07:44.836] Now finally what has to be done is this has to be indicated in the truss, as I have explained it in the previous two classes, that is either class number thirty or thirty-one. You can indicate the force in the member if it is tensile, it is acting towards the joint. [00:08:02.352] It is acting away from the joint. Or there is another way of indicating the forces in the members with respect to a table. That table is going to be looking something like this, where you will have serial number, member. [00:08:18.500] That is the name of the member and what is the force magnitude that is acting in that member and what is the type of force that is acting, whether it is compressive or tension. [00:08:28.676] So you can see here I mentioned all the members here, which there are seven members, so all the seven members are mentioned here. A B A E B A E Oh there is a mis equal to [00:08:45.796] AE is equal to E D, I think this should be E D. Then we have B C and B D and C D, and corresponding values of the forces that we have got it to while we have analysed it is what is mentioned along with the sign. [00:09:03.812] So if it is positive, as I told you, we have assumed positive sense for tension, so we write this as tension. If it is negative, we will write it as compression. So you can see wherever there is positive, we have mentioned it as tension, and wherever it is negative, we have mentioned it as compression. [00:09:19.328] So what is the advantage of doing this? Or why should we uh indicate whether it is tension and compression? Going further, these members have to be designed and we have to find out what is a type of [00:09:32.964] uh members that we are going to be adopting for manufacturing of this uh truss okay whether it can be angle or it can be a tube or it can be a I section or a C section or a Z section what is that and what should be the uh [00:09:50.816] flange width thickness of the flange thickness of the web what should be the depth of the web pre size section what should be the bottom flange top flange like that we have to determine all that and

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.244] That determination is different for a compression member and it is different for a tension member. That's why we should know whether the member is under compression or tension. However, in the first year syllabus, [00:10:15.648] which is common for all engineering branches, we just have to determine what is the magnitude of the force and what is the type of the force. We will be stopping at this point. Further design of the members will be taken by the civil engineering in their IR submission. Okay? [00:10:32.644] Right, I hope you have understood this. Let's go to the next uh problem. [00:10:37.540] This problem is problem number four bar ten. Again, we have taken these problems from the prescribed textbook that is Engineer Mechanics Statics by Miriam and Craig and adopted by Colden, 8th edition only, or eighth volume. Or eighth edition only is what we have adopted. [00:10:55.972] The problem statement goes something like this. Determine the force in each member of the loaded truss. Make use of the symmetry of the truss and of the loading. So here, [00:11:09.412] We have to find out the forces in all the members, but he has given you an hint. What is that hint? It says that make use of the symmetry of the truss and the loading. What does this mean? No, we say that a trust is symmetrical. When [00:11:29.060] Three conditions are satisfied. What are these three conditions? The three conditions is if you identify a truss having an symmetric axis, then the truss should be symmetric in [00:11:44.996] Geometry to the left of the section as well as to the right of the section. Okay, now in this problem, you can choose the symmetric axis as CG, that is this one, CG. Now if you observe you have the same number of

members to the left. [00:12:04.292] As to the light, right? Similarly, we have the same number of joints to the left of CG and to the right of CG also. I hope this is clear. [00:12:17.956] Okay. That is the first condition. The second condition is the loading also should be symmetrical with respect to the symmetric axis. That means we have a loading at turn. [00:12:30.564] As 30 kilometer, which is to the left of the symmetric axis Cg, which is at a distance of 5 meters from G. [00:12:39.748] Similarly, we also have a loading at F which is 30 kilometers which is to the right of G and the distance from G to F is also 5 meters. So it is symmetrical with respect to loading also. However, we have a loading at G itself and that is along the symmetric axis itself, so it is symmetrical. [00:12:59.140] Okay, so two conditions one is symmetry with respect to geometry, the second condition is symmetry with respect to the loading. [00:13:06.436] The last condition is it should be symmetrical with respect to the reactions also. Note, I'm telling it should be symmetrical with respect to reactions also. What is this? We get the reaction because of the supports. [00:13:21.712] Now if you observe here at A point we have a pin connection, whereas at E point we have a roller connection. [00:13:30.628] Now at E how many reactions are there? One vertical and one horizontal is what we take, but E we take it as one reaction which will be perpendicular to the line of the roller, which is in this case is line of roller is horizontal, so your reaction will be acting vertical. [00:13:46.628] If you absorb the entire truss, we don't have any horizontal load acting external. Therefore, horizontal reaction at A is all zero. Since the horizontal reaction at A is zero, we will have only vertical reaction at A as well as vertical reaction at E. [00:14:07.012] And this is symmetrical with respect to the symmetric axis. That means from G the reaction or the load at A is at a distance of ten meters. Similarly, the reaction at E is at a distance of ten meters from G. [00:14:22.468] So it is all it is symmetrical with respect to the distance as well as the support, it is vertical support. So we can say that this trust is symmetrical. Now what is the advantage of this? The advantage of a symmetrical truss is that [00:14:39.140] Whatever forces that we get to the left of the symmetric axis in the members, in all the members, the same forces will be acting to the right side also. That is the advantage. So if I want to determine the forces in all the members of the stresses. [00:14:56.452] Now it is sufficient if I find out the forces in the member A B then AH then BH B C then C and H G.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:11.908] Okay, so as well as CG. If I could find out the forces in this six members, one, two, three, four, five, seven members. One, two, three, four, five, six and seven members, the forces in the member GF will be equal to GH. [00:15:31.620] CF will be equal to CH, C D will be equal to C B, then your DE will be equal to PA, then your F E will be equal to H G. So that is what is the advantage of a symmetric process. So [00:15:50.628] The next thing. How do we solve this? That's explanation for a symmetrical process. Now how do we solve this? As for the method of joints, we have to identify a joint which has got only two unknowns. But if you observe here, we cannot find out any. [00:16:09.060] Joint which has got only two unknowns, there are more than two unodes. But if I could somehow determine the reaction at A and reaction at E, then I can start with joint A or joint E. So since the truss is symmetrical, [00:16:27.460] The vertical reactions will also be symmetrical. Therefore, I can easily say that reaction at A is equal to reaction at E, which will be equal to total vertical forces that is acting divided by 2. What is the total vertical force that is acting? [00:16:43.588] So total vertical force is equal to 30 plus 60 plus 30, which is 120, divided by 2 will be 60. That's why we get the reaction at A along the vertical and the reaction at E to be equal to 60 kilowattons. And as

I have already explained, if you take sigma f y equals 0, you get the reaction at [00:17:03.684] AE at that is the horizontal reaction at A will be equal to zero. [00:17:09.316] Next thing is we have to determine what is the angle made by the member AB with respect to AH or the horizontal, that is the angle PAH, which we can take this triangle and find out this angle, which I if I take it as theta, then it will be done inverse of [00:17:29.476] 4 by 3 which is indicated here and that angle will be 38 point so. So you take the joint E, write down that joint separately. Now when I write down the joint separately, I am assuming the force in the member AB as [00:17:45.540] Compressive and the force in the member AH as tensile, then that's why you can see here force in the member AH is moving away from the joint, force in the member AB is moving towards the joint. [00:17:57.508] So I can resolve this force in vertical and horizontal and use the equations of equilibrium. Sigma F equals 0, sigma f x equals 0. So when I take f y equals 0, I get the force in the member AB. If I take sigma f x equals 0, I will get the force in the member AH. [00:18:14.640] Now I hope this is clear. You can look at the value of this in the PPT that has been attached in PSO Academy. Okay? So continuing this, next we will take joint B. In joint B again, we have assumed the force in the member. [00:18:31.460] uh A B as compressive, so here also we will be assuming the same thing to be moving toward the joint. Whereas at B the force is moving away from the joint, we are assuming it to be under tension, this also the force in B H also as tensile force. [00:18:46.180] We had got the value of the force in the member AB as 96 kilonewton, which is compressive, and that is positive, we have got. So we will take that as it is. And then now we will apply the [00:19:00.176] apply the equations of equilibrium after resolving this 96 kilometer force along vertical and horizontal. So applying that we get the forces in the member B C and BH as indicated here. Next you take joint H in joint H [00:19:17.508] We have got the value of P C which is seventy-five. We have got it as negative. [00:19:23.524] So, however, we will keep that force as it is and assume it to be under tension itself. So, 75 we will get it as it is and it is acting this way. Then 60 kilon is the force acting upwards, 30 kilonewton is acting downwards. How did we get this 60? [00:19:40.160] We have got it as BHS sixty tensile, so that is as it is. Then these two are to be determined. So this angle will also be theta, so I can resolve the force in CH vertical and horizontal and again use the equations of equilibrium by doing so. [00:19:56.048] I get the value of CH and GH and this is what the values that we have got. Okay? Now CH is assumed to be compressive, so that is why you can you can write as C here even though it is positive. Then CG is assumed to be tensile, we get it as positive and that is why it is tensile. Next you can take joint at G

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:15.588] Joint G we have four members. So use all the four members are either vertical or horizontal. So directly use uh the equations of equilibrium. So sigma f y equals zero, I get C G to be equal to sixty kilonotten. And it will be under tension. [00:20:33.156] So by symmetry as I explained earlier, you can write the forces in the member which is to the right of the symmetric axis CG, which will be the same as the left side. Okay. I hope this is clear. Okay. We will continue uh [00:20:49.316] in the next class. Okay? So for the time being, thank you very much.

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Full transcript: 3380 normalized words, 83 timestamped segments, 1270.379 seconds duration. Low-confidence segments flagged: 0.

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16 UE26ME142A_Unit 2_Class 24_AV Summary

This lesson is indexed from the source video titled UE26ME142A_Unit 2_Class 24_AV Summary. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000242 · EV0000243

Graphs and diagrams



Figure G000107. UE26ME142A Unit 2 Class 24 Lecture PPT · source slide page 1.

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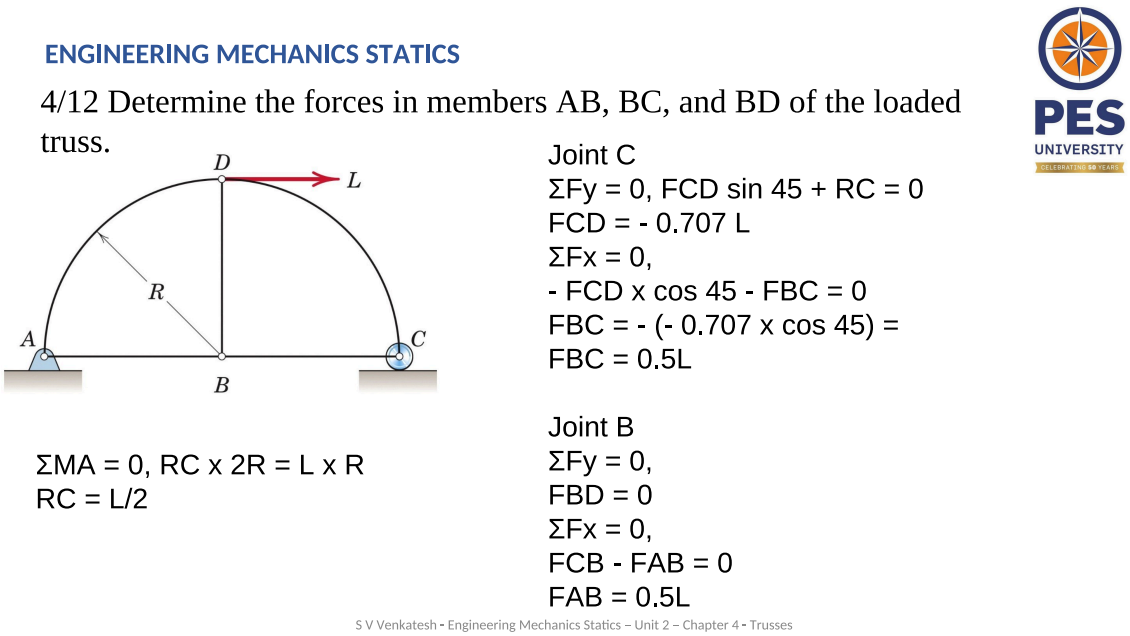


Figure G000108. UE26ME142A Unit 2 Class 24 Lecture PPT · source slide page 2.

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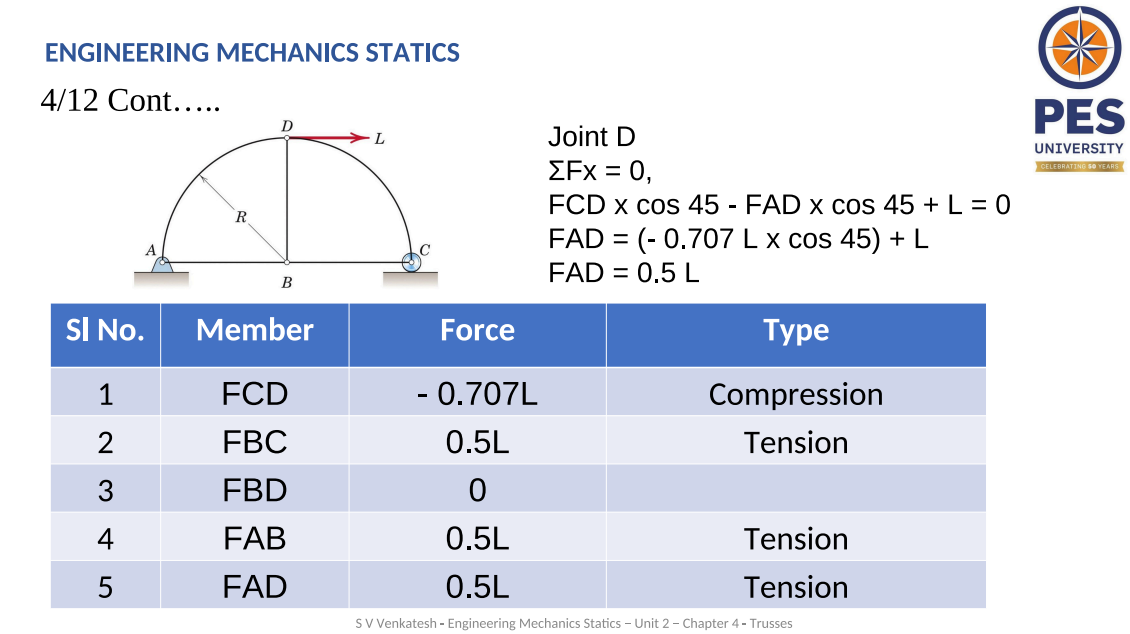


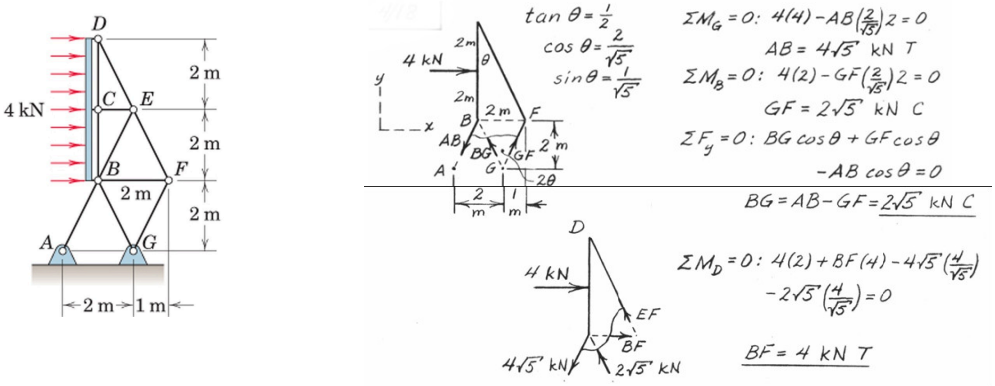
Figure G000109. UE26ME142A Unit 2 Class 24 Lecture PPT · source slide page 3.

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ENGINEERING MECHANICS STATICS

4/21 The signboard truss is designed to support a horizontal wind load of 4 kN. A separate analysis shows that 5/8 of this force is transmitted to the center connection at C and the rest is equally divided between D and B. Calculate the forces in members BE and BC.



S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses

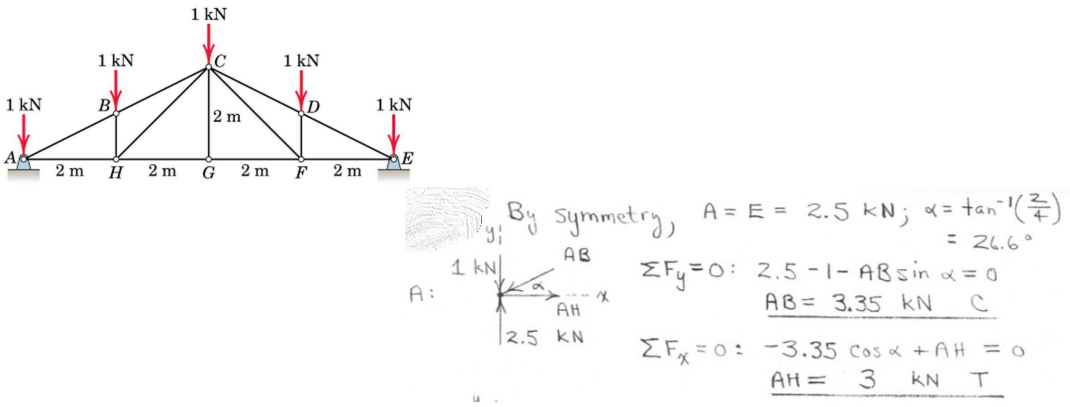
Figure G000110. UE26ME142A Unit 2 Class 24 Lecture PPT · source slide page 4.

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ENGINEERING MECHANICS STATICS

4/23 A snow load transfers the forces shown to the upper joints of a Pratt roof truss. Neglect any horizontal reactions at the supports and solve for the forces in all members.



S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses

Figure G000111. UE26ME142A Unit 2 Class 24 Lecture PPT · source slide page 5.

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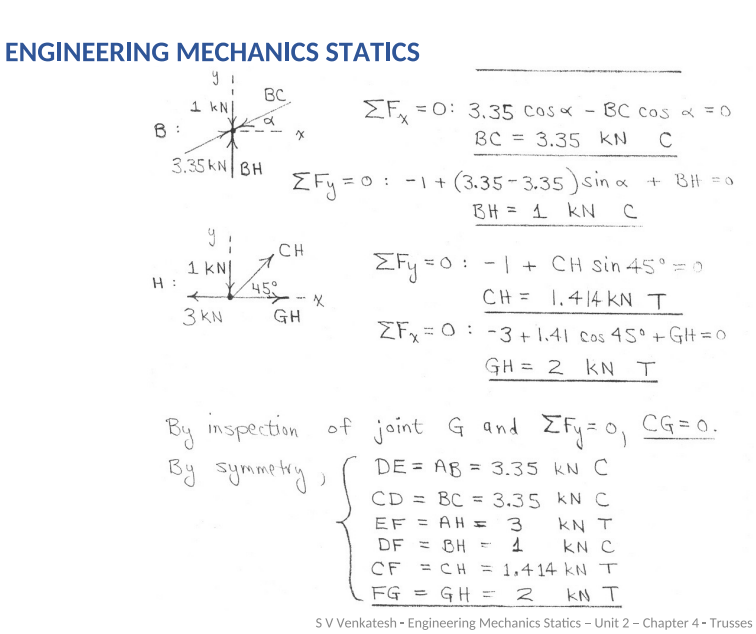


Figure G000112. UE26ME142A Unit 2 Class 24 Lecture PPT · source slide page 6.

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Figure G000113. UE26ME142A Unit 2 Class 24 Lecture PPT · source slide page 7.

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Important points and why

1. Source slide page 1: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000498 · EV0000499 · EV0000500
2. Source slide page 2 (extracted text): ENGINEERING MECHANICS STATICS 4/12 Determine the forces in members AB, BC, and BD of the loaded truss. Joint C ΣFy = 0, FCD sin 45 + RC = 0 FCD = - 0.707 L ΣFx = 0, - FCD x cos 45 - FBC = 0 FBC = - (- 0.707 x cos 45) = FBC = 0.5L Joint BΣMA = 0, RC x 2R = L x ΣFy = 0,R FBD = 0RC = L/2 ΣFx = 0, FCB - FAB = 0 S V Venkatesh - Engineering MechanicsFABStatics= –0.5LUnit 2 – Chapter 4 - Trusses EV0000501 · EV0000502 · EV0000503
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS STATICS 4/12 Cont..... Joint D ΣFx = 0, FCD x cos 45 - FAD x cos 45 + L = 0 FAD = (- 0.707 L x cos 45) + L FAD = 0.5 L Sl Member Force Type No. 1 FCD - 0.707L Compression 2 FBC 0.5L Tension 3 FBD 0 4 FAB 0.5L Tension 5 FAD S V Venkatesh - Engineering0.5LMechanics Statics – Unit 2 – ChapterTension4 - Trusses EV0000504 · EV0000505 · EV0000506
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS STATICS 4/21 The signboard truss is designed to support a horizontal wind load of 4 kN. A separate analysis shows that 5/8 of this force is transmitted to the center connection at C and the rest is equally divided between D and B. Calculate the forces in members BE and BC. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000507 · EV0000508 · EV0000509
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS STATICS 4/23 A snow load transfers the forces shown to the upper joints of a Pratt roof truss. Neglect any horizontal reactions at the supports and solve for the forces in all members. S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses EV0000510 · EV0000511 · EV0000512
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS STATICS S V Venkatesh - Engineering Mechanics Statics – Unit 2 – Chapter 4 - Trusses ## Page / slide 7 (OCR) THANK YOU Dr. S V PES Venkatesh Civil Engineering Department of S & H UNIVERSITY CELEBRATING 5O YEARS EV0000513
7. Source slide page 7: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000514

short-note recap

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.676] Dear students, welcome to Engineering Mechanics Statics. My name is Dr. Sv Venkitish from Civil Engineering under the Department of Science and Humanities, PS University. This is class number 33. [00:00:16.804] In the previous classes we had discussed about chapter four, which is on trusses under unit two. We had solved a few problems, so we'll be continuing solving a few more problems in today's class. [00:00:33.092] So this is problem two bar twelve, which is taken from the prescribed textbook that is Engineer Mechanics Statics, eighth edition by Miriam Craig and Bolton. The statement of the problem goes like this determine the forces in members A B, B C and B D. [00:00:53.060] of the loaded truss. So this is a unique problem where you have a truss which is supported at A by a pin connection, at C by a roller connection, and there is a horizontal load applied at point D. [00:01:11.492] So if you observe carefully, this is not a symmetric process. Why it is not symmetric?

Because we have a horizontal force here at D. Okay? And if you take the symmetric axis as B D. [00:01:29.444] Of course it is symmetric with respect to geometry because you have the same number of members to the left as well as the two right of this line B. [00:01:38.532] However, the loading you have only at T which is acting towards the right, we don't have a loading towards the left. So it is not symmetrical with respect to the load. Also, [00:01:53.956] Supports that we have here is an hip support and a roller support. So since it is an hip support, you will have two reactions, one vertical and one horizontal. [00:02:06.532] So since there is one vertical and one horizontal, we will also have a horizontal reaction because we have an horizontal force there, yeah. Okay. Next, we have to determine the forces in few of the members, whatever is asked according to the problem. So [00:02:26.852] We will first have to determine the support reactions because we cannot start with any of the joints, which has got only two unknowns, where we can determine the forces. So we need to first find out the support reactions. So if you take the moment about A, we get the reaction at C into this distance, which is 2R. [00:02:46.340] Okay, in the anti-clockwise direction, which should be equal to L into R which is in the clockwise direction. So we get the reaction at C as L by 2, which is acting upwards. So if I take sigma F y equals you know I get the reaction at A which should be acting downwards. [00:03:05.124] Which is also equal to L by 2. Then if you take the algebraic sum of all the forces along X direction, we get the reaction at A, which will be acting towards the left, which is equal to L. Next we can take any of the joint, either joint C or A. So we will take joint A. [00:03:23.524] Okay. Now the speciality here is when we have a curved member like this. [00:03:29.476] The force in the current member will be or should be taken to be same as the force in the member where you join that two points or the joints of this member. [00:03:44.740] Okay, which is in this case is A and D, you join them by a straight line, whatever will be the force that is going to be acting on that straight line will be the force that is going to be acting in the curved member also. Similarly for this member C D, you join the line from D to C. [00:04:01.008] Find out what is the force that is going to be acting along that line C D. The same will be the force that is going to be acting in the current member C D as well. Okay. So you take this joint. [00:04:14.020] So the angle made by DAB will be 45 degrees because this is also R, this is also R so I can resolve the force in the member A D along the vertical and horizontal and use the equations of equilibrium. So we get the force in the member AB [00:04:30.848] to be equal to L, which is going to be uh intermember which is acting towards each other, so that will be under tension. Then you take the uh force in the uh sorry you will have the [00:04:44.516] I am sorry about that. First you resolve this force along vertical and horizontal. You take the algebraic sum of all the forces along vertical. So we get the force F AD [00:04:56.644] Sine 45 to be equal to the reaction that's going to be acting here, which will be acting downwards, which is equal to L by 2. So we get the force in the member F AD. Then you take all the forces along the x-axis, so we get the force in the member AB.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:14.500] plus the force, the component of the force in the member F A D sine forty five that will be this one equal to the force that is L acting along the horizontal reaction at A. [00:05:30.372] So from that we will get the force in the member A D. Then we take this joint to B [00:05:35.812] The force in the member B D will be zero because there is no force acting at B. So the force in the member B D will be zero. Then the force in the member B C will be equal to the force in the member AB itself. So we get that. Then you take joint D. [00:05:53.444] Then we will get from joint D the force in the member B C.

Okay so that is what is explained here. So we get the force in the member A D because here we have determined the force from joint C, then B, then we have taken D. [00:06:11.172] Now I have explained it how to do it from joint A and then go to joint B and then joint D. But here it is joint C, then joint B and joint D. However you get these values. So those values have been tabulated here, F C D. [00:06:28.580] You get it as negative minus 0.707 L that is in this member. Okay. Since it is negative, it is going to be compressive. Then force in the member B C, we will get it as 0.5 L, which is positive, so it is tension. [00:06:45.156] Force in the member B D will be 0 because there is no external force acting at B. So directly we get that. Then force in the member AB will be the same as the force in the member B C, which is 0.5 L, which is tension. Then [00:07:00.676] Force is the member, A D is also tensile, so that we will get. Now, how do we analyze whether what we have done is correct or wrong? Now, when I apply a force L here at D, how does this rotate? [00:07:18.500] Truss would try to rotate in the clockwise direction. [00:07:22.948] So obviously when this rotates try to rotate in the clockwise action, you have a reaction here upwards, you have a force here acting like this. So this member that is C D would reduce its length. So definitely it is under compression, and that's what we have got here. Okay similarly, if I take AD. [00:07:42.628] So, since this is going to rotate, then the member AD will get elongated because the force or the reaction acting here is acting downwards, and this also will be acting towards the left. So this member AD will get stretched, so that is under tension. So we have got AD as tensile itself. [00:08:02.788] Then when this tries to rotate along this, this member and this member would try to go down because here these two will not move, only this can move. So when B tries to [00:08:18.404] Come down because of that, your AD has to elongate and AB also has to elongate. Both of them has to be intentional and that's what we have got here. [00:08:27.844] I hope this you have understood. Okay? So how to determine whether the force is in under compression or tension is what I have tried to explain here. Of course I have explained how to determine the forces in all the members and as well as how to determine the reactions at A and C. [00:08:45.668] And I have also told you why this truss is not a symmetrical truss. Moving on, I have chosen this problem, which is 4 bar 21. [00:08:59.652] The statement of the problem goes like this. The signboard truss is designed to support an horizontal wind load of 4 kilometer. A separate analysis shows that 5 by 8 of this force is transmitted to the center connection at sea. [00:09:19.716] And the rest is equally divided between B and B. Calculate the forces in member BE and BC. From the statement, what we can understand is there is a truss which is designed for supporting a sign mode. [00:09:39.844] The total force that is going to be acting on the signboard because of the wind load. Okay. Wind load will be acting on an area, but they are given what is the total load that is going to be acting on this signboard as well. [00:09:55.364] They are told what will be the portion of this total load which will be acting at each of the joints at that is at P, C and D.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:05.956] And for that, there is a separate analysis that they have done, which is not included in the in this in this syllabus. So let's not worry about what is that analysis. Let's look at what are the forces that is acting. He says that five by eight of the total load will be acting at C. [00:10:23.620] So straight away total load is acting is 4 kilometers, but if I take the value that is going to be acting at C, it is going to be 5 by 8 of 4, that will be the load that is going to be acting at C. [00:10:38.212] Then the remaining load, that is 3 by 8, which is remaining of the four kilometers, is to be equally shared at D and C is what is given. So at D [00:10:49.764] The four that is going to be acting will be three by eight into half

of that into four will be acting at D and the same load will also be acting at B. [00:10:59.300] Okay, so that is what is the load distribution that we have to look at it. And also what is asked is to determine the force in the member PE and BC. That is force in the member BE and BC. [00:11:14.148] So to get these values we can start from joint B [00:11:19.940] Get the value of in these two members, then take joint C, get the value in these two members, then take joint B, sorry, joint E, and then get these two members. From this we will be able to get the value of B and B C. Now however, [00:11:37.540] If you take the entire truss as it is, it is supported at A by a pin connection, at G also it is supported by a pin connection. So here you will have two reactions, one vertical and one horizontal. Here also will have two reactions, one vertical and one horizontal. [00:11:54.340] So there are four support reactions, but we have only three equations of equilibrium. So which means that it is an indeterminate structure, we require one more equation. However, to determine the forces in the member BE and BC, we don't have to. [00:12:12.996] Find out the value of the support reaction first thing. Okay. So we don't have to determine the support reaction at E and G, even though it is indeterminate, we don't have to find that. The second thing is [00:12:26.468] Since we are he has asked us to find out only the forces in B and B C, we have to solve only the joint B, joint C, and joint E only. We don't have to worry about solving other joints. So let us not worry about that. Okay? So what is given here is [00:12:46.148] How the forces is going to be acting here. So they have taken sigma m at g equals to zero. They have explained how to get the forces in AB obviously and sigma m b equal to zero by using alternate equations of equilibrium. [00:13:05.892] Okay and get the force in G f then use sigma f y equal to 0 and get the force in the member uh B G, which will be equal to uh AB minus A G. So you will get that force also. Then use the [00:13:23.716] Equation sigma m d equal to 0. Okay, sigma m b equals to 0 and then find out the force in the member Bf. Now this method that they have used is method of section. Let us not worry about this. We will be using as I described by taking [00:13:43.908] uh the joint D analyzing this, getting these two equations. [00:13:49.540] or getting these value of these two, then analysing joint C, getting the value of C E and C B, and then using the joint E to get the force in the member B. Let us restrict to that, let us not worry about how to get the [00:14:06.052] forces by using the method of section. Okay. Now the next uh problem that is chosen is 4 bar 23 [00:14:17.412] Which says that if snow load transfers the forces shown to the upper joint of a prat roof truss, neglect any horizontal reactions at the supports and all for the forces in all the members. Now here if you observe. [00:14:37.316] What we can understand from the statement is that the load coming on the truss, which is a Pratt truss, is because of the snow load, and they have given what is the load that is going to be acting at each of the joints, that is A, B, C, D, and E. [00:14:53.188] And if you observe all the forces are vertically acting downwards and it is equal to one kilometer. And we have the reaction acting at A, which is a pin connection, we have the reaction acting at E, which is also a pin connection. But since there is no horizontal

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:10.948] Force that is acting, we can say that the horizontal reaction at A and E will be equal to 0. So what is left is only the vertical reaction at A and vertical reaction at E. [00:15:21.188] And since the trust is symmetrical with respect to geometry as well as loading as well as the support reactions, we can say that the vertical reaction at A and E will be equal, and which will be equal to 2.5 kilonuton, which is nothing but 1 plus 1 plus 1 plus 1 plus 1, which is 5 kilonuton divided by 2. [00:15:41.444] And that is what is given here. Next, consider the angle made by the line B C with the horizontal, that

is angle CAG to be equal to alpha. [00:15:57.508] We get the value of alpha standing inverse of 2 by 4, and that's what is indicated here. So we get the value of alpha. Using that and considering the joint A, we get the force in the member AB under AH. [00:16:12.836] By using the equations of equilibrium, we can get the value of AB and AH. Next, to solve this, first we will solve joint A as I've explained just now, then go to joint B. There are only two members, we can get the value in BG as well as B H. [00:16:32.676] Okay. Then we can take the joint H, we can get the value in H G and H C. Then we can take the joint G. [00:16:46.180] Where we can easily identify that the force in the member CG is zero because there is no vertical external force that is acting at G. Since these two is going to be equal, the force in the member CG will be zero. Only these four joints is what we have to solve. So I've explained about joint A. [00:17:05.188] Next, you take joint B. What are the forces acting? One is external force that is acting because of the snow load one kilon. Then we have BH acting. Then we have the force in this which we have determined as 3.35 kilon tons, then the force in BC. So if you take sigma fx equal to 0. [00:17:24.036] We can find out what is the force in B C, which we will get as 3.35 kilonewtons. Then you take sigma f y equal to 0, we get the force in the member bh as 1 kilowton. So along X you have to resolve this force and resolve this force. [00:17:41.444] Both of them are acting at the same angle, alpha, so these two has to be equal, so we get as three point three five. When I resolve this along vertical and resolve this as along vertical, they both will cancel, so we get VG as one kilometer straight away. [00:17:55.652] Then you take the joint h and then determine what is CH and CG, resolve CH along vertical and horizontal, take sigma y equal to 0, we get ch as 1 point four one four. Then sigma f x equal to 0, if you take I get the force in the member gh. [00:18:13.412] Okay and as I have explained the force in the member Cg will be equal to 0. So since the truss is symmetrical above the line Cg, we can say that the force in the member D is equal to AB, which is 3.35 kilo ton, which is compressive. [00:18:33.028] Then Cd is equal to B C, which is equal to 3.35 kilonewton, which is also compressive. Then EF is equal to [00:18:41.956] A H which is equal to 3 kilonut time which is tensile, then Bf is equal to BH which is equal to 1 kilonut time which is compressive. Cg is equal to CH which is equal to 1.414 kilonut time which is tension. [00:18:57.956] Fg is equal to G H, which is equal to two kilone, which is tension, and of course CG force in the member CG is equal to zero. Okay. Now because of symmetry. [00:19:10.308] We are able to get the forces in all the members by solving only three joints, and of course the joint G, if you take you get the force in the member CG. Okay. So if it is symmetrical, it is advantageous. [00:19:24.836] Because we have the same forces acting to the left of the symmetrical axis and right of the symmetrical axis. Okay. So we will stop today's class here. I am Dr. Steven Tesh from Civil Engineering, the Department of Science and Humanities, PS University. [00:19:41.924] Let's meet in the next class, that is class number thirty-four, and discuss more about. Thank you very much.

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Full transcript: 3208 normalized words, 77 timestamped segments, 1190.189 seconds duration. Low-confidence segments flagged: 0.

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17 Source slide deck: UE26ME142A Unit 2 Class 25 Lecture PPT

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Graphs and diagrams

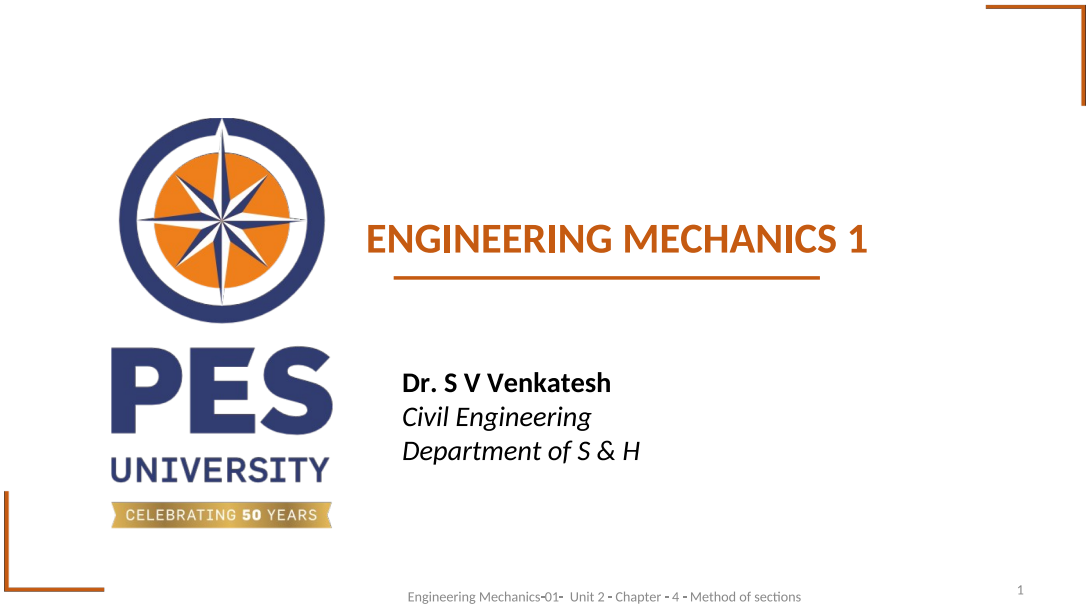


Figure G000114. UE26ME142A Unit 2 Class 25 Lecture PPT · source slide page 1.

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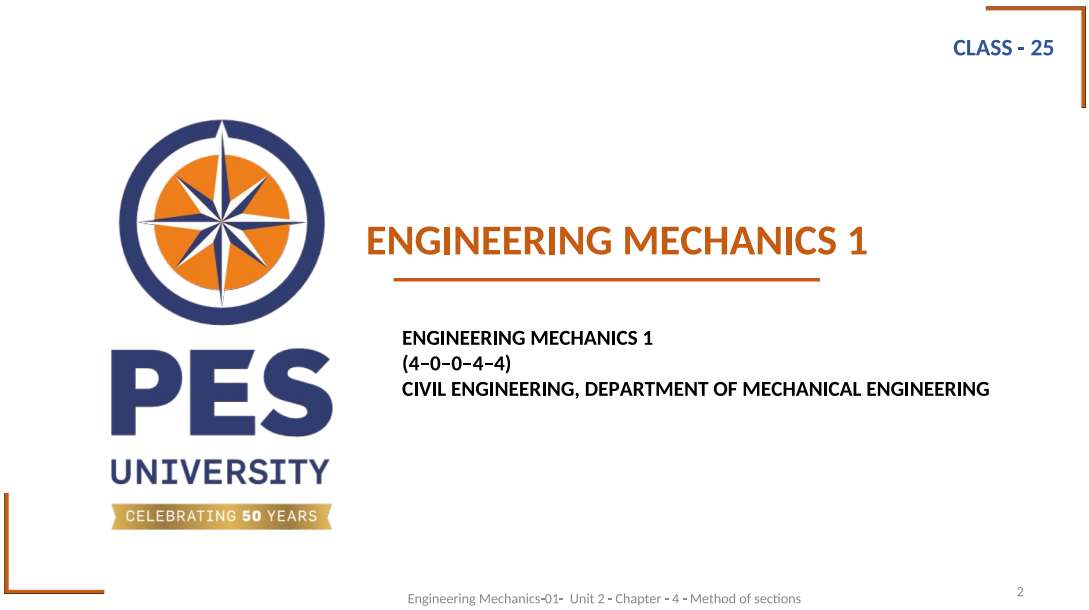


Figure G000115. UE26ME142A Unit 2 Class 25 Lecture PPT · source slide page 2.

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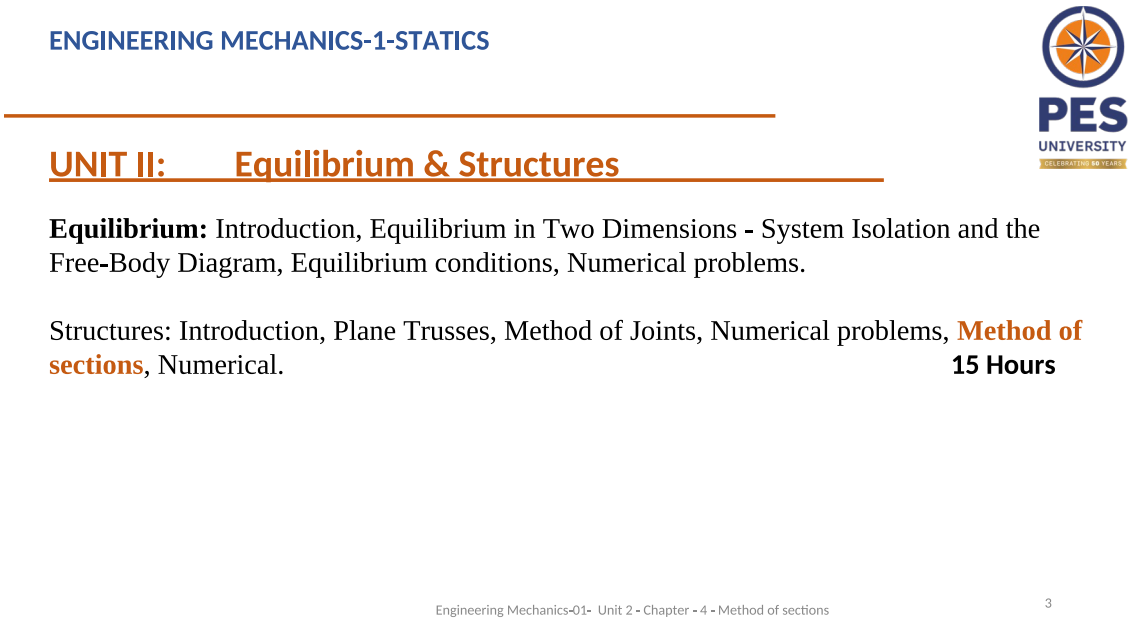


Figure G000116. UE26ME142A Unit 2 Class 25 Lecture PPT · source slide page 3.

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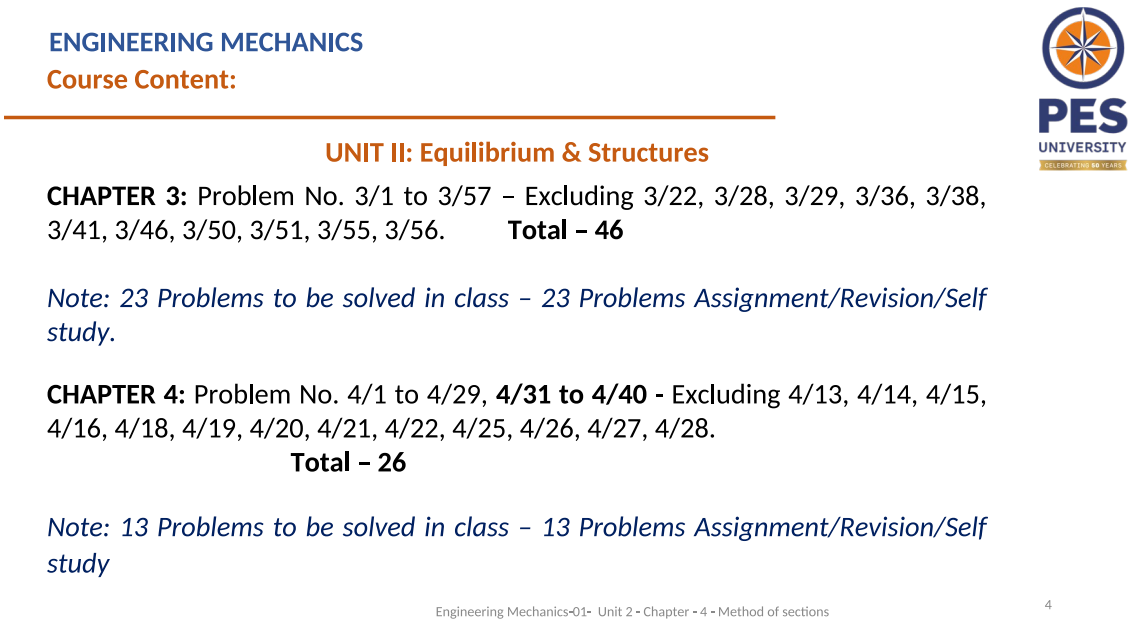


Figure G000117. UE26ME142A Unit 2 Class 25 Lecture PPT · source slide page 4.

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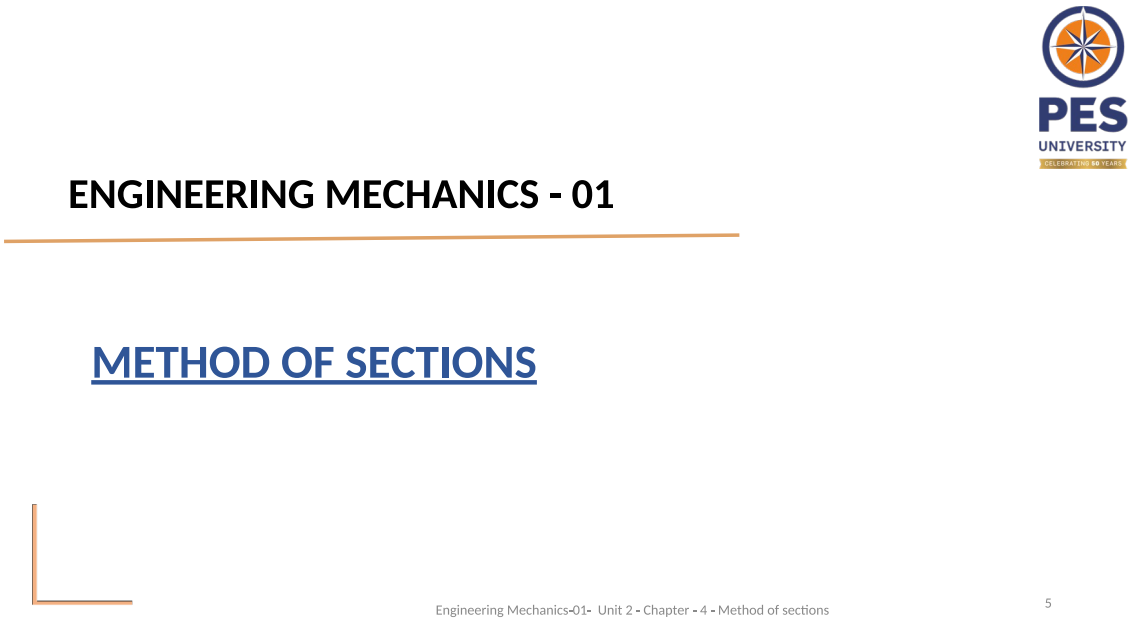


Figure G000118. UE26ME142A Unit 2 Class 25 Lecture PPT · source slide page 5.

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ENGINEERING MECHANICS - 1
Method of sections

IIlustration of the method of sections

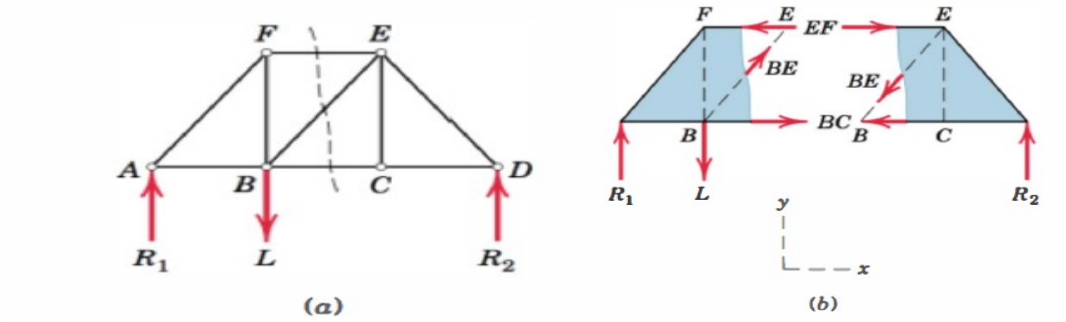


Figure 4/12

Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections 6

Figure G000119. UE26ME142A Unit 2 Class 25 Lecture PPT · source slide page 6.

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ENGINEERING MECHANICS - 1
Method of sections (Procedure)

Solve for unknown forces using the method of sections is: (*Source: Engineering Statics-Daniel* W. Baker, William Haynes)

- Determine if a truss can be modeled as a simple truss.
- Identify and eliminate all zero-force members.
- Solve for the external reactions, if necessary.
- Every cut member exposes an unknown internal force(not morethan 3 unknowns)
- Select the easier of the two halves of the truss and draw its free-body diagram.
- Assume that all unknown forces are in tension, then solve for chosen section using Equilibrium equations later procedures are same as Method of Joints

Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections 7

Figure G000120. UE26ME142A Unit 2 Class 25 Lecture PPT · source slide page 7.

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Figure G000121. UE26ME142A Unit 2 Class 25 Lecture PPT · source slide page 8.

EV0000537

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS 1 Dr. S V Venkatesh CivilEngineering DepartmentofS&H 1Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections EV0000515 · EV0000516 · EV0000517
2. Source slide page 2 (extracted text): CLASS - 25 ENGINEERING MECHANICS 1 ENGINEERING MECHANICS 1 (4–0–0–4–4) CIVIL ENGINEERING, DEPARTMENT OF MECHANICAL ENGINEERING 2Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections EV0000518 · EV0000519 · EV0000520
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS-1-STATICS UNIT II: Equilibrium & Structures Equilibrium: Introduction, Equilibrium in Two Dimensions - System Isolation and the Free-Body Diagram, Equilibrium conditions, Numerical problems. Structures: Introduction, Plane Trusses, Method of Joints, Numerical problems, Method of sections, Numerical. 15 Hours 3 Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections EV0000521 · EV0000522 · EV0000523
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS Course Content: UNIT II: Equilibrium & Structures CHAPTER 3: Problem No. 3/1 to 3/57 – Excluding 3/22, 3/28, 3/29, 3/36, 3/38, 3/41, 3/46, 3/50, 3/51, 3/55, 3/56. Total – 46 Note: 23 Problems to be solved in class – 23 Problems Assignment/Revision/Selfstudy. CHAPTER 4: Problem No. 4/1 to 4/29, 4/31 to 4/40 - Excluding 4/13, 4/14, 4/15, 4/16, 4/18, 4/19, 4/20, 4/21, 4/22, 4/25, 4/26, 4/27, 4/28. Total – 26 Note: 13 Problems to be solved in class – 13 Problems 4 Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections Assignment/Revision/Selfstudy EV0000524 · EV0000525 · EV0000526

5. Source slide page 5 (extracted text): ENGINEERING MECHANICS - 01 METHOD OF SECTIONS 5 Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections EV0000527 · EV0000528 · EV0000529
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS - 1 Method of sections IIlustration of the method of sections Figure 4/12 6 Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections EV0000530 · EV0000531 · EV0000532
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS - 1 Method of sections (Procedure) Solve for unknown forces using the method of sections is: (Source: EngineeringStatics-Daniel W. Baker, William Haynes) •Determine if a truss can be modeled as a simple truss. •Identify and eliminate all zero-force members. • Solve for the external reactions, if necessary. •Every cut member exposes an unknown internal force(not morethan 3 unknowns) •Select the easier of the two halves of the truss and draw its free- body diagram. 7 Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections•Assume that all unknown forces are in tension, then solve for EV0000533 · EV0000534 · EV0000535
8. Source slide page 8 (extracted text): THANK YOU Dr. S V Venkatesh CivilEngineering DepartmentofS&H 8Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections EV0000536 · EV0000537 · EV0000538

SHORT-NOTE RECAP

Source deck preserved: UE26ME142A Unit 2 Class 25 Lecture PPT. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000515

18 Source slide deck: UE26ME142A Unit 2 Class 26 Lecture PPT

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Graphs and diagrams



Figure G000122. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 1.

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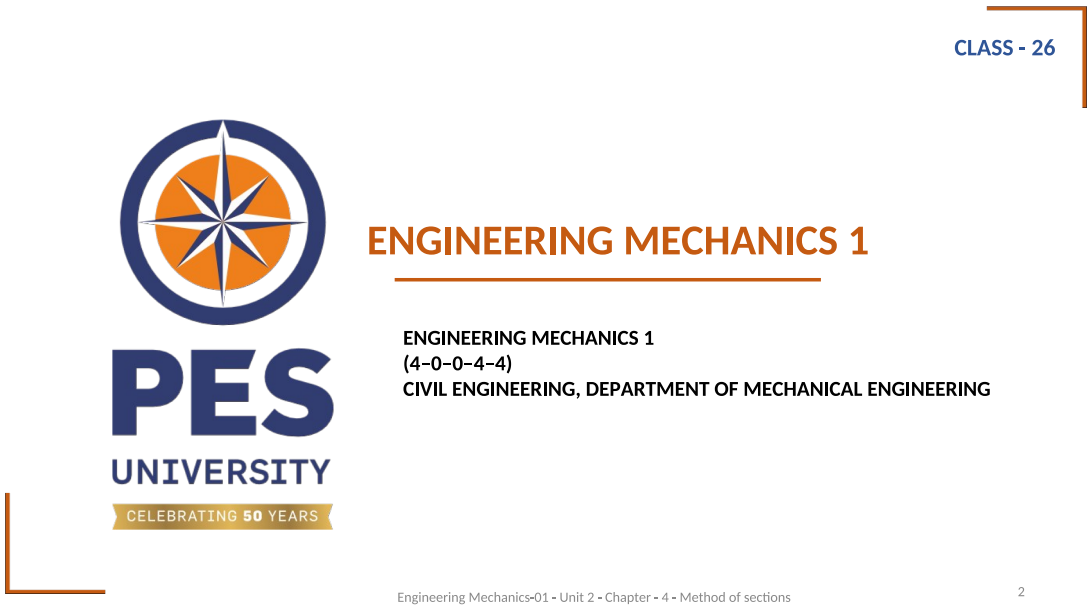


Figure G000123. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 2.

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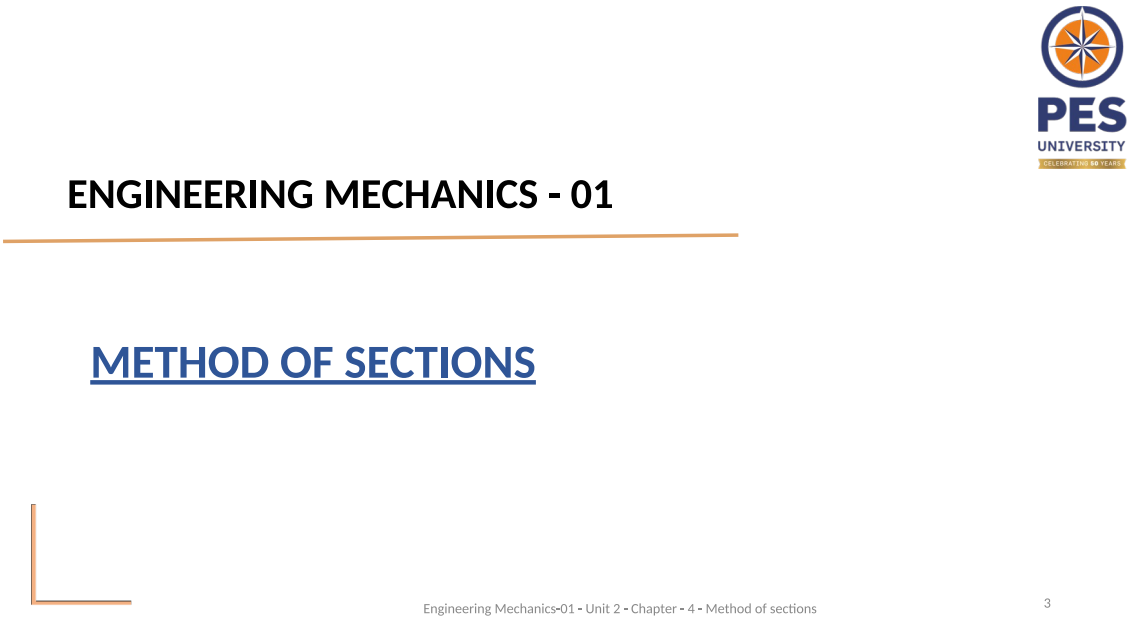


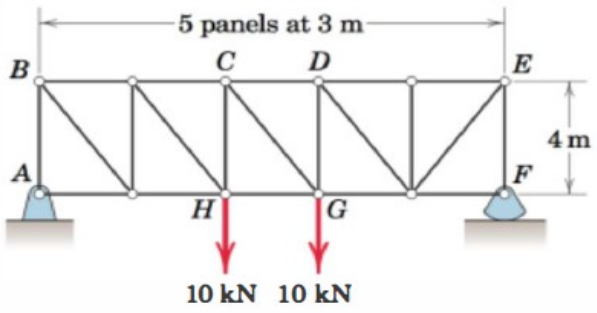
Figure G000124. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 3.

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ENGINEERING MECHANICS-1
Method of sections

4/31 Determine the force in member CG.



Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 4

Figure G000125. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 4.

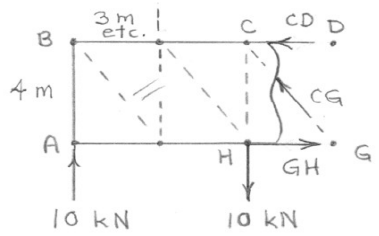
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ENGINEERING MECHANICS-1
Method of sections

4/31

4/31 By inspection, the reaction at each support is 10 kN up.



$+\uparrow \Sigma F = 0: \quad CG\left(\frac{4}{5}\right) + 10 - 10 = 0, \quad CG = 0$

Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 5

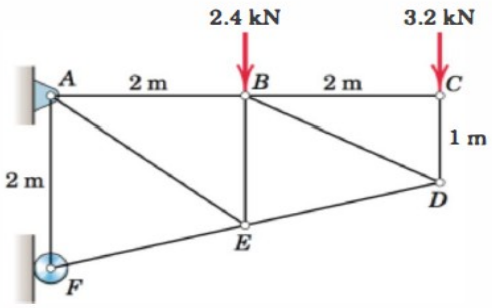
Figure G000126. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 5.

EV0000552

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/32 Determine the force in member AE of the loaded truss.



Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 6

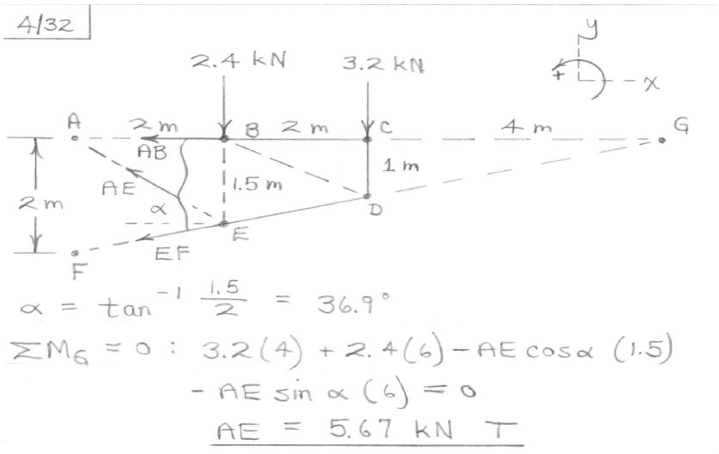
Figure G000127. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 6.

EV0000555

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/32.



Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 7

Figure G000128. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 7.

EV0000558

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/33 Determine the forces in members BC and CG.

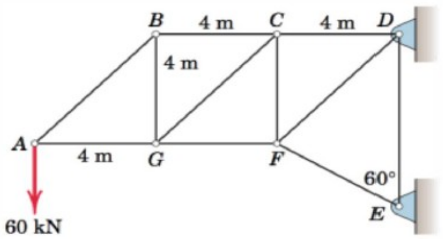


Figure G000129. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 8.

EV0000561

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/33.

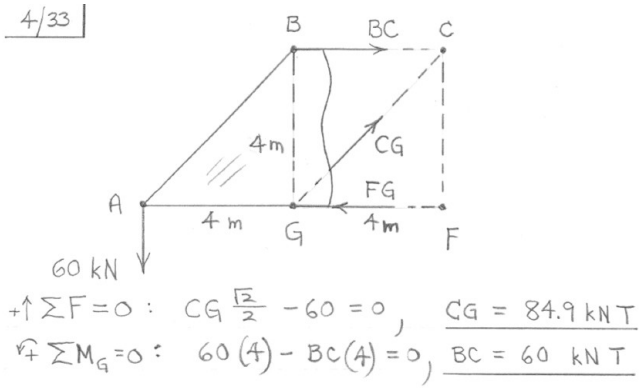


Figure G000130. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 9.

EV0000564

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/34 Determine the forces in members CG and GH of the symmetrically loaded truss.

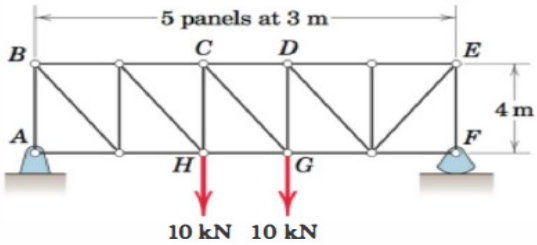


Figure G000131. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 10.

EV0000567

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/34

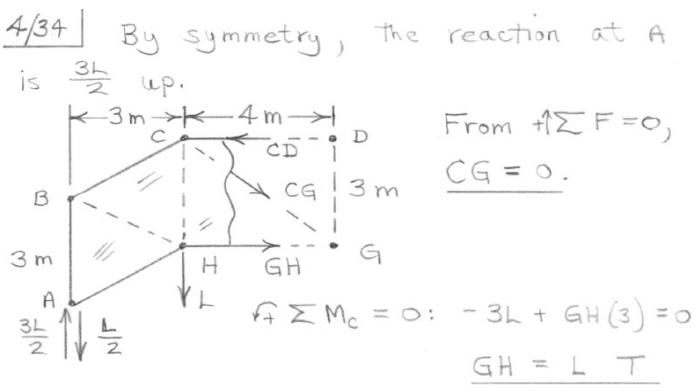


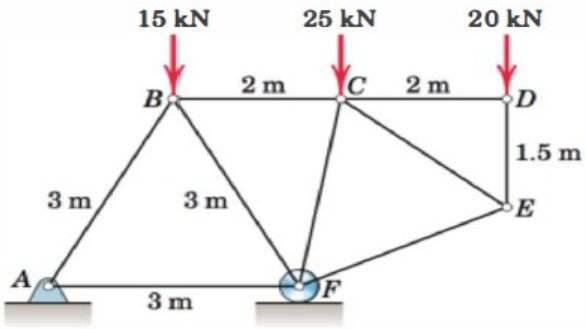
Figure G000132. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 11.

EV0000570

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/35 Determine the force in member BC of the loaded truss.



Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 12

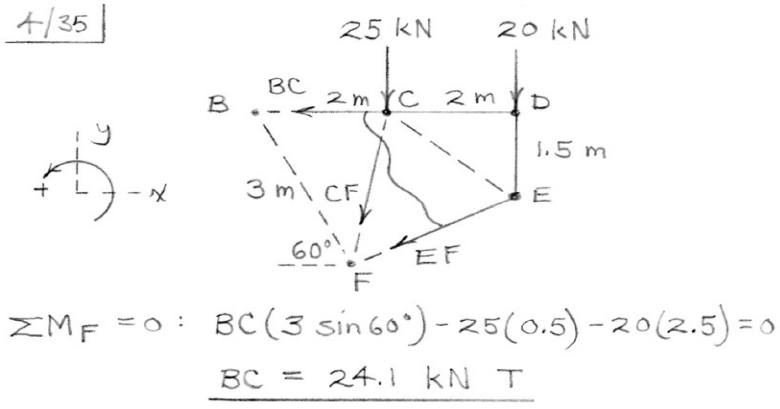
Figure G000133. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 12.

EV0000573

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/35



Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 13

Figure G000134. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 13.

EV0000576

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000135. UE26ME142A Unit 2 Class 26 Lecture PPT · source slide page 14.

EV0000579

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS 1 Dr. S V Venkatesh CivilEngineering DepartmentofS&H Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 1 sections EV0000539 · EV0000540 · EV0000541
2. Source slide page 2 (extracted text): CLASS - 26 ENGINEERING MECHANICS 1 ENGINEERING MECHANICS 1 (4–0–0–4–4) CIVIL ENGINEERING, DEPARTMENT OF MECHANICAL ENGINEERING Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 2 sections EV0000542 · EV0000543 · EV0000544
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS - 01 METHOD OF SECTIONS Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 3 sections EV0000545 · EV0000546 · EV0000547
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/31 Determine the force in member CG. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 4 sections EV0000548 · EV0000549 · EV0000550
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/31. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 5 sections EV0000551 · EV0000552 · EV0000553
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/32 Determine the force in member AE of the loaded truss.. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 6 sections EV0000554 · EV0000555 · EV0000556

7. Source slide page 7 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/32.. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 7 sections EV0000557 · EV0000558 · EV0000559
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/33 Determine the forces in members BC and CG.. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 8 sections EV0000560 · EV0000561 · EV0000562
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/33.. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 9 sections EV0000563 · EV0000564 · EV0000565
10. Source slide page 10 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/34 Determine the forces in members CG and GH of the symmetrically loaded truss.. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 10 sections EV0000566 · EV0000567 · EV0000568
11. Source slide page 11 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/34 . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 11 sections EV0000569 · EV0000570 · EV0000571
12. Source slide page 12 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/35 Determine the force in member BC of the loaded truss. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 12 sections EV0000572 · EV0000573 · EV0000574
13. Source slide page 13 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/35 . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 13 sections EV0000575 · EV0000576 · EV0000577
14. Source slide page 14 (extracted text): THANK YOU Dr. S V Venkatesh CivilEngineering DepartmentofS&H Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 14 sections EV0000578 · EV0000579 · EV0000580

SHORT-NOTE RECAP

Source deck preserved: UE26ME142A Unit 2 Class 26 Lecture PPT. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.
EV0000539

19 Source slide deck: UE26ME142A Unit 2 Class 27 Lecture PPT

This entry preserves the complete slide deck titled UE26ME142A Unit 2 Class 27 Lecture PPT as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mechanics 1, Unit_02, to avoid an unsupported match. EV0000581 · EV0000582

Graphs and diagrams

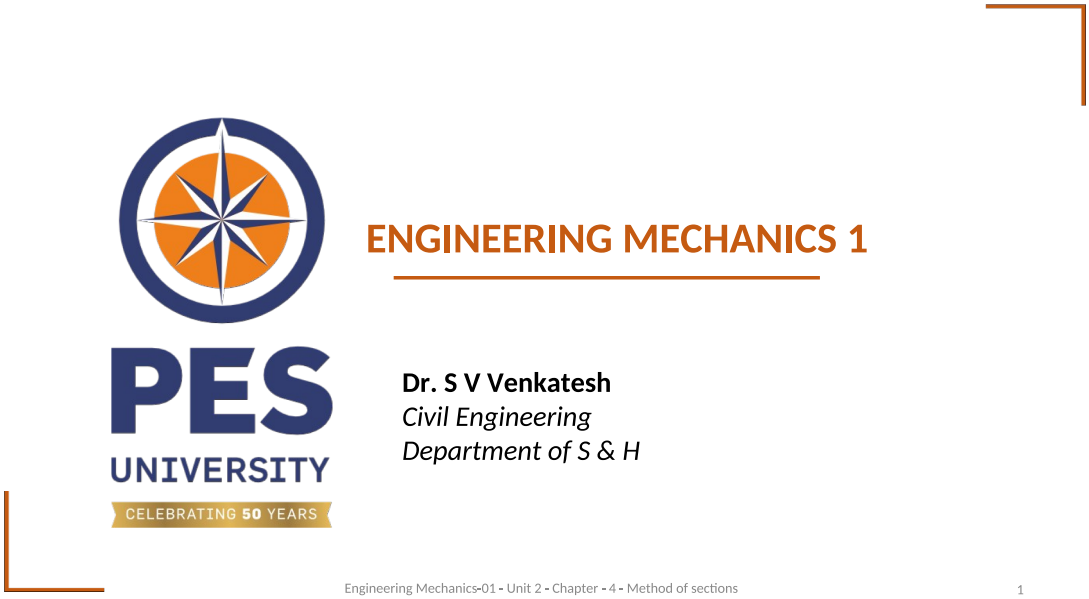


Figure G000136. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 1.

EV0000582

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

CLASS - 27

PES UNIVERSITY
CELEBRATING 50 YEARS

ENGINEERING MECHANICS 1 – STATICS

ENGINEERING MECHANICS 1 – STATICS
(4–0–0–4–4)
CIVIL ENGINEERING, DEPARTMENT OF MECHANICAL ENGINEERING

Figure G000137. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 2.

EV0000585

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

PES UNIVERSITY

4/36 Determine the force in member BE of the loaded truss.

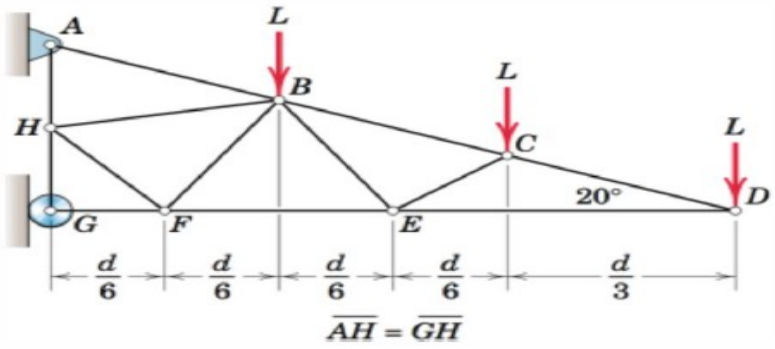


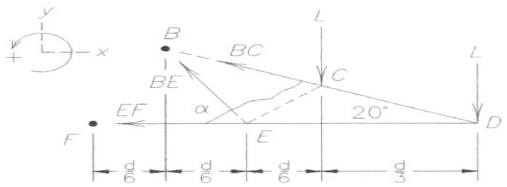
Figure G000138. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 3.

EV0000588

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/36



$$\alpha = \tan^{-1}\frac{\frac{4}{6}d\tan 20°}{\frac{d}{6}} = 55.5°$$

$$\Sigma M_D = 0: \; L\left(\frac{d}{3}\right) - BE\left(\frac{d}{2}\right)(\sin 55.5°) = 0$$

$$BE = 0.809L\ T$$

Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 4

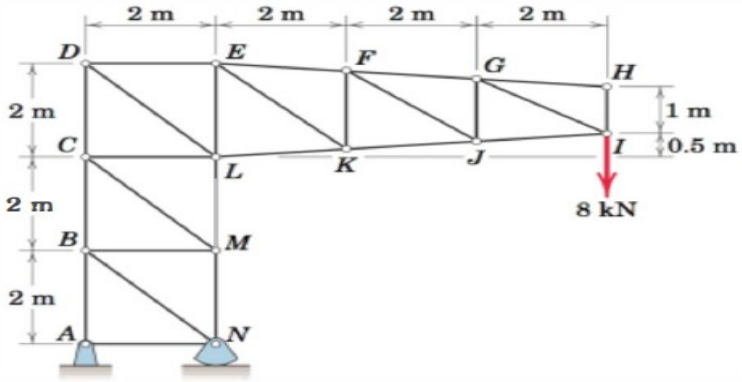
Figure G000139. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 4.

EV0000591

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/37 Determine the forces in members DE and DL.



Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 5

Figure G000140. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 5.

EV0000594

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

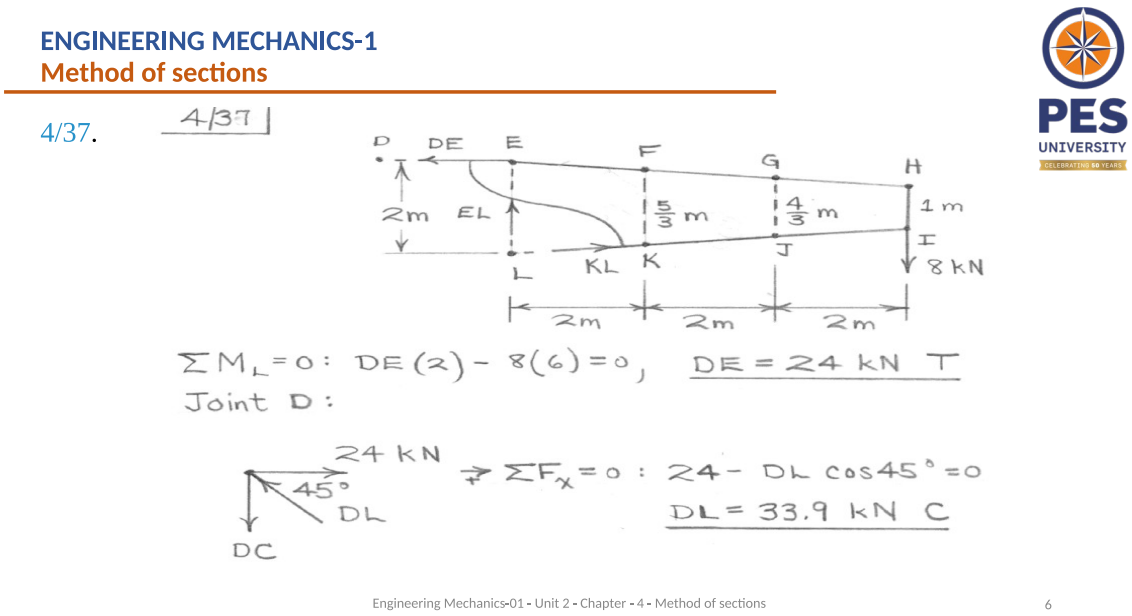


Figure G000141. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 6.

EV0000597

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

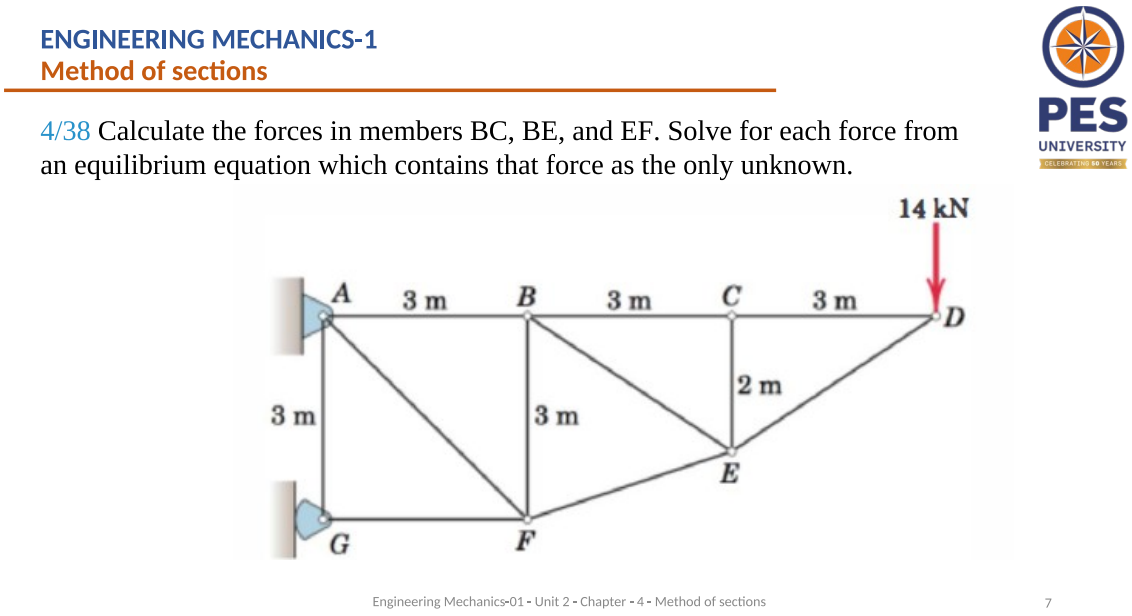


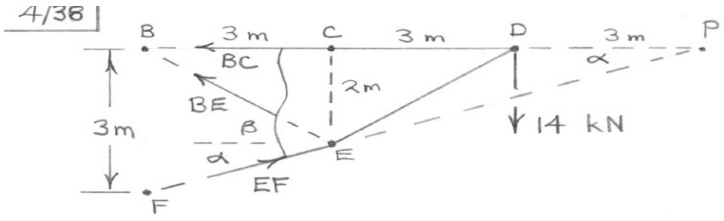
Figure G000142. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 7.

EV0000600

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/38



$\alpha = \tan^{-1}\left(\frac{2}{6}\right) = 18.43°, \quad \beta = \tan^{-1}\frac{2}{3} = 33.7°$

$\curvearrowleft + \Sigma M_E = 0: \quad BC(2) - 14(3) = 0, \quad BC = 21 \text{ kN T}$

$\curvearrowleft + \Sigma M_P = 0: -BE\sin\beta(9) + 14(3) = 0, \quad BE = 8.41 \text{ kN T}$

$\curvearrowleft + \Sigma M_B = 0: \quad EF\cos\alpha(3) - 14(6) = 0, \quad EF = 29.5 \text{ kN C}$

Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 8

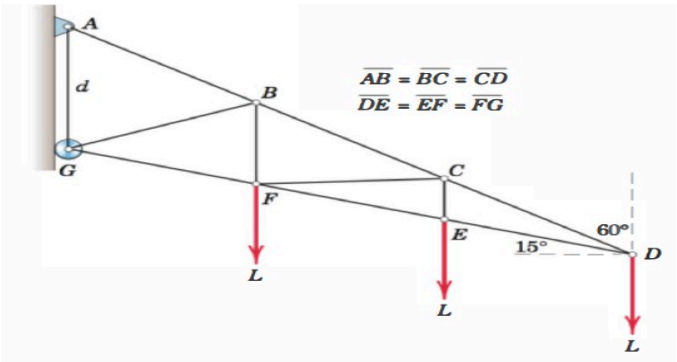
Figure G000143. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 8.

EV0000603

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/39 Determine the forces in members BC and CF of the loaded truss, repeated here from Prob. 4/19.



Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 9

Figure G000144. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 9.

EV0000606

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

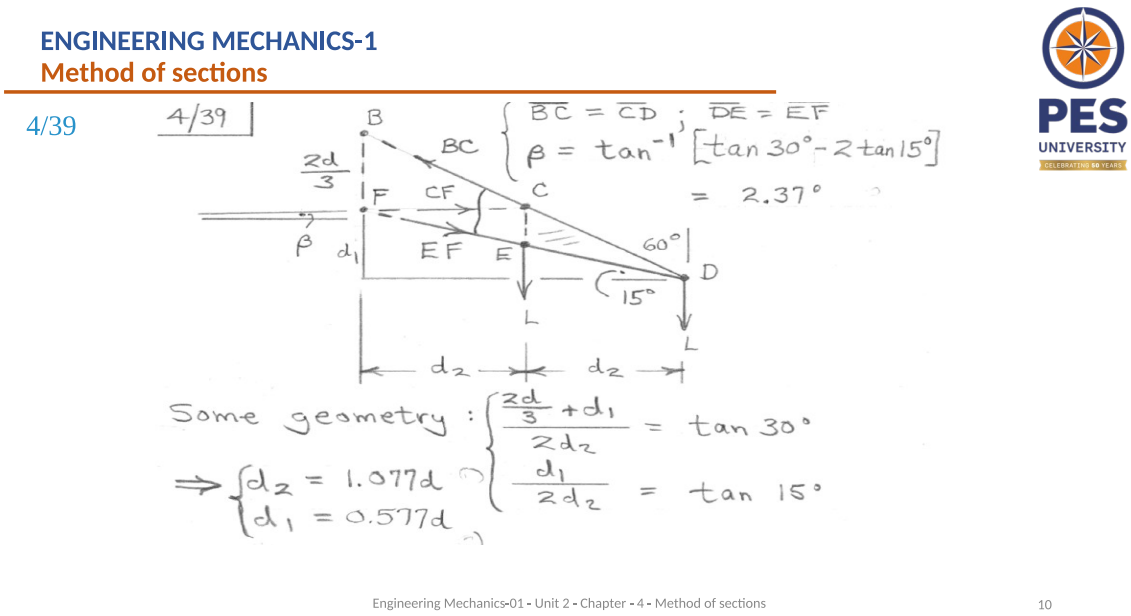


Figure G000145. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 10.

EV0000609

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

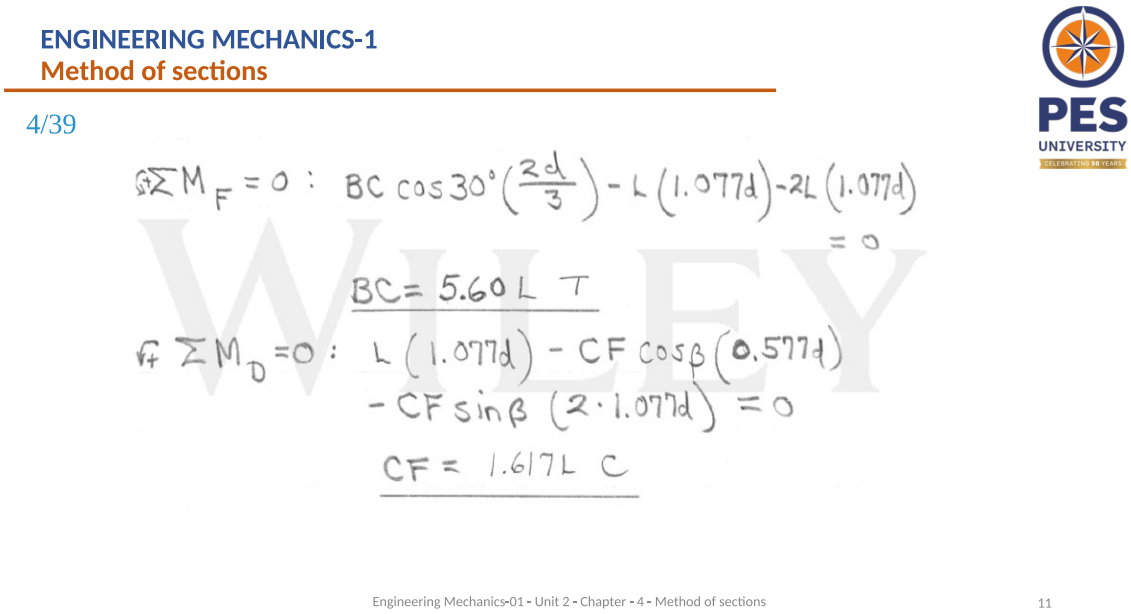


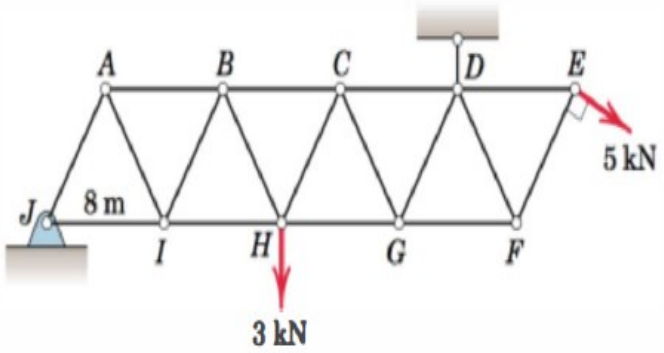
Figure G000146. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 11.

EV0000612

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/40 Calculate the forces in members BC, CD, and CG of the loaded truss composed of equilateral triangles, each of side length 8 m.



Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 12

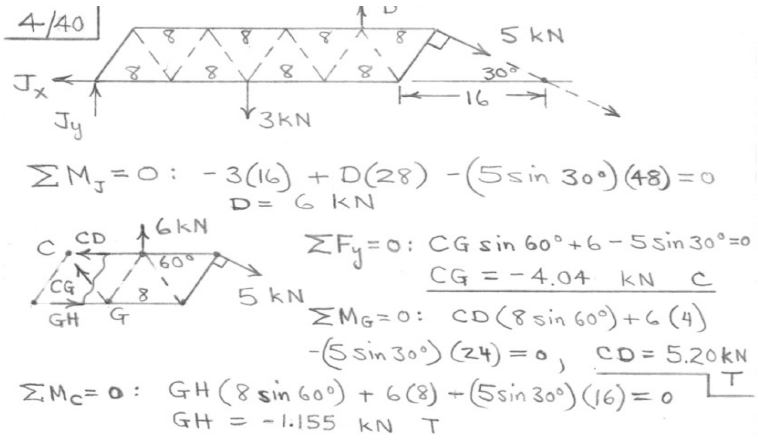
Figure G000147. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 12.

EV0000615

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MECHANICS-1
Method of sections

4/40



Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of sections 13

Figure G000148. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 13.

EV0000618

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

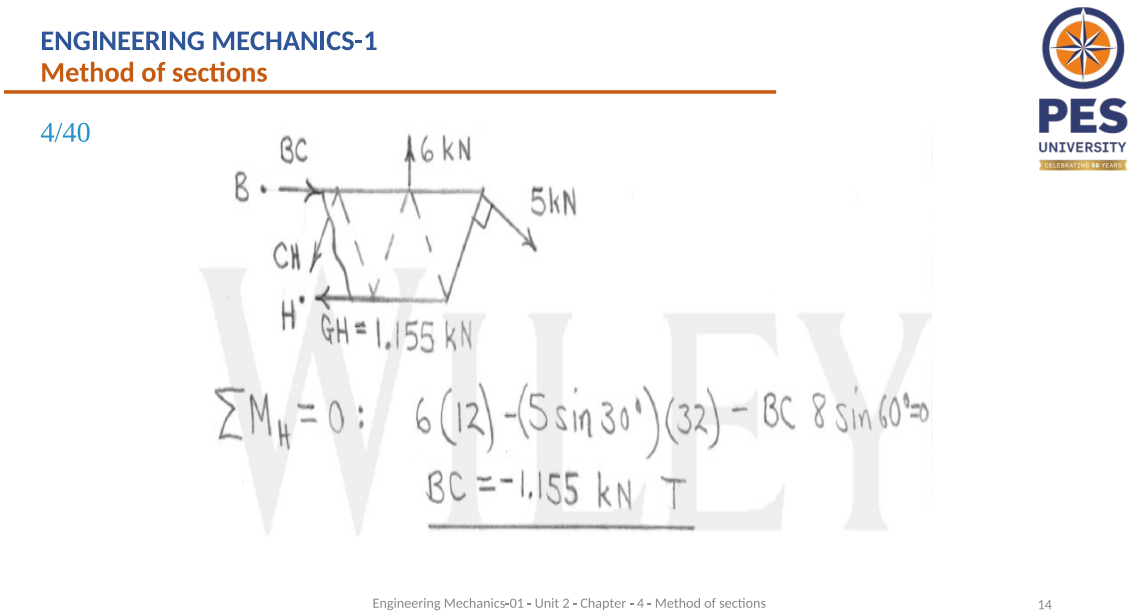


Figure G000149. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 14.

EV0000621

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Figure G000150. UE26ME142A Unit 2 Class 27 Lecture PPT · source slide page 15.

EV0000624

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MECHANICS 1 Dr. S V Venkatesh CivilEngineering DepartmentofS&H Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 1 sections EV0000581 · EV0000582 · EV0000583
2. Source slide page 2 (extracted text): CLASS - 27 ENGINEERING MECHANICS 1 – STATICS ENGINEERING MECHANICS 1 – STATICS (4–0–0–4–4) CIVIL ENGINEERING, DEPARTMENT OF MECHANICAL ENGINEERING Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 2 sections EV0000584 · EV0000585 · EV0000586
3. Source slide page 3 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/36 Determine the force in member BE of the loaded truss. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 3 sections EV0000587 · EV0000588 · EV0000589
4. Source slide page 4 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/36 . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 4 sections EV0000590 · EV0000591 · EV0000592
5. Source slide page 5 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/37 Determine the forces in members DE and DL. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 5 sections EV0000593 · EV0000594 · EV0000595
6. Source slide page 6 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/37. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 6 sections EV0000596 · EV0000597 · EV0000598
7. Source slide page 7 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/38 Calculate the forces in members BC, BE, and EF. Solve for each force from an equilibrium equation which contains that force as the only unknown. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 7 sections EV0000599 · EV0000600 · EV0000601
8. Source slide page 8 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/38 . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 8 sections EV0000602 · EV0000603 · EV0000604
9. Source slide page 9 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/39 Determine the forces in members BC and CF of the loaded truss, repeated here from Prob. 4/19. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 9 sections EV0000605 · EV0000606 · EV0000607
10. Source slide page 10 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/39 . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 10 sections EV0000608 · EV0000609 · EV0000610
11. Source slide page 11 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/39 . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 11 sections EV0000611 · EV0000612 · EV0000613
12. Source slide page 12 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/40 Calculate the forces in members BC, CD, and CG of the loaded truss composed of equilateral triangles, each of side length 8 m. . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 12 sections EV0000614 · EV0000615 · EV0000616

13. Source slide page 13 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/40 . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 13 sections EV0000617 · EV0000618 · EV0000619
14. Source slide page 14 (extracted text): ENGINEERING MECHANICS-1 Method of sections 4/40 . Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 14 sections EV0000620 · EV0000621 · EV0000622
15. Source slide page 15 (extracted text): THANK YOU Dr. S V Venkatesh CivilEngineering DepartmentofS&H Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 15 sections EV0000623 · EV0000624 · EV0000625

SHORT-NOTE RECAP

Source deck preserved: UE26ME142A Unit 2 Class 27 Lecture PPT. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000581

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
UE26ME142A_Unit 2_Class 14_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 14_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 14_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 14_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 15_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 15_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 15_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 15_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 16_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 16_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 16_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 16_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 17_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 17_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 17_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 17_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 19.0_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.0_AVSummary__bd8cd9929f8bd715.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.0_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.0_AVSummary__bd8cd9929f8bd715.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.0_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.0_AVSummary__bd8cd9929f8bd715.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.0_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.0_AVSummary__bd8cd9929f8bd715.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.1_AVSummary__877d29f4ad0c369d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.1_AVSummary__877d29f4ad0c369d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 19.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.1_AVSummary__877d29f4ad0c369d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.1_AVSummary__877d29f4ad0c369d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.2_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.2_AVSummary__ee992a99bb75e05b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.2_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.2_AVSummary__ee992a99bb75e05b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.2_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.2_AVSummary__ee992a99bb75e05b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.2_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.2_AVSummary__ee992a99bb75e05b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 19.3_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.3_AV_Summary__d60756405f7bf7e2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.3_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.3_AV_Summary__d60756405f7bf7e2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.3_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.3_AV_Summary__d60756405f7bf7e2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.3_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.3_AV_Summary__d60756405f7bf7e2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.4_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.4_AV_Summary__5e4cbdbb6d761857.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.4_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.4_AV_Summary__5e4cbdbb6d761857.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 19.4_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.4_AV_Summary__5e4cbdbb6d761857.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.4_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.4_AV_Summary__5e4cbdbb6d761857.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 21_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 21_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 21_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 21_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 22_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 22_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 22_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 22_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 23_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 23_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 23_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 23_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 24_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 24_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 24_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 24_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: UE26ME142A Unit 2 Class 14 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 14 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 14 Lecture PPT (converted PDF; original hash 15da602433d67b8f691b145df45757178b9b3848800a2a73732c9738e0f13ae4)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 15 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 15 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 15 Lecture PPT (converted PDF; original hash 89ba5293343d1098702f71e08aad13f40b1746763b94a71a51431b22641f12f0)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 16 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 16 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 16 Lecture PPT (converted PDF; original hash b92d5b6449d2b1d0e7939ef64985be8ceee8a6dcc6e9b2b4d9c54b8e8cda12cf)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 17 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 17 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 17 Lecture PPT (converted PDF; original hash 17bd7c62545d7149617f64fd0d4cf7a538e17facccb990bc698a31ce012ad248)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 18 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 18 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

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Page-render source for UE26ME142A Unit 2 Class 18 Lecture PPT (converted PDF; original hash 85c8ad866d36397419056cc6aa0adf9538c8032a3f4d265f9ff53ea40cb9c317)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 19.0 Lecture PPT (converted PDF; original hash a8da7337867f3a999244592eb71afafb59434ce313cf1b11c4d6f5f59348ed93)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

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Page-render source for UE26ME142A Unit 2 Class 19.1 Lecture PPT (converted PDF; original hash 4688882de86918464ab39ce7074721d5300631f00b354558a4dd8f850a28bbeb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.2 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.2 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 19.2 Lecture PPT (converted PDF; original hash a5ba8cda3b81747db031cb68ac3d86e434f1f076393e4f47c1c1b1aebe018e0c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.3 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.3 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 19.3 Lecture PPT (converted PDF; original hash 9c0be305589bdb3099cdb1dd33474c3d8487f218b47a12992ccc419a2fd76e22)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.4 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.4 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 19.4 Lecture PPT (converted PDF; original hash 6ee58b950fa60fb1cf9861566536c6dfcde27c27b3b7d9765b64ffdabd263eef)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.5 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.5 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

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Page-render source for UE26ME142A Unit 2 Class 19.5 Lecture PPT (converted PDF; original hash c4c0b2c1da6ec17d1df4917d1457210a2f14451595cb0834e64b921b6245c826)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 21 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 21 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 21 Lecture PPT (converted PDF; original hash 654ff6d174c162369b8262db13c6dfdebeb6e04b9ef5ff4d0e59bba981cc4734)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 22 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 22 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 22 Lecture PPT (converted PDF; original hash 94a72dcc07893b420ffe9ef97557283604675f4a32e93c57b79637e014454b0c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 23 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
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Page-render source for UE26ME142A Unit 2 Class 23 Lecture PPT (converted PDF; original hash 416e21f70149398de40b961e40fe20526efc7233cc456d68b7461737cfb9ee15)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 24 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 24 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 24 Lecture PPT (converted PDF; original hash 7f455b5442d047c16abcc290ac99dc476657492e34a722aeb825b1fbc964f6c2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 25 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 25 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 25 Lecture PPT (converted PDF; original hash 3f7038dca9063fae74f933716599fb7871a17fcb789a660a382bf2632613532c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 26 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 26 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 26 Lecture PPT (converted PDF; original hash e912704c214b466e06ae34f25b61ac6dbfc415fa8d05d19c2148b6b9c5e9a360)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 27 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 27 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 27 Lecture PPT (converted PDF; original hash 54f8351c620edee63d6f52420621272348097c94fa3bf919fd13226ed2925b29)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.

4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.
9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV00000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV00000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000014	SRC00002/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000015	SRC00003/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000016	SRC00004/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0072. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00005/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00002/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0073 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00003/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0073 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00004/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0073 to s0076. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000021	SRC00005/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0073 to s0076. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000022	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000023	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000024	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000025	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000026	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000027	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000028	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000029	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000030	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000032	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000034	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0068. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000035	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0068. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000036	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0068. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000037	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0068. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000038	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000039	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000040	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000041	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000042	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000043	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000044	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000045	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000046	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000047	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000048	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000049	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000050	SRC00010/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0070. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000051	SRC00011/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0070. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000052	SRC00012/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0070. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000053	SRC00013/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0070. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000054	SRC00010/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0071 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000055	SRC00011/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0071 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000056	SRC00012/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0071 to s0077. Sidecar variants compare at similarity 0.9996; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000057	SRC00013/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0071 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000058	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000060	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000063	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000064	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000065	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000066	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000067	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000068	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000069	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000070	SRC00014/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0068. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000071	SRC00015/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0068. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000072	SRC00016/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0068. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000073	SRC00017/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0068. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000074	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000075	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000076	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000077	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000078	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000079	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000080	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000081	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000082	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000083	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000084	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000085	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000086	SRC00018/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000087	SRC00019/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000088	SRC00020/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0072. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000089	SRC00021/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0072. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000090	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000091	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000092	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000093	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0039. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000098	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000099	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001000	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001001	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001002	SRC00022/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001003	SRC00023/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001004	SRC00024/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0069. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000105	SRC00025/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000106	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000107	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000108	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000109	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000110	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000111	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000112	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000113	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000114	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0035 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000115	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0035 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000116	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0035 to s0050. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000117	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0035 to s0050. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000118	SRC00026/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0051 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000119	SRC00027/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0051 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000120	SRC00028/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0051 to s0066. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000121	SRC00029/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0051 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000122	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000123	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000124	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000125	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 26	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 27	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 28	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 29	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 30	SRC00030/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 31	SRC00031/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 32	SRC00032/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000133	SRC00033/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000134	SRC00030/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000135	SRC00031/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000136	SRC00032/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0064. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000137	SRC00033/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0055 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000138	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000139	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000140	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000141	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000142	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000143	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000144	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000145	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000146	SRC00034/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000147	SRC00035/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000148	SRC00036/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000149	SRC00037/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000150	SRC00034/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0068. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000151	SRC00035/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0068. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000152	SRC00036/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0068. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000153	SRC00037/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0068. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000154	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000155	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000156	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000157	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000158	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000159	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000160	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000161	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000162	SRC00038/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000163	SRC00039/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000164	SRC00040/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000165	SRC00041/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000166	SRC00038/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000167	SRC00039/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000168	SRC00040/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0069. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000169	SRC00041/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0054 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000170	SRC00042/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000171	SRC00043/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000172	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000173	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000174	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000175	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000176	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000177	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000178	SRC00042/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000179	SRC00043/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000180	SRC00044/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000181	SRC00045/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001182	SRC00042/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001183	SRC00043/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001184	SRC00044/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0071. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001185	SRC00045/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0071. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001186	SRC00046/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001187	SRC00047/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001188	SRC00048/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000189	SRC00049/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000190	SRC00046/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000191	SRC00047/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000192	SRC00048/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000193	SRC00049/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000194	SRC00046/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000195	SRC00047/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 96	SRC00048/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 97	SRC00049/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 98	SRC00046/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 99	SRC00047/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 00	SRC00048/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0066. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 01	SRC00049/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 02	SRC00050/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002 03	SRC00051/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 04	SRC00052/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 05	SRC00053/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 06	SRC00050/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 07	SRC00051/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 08	SRC00052/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 09	SRC00053/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000210	SRC00050/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000211	SRC00051/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000212	SRC00052/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000213	SRC00053/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000214	SRC00050/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000215	SRC00051/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000216	SRC00052/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0080. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000217	SRC00053/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000218	SRC00050/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000219	SRC00051/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000220	SRC00052/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0087. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000221	SRC00053/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0087. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000222	SRC00054/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000223	SRC00055/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000224	SRC00056/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000225	SRC00057/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000226	SRC00054/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000227	SRC00055/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000228	SRC00056/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000229	SRC00057/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000230	SRC00054/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000231	SRC00055/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000232	SRC00056/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000233	SRC00057/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000234	SRC00054/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000235	SRC00055/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000236	SRC00056/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000237	SRC00057/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000238	SRC00054/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000239	SRC00055/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000240	SRC00056/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0083. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000241	SRC00057/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0081 to s0083. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000242	SRC00058/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000243	SRC00059/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000244	SRC00060/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000245	SRC00061/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000246	SRC00058/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000247	SRC00059/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000248	SRC00060/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000249	SRC00061/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000250	SRC00058/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000251	SRC00059/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000252	SRC00060/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000253	SRC00061/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000254	SRC00058/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000255	SRC00059/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000256	SRC00060/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000257	SRC00061/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000258	SRC00062/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 59	SRC00064/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 60	SRC00063/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00002 61	SRC00062/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 62	SRC00064/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 63	SRC00063/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 64	SRC00062/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 65	SRC00064/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 66	SRC00063/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 67	SRC00062/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 68	SRC00064/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 69	SRC00063/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 70	SRC00062/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 71	SRC00064/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 72	SRC00063/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 73	SRC00062/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 74	SRC00064/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 75	SRC00063/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 76	SRC00062/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 77	SRC00064/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 78	SRC00063/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 79	SRC00062/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 80	SRC00064/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 81	SRC00063/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 82	SRC00064/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00002 83	SRC00064/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 84	SRC00065/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 85	SRC00067/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 86	SRC00066/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00002 87	SRC00065/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 88	SRC00067/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00002289	SRC00066/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00002290	SRC00065/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002291	SRC00067/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002292	SRC00066/p3	covered	Alternate extracted slide text page 3; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00002293	SRC00067/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002294	SRC00067/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00002 95	SRC00067/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 96	SRC00067/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 97	SRC00068/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 98	SRC00070/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00002 99	SRC00069/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00003 00	SRC00068/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 01	SRC00070/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 02	SRC00069/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 03	SRC00068/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 04	SRC00070/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 05	SRC00069/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 06	SRC00068/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 07	SRC00070/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 08	SRC00069/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 09	SRC00068/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 10	SRC00070/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 11	SRC00069/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 12	SRC00070/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000313	SRC00070/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000314	SRC00071/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000315	SRC00073/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000316	SRC00072/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000317	SRC00071/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000318	SRC00073/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00003319	SRC00072/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003320	SRC00073/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003321	SRC00073/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003322	SRC00073/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003323	SRC00073/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003324	SRC00074/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 25	SRC00076/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 26	SRC00075/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00003 27	SRC00074/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 28	SRC00076/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 29	SRC00075/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 30	SRC00074/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000331	SRC00076/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000332	SRC00075/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000333	SRC00076/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000334	SRC00076/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000335	SRC00076/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000336	SRC00077/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000337	SRC00079/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000338	SRC00078/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000339	SRC00077/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000340	SRC00079/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000341	SRC00078/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000342	SRC00077/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000343	SRC00079/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000344	SRC00078/p3	covered	Alternate extracted slide text page 3; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000345	SRC00079/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000346	SRC00079/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000347	SRC00079/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000348	SRC00080/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000349	SRC00082/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000350	SRC00081/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000351	SRC00080/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000352	SRC00082/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000353	SRC00081/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000354	SRC00080/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 55	SRC00082/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 56	SRC00081/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 57	SRC00080/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 58	SRC00082/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 59	SRC00081/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 60	SRC00082/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00003 61	SRC00082/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 62	SRC00083/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 63	SRC00085/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 64	SRC00084/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00003 65	SRC00083/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 66	SRC00085/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00003367	SRC00084/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003368	SRC00085/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003369	SRC00085/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003370	SRC00085/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003371	SRC00085/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003372	SRC00086/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000373	SRC00088/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000374	SRC00087/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000375	SRC00086/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000376	SRC00088/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000377	SRC00087/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000378	SRC00086/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003379	SRC00088/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003380	SRC00087/p3	covered	Alternate extracted slide text page 3; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00003381	SRC00086/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003382	SRC00088/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003383	SRC00087/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003384	SRC00088/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00003 85	SRC00088/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 86	SRC00088/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 87	SRC00089/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 88	SRC00091/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 89	SRC00090/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV00003 90	SRC00089/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003391	SRC00091/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003392	SRC00090/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003393	SRC00089/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003394	SRC00091/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003395	SRC00090/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003396	SRC00089/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 97	SRC00091/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00003 98	SRC00090/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 99	SRC00089/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 00	SRC00091/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004 01	SRC00090/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 02	SRC00089/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000403	SRC00091/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000404	SRC00090/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000405	SRC00091/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000406	SRC00091/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000407	SRC00092/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000408	SRC00094/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000409	SRC00093/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000410	SRC00092/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000411	SRC00094/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000412	SRC00093/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000413	SRC00092/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000414	SRC00094/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000415	SRC00093/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004016	SRC00092/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004017	SRC00094/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004018	SRC00093/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00004019	SRC00094/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004020	SRC00094/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004021	SRC00095/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000422	SRC00097/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000423	SRC00096/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000424	SRC00095/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000425	SRC00097/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000426	SRC00096/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000427	SRC00095/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000428	SRC00097/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000429	SRC00096/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000430	SRC00095/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000431	SRC00097/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000432	SRC00096/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000433	SRC00095/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000434	SRC00097/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000435	SRC00096/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000436	SRC00095/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000437	SRC00097/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000438	SRC00096/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000439	SRC00095/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000440	SRC00097/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000441	SRC00096/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000442	SRC00095/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000443	SRC00097/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000444	SRC00096/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000445	SRC00095/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000446	SRC00097/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000447	SRC00096/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000448	SRC00095/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000449	SRC00097/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000450	SRC00096/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000451	SRC00095/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000452	SRC00097/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000453	SRC00096/p11	covered	Alternate extracted slide text page 11; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000454	SRC00095/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000455	SRC00097/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000456	SRC00096/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000457	SRC00095/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000458	SRC00097/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000459	SRC00096/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000460	SRC00095/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000461	SRC00097/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000462	SRC00096/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000463	SRC00095/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000464	SRC00097/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000465	SRC00096/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000466	SRC00097/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000467	SRC00097/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000468	SRC00097/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000469	SRC00098/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000470	SRC00100/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000471	SRC00099/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000472	SRC00098/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000473	SRC00100/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000474	SRC00099/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000475	SRC00098/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000476	SRC00100/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000477	SRC00099/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000478	SRC00100/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000479	SRC00100/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000480	SRC00101/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000481	SRC00103/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000482	SRC00102/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000483	SRC00101/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000484	SRC00103/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000485	SRC00102/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000486	SRC00101/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000487	SRC00103/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000488	SRC00102/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000489	SRC00101/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000490	SRC00103/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000491	SRC00102/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000492	SRC00101/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000493	SRC00103/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000494	SRC00102/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000495	SRC00103/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000496	SRC00103/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000497	SRC00103/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000498	SRC00104/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000499	SRC00106/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000500	SRC00105/p1	covered	Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 01	SRC00104/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 02	SRC00106/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 03	SRC00105/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 04	SRC00104/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 05	SRC00106/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 06	SRC00105/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 07	SRC00104/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 08	SRC00106/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 09	SRC00105/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 10	SRC00104/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 11	SRC00106/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 12	SRC00105/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000513	SRC00106/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000514	SRC00106/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000515	SRC00107/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000516	SRC00109/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000517	SRC00108/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000518	SRC00107/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000519	SRC00109/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000520	SRC00108/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000521	SRC00107/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000522	SRC00109/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000523	SRC00108/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000524	SRC00107/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005 25	SRC00109/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 26	SRC00108/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 27	SRC00107/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 28	SRC00109/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 29	SRC00108/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 30	SRC00107/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000531	SRC00109/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000532	SRC00108/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000533	SRC00107/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000534	SRC00109/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000535	SRC00108/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000536	SRC00107/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000537	SRC00109/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000538	SRC00108/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000539	SRC00110/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000540	SRC00112/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000541	SRC00111/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000542	SRC00110/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000543	SRC00112/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000544	SRC00111/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000545	SRC00110/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000546	SRC00112/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000547	SRC00111/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000548	SRC00110/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000549	SRC00112/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000550	SRC00111/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000551	SRC00110/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000552	SRC00112/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000553	SRC00111/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000554	SRC00110/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000555	SRC00112/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000556	SRC00111/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000557	SRC00110/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000558	SRC00112/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000559	SRC00111/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000560	SRC00110/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000561	SRC00112/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000562	SRC00111/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000563	SRC00110/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000564	SRC00112/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000565	SRC00111/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000566	SRC00110/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000567	SRC00112/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000568	SRC00111/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000569	SRC00110/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000570	SRC00112/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000571	SRC00111/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000572	SRC00110/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000573	SRC00112/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000574	SRC00111/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000575	SRC00110/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000576	SRC00112/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000577	SRC00111/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000578	SRC00110/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005 79	SRC00112/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 80	SRC00111/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 81	SRC00113/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 82	SRC00115/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 83	SRC00114/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 84	SRC00113/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000585	SRC00115/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000586	SRC00114/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000587	SRC00113/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000588	SRC00115/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000589	SRC00114/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000590	SRC00113/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000591	SRC00115/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000592	SRC00114/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000593	SRC00113/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000594	SRC00115/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000595	SRC00114/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000596	SRC00113/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00005 97	SRC00115/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00005 98	SRC00114/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 99	SRC00113/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006 00	SRC00115/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00006 01	SRC00114/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00006 02	SRC00113/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000603	SRC00115/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000604	SRC00114/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000605	SRC00113/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000606	SRC00115/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000607	SRC00114/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000608	SRC00113/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000609	SRC00115/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000610	SRC00114/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000611	SRC00113/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000612	SRC00115/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000613	SRC00114/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000614	SRC00113/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000615	SRC00115/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000616	SRC00114/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000617	SRC00113/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000618	SRC00115/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000619	SRC00114/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000620	SRC00113/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00006021	SRC00115/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00006022	SRC00114/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV00006023	SRC00113/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006024	SRC00115/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00006025	SRC00114/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

C Revision and retrieval practice